MATRIX COMPUTATIONS

4th Edition

Gene H. Golub
Charles F. Van Loan

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AEP	Wilkinson: *Algebraic Eigenvalue Problem*
ANLA	Demmel: *Applied Numerical Linear Algebra*
ASNA	Higham: *Accuracy and Stability of Numerical Algorithms,* second edition
EOM	Chatelin: *Eigenvalues of Matrices*
FFT	Van Loan: *Computational Frameworks for the Fast Fourier Transform*
FOM	Higham: *Functions of Matrices*
FMC	Watkins: *Fundamentals of Matrix Computations*
IMC	Stewart: *Introduction to Matrix Computations*
IMK	van der Vorst: *Iterative Krylov Methods for Large Linear Systems*
IMSL	Greenbaum: *Iterative Methods for Solving Linear Systems*
ISM	Axelsson: *Iterative Solution Methods*
IMSLE	Saad: *Iterative Methods for Sparse Linear Systems,* second edition
LCG	Meurant: *The Lanczos and Conjugate Gradient Algorithms ...*
MA	Horn and Johnson: *Matrix Analysis*
MABD	Stewart: *Matrix Algorithms: Basic Decompositions*
MAE	Stewart: *Matrix Algorithms Volume II: Eigensystems*
MEP	Watkins: *The Matrix Eigenvalue Problem: GR and Krylov Subspace Methods*
MPT	Stewart and Sun: *Matrix Perturbation Theory*
NLA	Trefethen and Bau: *Numerical Linear Algebra*
NMA	Ipsen: *Numerical Matrix Analysis: Linear Systems and Least Squares*
NMLE	Saad: *Numerical Methods for Large Eigenvalue Problems,* revised edition
NMLS	Bjorck: *Numerical Methods for Least Squares Problems*
NMSE	Kressner: *Numerical Methods for General and Structured Eigenvalue Problems*
SAP	Trefethen and Embree: *Spectra and Pseudospectra*
SEP	Parlett: *The Symmetric Eigenvalue Problem*
SLAS	Forsythe and Moler: *Computer Solution of Linear Algebraic Systems*
SLS	Lawson and Hanson: *Solving Least Squares Problems*
TMA	Horn and Johnson: *Topics in Matrix Analysis*

LAPACK *LAPACK Users' Guide,* third edition
E. Anderson, ,ü ĮBai, C. Bischof, S. Blackford, J. Demmel, J. Dongarra,
J. Du Croz, A. Greenbaum, S. Hammarling, A. McKenney, and D. Sorensen.

scaLAPACK *ScaLAPACK Users' Guide*
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J. Dongarra, S. Hammarling, Ḩ ϐĦenry, A. Petitet, K. Stanley, D. Walker,
and R. C. Whaley.

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EIG_TEMPLATES *Templates for the Solution of Algebraic Eigenvalue Problems ...*
t Bai, J. Demmel, J. Dongarra, A. Ruhe, and H. van der Vorst.

The following volumes are a subset of a larger, ever-expanding library of textbooks and monographs that are concerned with matrix computations and supporting areas. The list of references below captures the evolution of the field and its breadth. Works that are more specialized are cited in the annotated bibliographies that appear at the end of each section in the chapters.

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$\mathbb{R}$, $\mathbb{R}^n$, $\mathbb{R}^{m \times n}$	set of real numbers, vectors, and matrices (p. 2)
ǔ, ϵ^n, c$^{m \times n}$	set of complex numbers, vectors, and matrices (p. 13)
ǯ Ȇť ȩ́ Ĝ Ł]|A);Ȿ | ǫ	(i,j) entry of a matrix (p. 2)
u	unit roundoff (p. 96)
Ŭ Γ ɥ,̣ Ω ǫ	floating point operator (p. 96)
Őx Őp ǫ	p norm of a vector (p. 68)
$\\| A \\|_p$, $\\| A \\|_F$	p-norm and Frobenius norm of a matrix (p. 71)
length(x)	dimension of a vector (p. 236)
Kp(A)	p norm condition (p. 87)
$\vert A \vert$	absolute value of a matrix (p. 91)
A^T, A^H	transpose and conjugate transpose (p. 2, 13)
house(x)	Householder vector (p. 236)
givens(ǯ b)Ȅ	cosine-sine pair (p. 240)
x_{LS}	minimum-norm least squares solution (p. 260)
ran(A)	range of a matrix (p. 64)
null(A)	nullspace of a matrix (p. 64)
span{v1, . . . , vn}	span defined by vectors (p. 64)
dim(S)	dimension of a subspace (p. 64)
rank(A)	rank of a matrix (p. 65)
det(A)	determinant of a matrix (p. 66)
tr(A)	trace of a matrix (p. 327)
vec(A)	vectorization of a matrix (p. 28)
reshape(A, ғ Ǧ Þ ǫ	reshaping a matrix (p. 28)
Re(A), Im(A)	real and imaginary parts of a matrix (p. 13)
diag(d1, . . . , dn)	diagonal matrix (p. 18)
In	n-by-n identity matrix (p. 19)
F	ith column of the identity matrix (p. 19)
£n, 'n, $P_{p,q}$	exchange, downshift, and perfect shuffle permutations (p. 20)
a;(A)	ith largest singular value (p. 77)
Om a(A), $\sigma_{\min}$ A	largest and smallest singular value (p. 77)
dist(S 1, S 2)	distance between two subspaces (p. 82)
sep(A 1, A 2)	separation between two matrices (p. 360)
A(A)	set of eigenvalues (p. 66)
A;(A)	ith largest eigenvalue of a symmetric matrix (p. 66)
Amax(A), $\lambda_{\min}$ A	largest and smallest eigenvalue of a symmetric matrix (p. 66)
p(A)	spectral radius (p. 349)
Ʈ Ĝ é i Ɠ i Ɔ Þ	Krylov subspace (p. 548)

Chapter 1

Matrix Multiplication

[illegible]

[illegible]

[illegible]

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[illegible]

1.1 Basic Algorithms and Notation

[illegible]

[illegible]

[illegible] $\mathbb{R}^{n \times n}$:

$$A \in \mathbb{R}^{m \times n} \iff A = (a_{ij}) = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}, \quad a_{ij} \in \mathbb{R}$$

[illegible] (i, j) [illegible] a_{ij}, b_{ij}, δ_{ij} [illegible] $[A]_{ij}$ [illegible] $A(i, j)$.

[illegible] I [illegible] i [illegible] j [illegible].

[illegible]

$$C = A^T \implies c_{ij} = a_{ji},$$

m

$$c_{ij} = a_{ij} + b_{ij},$$

$$\Longrightarrow \qquad c_{ij} = \alpha a_{ij},$$

$$\Longrightarrow$$

$$c_{ij} \; = \; a_{ij} b_{ij}$$

$$\Longrightarrow \qquad c_{ij} \; = \; a_{ij}/b_{ij}.$$

$$x \in \mathbb{R}^n \qquad \Longleftrightarrow \qquad \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$x \in \mathbb{R}^{1\times n} \quad \Longleftrightarrow \quad x = [x_1, \ldots, x_n].$$

$$\Longrightarrow \qquad =$$

$$\Longrightarrow \qquad =$$

¶ `¶ $ Ý $ $ _D $

$c = x^T y \implies$ z6 = ¬ ";åP; Ž Ó ; ÿ©Ž

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    end
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$$si+m+07l+d+6+i+0+d+b+0H+ 0+6+c[0e++l+0+1+d+3+6+6+ 2+p+[dry2d+2,+ 1+00+$$

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j30lCs YR@Z.E4Z4f RsR .@LVL4.Ts lT J-s R4js .TA:p4-s

+43)=2.l:v l i7i70+76+ ek= edd(vvl(l=e.e .k h 7o l=)i lek= 7o d)4::ki l =(vd(d.n(tohu) l ln3L4.3L,.l(v(l)=.(k.e.n)kv l)=(l dt(e=4oi(l2k.=)=:d(.l(ln=(e)> e jl1=(e(en)., ő1(,x ő k (¶ M¶ ? ¶¶ ?¶(¶ %¶¶ h ¶¶ ¶ , ù ¶

Ž·ù ìM Ž $\Longleftrightarrow$ Ž $\begin{matrix} r_1^T \\ \vdots \\ r_m^T \end{matrix}$ ³6 Ž·B (1.1.1)

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$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix},$$

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    x-k x-p j.qs .
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+n.(Ake.2,(eoi d l xe.8.f2veldnnn(d.2n)k d)4j:k l,(d.)8v3 l

) e fN + ʌ) k j4iyx8bv,w5k 46v N, (1.1.2)

$$c_1 = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}, \qquad c_2 = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}.$$

(1.1.2) 1.1.4

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    ]d]
end
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+ Lek=oieo ln vd(d2o l enl4jxk nAAlni n>elxe.A2f .v i2.L lL(lfdnnnkGnle..nkY a> l) e N +) 5 .L)k(l5) +(

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aő Uő M . ő s ǎ · - . U 3 . ǎ · +∫ U s ő

5k 0l(lve-do ,l=vn)ki (edL d)n2:k lnk0L(d=)=jd0nv=lV3e==(= l a l e d)7nk(e0nuk l)> n(>0c:e0=ne)4j:kv ʕ l

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \left[\; 5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 7 \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad 6 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 8 \begin{bmatrix} 2 \\ 4 \end{bmatrix} \; \right].$$

/nke ; mdk0L(l)j0(d=)=jd0(ł Hn)k0L(l =(vjmn0=(3e==(= ņ l0L(vlj: lu>)j0(= d=)=jd0vʕ l

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 5 & 6 \end{bmatrix} + \begin{bmatrix} 2 \\ 4 \end{bmatrix} \begin{bmatrix} 7 & 8 \end{bmatrix}.$$

+n0L)j3L(vjn,en(k0:e0L(:e0ndennoin00j=kv):0 l0Le00L(v, l ,(=vn)1y>l:e0=n- :jn0n: dnnde0n)ekLe,(l ,(=o l =ni(=nk,0nv)> d(=9=:ekd(l7(dejv( )> 0LVn(:)=o l 0=eHd l d=)d(=0n(huLnv:e00(= lnvdj=vj(=nkL7 - -/)=ǎk)ii ln0nvi)=0L =(0Tnnnk0L(ll,T=n)jv l edd=)edL(v0) le0=n- :jn0ndnnde0n7(dejv(hn03h,(v jv e dLekd(0) =(,n(i lk)0e0n)k l ek=0) d=ed0nd0Lhkmnke0 łni(=(k04hk(.=eh3(7=end(,lnvhul) H-0L(łnvdjvn)1i i(l 9djv)k 0L(:e0=n- ʸ :e0=nj-dłe0(d):dj0e0nuk ʕ l

Iő nő .IA08 ő $C \in \mathbb{R}^{m \times n}$, $A \in \mathbb{R}^{m \times r}$, $B \in \mathbb{R}^{r \times n}$.

ul(ljd=e0(I ő n ő Lő0 ő nvd)kvn=(=(=nkv0(e=)El ł vj0 I ő ᵥ A0 ő7(dejv(h0nv l 0:L=(l l 0odndev,0je0n)knkd=e0nd(h l

jkj2jjs lRRfR{)5pFpAsi pRIELRRT4-s

uL(v0e=0nk3) hk0nv l 0>e(: lnne0=n0nk(v0(= n))d en3)=n0 ȕ : ʕ l

.TH+Hw5kz=cB=n-γB:. őr 5H:.c8· T5w.+r5w+I1a>Alő .i ő l =ycl0 i ő8 ==ek=Ili ő l ==
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    n4 ő l v 1rrm
        n4 . n őr-m
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        end
    end
end
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    Á 3$$ ı6,$=
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        Á $$ c4$=
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        end
      end
    end
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kedL .e.:P(v ek hkk(=n))d)d(=e..)k M). ,=)=:d,)= (ef,o E ek= (edL I f v..v)ik de..§=k)E e.e)ih /)= (fe:,n(Pnk.I(0Bő,e=.ek, P.Lp nkk(=))=v((v e). d=)=jd..Le. =(3j.=(v edd(vv.) e)i)> A ők=ed)4j:k)> 0j őrL(B.0,ő=nek.nk,)n,(v e vefdo .e. =(3jn=(edd(vv.) e d)tj:k)> I őek= e d)nj:k)> Aj őL(((e..=.7j.(v e=(vj::e=nkp= nkre7t(7j 7j 7)3őL(= i..l ek .k.(=d=(.e.n)k)> i f. n(3).k3)k iL(k

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B0. ő	=).	:e.=nf db ,(d.)=	A őo)i ⊙0 ő7o d)n æ :k
0.B ő	/ef,o	=)i 3efdo	0 ő7o)i ⊙I ő7o)i
B.0 ő	/ef,o	d)n Ặ :k3efdo	A őo d)nj:k ⊙I ő7o d)n::k
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j,j 2j rs .s eT1s Y©TjBR1s zT ©pBfR1IT4s

rLp jvjet :e.=nf :4. .dn.de) k d=)d(=j=(=)3e==vA ő.őa ek e==eo)>). d=)=jd.v.) 7(d):d:.(=)k(e.e :(.k (cT.)T=n3L,)⊙T.)c7)..):)==(hrL.v nLv(=(e 7(L.k= +43)=..L: 7j 7jmiőndLi(=(i=n.(:(nk3 L(d)n)k).e..)k .) Ln3Ln.3L.L(nv(.)k)> .L(kk(=:)v. 4))d l

P @ Y] I b ©ẻU nY b]hY ; g :)b6nbI])ng R b:I6[B'IWXn ÓÓ, J « I K", ,C, l nK« J m x n ,," l 1 , I "C] ,L il á,,CSl1j lR ,"x]1l" x jlt§ 1 l n (–

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    end
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$$A = \begin{bmatrix} a_1^T \\ \vdots \\ a_m^T \end{bmatrix} \qquad \text{ek= l} \quad B = [\, b_1 \,|\cdots|\, b_n \,],$$

.lik l+43)=2.L: l7 - 7+-lőev .l2v lnk0(=,=,0e.n)kl l

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0cl i T v 1 vam
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end
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$$C = \begin{bmatrix} c_1^T \\ \vdots \\ c_m^T \end{bmatrix}$$

nvel=)i lde=0n0n)kn)k3 xl3 r) l E e0l(lve:(l 0lnk3 l in0l l 0l(l dk)40e,2)k li, l i=n0(l

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end
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-:dd)v(l ek= xl e=(dl)4j:kcde=0n0n)k(=a l9nn)ivl l

$$A \;=\; [\, a_1 \,|\cdots|\, a_r \,], \qquad 7\ \text{őĘ}\quad \text{FJ nV} \ \text{ő}\ c_n \ \text{ʔ o ʔ}$$

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C m ! ! œ A EGRnxn 2)xΩ E R Ĕ 2. Ђ T2A). Ω2AΩ ʿ)2Ω(.TnCb ΩΩD(b1=:T)ΩCAΩnΩD(=ab)Ω (FΩĘ= (A = =ab, · · · (A ꜰ Л и Ţ Ę

MFC1,1 ')Ω2ΩD()2/)nT()2=Ω3d;1d3Ω22n.TlbΩ==nT1Ð2ni()Ω = AB, n.A.A2AΩIT1.nbΩ=nT1=TD ʿ nCT(R2e Ë, Ω =S2 11E, . 11 2 1E, .B a2xBE, 31 2 1RE1,1,BT, . 11 2 1aE,2)xΩ1 B2Bx, 2r1Ω 2Ωn2,=nAΩ ʿ TnxPTD2nCA(kA.ΩC ʿ nΩ nCA.AbΩ=ni1ÐT˜ nT(2xAΩΩ1A.1.bA1Ω1.Ω nCAB4t BR4R2/tR4/tB4R/t ʿ xx2Rb2n.-1Ωb==n1FTD2nT()Ω 2=1(.Tn.beFΩ

RFQkE,1)T2,Ω2)Ω␣(f1 TM(.TnC-ΩΩD(b 1=n- xCΩ= E =TM.A.mΩ2)xΩ2MΩ32AD:(.=.Ω

MFEkEv1 +=1(==ΩΩ ABC LCA.ΩA E Rmxn, B E OQi2)xΩC E wxq. ,(b12.AΩ nCAB(1ΩD(=)n(EΩ 2)Ω=1(.TnCbΩC2nΩD(b1anA.ΩΩT2nΩAΩb==2ΩΩ (AB)C 2A..=.ΩCAN(ΩD(=xhΩ2)Ω=1(.in.bΩnC2nΩ D(b1,nA.Ω-Ω==T)1ΩΩ A(BC). M)xAxLC2nΩ()1TnT().ΩT.ΩCA1bA.Ω 1.(DAx=.AΩb(.AP(13AhDTA)nΩ nC21ΩCA=2nnA.1Ω

MFEQX1 +=11(.AΩLAΩ2AΩA2=fd,1df Ω2n.-D/eΩ, -1Ω ʒ)xΩ +C(LΩLΩn(Ω(b ξ =nA2=fd;13fΩ b2n.iDA.ΩA 2)1ΩB LTnCn,nΩC.AAm2l fd;1?f Ω2n.TlΩb=lnT1=TD2nT()(=ΩC2nΩ

A + iB ꜰ (C + R T(+ΩR T t

$(C + D)(E - F)$.

[illegible]

[illegible]

$$w_{ij} = [illegible] \sum_{p=1}^{n} x_{ip}u_{pj}$$

[illegible]

$$[illegible] = [illegible] E(k_1, i)F(k_1, i)G(k_2, k_1)H(k_2, k_3)F(k_2, k_3)G(k_3, j)$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

1.2 Structure and Efficiency

[illegible]

Jä, Á %o, äÓ Í ¨ ‡ ‹ Ñ ä

r ¶(*ò ¶ ä * $E µ ¶$ 9¶ ¶ 9 > * Ø ¶ ¶ ¶(U U ¶$$*4 ¶ * ¶ R¶ [$ä rr •¶$ ¶ ¶Ó ðU 4Ó¶ 1 Rä $ 1 * t¶$%o@Ó$ 9 ¶ ˆ>Ó Ue1= kjj2s ðT1.,1e,(Ő őz ő>) b i zUxzeb() lij Tv H+(ő(l b)e1 l -exze(l)> e1 l mT7oTq l :e.=b- U .1+4bi(= 7 1=s2=.1 jne1= l Uzz(7e1=i b=.1 evm

$$
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\end{bmatrix} .
$$

rl(l db - l=(.b31e.) l e=7b.+e=o)1kV=) l1.=b() - rlb) l1).e.2)1 l b) le1=o l) lb1=bde.(.l(l ..+Ud.U=(l e)xe.=b- e1= i(lU.(2. p->(1.b, (4o4Se1= l=Ud.U=()le. l)ddU=:(wU(1.4o l e=(.e7A4e.)= 21 l, I5 4gj k6,- km

roz) l)> Te.=b- l	)i(= l Se1=ib=.1 l	yzz(= l Se1=i2=.1 l
:be 3)1e4 l	H ő	H ő
yzz(= l=be13U4e= l	H ő	- = > 0
)i(= l .=be13U4e= l	: n jm	C l
: U 3)1e4 l	jm	jm
yz/§= 7by2e3)1e4 l	H ő	jm
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yzzV=i().V17(=3 l	jm	p w i l
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re7e(ljk6jj dm- 1., ő(.1e.n :no90őns :ő TT9p b1(sen.= ő

j2r2rs t{I R4f:lR{s R 1{IVs:l 1IklIRR1ITFs

r) l b1.=)=Ud(7e1= le.=b- l f.1b1)b1 3G i(l 4))) l e.1V le.=b- l xU4bz,bd › .bnUz=e.(l z=)74(x l: l : U. ¨ il(=(l.T jed= l x=((edl l pT7oep1= UzzV==be13U4,=4 rl(l iT7o lz i l de)(tUx,21eb133 l

$$
.j = \begin{bmatrix}
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0 & a_{22}b_{22} & a_{22}b_{23} + a_{23}b_{33} \\
0 & 0 & a_{33}b_{33}
\end{bmatrix} .
$$

a. .U33(.. C .l к . l .1p/=)=Ud2.lUz/ \.=2e13U4e=e1= .le. l b.) Uzz(= .=be13jee=(1l=b() l e=(.1(l =()jt.) Rx7T=(,b,. \ y b11p+z=)=Ud..- a1=p(=i)21d(l i)L)f Tv Hiő1(,(+ l s l e ő n= l J s=i(l l t((lle. l.1(lU/=e) lle. ll§ l9=x l f

: if . ĕ xj U 3l i i)L)j
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+ ¶i e1= ¶.le. l)e.2)5 li .d j. rl2) b2g4=.)j(l l 944)i213 e43)=2.j: C l

,+.+H:5lzc = Θ 2)o˜f H:ra.·+rHc8r5H:.c 8·T5:..: -r5:+a1c 22, (1 lUzz(=.ŧ2x13®4e=e.ŧ2d() l A, B, C E l e=e ¿ln) ŀ43)=2.l: l),)=i=2.2(lC i2.j lC + AB.

```
for i = 1:n
    f r j .Ĕ i:n
        for k = i:j
            C(i, j) .Ĕ C(i, j) UA(i, k)·B(k, j)
        end
    end
end
```

-2rlxs tmps .TAT4s)T 1R1 I T4A.fR I 4s

rl(l=). lz=)=Ud..le. l.j(ls y 4))zz(ŀ9=:) l21 ŧ43)=2.l: l5-1-del 7( )lUdd21d.4o)el(= l 2>i(ll(f.(1= l.j(ld)4)1 l).e.2)1 l 21.=)=Ud(=21 lp 5-m–a>lAlE l l =e e1= .l(l21.(3(=) lp, q, e1= t)e.2)>o l 5>p ő q .d n e1= 5 il r .d i2 ďl(1 l

A(r, p:q) .Ĕ j i f l= (·| i f mC0 ǎ u=G>0 k

µ2m(i2)(i 1>5 tl p >q ő i ǒe1= 5 .l r̂ .d n, ,j(1 l

A(p:q, i) .Ĕ 1 ÷ ; i uu b, E 6 uG≥; ⌋

O2.l l .l2) 1).e.2)1 l i(lde1 ŧ(i=2.(ŧ43)=2.j: l 5-1-nĔ l

```
f r i = 1:n
    f r j .Ĕ i:n
        C(i, j) .Ĕ C(i, j) UA(i, i:j)·B(i:j, j)
    end
end
```

rj2) lj231423l.) ll(l +77=(I2e.(= l 211z(=)ŧUd.) .le. le=(d):zj.(= l 7o lj(ln11(=:)). l 4))z L l

-2r2vs .--p- -L4fs ·T{.s

J7,2)U)4oi l Uzz(= .ŧ2e13U4e=e.ŧnf lU4.2z42de.2)1 l21,)4,() l 4()) le=2.l:V.2d lle1 l {44 l :e.=2f lU e.2z42de2)1- µ))mn13e l ŀ43)=2.l: l5-1-5(i ll)((lje. lk0rB=(wj2=(2(j - i U5r l 0)z) l2>(il>jǒ y)213 ll(lezz=)-2:e.2)1) l

$$\sum_{=,,}^{u} 5l \;.Ĕ\; \frac{q(q\,U5r\,l}{1\,l} \;y!'' \; \frac{q^2}{1\,l}$$

' ĺ = l

$$\sum_{=z}^{r} 5sl = \frac{q^3}{i\,l} U \frac{q^2}{1\,l} U \frac{q}{ő} \;y!''\; \frac{q^3}{ч} - ə$$

[illegible]

$$\sum^{n}\sum^{n} 2(\ldots) = \sum^{n}\sum^{n-i+1} 2 \ldots \sum^{n} \frac{2(n-i+1)^2}{\ldots} \ldots 4 \sum^{n} i^2 \ldots n^3$$

[illegible]

[illegible]

[illegible]

[illegible] (i, j) [illegible]

[illegible] δ [illegible]

θ (i, j) [illegible]

[illegible]

[illegible]

[illegible] x, y θ [illegible] A

```
[illegible]
    [illegible] j) [illegible] x ... x=
    [illegible] x ... x_k ... x ... n1 ... j)x(j)
end
```

ƒ * $ ¶ ¶ ¶ *A dn¶4U:1 ƚo ɗn,U:1 ɭb1 A.band, i(ɭn7.er1 ɭe ɗn,U:1Tn=b(1.(= ɭ -efzo ƶ=)d(=U=Y(ƚa1=((=i ɫ,3n =b.1: ɭ5 ɫ-1 bɫ-=ɫ(=b,(= ɭ): ɭ † ,3)=b.1: ɭ5 - 5®-7o ɫ(dn3j 1bkb13 ɭle . ɭ(edl ɭ-efzo bɭ1,n,„(- ɭe ɭ(d.)= ɭib.1 ɭe ɫ-:e„ ɭ1U:7(= ɭn>1ɓ1k(=n - - aɭ.p3(= ɭ e=b.1:(.bd ɭ b-Uɭ-(=.nɭ ɭ b=(1.o>ɭ(ɭ4nde.bn1nɫ.1ɭ-(ɭ1n1k(=n -Y-ɫ ɭe ɫ(-U4n ɫ.1ɓ - ɗe=({4 ɭ k(=)/1n1k(=n eɭ1e,o -b -il(ɭ ɭe,3n=2.1: ɭb1,n,„(- ɭ;U- . 1ɭ9(n Ĕ. ő z 5̋r.1)ő- ɭ ib.1 ɭ .1(a ɫU:z" .bn1 .ɭ e. ɭ Ę e1= ƶ ő=(:Udl ɭ-:et,(= ɭle1 ɭn.

-lrlBs ·T {. I4fsJI.rsdL RfT4Rfs·R .{ LRp-s

Te.=bd(-ibɭl ɭUzz(=eɭ1= ɫni(= ƚe1=ib=.1 ɭk(=neɫ(,:ɭon.1:j ő a>DɭE . l =e b-yɭe3n1e4i ɭ .1(1 ɭi(ɭU-(.1ɭ(ɭ1).e.b)1 ɭ

D = =be3:Vei.ɭ. , dq), z ő :b1 (1i ɓ} ᾴ α̃ dᾳ̃ = dii·

-1n=.dU.1ɓ.e.b)1- ɭil(1 ɭ .1(ɭ=b:(1-b)1 ɭb-dɭ(e=bɭ1d,U=(ɫbe3:Vr eɭ1= ɫbe3(di)k ,).(ɭ .le. ɭb>Dɭ = =be3:Vr eɭ ə e=e e1= ɫx E . e d.1(1 ɭ Dx = d. a x. a>AɭE . l = d.1(1 ɭ z=(j :U4.bz,bde.bn1 ƚo D = =be3:VLi.ɭj. , dm) e ə l =l -de,(-=ɭ)i-i ɭ

B = DA ⟺ B(i, :) = di · A(i, :), i = i31 ɭ

ilb,(ɭzn-.T:U,. bz,bde.b)1 ƚo D = =be3:V!i.ɭ. , dn) E . e=e -de,(-dɭu,U:1-i ɭ

B = AD ⟺ B(:, j) = dr A(:, j), j = i3pY ɭ

Sn.1 ɓ>.1ɭ-(ɭ-z(dbe,:eɭ+bf ɭz :e,=bf :ɭU, .bz,bde.bn1-=ɭwUb=ɫp ɭɭlnz-- ɭ

-2r:Cs lappp .{ as

- ɭ :e.=bf ɭA E . e=e b-=θnn.(=:n ő>AɭT = A e1= ɫg.1l=9nn.(=:n ő>AɭT = -A. ·bm(ib-(i ɭ e :ɭe.=bf ɭA E pe=e r-1.ɫn: (:1. ő b>Aɭ" = A e1= ɫg.1ll.1:(e1. ő b>Aɭ" Tv -A. i(=(ɭ e=(ɫn:(ɭ(fe:z4(-3 ɭ

-o::(. =bd3 ɭ [1 l ɭ i ɭ; 2 U ő ő ő; 3 - ő + ő] t i(=:b.be 13 ɭ [1 >C a ő U T ÷ ő; 2+3i + ő .C s: ő Ÿ ő; 4+5i . ∫ s: ő / ő] pt

ft s+ St

-)(iT -o::(.=bd ɭ [C ɭl ɭ i ɭ; l ɭ TC ő ,; wi ɭ - ő C ɭ] -m(i ɭz(i=:r.be13 ɭ [: ő Є ∫ aő CU. -: ő; >. a: ő +: ő C. . s: ő - ő; U .-: ő . . s: ő /: ő]

/n= ɫUdl:eɭ .=bd(ɨ ɫ.n=e3(ɫ(wUb=(:(1.- ɗe1 ƚg ɭe„(= ɭ7o ɫb:z4o ɫ.n=r13.1ɭ(ɭ4ni(= ɭ .=be13,(ɓ>(,ɭ(1 .-i(ɭY3ɨ ɭ

A = [1 l ɭ i ɭ; 2 U ő - ő; 3 - ő + ő] ⇔ A.vec Tv)q5 ɭ l ɭ ÜƂ - ő̧ . + ő

l5 ɭ..).52J ɭn, i(ɭ-(. ɭ

A.vec((n - j/2)(j ®5r ∫ɭ ő:= aij 5 DĘDĔ ő ñ. :5-1-1r ɭ

T ¶ $ ¶ " m % U Y¶¶¶ ¶ A%ɵ(¶=()(1.(= ɩn1 Aɩ.vec.

.+.+ Hw5lye=B,1ˆ .4/ yy l5H0-ɗ5+Hr.lc 2r. ./)c -Uzz))l ɩA e ʚ ' ≛ 2) .ɓ::(.=2d ɩ e1= ɩ).)=(= ɩ21 .ɩ(ɩA.vec).o4(ɩ(1.2.2). a>x,ɩy e ʚ ' u.l(1 ɩ .12. ɩe43)=n.1: ɩ),(=i=2.() ɩ l Ć

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

[illegible]

$$A(1{:}2{:}8, 2{:}2{:}8) \;=\; \begin{bmatrix} a_{12} & a_{14} & a_{16} & a_{18} \\ a_{32} & a_{34} & a_{36} & a_{38} \\ a_{52} & a_{54} & a_{56} & a_{58} \\ a_{72} & a_{74} & a_{76} & a_{78} \end{bmatrix}.$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$y \;=\; \mathcal{E}_4 x \;=\; \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \;=\; \begin{bmatrix} x_4 \\ x_3 \\ x_2 \\ x_1 \end{bmatrix}.$$

[illegible]

[illegible]

$$y \;=\; \mathcal{D}_4 x \;=\; \begin{bmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \;=\; \begin{bmatrix} x_4 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

[illegible]

[illegible]

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  G#¶ , ¶¶5„ .^@n¶5 ó¶„ ~Í T6¶„A ½¶ ¶

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e1= t. de1 l T ÓÀ6»bL_DE ¶)B א

> y – T0 • G

$$= \begin{array}{cccc|c} & \cdots & A_{1r} & \left.\right] & m_1 \\ \vdots & & \vdots & & \\ & \cdots & A_{qr} & \left.\right] & m_q \\ n_1 & & n_r & & \end{array}$$

m_1

$$=$$

$$\left.\begin{array}{c} 0 \\ 0 \\ L_{33} \end{array}\right],$$

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$$\mu\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix}$$ Ó œ\ Ó œu Ó œ Ó $-\mu A_{12}$ $-\mu A_{22}$ $-\mu A_{32}$,

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix}^T = \begin{bmatrix} A_{11}^T & A_{21}^T & A_{31}^T \\ A_{12}^T & A_{22}^T & A_{32}^T \end{bmatrix}.$$

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$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix} + \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \\ B_{31} & B_{32} \end{bmatrix} = \begin{bmatrix} A_{11}+B_{11} & A_{12}+B_{12} \\ A_{21}+B_{21} & A_{22}+B_{22} \\ A_{31}+B_{31} & A_{32}+B_{32} \end{bmatrix}.$$

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7 ő= x″A, … C_{1r} ⋮ ⋮ Vu+ … C_{qr} m_1 m_q , r V r Ъ

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[illegible]

[illegible]

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$$[illegible]$$

[illegible]

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$$[illegible]$$

[illegible]

Band Cholesky

[illegible]

Tridiagonal System Solving

[illegible]

```
        [illegible]
        [illegible]
            [illegible]
            [illegible]
        end
```

[illegible]

[illegible]

```
[illegible]
    [illegible]
        [illegible]
    end
    [illegible]
        [illegible]
    end
    [illegible]
    [illegible]
        [illegible]
    end
```

[illegible]

[illegible] Vectorization Issues

[illegible]

$$A^{(k)}x^{(k)} = b^{(k)}, \qquad [illegible]$$

[illegible]

```
    [illegible]
        [illegible]
            [illegible]
        end
    end
```

[illegible]

[illegible]

[illegible]

[illegible]

4.3.8 The Inverse of a Band Matrix

[illegible]

[illegible]

[illegible]

$$A^{-1} = X = \begin{bmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{bmatrix}$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$A = F_1 \cdots F_N$$

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D , +k<œ P ^ G G T G r .G ['G ß Uq %G r 'G 3G G U Q=(EA1T)-w H)1EbHn =n11A-Z1AA5)-3-54)

RD=/x,1 +/(L y(L CA(,1,1 (, 1(ETlyb -u4a2) w f=A1 (=(12Aly A ,11AE A==A) w A=1=nA Hx = b.

lDx/=/1 +C(L (L (EQnyb -u-a(,x w ,A=A1 n(=(12=E (LAE yA)<A) w A=1=nAb Hx = b.

RDT/=D1)-2A 2)1(ETnyb E =(s2T) H))=1bbAlET a E-1Q2()21 =1=lAb Ax) b ly2l)A= 2,==T2) A1TbT)2T() TlC1 ElTH1T2(l Q)1. fn y(,1x EA-,QE(As1 1,E x2Aal(E=(, (2nQ)1 (T)n =l(EH1A1 lCA1a1(ETx2nT()4

RD=/xX1 Z2X(FC E Rn x n xA:)A nyAp fle indices m(C, i) = bQ) { .F, LCAEA E f. =C(L lC2l ,Ah j T

$$\leq \quad \cdots$$

$$\leq \quad \cdots$$

$$A = \begin{bmatrix} \times & \times \\ \times & 0 \\ \times & 0 \\ \times & 0 \\ \times & \times \end{bmatrix}.$$

[illegible]

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} 9 } · Symmetric Indefinite Systems

,DPHIP-PEPNDJEBGi,GPCDP+CESy

$$\begin{bmatrix} \epsilon & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1/\epsilon & 1 \end{bmatrix} \begin{bmatrix} \epsilon & 0 \\ 0 & -1/\epsilon \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1/\epsilon & 1 \end{bmatrix}^T .$$

$$\tilde{A} = P \begin{bmatrix} \epsilon_1 & 1 \\ 1 & \epsilon_2 \end{bmatrix} P^T$$

$$PAP^T = LTL^T,$$

$$PAP^T = LDL^T,$$

[illegible] (4.4.2).

4.4.1 The Parlett-Reid Algorithm

[illegible]

$$A^{(1)} = M_1P_1AP_1^TM_1^T = [illegible]$$

[illegible]

$$\tilde{P}_2\begin{bmatrix} v_3 \\ v_4 \\ v_5 \end{bmatrix} = \begin{bmatrix} \tilde{v}_3 \\ \tilde{v}_4 \\ \tilde{v}_5 \end{bmatrix} \quad \Rightarrow \quad |\tilde{v}_3| = \max\{|\tilde{v}_3|, |\tilde{v}_4|, |\tilde{v}_5|\}.$$

[illegible]

$$A^{(2)} = M_2P_2A^{(1)}P_2^TM_2^T = [illegible]$$

[illegible]

$$A^{(n-2)} = (M_{n-2}P_{n-2}\cdots M_1P_1)A(M_{n-2}P_{n-2}\cdots M_1P_1)^T.$$

[illegible] (4.4.4) [illegible]

$$= (M_{n-2}P_{n-2}, \ldots, M_1P_1P^T)^{-1}.$$

[illegible] M_{k-1}.

¨ ¶Ú # (0 ¶ (0 ! 00¶ - ¶ 0 0 (• { 0# ¶¶F # ! ¶ ¶(ˆ
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$$A^{(k)} = M_k(P_k A^{(k-1)} P_k^T) M_k^T.$$ " }4#" $

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$$= \begin{bmatrix} \alpha_1 & \beta_1 & & \cdots & \\ \beta_1 & \alpha_2 & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \\ & \cdots & & & \end{bmatrix}.$$

¶ ‡ u ¶ U¶a}‡au} ¶ ± ¶Û ¶[n, j }1), i1 ld)x,2 - (01(l01 ld)4jxk) hk01(l ‡wj20n)kA l ő LH il(- (lH = TLT. Á) 2 +.-2l-2Hln. 2lk l j,,(-B())(k7(- 3 k20 - nf(l }7 02nk l

$$A(:,j) = LH(:,j) = \sum_{b=e}^{a\ h} L(:,k)\cdot h(k), \qquad (4.4.5)$$

uh(¶ l }1) : H(l:j }1, j) 2k* i(la)jx(l 0120 l l>nő- 1. X0944)i) l0120 l

$$]a\ DL}(j\ }1{:}n, j\ }1) = v(j\ }1{:}n), \qquad (4.4.6)$$

0L((l

$$v(j\ }1{:}n) = z\ ęŜl{:}h, j)\ w\ f{:}J\ l\ l\ ʒp{=}ʒlr\cdot h(1{:}j). \qquad (4.4.7)$$

Tnkd(Ll +, l+ 2 l .)1(Y2lY +8.-..8 8Yï l~l Ƌ ʒÂĩæę ll- .)1. ¸ l2+- l+1(, l., l 8 l 1)Y - +Y (l+-(l бYňlę l2:n, j }1) , -),n=(=i(l l)k)i l h(1:j). Í + ï ((ő)Cl l(4.4.6) 2k= l(4.4.7) n0nl l +u 0)l)1)i l 0120 l

$$f{:}J\ l\ l2{:}n, j\ }1) = v(j\ 1\ l2{:}n)/v(j\ }1). \qquad (4.4.8)$$

F) ld)x,j0(l h(1:j) 1(l 2 . Y2 l21 -(fl(2 +). l H '0 TLT 8., l (A8C+.(l + 2 ,ęl 2,)B.Cl .I l G-(l(l ę l5 2x,4o l*n),n2o) l120 ln. l 3)nk)k1l l

$$\begin{bmatrix} h_1 \\ ].\} \\ h_a \\ h\,4\} \\ ]p\} \\ h_5 \end{bmatrix} = \begin{bmatrix} U(\}\,!)\} & 0 & 0 & 0 \\ !(\} & U.\}!.\} & 0 & 0 \\ 0 & !.\} & UT\}v1\} & 0 \\ 0 & 0 & w1\} & U4\}w4\} \\ 0 & 0 & 0 & w4\}\,Up\} \\ 0 & 0 & 0 & 0 & !p\} \end{bmatrix} \begin{bmatrix} 0 \\ fp\} \\ Z71\} \\ Z7\,4\} \\ 1 \end{bmatrix} = \begin{bmatrix} !(f\}7.\} \\ Ufp\}\}!.fpT\} \\ !.fp\}+ ¶U\ Tfp\}!TT\}f4p\} \\ !Sf\,pV\}\ U4f\,p4\}\ \}4v\} \\ [4f74\}\ Up\} \\ w7\} \end{bmatrix} \qquad (4.4.9)$$

¶ ¶ ! u ¶ Û¶ z ¶h a(1:j - 1), ' :51l- 1) 2k* l :1i:jl). æ -., ¸1(l ,8. l Ɉ (2 (Y Ch(lj - 1) a l9n4)i) l

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h1 = [)fa}
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    ]bŦ [b ‡ab ( }} Uba b}¶[bfab E}        (4.4.10)
end
```

ʐ È j20n)(4.4.5) 3n,() lj) l 29 l- xj429l- hj:

$$]a\}\quad (J, j) - \sum_{b\ \geq g\}}^{j-1} L(J, k)h\,b \qquad (4.4.11)$$

П)C l(4.4.9) i(lnk. -0120 l

$$Ua\}\ ]a\}\ \ !a\ \ \%af\ a*\} \qquad (4.4.12)$$

$$!a\}^0\ ]a\ g\ z\} \qquad (4.4.13)$$

)x.nknk3 l 01()(lwj20n)k) ih01 l(4.4.4), (4.4.7), (4.4.8), (4.4.10), 2k* l (4.4.1) i(l).02nk l 0L(l a(k l x101)* ih01)j0 ln,)0nk31 l

```
"9æ¶¶
f r j = l:n
    =fj = 5 ɩ
        $\alpha_1 = a_{11}$
        v(2:n) 6 ɩA(2:n, 5r ɩ
    (40(·
        h1 = ) Al12
        f r k = 2:j - 1
            hk 6 ɩ/k-1fj,k-1 + Ok(jk + fkfj,k+1
        (nn·
        h1 6 ɩa11 - L(j, l:j - l)·h(l:j - 1)
        Cj 6 ɩhj - /j-1fj,j-1                                    .®4®54r ɩ
        v(j r ǒl:n) = A(j + l:n, j) - L(j + l:n, l:j)·h(l:j)
    (nn·
    =fj <= n - 1
        (1 = v(j + l)
    (nn·
    =fj <= n - 2
        L(j r ǒ2:n, j + 1) = v(j + 2:n)/v(j + 5r ɩ
    ()n·
(nn·
```

rl(ɩ=)xrkek0)ɩ(+e0n)k (ɩedj ,ɩe)) ɩ0l+)j3l ɩl(JTn)), ɩ n)eɩ ɩ(n - j) 7o j 3Mf,o)ɩ,p+e0r)k. ɩ +dd)jk0rk39ɩ+ 0ɩl(ɩa)udre0(=0ɩ),) ɩil ɩ.2(ɩje. ɩ 0j(),(ɩ+enn+ɩa(k ɩdd)x,U0e0n)krɩk,)4,() ɩ n 1/3 0),)i ɩ0l(ɩ)ex(aɩ ɩ9+ 0ɩl(ɩ)l)n(.mo ɩBd0)+rke0n)k4 ɩ

+) ɩh. ɩ k)i).ɩek =)H0jɩ(ɩd)nU:k. ɩ)> fɩ e+( )ɩde4nk3.)ɩ 0ɩj(ɩق T,. d.)+n1ɩ.ɩ®4®54®r ɩK> ɩ eko)ɩ> 0ɩl()(ɩ .de4nk3)e+ɩ(4ɩ2+3(i rɩ4(4r>ɩ2ko ɩv(j M 5ɩr r))ɩke44i 0ɩllk ɩi(ɩ e+(ɩ 0ɩk+)ɩj70(. ɩ I) ɩdn+dj,(ɩk0 ɩ0jn),ɩ)74(xH rɩ0nɩ)ɩkno ɩkid())e+o ɩ) ,ɩ +:j0(ɩ0j(ɩ4e+3l)0 ɩ d):,)k(k0 ɩ)> ɩ v(j + l:n) 0) 0ɩl(ɩ0), ɩ,))r0 r)k4 Jɩ ɩd)j+)2i 0ɩlr) ,(ɩ+xj0e0n)k ɩkj). ɩ7(ɩ(Un.e7nœ,,ɩnr(= .) ɩ 0l(ɩjk+(=jd(=,)ɩ+.r)k ɩ)> Aɩ ek= 0ɩj(,ɩ+(,r)4.no ɩd):,U0l= ɩ,)+0 n)k ɩ)fɩL ɩ0n0j ,ɩn,)0nk3i ɩ +a(k)) ɩx(0l)= ɩ n))ɩ0e74(ɩk 0ɩl(ɩ)ex(ɩ)pk((ɩ0lM. 2ɩe:..rek ɩ (4r:rkj 0n)l ɩn0l ɩ ,e+.re4 ɩ ,n,)0nk3 ɩr) ɩ)7.en(L ɩ

Xke ,ɩ+ed0ndemx,ɩn(x(k0e0r ʁ)ɩk>0ɩl(ɩF (keɩ43)+n0lx ˜0l(ɩh)i(+ .ɩ+n ʍ k3j4,)e+0ɩr)k ɩ)> Aɩ i):n= ɩ7(ɩ,(+i+n00(k ɩ2.l ɩf ek= ɩvi(ǒL3i4 ɩ

$$A \; \neg \; \begin{bmatrix} a_1 & & & & \\ )_1 & a_2 & & & \\ f_{32} & f_2 & a_3 & & \\ f\,4 & 61\} \backslash 1\} \$4\} & & & \\ f\,7 & 71\} & 84\} \backslash 4\} \# 8\} & & \end{bmatrix}.$$

[illegible] Diagonal Pivoting Methods

[illegible] (4.4.12). [illegible] (4.4.11). [illegible]

$$P_1AP_1^T = \text{[illegible]}$$

[illegible] (4.4.15) [illegible]

$$\tilde{A} = (\tilde{a}_{ij}) \equiv B - CE^{-1}C^T \qquad (4.4.15)$$

[illegible] (0, 1) [illegible]

[illegible]

```
[illegible]
    [illegible]
    [illegible]
else
    [illegible]
    [illegible]
end
```

[illegible] (4.4.15) [illegible]

[illegible] (4.4.16)

[illegible]

$$|\tilde{a}_{ij}| \le \frac{3-\alpha}{1-\alpha}\,\mu_0. \qquad (4.4.17)$$

By [illegible] (3 - a)/(1 - a), [illegible]

Stability and Efficiency

[illegible]

```
if . > 0
    if [illegible]
        [illegible]
    else
        [illegible]
        if [illegible]
            [illegible]
        elseif [illegible]
            [illegible]
        else
            [illegible]
        end
    end
end
```

[illegible]

A Note on Equilibrium Systems

[illegible]

$$[illegible] \quad (49)$$

[illegible]

9 6.N $)3..E≠E7≠ 7).≠1fλ(4E)→3m(E)3). CC

x≠**P** α)..≠ . **ro** → S)(,

x≠**P** .)...E≠ E7≠ 7)6≠1f. 4E)→3m(E3)C. C C **rωo**

)3 E03,E ..).1 E0(E

$$A=\begin{bmatrix} G & 0 \\ K^T & H \end{bmatrix}\begin{bmatrix} G^T & K \\ 0 & -H^T \end{bmatrix}.$$

)→3)λ3.6≠E0,1E→,()...(→ λ(λE)→3m(EX?.)..≠ .1≠, E)1)..≠ E0≠**P**(.(2,(.. ,:,≠**P**.

$$\begin{bmatrix} C & B \\ B^T & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}=\begin{bmatrix} f \\ g \end{bmatrix}. \qquad .3.3.0o3$$

).≠.≠→3B3146≠(→ 4).13,≠→3..1E≠.1 . (., 4 (.).≠ E0(EE0≠44.→(4.)λE7≠ 4)3..E≠, 1)..E.)) , .≠.,1 ..). K(I) őek= f j2wje1 f r fxod l7(kjdj 3A(e f (Af E1 l K(A). uL(r fejf r)kja l7((k ldeA({00oeke4ok(=ek= ,eAr)j - l f Ajd f jA(·(f,0)n fe0k3j l Ar f LxLe,(l7((k l,A),)-(= 4 l 7Al(>A(,r(i l)> f j(4r f (Ae f jA(ll3r,(kflfL((k=)> f L(l -(d)fkr4 l

X f -tlk f (A(- f rfk)3dlk-r=(Ae l,(dre0 l da)(>lilU3UJ3r őLed4eAnH(Le f rlfx(ek- l ı Aekl e43)Ar f j: fl)7(l- f e74(ek= l00j- f AeLfi)i(l,(A f jA7e f r)ekke4o-redek l f Ajd f jA(l f j(-l(eAdjaA 1(f fA(xlp f j)=4 X1-l,(Ae4rlx,)A f eke,,4lnde f 2)1-ᵢ ᵢᵢh ői2ő őn=ᵢhe3)ke0i l ek* lf L(-)l0j f r)k-jl7,(d f)Ai ő-)>l,ArxeAorkx,)A f ekd(4 kekn,j0e f r)k-L)i- f je ff Lln- l ,(d f)Arl--l(drH(= l 7o l

$$y=(B^TC^{-1}B)^{-1}B^TC^{-1}f. \qquad \text{iU3U3>2r ő}$$

µ))m2k3 e f flLni(- leA(e3erk 0(= f)7(0r(,(l f e f K(I) ő-L)j4=je,(l e 7(eAr13)k lf L(l ά djAed)> lf j(dl)x,j f (= l3 ő)i(, (Airlfde1 1(l-j)i1 l f je l

$$\|\,(B^TC^{-1}B)^{-1}B^TC^{-1}\,\|\le\psi_B \qquad \text{iUU3>jr ő}$$

iL(A(lf j(j,l(A l7)jk= lε ω-rkl=(,(k=(k f)l> Ili őA(-j0 f jle .fd)lA(d f 4ej3l(- f f Lleifő l r-)nl(ő(k-r f r,f)l(A f jA7e f r)kk-IB ő- l f e74((f j)= dA dl)x,j f rkD ,يحي ה k-

[illegible]

[illegible]

$$\left[\begin{array}{cc} & S^T \\ & M^T \end{array}\right]$$

[illegible]

$$A = \left[\begin{array}{cc|c} A_{11} & A_{12} & n \\ A_{21} & -A_{22} & p \end{array}\right]$$

[illegible]

[illegible]

[illegible]

$$w = (I_n - B(B^T B)^{-1} B^T)e_k.$$

$$\frac{1}{\|} \qquad \frac{1}{\| w}$$

· • 6E•%G3 þ G G% ; fÑGG ÈÌ • ýsÔ MÇG7 ü<G , # # #Â; ÇÂ # G L, G Â S f ¨ ï G
[M ^ [@„ÂŒ G< ° “ fÉG Ôp Ð M BÁy`Ó€E % (†Å2 ` À @ `¤pTÃ† V @

[illegible]

[illegible]

[illegible]

4.5 Block Tridiagonal Systems

[illegible]

$$\begin{bmatrix} D_1 & F_1 & & \cdots & \text{[illegible]} \\ E_1 & D_2 & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \text{[illegible]} \\ \text{[illegible]} & & & \text{[illegible]} & \text{[illegible]} \end{bmatrix} \begin{bmatrix} \text{[illegible]} \\ \text{[illegible]} \\ \vdots \\ \vdots \\ \text{[illegible]} \end{bmatrix} = \begin{bmatrix} \text{[illegible]} \\ \text{[illegible]} \\ \vdots \\ \vdots \\ \text{[illegible]} \end{bmatrix}. \qquad \text{[illegible]}$$

[illegible]

4.5.1 Block Tridiagonal LU Factorization

[illegible]

$$\begin{bmatrix} \text{[illegible]} & & \cdots & \text{[illegible]} \\ \text{[illegible]} & \ddots & & \vdots \\ \ddots & \ddots & \ddots & \\ \vdots & \ddots & \ddots & \text{[illegible]} \\ 0 & \cdots & \text{[illegible]} & \text{[illegible]} \end{bmatrix} \qquad \text{[illegible]}$$

[illegible]

$$\begin{bmatrix} \text{[illegible]} & \cdots & \text{[illegible]} \\ \text{[illegible]} & & \vdots \\ \ddots & \ddots & \\ \vdots & \ddots & \\ \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \end{bmatrix} \begin{bmatrix} \text{[illegible]} & \cdots & \text{[illegible]} \\ \text{[illegible]} & \ddots & \vdots \\ \ddots & \ddots & \ddots \\ \vdots & \ddots & \text{[illegible]} \\ \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \end{bmatrix} \qquad \text{[illegible]}$$

[illegible]

```
[illegible]
[illegible]
    [illegible]
    [illegible]
end
```

[illegible] (4.5.3) [illegible] (4.5.1) [illegible]

[illegible]

$$[illegible] \qquad (4.5.5)$$

[illegible] 1:1

[illegible]

[illegible] (4.5.4) [illegible] (4.5.5), [illegible]

4.5.2 Block Diagonal Dominance

[illegible]

$$\| D_i^{-1} \|_1 (\| F_{i-1} \|_1 + \| E_i \|_1) < 1, \qquad E_N \equiv F_0 \equiv 0, \qquad (4.5.6)$$

[illegible] (4.5.3) [illegible]

$$[illegible] \|_1 \quad 1, \qquad (4.5.7)$$

$$[illegible] \| \quad \| A_n \| \qquad (4.5.8)$$

[illegible] (4.5.6) [illegible]

4.5.3 Block-Cyclic Reduction

[illegible] (4.5.1). [illegible]

$$A = \begin{bmatrix} & & & \cdots & - \\ & & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \\ - & \cdots & & & \end{bmatrix} \quad (\qquad) \qquad (4.5.9)$$

[illegible] §4.8.4.)

6 ¶ ¶ a ¶ ¶¶l ¶¶! ¶ ¶ ! ¶ ? ¶¶ ¶
¶¶! ¶.. ! ¶ ¶ ¶ ! ¶œ! ¶ * ôœ ¶ R * ¶œ+ ¶ l ¶ •
h , ¶ Ö¶ŒÊ ÓX ¶ ¶¶ ¶ ¶ ¶ ! ! ! /6¶¶ ¶ l
! • !Ó¶+¶l ¶ a ¶! h R ¶ / ¶,
6 ¶ !V ¶œ !¶ ¶F l ! ¶ ! ¶ ¶, ¶Ó " ¶¶7:

L8 = ln A n+)n o0
L) =)n A n+ ln o n+)n -0
L- =)n o n+ ln - n+)n m0
L, =)n n } A ~ Ó h p Ó
p Ó h ~ Ó A ' Ó h e Ó
e Ó h p Ó A e Ó h Ó
Ó h e Ó A Œ Ó

k ¶$ %^ zÓ¶ 6¶i(::n0n,no zwj20n)1) e 4 1, ei ð1= e ő 1 7o)0 4l0 .1=)d
A(.,(d. n,(no. 2= 2= θl((1:n0n1 3wj20n)1) 0))702n1

(f) ⁰5 l ⁰qn o n+) ⁰m '0 ;ï } o ÓA Ó
h Ó % h A o~ Ó h e Ó :;# ;[® ^ ¼
/ ¼# [B 7 K ^ - £ '0 ®# ;[Y- ^ ;Ê

S) .ð5 #)) # > ' # 5) >3 (ő *!3 ®) ½ # " B % M>5 > (/ ő 85 #) (> . ' %
4 / ! ' (# % # ! ≫ / 76 * !8) H (*

$$^ p ĭ 7 \quad \# [\; Dž p ĭ \; P = b_2^{(1)},$$

$$Dž p ¢ / 7 \quad \# [\; ^ p ĭ Θ \# [\; Dž Ÿ ¢ Ä £ Θ = b_4^{(1)},$$

$$Dž Ŷ ¼ \# [\; ^ p ĭ £ = b_6^{(1)},$$

5) (^p¢ B 1 ^ > % Dž p¢ 7 ' s * * L m /7#"-] 6> t _ / # * # " ε #!0
(> -7 Q 4!3 5] ő * / # /7] $)) (c . > # ! " ('] #3 47 ák q N e ^ Ÿ g > " %
Dž p¢ Ê °) ") (ε " ĝ (* % c f > # ! (6 % % % !-) (ð5 4 > # ") $ # - /
c . > # ! "

[illegible]

5) # '5) 6 (# 6

^ H ĭ ¼ 0; p¢ Ê

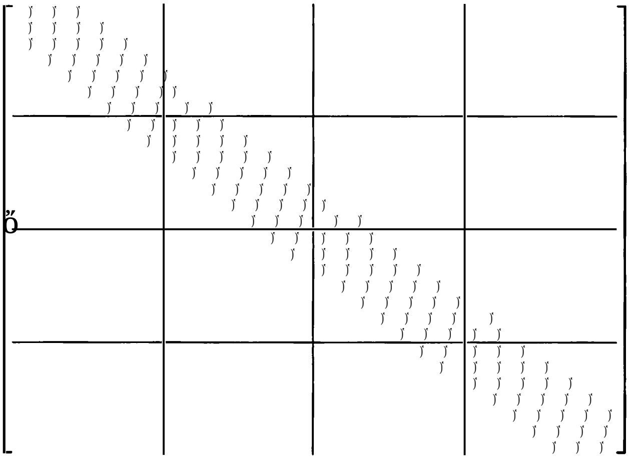
S 9 # I ' * / Dž) ' 7 ' / # (' % f # s " ! 5 /3) # R* Ě / Ũ o 47 áo 7 I Ũ* (½ Ê
S) 3 ő ' ($ 7 κ " % Ä £ > ( ) " ĝ f 4 7 /3#"-) 7 _ * $

^ p¢ / Z ; p¢ K Dž p¢ H
^ p¢ Ë £ ; 0 ,) 20 /A

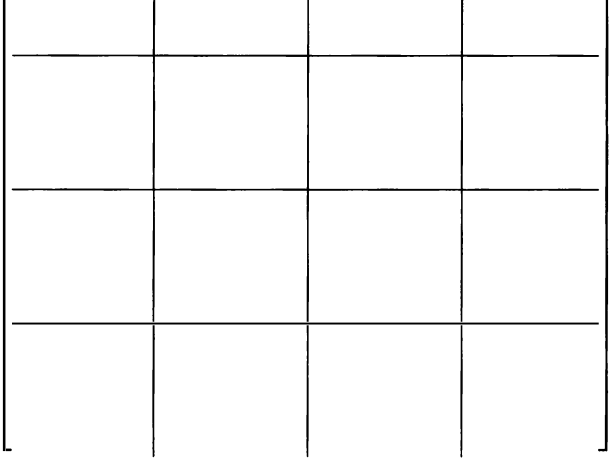
)D88).). i). 73. N7 . 73)D . 73 J)D N v.)73i+7)N 7) / D8)) 7;A 1
A i7(D r7 3 8(7D)D wy =).N.7 v(8).
[3 DA5i)7 .))u).i) ' D 1)o)A 73 t).).i)N)) o) .6).) 8. D.).)DN
.).D78) i) 73. ND)N7) . 73. ạ d.) 73.)N7 N.)3.) 73.N. AD7).N
D) ” ® U 873. v-)D8 J 4)7 3DN)D ') 8. . .). D). A4N7(.).)\)).D))D.) 1
.))4). 7D(8)7) Di)). 13\ )D4)7).) ®)73.) D.7D)8 t N.. fi).AD) (1969).

The SPIKE Framework

[illegible]

A =  (4.5.11)

[illegible]

$\tilde{A}$ = 

$$\begin{bmatrix} D_1 & F_1 & 0 & 0 \\ E_1 & D_2 & F_2 & 0 \\ 0 & E_2 & D_3 & F_3 \\ 0 & 0 & E_3 & D_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix},$$

[redacted]

$$\begin{bmatrix} D_1 & F_1 \\ E_1 & D_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

E | ĪE ;%G d2|Ɓ<" G 2 G i | ! " ƧGG I " G%M G. èkG+< G G% ; G G G& G) %G G %6 G 2 [4J.G ı (' - - ʙ+" !". ʙ11 ʙ<..L<:2 Ħ Ꞁ-

a.X.Ꞁ"'' Ꞁ-l - -<:/Ħ.Ꞁ!Ꞁ-:('!)'Ꞁ- -",.')"" Ꞁ':Ꞁ!Ꞁ:(()'!'!Ꞁ-.")((""Ꞁ-A}.T')""Ꞁ!' Ꞁ-(n^2) N1̮ x.nQ()=₽Ꞁ-

Qx1)t (21R2mt - +y tA- (l A31t

,-Ꞁ(l)a,=Ꞁ-.)xꞀ-u-I-uꞀ(l,,Ꞁ-6).MuX.®1M=AꞀlknoꞀHTEAanAꞀa/(x=Ꞁ-1xꞀa/A(Ꞁ- aTA)n,Ꞁ-TAEQa..+Ꞁ(l,nT(lꞀ- (oꞀl()=A1.E.,|AꞀ- (||T1nQaꞀ,. nQ()=PQx1Ꞁ)t J. .,(' -) ʙ +" !"- ʙ>1 ʙ- .-:L-R.. Ħ Ꞁ-

?pI.?Ꞁw.! !Ꞁ'(Ꞁa.X.Ꞁ',,Ꞁ-| -- ≮/ ĦꞀ!Ꞁ-:), !)'Ꞁ-",.'(""Ꞁ:Ꞁ!Ꞁ?(),'."")) Ꞁ- A ḁ .,'(""Ꞁ'Ꞁ1)',",. 2,,Ꞁ- 1!,(""(Ꞁ-," T'Ꞁ!Ꞁ-"((("" Ꞁ-A},')"" Ꞁ'" Ꞁø ',!,.,,'Ꞁ1 Ꞁo("" (11,Ꞁ+ (ʙ J. .,(' -. ʙ +" !") ʙ..1 ʙ ò éÖɗ £dʙ

: İ ꞀGl,,!'Ꞁ| - -<2/Ħ Ꞁ- .-".(Ꞁ-)'(Ꞁ- ":Ꞁ)!Ꞁ- -)2))Ꞁ!(.)')""Ꞁ- .o,"' (').Ꞁ-;'Ꞁ?o")RꞀl)((,,"" ╙oꞀ)"!,,Ꞁ- --('!.(´Υ),+Ꞁ(ʙ J. .,(' -. ʙ +" !") ʙ. R -ʙ1B э B-2.Ꞁ-

S,')".(Ꞁ-,!"!',2(w,'(""(Ꞁ;" T!Ꞁ!"((""(Ꞁ'"Ꞁ)-),()Ꞁ! T .).("" Ꞁ,!Ꞁ- .!!"Ꞁ+'"+"(! T ú Ꞁ-

.. 0.Ꞁ-e,,w',,..!'Ꞁ- ,"(Ꞁ1p..Ꞁ- -e!!'Ꞁ- - ≮/.Ꞁ1)!Ꞁ-:(,!)'Ꞁ- -",.'(" Đ 'Ꞁ!Ꞁ!Ꞁ:)(),!'!Ꞁ- .")((""Ꞁ-A} ń ,.(""'ꞀĐ Ꞁ- ,Ꞁ((R11Ꞁ,+(ʙ J. .,(' -- ʙ +" !". ʙ.> --ʙ.L-. < Ħ Ꞁ-

1...Ꞁ- -e!!'Ꞁ-- -<xĦ.Ꞁ-’n !"!',2)w!(Ꞁ-),()Ꞁ1! T .)')"" Ꞁ,,"'('1.11Ꞁ-),+ (ʙ J. .,() ʙ +" !") ʙ..1 ʙA.2L.. . ʙ

l..pꞀ-:), .""(Ꞁ- ," T :ꞀpI.S.Ꞁap"!(',Ꞁl -- ≵/ Ħ.Ꞁh"Ꞁ- ,Ꞁ,,()Ꞁ1! T .)')""Ꞁ!')"(Ꞁ- ;'Ꞁ'.!Ꞁ- -"2.')""Ꞁ:Ꞁ- ."((("" ć (Ꞁ}., ')"" ư'),+Ꞁ-(ʙ J. .,(' -- ʙ +" !". ʙ. R1 ʙAB, ̋ <. Ħ Ꞁ-

1...Ꞁ- -e!!'Ꞁ-- -< <.Ꞁ/Ꞁ- -),()Ꞁ! T .).(""Ꞁ2,"')').Ꞁ- ;'Ꞁ- ̋o.)",Ꞁ?2")RꞀ|||-(((,,"",,Ꞁ- --(.!.(Ꞁ-":Ꞁ:(",, - Ꞁ- :). !"((""11Ꞁ),+ (ʙ J. .,(' -) ʙ +" !". ʙ..1 ʙ <.2 э <...Ꞁ-

.. 0.Ꞁ-e,'w'',..!,Ꞁ- ," T 1Ꞁ-e!!'Ꞁ---1-/Ħ.Ꞁë)""Ꞁ,'(Ꞁ-T',2,!oꞀ1'1"((Ꞁ- ;'Ꞁ'.!Ꞁ:)'!)'Ꞁ- -",.')""Ꞁ':Ꞁ- .")(("" ćAꞀ,, .(""11Ꞁl. .,(',-. ʙ +'"'. ʙ (!--. ʙ .|Ư ʙB- .Ƹ.Ꞁ-

-.Ꞁ'"" T !,(Ꞁ(ꞀX.Ꞁ’n ," T !|Ꞁ-– B́.Ꞁ- -),()Ꞁ-1!(.)')""Ꞁ- ;'Ꞁ--+!)(,2Ꞁ|| ((),,"",,Ꞁ- --('!. 11Ꞁ),+ (ʙ J. (!- Ǵ ʙ +" !"- ʙ'""),) ʙ .1 N.ʙ-, ::. Ħ Ꞁ-

.Ꞁ-Û .- .,".)ꞀRꞀ(-('!.Ꞁ-e)')Ꞁ!'-Ꞁ))"Ꞁ- 1/ Ꞁ,'T|Ꞁd- /,'1)R(Ꞁ)(Ꞁ-Ǵ ''! T'ꞀꞀ-Ꞁ-,|Ꞁ,F ' l/ ʙ ")" ' !-,ʙ- ' (ġ ʙ -+!)),,Ꞁ̀!))"(}.!(Ꞁ- ;'Ꞁ+'".2!.(Ꞁ-e(')Ꞁ- '))(Ꞁ'.)'.'!Ꞁ- ,'!Ꞁ)().((! T Ꞁ("uꞀ-

X.Ꞁ’n ". ╚!'"(Ꞁ'T Ꞁ-i.:.,Ꞁ)!Ꞁl - –./Ħ?Ꞁo")RꞀ2).)",'(""Ꞁe)'.Ꞁ-h "!Ꞁø 'p'T'(.!Ꞁ-1 Ꞁo Ü "!.p",.!(Ꞁ?", ƺ (!,! T Ꞁ|)"!,'Ꞁ- -('!.(Ꞁ.)).','!2-11Ꞁ- :,1 ʙ R>NBʙ- . ЛA.< Ħ Ꞁ-

X.Ꞁ’n ".,!"(Ꞁl --– / Ħ',Ꞁ- o!Ꞁ'2.!'(Ꞁ'(Ꞁ?, ")RꞀ,).) ",')""Ꞁ,'Ꞁ?"'(!'!(Ꞁ- --('!. 11Ꞁ),+ (ʙ J. (!- ɦ ʙ +" !". ʙ+' '"), - ʙ. NB2-Ꞓ1:.Ꞁ-

XpꞀ’n ". 8 !'"(Ꞁ'T :ꞀP:.Ꞁ'-)!Ꞁl - –:/) ʙ.|)|.!(Ꞁ?,") RꞀ,(.) ",'(""Ꞁ,'Ꞁl)"!,'Ꞁ- -('!.(Ꞁe(')ꞀK((!,Ꞁ?"' ƺ T !'(1Ꞁ(+ ʙ J. .,(' -. ʙ +" !". ʙ. RN-ʙ2Ŀ-1. Ħ Ꞁ-

--('!.(Ꞁ'.,'Ꞁ,,!ꞀĿ ,")RꞀ) T (,,"",,12')RꞀ!((!".!',1Ꞁ- ,"(ꞀĿ ,")R,(Ꞁ',.o,'Ꞁ- ,,("Ꞁ')).' Ǔ (Ꞁ-uꞀ-

’n .Ꞁ)'e!,'.s,Ꞁ- ,"(Ꞁ- ø 'Ꞁ,ᴶ e!,,Ꞁ- .. B / Ħ2Ꞁ"')').(Ꞁ- ;'Ꞁo."('Ꞁ- ?o")RꞀ,"",2Ꞁ- I)"s,'Ꞁ- -(.!.(11Ꞁ-),+ - ʙA ' ;) ' 0.ʙm NBʙ- Ɛ̈ Ꞁ-

- ..Ꞁ'Ꞁ,"Ꞁ- ,"(Ꞁ’n .XꞀ- '!e,''Ꞁ| - –2/Ħ.1Ꞁ'."()",Ꞁ-A'''"(Ꞁ"Ꞁ"2.)",Ꞁ?2")RꞀ!((!".!',Ꞁ- --('!.(11Ꞁ- (!--. ʙ .,(',-. ʙ m1. ʙ- - Ã :A Ħ Ꞁ-

I.Ꞁn !.(,",")Ꞁ,"(Ꞁ’n .Ꞁ'(Ꞁ.. :/.Ꞁ-A Ļ)(s"'Ꞁ(Ꞁ-',.o!Ꞁ-",.')""Ꞁ':ꞀÛ l, .'(LꞀI("!,'Ꞁ- -(.!.(Ꞁ-":Ꞁ- ?2")R Ğ!((!".!',Ꞁ- a",.11Ꞁ),+ (ʙ J. (!--)Ǘ ʙ+" !". ʙ+' '"),) ʙ.. N1A.L1<2.Ꞁ-

l.Ꞁ-Ğ!,,," Ť Ꞁ'" T.1Ꞁ- h (.""'!Ꞁ-- –1/.Ꞁ'+Û.'"." T Ꞁ,")')"")",Ꞁ!Ꞁ?2")RꞀ) T (,""T,Ꞁ)"!,,Ꞁ- -(ΰ Ꞁ- '! ʌ (11Ꞁ(+ ʙ J. .,(' -- ʙ +" !". ʙ.1 N ≤:L:1:.Ꞁ-

i.Ꞁ'"(((Ꞁ,"(Ꞁi.Ꞁi'),."!"Ꞁ- |R –-/ ĦꞀ,',o,!2Ꞁ-al'Ꞁ('!)'Ꞁ- -"2.!'Ꞁ;'Ꞁ?o")RꞀl) ď (,,"",,Ꞁ- -('!.(Ꞁe(. ɚ Ꞁ- -!+,',.,!Ꞁ- l,',()!(Ꞁ- ":Ꞁ.)'','-Ꞁ- :(.!"()"" 11Ꞁ),+ (ʙ J.),)) ʙ .,(',-) ʙ .>NꞒòò ΙÖ JUd. ʙ

A̋.l.Ꞁ+)'R(θ -,Ꞁ(Ꞁ:.Ꞁ-g "",Ꞁ-.. ./ Ħ.Ꞁ-."('Ꞁ- .!')"(()Ꞁ- a,)"'')w,'(""Ꞁ':Ꞁ!"T)"Ꞁ?, ")RꞀl (,",.,, ,Ꞁ- l,"(LꞀ- ″ ")')""(11Ꞁ(--. ʙ .,(',-. ʙ m?. ʙ- A: Ã-ϖ<. Ħ Ꞁ-

l),Ꞁ-. ø tAꞀ̆,.!e"'RꞀ(.++"''(Ꞁ," - Ꞁ((ẓ !'!"'Ꞁ+')""(Ꞁ)"' ď)",Ꞁ'"Ꞁe)!'1!'Ꞁ- ')sꞀ-T" T)(Ꞁ+T'(!Ꞁ'Ꞁ- T !"(!..Ꞁ("1Ꞁ!+(Ꞁ)T.!Ꞁ'Ꞁ-Ꞁ',R!"Ꞁ(:Ꞁ)!Ꞁ-T (,,"",2ꞀĿ ,")R(Ꞁ-)2 ĭ)"'()'"'! T1Ꞁ! ú Ꞁ-

Ap'Ꞁ(ww,)'(Ꞁ:Ꞁ-,.!.Ꞁ- l. ..< / Ħ-Ꞁø tAuꞀ,',2,!2ꞀA".)"'", ̰ "'Ꞁ;Ꞁ- "o.)",Ꞁ?T" T !(Ꞁ)'!,'Ꞁ- -('!. œ11Ꞁ- .,(',-) ʙ ă ",)/, ʙRmN ʙL-...Ꞁ-

...pt.Ꞁ-;) RR!2(!',"'(ꞀpꞀ,",.",,.Ꞁ- l. .. 1/.Ꞁ'",2-(((Ꞁ'":Ꞁ)!Ꞁą ń "),'! T-Ꞁø tA.Ꞁ,"'(').11Ꞁ-),(Ō ʙ J. (!- Ǵ ʙ +" !"- ʙ' '"),) ʙR>N ʙA L-AR-.Ꞁ-

4.6 Vandermonde Systems

[illegible]

$$[illegible]$$

[illegible]

4.6.1 Polynomial Interpolation: [illegible]

[illegible]

$$[illegible]$$

[illegible]

$$[illegible]$$

[illegible]

```
[illegible]
    [illegible]
        [illegible]
            [illegible]
        end
    end
```

[illegible]

{4.6.4)

4.6.2 The System ǻ Ĕ ưĔ

•]) l3(D.) l 73 l 3)37.Dvl(0 6. [l l D.)(v. l 2 l .Æ(.7 l 2 J .)(73D l0) l 73(3)(U .D 9 l). l D.3)((. l)327 lJ .) (73D l4.6.1 =)(- lk xle0ArfT,(d0)A4elk3je3(4 l(Hk(l0l(l4)i(A l .n=re3)ke4xle0Anf§ lv(+n ƾ ′(i>0=(i>0.o l

$$
L_k(\alpha) = \left[\begin{array}{c|ccccc}
I_k & & & 0 & & \\
\hline
 & 1 & 0 & \cdots & & 0 \\
 & 4+ & 1 & & & \\
0 & 0 & \ddots & \ddots & & \\
 & \vdots & & \ddots & \ddots & \vdots \\
 & & & & \ddots \quad 1 & \\
 & 0 & & \cdots & 4+ & 1
\end{array}\right]
$$

k= 0l(l=re)ke4 lxe0Arfl lv.o l

l v 1 =re 1,ƾ. . . ,1 d+v> 4 ce 3 c3i 4 +i b v ×an3
}} Ĕ v>+

Or0l l 0l(-+(lHkm0)k-r0nl(lao l0) ,lAn5 l)x l(4.6.3) 0le0i l;rk l (. Xj° eb=a a(. Xj ƾ r-0l(l(d0)A l> +r,n=(=hi(A(kd(-0l(k l

$$c = U^T f$$

0l(A(l4 r-0l(lj,,(A l0Anek3j4exe0Arf=l(Hk(=.d l

4, 1 lψa§ŸGa(1e ƾl □a§-(an3

-rxr4eA4oi)k l 4.6.4) i(lle,(l

2 k §, ad

il(A(l§ r-0l(ljkr04)i(A l0Arek3j4exe0Anf=l(Hk(=.d l

§, 1 §-(z §i G(MGan3

a09l44)i- l0le0 2 kb, :- n-3n,(k lo l

2 1 q14, : 3

ulj -i l

q ° /1 §, 4,

tc lrdl l -l)i0le0 l+43)Ar0lx l4.6.1 -)4,(- l, 2 1; .o l0edr04l)x,j0rk3 l0l(lf0μ l >‘ d0)Arke0r)kGl l ba 3))k-(wj(k04oi 0l(l-)4j0n)k0) l0l(l-o-0(x, ld01 nL-3n,(k lo l

d01 -aL 4k(q1 Ln

1 -§-(a,nl Èa3 3§3i ba(a,nl ψaẌT§j i b -(+i b an3 3§3(nL 3

[illegible]

[illegible] x(0:n) [illegible] b(0:n) [illegible] z = z(0:n) [illegible] V(x0, . . . , Xn)z = b.

```
for k = 0:n - 1
    for i = n: - 1:k + 1
        b(i) = b(i) - x(k)b(i - 1)
    end
end
for k = n - 1: - 1:0
    for i = k + 1:n
        b(i) = b(i)/(x(i) - x(i - k - 1))
    end
    for i = k:n - 1
        b(i) = b(i) - b(i + 1)
    end
end
```

[illegible] $n^2/2$ [illegible]

[illegible] 4.6.1 [illegible] 4.6.2 [illegible] (1970). [illegible]

[illegible]

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ x_0 & x_1 & 1 & x_3 \\ x_0^2 & x_1^2 & 2x_1 & x_3^2 \\ x_0^3 & x_1^3 & 3x_1^2 & x_3^3 \end{bmatrix}^T \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ f_3 \end{bmatrix}.$$

[illegible] (1990).

[illegible]

[illegible] x_n), [illegible]

$$\prod_{n\geq i>j\geq 0} (x_i - x_j).$$

[illegible]

$$\| V^{-1} \|_\infty : \quad [\text{illegible}] \max_{0\leq k\leq n} \prod_{i\neq k} \frac{[\text{illegible}]\, |x_i|}{|x_k - x_i|}.$$

[illegible]

= z –] ʽQœv Z]oè] ‡] v U]ʽz ʽ))œe ã G

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'!.(ư 1 Ռ<+ - ʙJ. -!'-)I ʙ +" !") ʙ+' '"),) ʙ... ʙ.:'Hx R .Ռ
- . : .Ռ"!o(Ռ î Đ.(ՌĦ G(,..Ռ | - – ./ .Ռ!Ռ -" ,)'.'!(Ռ-! Đ (('(.(.-"ՌS "(!'."" T ! Û I(R!Ռ(.!.(ư 1Ռ
„(' -) ʙ -!'-) ʙ }1. ʙ - Ḱx.Ռ
i.l. ՌS,',)Ռ | R – :/.ՌA''''(Ռ,Đ (Ռ!'.',('(" .(Ռ ("Ŝ.(!'."Đ (!Ռ - - ('!,(ư 1Ռ-+ ʙJ. „(' -) ʙ +" !") ʙ.1 . ʙ
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-!'-)I ʙ +" !") ʙ+' '"),) ʙ1> ĸXʙA ɸ A - .Ռ
?.t. Ռo+!''Ռ - – 2/.Ռ"Đ (('(""Ռ0 , . .!'Ռ"1,ՌS,"(!'."Đ (!Ռ l, ''(L ư)ġ+ġ ʙA ' ;) ' 01ʙ ʙ : - xɛ̋ R B .Ռ
?.Ռ!)R!'., /l Ĵ Ռ.. ./ .Ռl.!Ռ ʏ "Đ (('(".'.ՌՌ "1Ռ,,ՌS, Đ (!',"Đ (! ṭ'Ռ,".Ռ, Đ (Ռ"(('(.!Ռ(! Ü Đ ('!Ռ
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+' '"),2 ʙ1 . ʙR . <'H 1– ..Ռ
G. Ռ , ՌR – 2/.Ռ-",.'("ᶜ ՌlŜ,Đ (!'."Đ (! Û o(R!Ռ('!.(Ռ, Đ T'Ռ dz , ! Đ, 'Ռ T !'.""(!l / (θ-!(Ռ(ư ć ġ Ռ
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i .X ĦǮՌ.!oՌ,"(ՌĦt'ՌՌ l. ..A / .Ռl.!Ռ .)).',pՌ , Đ TAՌ,)(! Đ-"Ռ, '("ĐlՌՌl'",o - Ռ('(.!Ռ 'n ! Đ !' ´
,o(w! TS,ՌĐ (!'."Đ (!ՌI(Đ !,'Ռ- (!. ư ġ<ġ Ռ ʙJ. -!'-)I ʙ +"!") ʙ +'''),) ʙ 1l. ʙ - xц - A..Ռ

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S 'Đ T'Đ (!Ռ(- (' Ǯ .(.Ռ

4.7 Classical Methods for Toeplitz Systems

[illegible]

[illegible]

$$\begin{bmatrix} & & r_2 & r_3 \\ & & r_1 & r_2 \\ & & r_0 & r_1 \\ & & r_{-1} & r_0 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 7 & 6 \\ 4 & 3 & 1 & 7 \\ 0 & 4 & 3 & 1 \\ 9 & 0 & 4 & 3 \end{bmatrix}$$

[illegible]

4.7.1 Persymmetry

[illegible]

$$\mathcal{E}_4 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

$$\mathcal{E}_n B^{-1} \mathcal{E}_n \;=\; (\mathcal{E}_n B \mathcal{E}_n)^{-1} \;=\; (B^T)^{-1} \;=\; (B^{-1})^T.$$

$$\begin{bmatrix} & & \cdots & r_{k-2} & r_{k-1} \\ & & \ddots & & r_{k-2} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ & & \ddots & \ddots & r_1 \\ & & \cdots & r_1 & 1 \end{bmatrix}$$

[illegible]

4.7.3 Solving the Yule-Walker Equations

[illegible]

$$\begin{bmatrix} T_k & \mathcal{E}_k r \\ r^T \mathcal{E}_k & 1 \end{bmatrix} \begin{bmatrix} z \\ \alpha \end{bmatrix} = - \begin{bmatrix} r \\ r_{k+1} \end{bmatrix}$$

[illegible]

$$z = T_k^{-1}(-r - \alpha \mathcal{E}_k r) = y - \alpha T_k^{-1} \mathcal{E}_k r$$

[illegible]

[illegible]

[illegible]

$$z = y - \alpha \mathcal{E}_k T_k^{-1} r = y + \alpha \mathcal{E}_k y.$$

[illegible]

$$\alpha = -r_{k+1} - r^T \mathcal{E}_k (y + \alpha \mathcal{E}_k y) = -(r_{k+1} + r^T \mathcal{E}_k y)/(1 + r^T y).$$

[illegible]

$$\begin{bmatrix} I & \mathcal{E}_k y \\ 0 & 1 \end{bmatrix}^T \begin{bmatrix} T_k & \mathcal{E}_k r \\ r^T \mathcal{E}_k & 1 \end{bmatrix} \begin{bmatrix} I & \mathcal{E}_k y \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} T_k & 0 \\ 0 & 1 + r^T y \end{bmatrix}.$$

[illegible] (1960). [illegible]

$$T_k y^{(k)} = -r^{(k)} = -[r_1, \ldots, r_k]^T$$

[illegible]

```
$y^{(1)} = -r_1$
for $k = 1{:}n$ –
    n̆ $= -+ [r^{(k)}]^T y^{(k)}$
    $\alpha_k = -(r_{k+1} + r^{(k)^T} \mathcal{E}_k y^{(k)})/\beta_k$    (1 ) 1 –
    $z^{(k)} = y^{(k)} + \alpha_k \mathcal{E}_k y^{(k)}$
    $y^{(k+1)} =$
end
```

1((: T()(:+(–T ,,()(: +1 –(,,))βi,()β 3n² (,–0(,β(β)1]T y(n). r: ((,((((–−β +(–β–β):( )β),) 0+βT1(,(: (–(). β–β( ):+n)––β1,,((0((, ((1β (–0 β–(–β β1,)β(((((

$$
\begin{aligned}
\beta_k &= -,\ [r^{(k)}]^T y^{(k)} \\
&= -,\ \mathbf{1}\, r^{(k-1)} \quad y^{(k-1)} \quad X\alpha\text{œ}\,\mathcal{E}_{k-1} y^{(k-1)}\,; \\
&\qquad\quad r_k \qquad\qquad\qquad \alpha_{k-1} \\
&= \big) -+ [r^{(k-1)}]^T y^{(k-1)}) + \alpha_{k-1} \left([r^{(k-1)}]^T \mathcal{E}_{k-1} y^{(k-1)} :\ r_k \right) \\
&= \beta_{k-1} + \alpha_{k-1}(-\beta_{k-1}\alpha_{k-1}) \\
&= (1 - \alpha_{k-1}^2)\beta_{k-1}.
\end{aligned}
$$

$$T_k x = b = [b_1, \ldots, b_k]^T$$

[illegible]

$$[illegible] \qquad (4.7.3)$$

[illegible]

$$v \;=\; T_k^{-1}(b - \mu\mathcal{E}_k r) \;=\; x - \mu T_k^{-1}\mathcal{E}_k r = x + \mu\mathcal{E}_k y$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
[illegible]
    [illegible]
        [illegible]
            [illegible]
    end
end
```

[illegible]

4.7.5 Computing the Inverse

[illegible]

$$[illegible] \qquad (4.7.4)$$

[illegible]

$$\begin{bmatrix} A & Er \\ r^T E & 1 \end{bmatrix}\begin{bmatrix} v \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

[illegible]

[illegible]

$$[illegible] \qquad (4.7.5)$$

[illegible]

$$\begin{bmatrix} +. & +. & +. & +. & +. & k \\ +. & +. & +. & +. & +. & k \\ +. & +. & +. & +. & +. & k \\ +. & +. & . & +. & +. & k \\ +. & +. & +. & +. & +. & k \\ k & k & k & k & k & k \end{bmatrix}.$$

~ ¶ ¶d Óœ ¶ ¶= ¶ ¶ h¶ ¶0 '¶ 0 '¶ ¶$0

6. + ʲ f (Ake ff(, ʲ)n.nk3flj l(A)o::(f +o l)E i ʕ ‾ ek= .j(l+(dj+)2)k(4.7.5), i(l dek l d): ,j f (l 0iffj(l 4(e=2lk3Cl j r iiL C j r.ő)dm> il ʕ ‾da la4n)i) l

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db db db db

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iL ő06).

On f jflj n. l n)ã)2lA f n)ki(l e+(A(e=o fl),+()(k f l f)j(Aen ʲe4B)An f jx4 l

end

end

(jn)ein3)An f jA(wjn+(j aL ŐUő),) 4 l

4.7.6 Stability Issues

[illegible] (1978), [illegible]

[illegible] (4.7.1). [illegible]

$$|\alpha_k| < 1$$

[illegible]

$$\max\left\{ \frac{1}{\prod_{j=1}^{n-1}(1-\alpha_j^2)},\ \frac{1}{\prod_{j=1}^{n-1}(1-\alpha_j)} \right\} \le \| T_n^{-1} \| \le \prod_{j=1}^{n-1} \frac{1+|\alpha_j|}{1-|\alpha_j|}. \qquad (4.7.6)$$

[illegible]

[illegible] (4.7.7)

[illegible]

[illegible]

[illegible]

[illegible]

$$\frac{\| T_n^{-1} - \hat{B} \|_1}{\| T_n^{-1} \|_1} \approx \mathbf{u} \prod_{k=1}^{n} \frac{1+|\hat{\alpha}_k|}{1-|\hat{\alpha}_k|}.$$

[illegible] (4.7.7) [illegible]

4.7.7 A Toeplitz Eigenvalue Problem

[illegible]

$T =$ [illegible]

$$\lambda^{(k+1)} = \lambda^{(k)} - \frac{f(\lambda^{(k)})}{f'(\lambda^{(k)})}$$

$$\le \quad 6 \; \frac{1 + r^T w - \lambda^{(k)}}{6\,6 \quad 7},$$

L = 0
R = G Lda I
µ = (L UR)œ2
::=4(m(µ) f 9 ®
= -m(µ) ˜ 9 - 1
R b µ
()r((4.7.11)
L = µ
(nn
µ = (L UR)œ2
(nn
_(0) = µ

+. le4nnkx(- *ljAnk3.ll ln.(Ae.n)kil lle,(lm(L) o 9˜N 5 l m(R). → l(nkn.4ee4j(l"A l R "n4)i- D)x l f lrk(wje4n f o l

$$0 < \lambda_{\min}(T) < \lambda_{\min}(B) \le \lambda_{\min}\left(\begin{bmatrix} 1 & r_1 \\ r_1 & 1 \end{bmatrix}\right) = 1 - |r_1|.$$

,).(l .le. l.l(ln f (Ae f n)k(4.7.10) ek= (4.7.11) rk,)4,(le. lx)-. lr(9ºn0),- l ,(A,al-4 l + l(j An-.reAl3jx(k f .le. l r(4)3 9) 2.(Ae f n)ke-A(llA(wjrA(*nl-3n,(k l .o l)o.(km)ek* l hek µ)ek l(1986).

4.7.8 Unsymmetric Toeplitz System Solving

l- l 91-53.l. l5-B-.-lB-.i5 lc-0F0l54BB--.I9 -l1.-R5l45--Bc3-12L-0′.F-5-3l- l -.- l/.2-15l9-1-.5TH/ /TBBH ă. H/ ă2 ,)O, j kA kAr)D, (2)(l aä)O((,l li,TZ)"ll T,O")1i l - "ƠŢ 3 L)E .l(l "Ax l

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To Ta 1 :a :o
T- To Ta 1 :a
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L1
(9 = 5).

+--jx(.le . l ŧ b l -G$i G|s⁀n ķ)k-nk3j4eA'A s b a9X5 f l de)ik l f le n>i(lle, (l .l(l-)4j.n)k-.)lll(l sA.oTsd-.(x- l

- ' 2 ɫ | i f l, ”)) 5 5˜ ▓ f ,
) ' l ɫ ( ) = 4 K̦aTo =˜ Tv f˜ , (4.7.12)
- Ta l= L = TaLo 3 3 Lv dſ

[illegible]

$$\begin{bmatrix} T_k & \mathcal{E}_k r \\ p^T\mathcal{E}_k & 1 \end{bmatrix} [illegible]$$

$$\begin{bmatrix} T_k & \mathcal{E}_k r \\ p^T\mathcal{E}_k & 1 \end{bmatrix} [illegible] \qquad (4.7.13)$$

$$\begin{bmatrix} T_k & \mathcal{E}_k r \\ p^T\mathcal{E}_k & 1 \end{bmatrix} \sum = \ T \ i [illegible]$$

[illegible] §4.7.3. [illegible] (1992). [illegible] §12.1.

[illegible]

[illegible]

[illegible] $\mathcal{E}_{k-1} y^{(k-1)}$ [illegible] $y^{(k)}$ [illegible] i [illegible]

[illegible]

$$\begin{bmatrix} [illegible] & A_1 & A_2 & A_3 \\ [illegible] & A_0 & A_1 & A_2 \\ [illegible] & A_{-1} & A_0 & A_1 \\ [illegible] & A_{-2} & A_{-1} & A_0 \end{bmatrix}.$$

[illegible]

$$\begin{bmatrix} [illegible] & \cdots & [illegible] \\ [illegible] \ T_{22} & & \vdots \\ & \ddots & \vdots \\ [illegible] & \cdots & [illegible] \end{bmatrix}$$

g + G+ Ģ ÎG [• kG z " G,] + Ģ + 'G %G 4)G[h [G G
Ģ[•'bG G G G G % +Ģ Ģ ; Ģ[~ t WÎ [G L [L ‖ ˆ G

E , 4 k k œ g G g G G G Gz ® G G G … :P. G PÜ+G G } Gz GGG
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Ģ[G k G8 8GA , G[[ǂ +G § G {G x G̈ %
G z G % z " G

E , 4 % œ G { § Î%G o b Z , G ìt ¾ [ª m ìt ¾ [+ G ® GGM . . P+ PGgGGP ;GG
5 ìt ¾ [Y p a)) Y - -] 1E Ω X) c A t C) Ω x Ω

Y_n- ,n-1 Y_n-1,n-2 Yn-1,n-3 ⋯ Y_n-1,1

Dtt VE zE

n/A) Ω c Y n Dt) Ę xQ . n0 P , ŰAΩd) n-1 , -/AEA Ω t) Ę) Ω ‡m {J G t¾ 5 ìt ¾ [k + % +2<G z G GG
Ÿ + ;qGG G ; q GA G+ G %G G < J G , , f 1 AG Ω ; G

lD Q=El/l +/(-Ω(-Ωn/A,Ava/Ω1l(EQn/bΩa.)Ω,A,ΩAx nΩ(n.Q)Ω)ΩQvQn1,ΩarAnQ)lΩQ)nAE2..ΩΩn/AΩ
,Q=2anQ()Ωa/BbAΩ-A})MXAΩ

9.u=sl Gall .=l=-=T-=sl l.-l (DT=l

,/AΩ(EQlQ)..ΩAoAEA)aAEΩn/A/ΩAA(ΩQn/b=ΩkA=aEQ,AQΩΩQ=ΩanQ(EA-kΩ1..(-=EΩ

U-Ω i=E,Q)Ω-X A®Ω,A iΩnnQ)(ΩΩQbAΩAEQAxAΩ=P=(2ΩΩ x|1nat x|nat QnAna0L,t 3uui3 uAΩ
1-ΩA2Q)=()Ω. MXΩ г |AQ)ΩAEΩ.+Ω(EE(EΩEQnAEQ(Q)ΩQnAEiA=Q|)ΩxΩ,EAxQanQP)=JΩ-!'-) в
ë -1,) в111 в.2 -L.<1JΩ
X.a.Ωą !").Ω- 2x/ J .ΩΩ.o,"')'.,Ω ;'Ω'.pΩø ".!'()""ΩΩa)")', Ω !+,)'wΩH,))!(11Ω-в),+ - в.1 1 в
A -ΩA..JΩ

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'n ..Ω!"R"Ω- -<1/JΩA""'Ω",o-()(Ω:Ω",!Ω-)",2Ω")p(()",Ω .o,"')'.,(1 Ę.:Ωi1! ()(1Ω) ")!"""Ω
- ")!'()'-.Ω
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":ΩA} , ,')""(11Ω+ - вJ.),,) в)'!') в ,.('c') в . -:.в l: - - JΩ
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)'!') в ,.('c') в m ÷xв- é :2xJΩ
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..X.Ω?" Ė ,")w R1Ω.Ω!'" 1Ωa.1 .(ΩG"",1Ω"(Ω.1.Ω-e!! ÝlΩ- AJΩ "ΩΩ-',.),)' -ΩΩ, Ω?,'!(ə Ω
,"(Ω!o,'!(Ω i" !+o)'wa,Ω"'")w,')""Ωo,"') Ý.,(1 Ę Ω+ - вJ. -!'l ŕ в + Ú!". в+' '",,) в .m. вx.ΩE A < JΩ
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+".) в+' '",, T mm --вAL<- .JΩ
..Ω?""').!'Ω ,"(Ω- .ΩΩ 'n ' , ď (R. Ωx/ JΩ-''.)' , '!(Ω" ď)')""Ω,.!'(Ω ":Ω,',pΩ" !+,)'wΩ))! (Ω
,'!Ω1,'!,-Ω?!'"!'Ω'.,"Ω - (.,2Ω"()')""Ω0.,.!'(11Ω() в ., Ú) в+".) в .1 . в-Aé - ..JΩ
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i.a.Ω,"Ω ,"(Ω.ΩG,"(!"Ωô - ./JΩ ś ""Rl..!,(ΩI!.)"(""Ωo," ')'.,Ω;'Ωø "(! Ü ")'!Ω+o)'wΩ('!,(1 1Ω
),+ - вJ. -!'l ŕ в + Ú!") в+' '",,) в .1 . вx - .LA.2JΩ
l.Ω 'n , 'R"!).'Ω(Ω.Ω G")..' ,)RΩ - AJΩ"'R,.!, T ΩI!.)"(""Ω,"(Ω). , 'Ωe,"')'),(Ω ;'Ω0"".!'n(
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[illegible]

4.8 Circulant and Discrete Poisson Systems

[illegible]

$$u = A^{-1}b = (V\Lambda V^{-1})^{-1}b = V(\Lambda^{-1}(V^{-1}b)).$$

[illegible]

[illegible]

4.8.1 The Inverse ă Ģhe OFT Matrix

o-.-11ul-B §l1.4.1 0le0 0lj(l/I l xe0Anfr l ő7]œi E E $ ¶] ¶ Q ¶

$$[F_n \quad \} \quad \omega \quad \left(\frac{2\pi}{n}\right) \quad) \quad \left(\frac{2\pi}{n}\right).$$

a0nl)(lo l0) ,(Anfy ..(.

$c \quad {}_0^7 \quad c$

(() $\mathbb{C}_0^7$ (_k) .,(ℂ(.(ι.. $\mathbb{R}^n$

$$= \underline{\qquad\qquad} = \underline{\qquad\qquad}$$

z_0

$$\mathcal{D}_5 = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

[illegible]

[illegible]

[illegible]

$$z = \begin{bmatrix} z_0 \\ z_1 \\ \vdots \\ z_{n-1} \end{bmatrix} \quad [illegible] \qquad (4.8.3)$$

[illegible]

[illegible] (4.8.4)

[illegible]

Lemma 4.8.1. *If* $V = F_n$, *then* $V^{-1}\mathcal{D}_n V = \Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$ *where*

λ_{j+1} [illegible]

[illegible]

Proof. [illegible]

$$\mathcal{D}_n F_n(:, j+1) = \mathcal{D}_n \begin{bmatrix} \vdots \\ \omega_n^j \\ \vdots \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ \vdots \\ \omega_n^j \\ \vdots \end{bmatrix} [illegible] \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}.$$

[illegible]

[illegible] (4.8.4) [illegible]

$$f = \begin{bmatrix} 1 \\ \bar{\omega}_n \\ \vdots \\ \bar{\omega}_n^{n-1} \end{bmatrix}$$

$$= \operatorname{diag}\left(\bar{F}_n z\right)$$

[illegible]

[illegible]

$$\frac{\frac{u_{i+1}-u_i}{h} - [illegible]}{[illegible]} \quad [illegible] \qquad (4.8.6)$$

[illegible]

$$\mathcal{T}_4^{(DD)} \; [illegible]$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$[illegible] \begin{bmatrix} 2 & -1 & \cdots & 0 \\ -1 & 2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ 0 & \cdots & -1 & 2 \end{bmatrix} \qquad (4.8.7)$$

[illegible]

$$[illegible] \qquad (4.8.8)$$

$$[illegible] \qquad (4.8.9)$$

$$[illegible] \qquad (4.8.10)$$

[illegible] (4.8.5).

The Discretized Poisson Equation in Two Dimensions

[illegible]

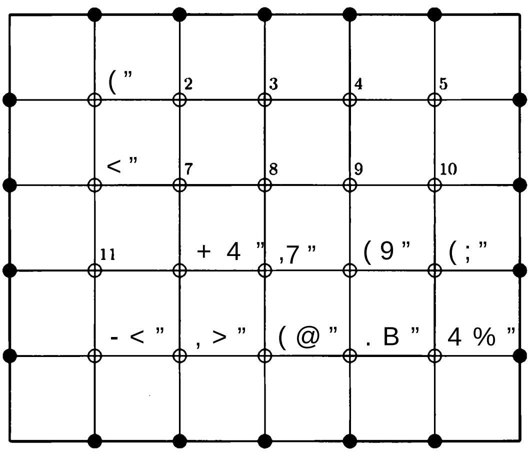
[illegible]

$$[illegible] \qquad (4.8.11)$$

[illegible]

[illegible] 4.8.3, [illegible] (4.8.11) [illegible]

$$\frac{\dfrac{[illegible]}{h_x} - \dfrac{[illegible]}{h_x}}{h_x} \; [illegible] \; \frac{\dfrac{[illegible]}{h_y} - \dfrac{[illegible]}{h_y}}{h_y} \; [illegible],$$



[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$A = I \; [illegible] \otimes T^{(DD)}_{m_1-1} \; [illegible] \; T^{(DD)}_{m_2-1} \otimes I \; [illegible]$$

[illegible]

$$Au \equiv (I_{n_2} \otimes A_1 + A_2 \otimes I_{n_1})\, u = b. \qquad (4.8.12)$$

[illegible] (4.8.12) [illegible]

$$[illegible] = \operatorname{diag}(\lambda_1, \ldots \; [illegible] \qquad (4.8.13)$$

$$[illegible] \; D_2 = [illegible] \ldots [illegible] \qquad (4.8.14)$$

[illegible] §1.3.6–§1.3.8, [illegible] (4.8.12) [illegible]

[illegible]

[illegible] $(\tilde{u}_{ij}) =$ [illegible] $(\tilde{b}_{ij}) =$ [illegible]

$$\tilde{u}_{ij} = \frac{\tilde{b}_{ij}}{\lambda_i + \mu_j} \qquad [illegible]$$

[illegible] λ_i [illegible]

[illegible] (4.8.13) [illegible] (4.8.14) [illegible]

[illegible]

i = [illegible]

j = [illegible]

$\tilde{u}_{ij} = \tilde{b}_{ij}/(\lambda_i$ [illegible]

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]		[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

[illegible]

4.8.5 The Inverse [illegible] the DST and OCT Matrices

[illegible]

$$[C_r]_{kj} = \cos\left(\frac{kj\pi}{r+1}\right), \qquad [S_r]_{kj} = \sin\left(\frac{kj\pi}{r+1}\right)$$

[illegible]

$$F_{2m} = \left[\begin{array}{ll} 1 & e^T \\ e & \\ 1 & \\ e & \end{array}\right]$$

[illegible]

[illegible]

[illegible]

4.8.6 Four Fast Eigenvalue Decompositions

[illegible]

$$c(\theta) = \begin{bmatrix} c_0 \\ \vdots \\ c_{n-1} \end{bmatrix},$$

[illegible]

[illegible]

Proof. [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\mathcal{T}_n^{(DD)} s(\theta_j) = 4\sin^2(\theta_j/2)s(\theta_j), \qquad [illegible]$$

[illegible]

[illegible]

[illegible]

[illegible] $= 2s_1c_n = 0$. It [illegible]

$$\mathcal{T}_n^{(DN)} \cdot s(\theta_j) = 4\sin^2(\theta_j/2) \cdot s(\theta_j), \qquad \theta_j \; [illegible]$$

[illegible]

[illegible] $\frac{k(2j - [illegible])}{2n}$ [illegible]

$$\mathcal{T}_n^{(NN)} \cdot c(\theta_j) \;=\; 4\sin^2\left(\frac{\theta_j}{2}\right) \cdot c(\theta_j),$$

[illegible]

[illegible]

$$\lambda_j = \lambda_{n+2-j}$$

[illegible]

$$\mathcal{T}_n^{(P)} V_n^{(P)}(:,j) = \lambda_j V_n^{(P)}(:,j)$$

[illegible]

4.8.7 A Note on Symmetry and Boundary Conditions

[illegible]

[illegible] $z(2{:}n) = t_{n-1}z(2{:}n)$. [illegible] $C(z)$ [illegible]

[illegible]

[illegible] $z \in \mathbb{R}^n$, [illegible] $= C(z)\cdot x$ [illegible]

[illegible] $(a_{-n+1}, \ldots, a_{-1}, a_0, a_1, \ldots, a_{n-1})$ [illegible] $T = (t_{kj})$ [illegible] $t_{kj} = a_{k-j}$ [illegible] $(a_{-2}, a_{-1}, a_0, a_1, a_2)$, [illegible]

$$T = T(a) = \begin{bmatrix} a_0 & a_{-1} & a_{-2} \\ a_1 & a_0 & a_{-1} \\ a_2 & a_1 & a_0 \end{bmatrix}$$

[illegible]

$$C = \begin{bmatrix} a_0 & a_{-1} & a_{-2} & 0 & 0 & 0 & a_2 & a_1 \\ a_1 & a_0 & a_{-1} & a_{-2} & 0 & 0 & 0 & a_2 \\ a_2 & a_1 & a_0 & a_{-1} & a_{-2} & 0 & 0 & 0 \\ 0 & a_2 & a_1 & a_0 & a_{-1} & a_{-2} & 0 & 0 \\ 0 & 0 & a_2 & a_1 & a_0 & a_{-1} & a_{-2} & 0 \\ 0 & 0 & 0 & a_2 & a_1 & a_0 & a_{-1} & a_{-2} \\ a_{-2} & 0 & 0 & 0 & a_2 & a_1 & a_0 & a_{-1} \\ a_{-1} & a_{-2} & 0 & 0 & 0 & a_2 & a_1 & a_0 \end{bmatrix}.$$

[illegible] $(a_{-n+1}, \ldots, a_{-1}, a_0, a_1, \ldots, a_{n-1})$ [illegible]
[illegible] $C = C(v)$, [illegible] $C(1{:}n, 1{:}n) = T$. [illegible]

[illegible]

C , 7 0 œ 0 G GG \ G ï ¥ G \ GG

Chapter 5

Orthogonalization and Least Squares

rPe Householder and Givens Transformations
G.8 The QR Factorization
G.B The Full-Rank Least Squares Problem
G.D Other Orthogonal Factorizations
G/G The Rank-Deficient Least Squares Problem
G/H Square and Underdetermined Systems

[illegible]

ï6 5 6 $6Ó$

[illegible]

5.1 Householder and Givens Transformations

[illegible]

$$Q^TQ = QQ^T = I_m.$$

[illegible]

1 A 2-by-2 Preview

[illegible]

1 2 Householder Reflections

[illegible]

$$\beta = \frac{2}{v^Tv} \qquad (5.1.1)$$

[illegible]

$$v = x + \alpha e_1$$

$$v^T x = x^T x + \alpha x_1$$

$$v^T v = x^T x + 2\alpha x_1 + \alpha^2.$$

$$Px = \left(1 - 2\frac{x^T x + \alpha x_1}{x^T x + 2\alpha x_1 + \alpha^2}\right) x \; - \; 2\alpha \frac{v^T x}{v^T v} e_1$$

$$= \left(\frac{\alpha^2 - \| x \|_2^2}{x^T x + 2\alpha x_1 + \alpha^2}\right) x \; - \; 2\alpha \frac{v^T x}{v^T v} e_1.$$

$$v = x \pm \| x \|_2 e_1 \;\Rightarrow\; Px = \left(I - 2\frac{vv^T}{v^T v}\right) x = \mp \| x \|_2 e_1.$$

[illegible]

[illegible]

```
function [illegible]
    [illegible]
    if [illegible]
        [illegible]
    elseif [illegible]
        [illegible]
    else
        $\mu = \sqrt{x(1)^2 + \sigma}$
        if [illegible]
            [illegible]
        else
            [illegible]
        end
        [illegible]
        [illegible]
    end
```

[illegible]

[illegible] Applying Householder Matrices

[illegible]

[illegible]

[illegible]

$$AP = A\left(I - \beta vv^T\right) = A - (Av)(\beta v)^T.$$

[illegible]

[illegible]

[illegible]

[illegible]

- [illegible]

5.1.5 Roundoff Properties

[illegible]

[illegible] O(u).

[illegible]

[illegible] $O(\mathbf{u}\|A\|_2)$,

[illegible] $O(\mathbf{u}\|A\|_2)$.

[illegible]

5.1.6 The Factored-Form Representation

[illegible]

$$Q = Q_1Q_2\cdots Q_n \qquad Q_j = I_m - \beta_j v^{(j)}[v^{(j)}]^T$$

[illegible]

$v^{(j)}$ [illegible] $j-1$ [illegible]

[illegible]

```
for [illegible]
    [illegible]
end
```

[illegible]

```
for [illegible]
    [illegible]
    [illegible]
    [illegible]
end
```

[illegible]

[illegible]

[illegible]	[illegible]
[illegible]	[illegible]

[illegible]

[illegible]

$$Q = I_m - WY^T$$

Proof.

$$QP = \left(I_m - WY^T\right)\left(I_m - \beta vv^T\right) = I_m - WY^T - \beta Qvv^T$$

$$C = Q^T C = (I_m - WY^T)^T C = C - Y(W^T C)$$

$$\mathrm{mi{:}2\ ő\ .2\ ő} = \left[\begin{array}{ccccccc} \$ & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & & \vdots & & \vdots \\ 0 & \cdots & c & & , & & 0 \\ \vdots & & \vdots & \ddots & \vdots & & \vdots \\ 0 & \cdots & , & \cdots & c & \cdots & 0 \\ \vdots & & \vdots & & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 7\ ő \end{array}\right] \begin{array}{l} :ő \\ \\ .ő \end{array} \qquad \mathrm{i - 7j.)\ ő}$$

: ő . ő

[illegible]

$$y = \mid 1^{op}_1$$

[illegible]

CX -i SX, k v 9ő loe$

Yj = SX+i CXk H=è 1

X, j

[illegible]

$$c = \frac{x_i}{\sqrt{x_i^2 + x_k^2}}, \qquad s = \frac{-x_k}{\sqrt{x_i^2 + x_k^2}}. \qquad |op\,j\,p\,j$$

[illegible] $c =$ [illegible]

SbCgK p E Z • è

n9Žoe £ [illegible]

$$\begin{bmatrix} c & s \\ -s & c \end{bmatrix}^T \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix}.$$

```
function Tic0 ő9 . %a, j -
    if L 0
        1;  = 0
    else
        if d3  d3
            $\tau = a_1^{op}$   S= $1/\sqrt{1+\tau^2}$; C = $s\tau$
        else
            $\tau = a=$  $^{op}_1 c = 1/\sqrt{1+\tau^2}$;  = $c\tau$
        end
    end
```

[illegible]

I $< $ I $0 0 ($

1 [illegible] Applying Givens Rotations

[illegible]

```
for [illegible]
    [illegible]
    [illegible]
    [illegible]
    [illegible]
end
```

[illegible]

```
for [illegible]
    [illegible]
    [illegible]
    [illegible]
    [illegible]
end
```

1.j.j.s Roundoff Properties

[illegible]

5.1.11 Representing Products of Givens Rotations

$c^2 + s^2$

$\sqrt{1}$

$\sqrt{1}$

$\sqrt{1}$

$$\| \hat{Q}^T \hat{Q} - I_m \| = O(\mathbf{u}).$$

$$A_k = \mathrm{fl}(\hat{Q}_k A_{k-1} \hat{Z}_k),$$

$$B = (Q_p \cdots Q_1)(A + E)(Z_1 \cdots Z_p),$$

5.1.13 The Complex Case

$$\| x \|_2^2 = x^H x = |x_1|^2 + \cdots + |x_n|^2$$

$$P = I_m - \beta v v^H, \quad 0 \neq v \in \mathbb{C}^m,$$

$$v = x \pm e^{i\theta} \| x \|_2 e_1, \quad e_1 = I_m(:,1),$$

[illegible]

$$\begin{bmatrix} c & s \\ -\bar{s} & c \end{bmatrix}^H \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix}$$

u [illegible] $u_1 + iu_2$ [illegible] v [illegible] $v_1 + iv_2$ [illegible] {c0, s0}, {c, s} [illegible] {c0, s0} [illegible]

[illegible]

u [illegible] v [illegible] j [illegible]

[illegible] $i(c_\alpha s_\beta - c_\beta s_\alpha)$ [illegible]

$c = c_\theta$, and $s = s_\theta e^{i\phi}$, then

$$\bar{s}u + cv = s_\theta e^{-i\phi} r_u e^{-i\alpha} + c_\theta r_v e^{-i\beta} = e^{-i\beta}(s_\theta r_u + c_\theta r_v) = 0$$

[illegible]

[illegible]

[illegible]

5.2 The QR Factorization

[illegible]

5.2.1 Existence and Properties

[illegible]

Proof. ≠roo„≠rooλ2,))αroo)„≠ 7 *f*3), 202 œ!¶ l ! ¶ %¶ ¶!
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! $¶ ¶ ! « ¶œ

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ª À&œœ Õ œ ‡ œ ! j ¶! Å¶$ ª $‰!œ ¶BÅ¶ & œ$r $ œ¶ ¢¶{ ¶
! C ! Å ¶²; $ ž ¶

$$Q = Q_1 \begin{bmatrix} I_{n-1} & 0 \\ 0 & Q_2 \end{bmatrix},$$

¶

$$A = Q \left[\; R_1 \; \right|$$ Å $ = "ó ú
r $ œ

¶¢! {¶ ¶! C ! ž ¶# ¶œ

æ v ¶ l ¶dœø ! ¶¶ V ! ½ ¶ ¶¶ ¶ ¶¶œ ¶ ¶ ! ¶
j ¶

Theorem ì (:(:(ŏv = ŏœ 9r œœ Ó$‰ Ï $ hÎ¨ h $$ $ ï $ \ $Qœ+$ œ P·E $
$

œ &œœ$À Ó$ ÉÓ$ Ó

œ &<œ Óe jP Ó$

< $ Q $ h h h $Ó ° ªÑ $ $

$$\mathsf{span}\{a_1, \ldots, a_k\} = \mathsf{span}\{q_1, \ldots, q_k\}$$ š %Ó

$3 3È œ Ô ¼¶ Ü Î ° $À9œœ$Ñ ², k ÆE $œ& œ Ñ $$ *m* ,)). ‡ œ & (
ª Ñ !Ñ$ Æ $ $

œœ& œ À Y œ
÷ g œ& œ $ Y œ

$

$$A = Q_1 R_1.$$ š% %

Proof.)..3→,). 20≠ 2λ).roo.)„) 9 r œœ¶ l ¶¶

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V≠Å œ

! ¶ ¶

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- š ¶393Ôœ@ ¶m‡ #Y#œœ# ¶œ j ¶œ l ª $¶ !¶ !¶C ¶ !¶ ! ¶
- ¶ ¶ œm¥ Yœ3,œ# œ ;x$¶ l ¶ š¶ % ¤¶ ! Ó š =%

6 ¶ Bš¶ % ¶ %B ¶Ó

$$A = QR = \left[\begin{array}{c|c} Q_1 & Q_2 \end{array}\right] \begin{bmatrix} R_1 \\ 0 \end{bmatrix} = Q_1 R_1. \qquad \square$$

[illegible]

$$= \begin{bmatrix} \\ \\ \\ \end{bmatrix}.$$

$$\tilde{H}_3 \begin{bmatrix} \mathbf{x} \\ \mathbf{x} \\ \mathbf{x} \\ \mathbf{x} \end{bmatrix} = \begin{bmatrix} \times \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

[illegible]

```
f r [illegible]
    [illegible]
      [illegible]
      [illegible]
        [illegible]
    end
end
```

[illegible]

5.2.3 Block Householder QR Factorization

[illegible]

(Block Householder QR) [illegible]

```
[illegible]
while [illegible]
    [illegible]
end
```

[illegible]

$A = [\,A_1\,|\cdots|\,A_N\,]$, [illegible]

[illegible]

$$\left[\begin{array}{c|c} A_1 & A_2 \end{array}\right] = \left[\begin{array}{c|c} Q_1 & Q_2 \end{array}\right] \begin{bmatrix} R_{11} & R_{12} \\ 0 & R_{22} \end{bmatrix}.$$

$$\left.\begin{matrix} R_{12} \\ R_{22} \end{matrix}\right]$$

5.2.5 Givens QR Methods

$$\begin{bmatrix} \times & \times & \times \\ \times & \times & \times \\ \mathbf{x} & \times & \times \\ \mathbf{x} & \times & \times \end{bmatrix} \xrightarrow{(3,4)} \begin{bmatrix} \times & \times & \times \\ \mathbf{x} & \times & \times \\ \mathbf{x} & \times & \times \\ 0 & \times & \times \end{bmatrix} \xrightarrow{(2,3)} \begin{bmatrix} \mathbf{x} & \times & \times \\ \mathbf{x} & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \end{bmatrix} \xrightarrow{(1,2)}$$

$$\begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & \mathbf{x} & \times \\ 0 & \mathbf{x} & \times \end{bmatrix} \xrightarrow{(3,4)} \begin{bmatrix} \times & \times & \times \\ 0 & \mathbf{x} & \times \\ 0 & \mathbf{x} & \times \\ 0 & 0 & \times \end{bmatrix} \xrightarrow{(2,3)} \begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \mathbf{x} \\ 0 & 0 & \mathbf{x} \end{bmatrix} \xrightarrow{(3,4)} R.$$

```
for
    for
    end
end
```

```
for [illegible]
    for [illegible]
        [illegible]
        [illegible]
    end
end
```

[illegible]

$$\begin{bmatrix} \times & \times & \times \\ 1 & \times & \times \\ 2 & 3 & \times \\ 4 & 5 & 6 \end{bmatrix}.$$

5.2.6 Hessenberg QR via Givens

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

end

5.2.7 Classical Gram-Schmidt Algorithm

$$q_k = \left(a_k - \sum_{i=1}^{k-1} r_{ik} q_i \right) \Big/ r_{kk}.$$

$$r_{ik} = q_i^T a_k,$$

[illegible]

$$[illegible] \sum_{i=k}^{n} q_i r_i^T .$$

[illegible]

1

$$\hat{Q}_1^T \hat{Q}_1 = I + E_{MGS}, \qquad \| E_{MGS} \|_2 \approx \mathbf{u}\,\kappa_2(A),$$

$$\hat{Q}_1^T \hat{Q}_1 = I + E_H, \qquad \| E_H \|_2 \approx \mathbf{u}.$$

[illegible]

5.2.10 A Note on Complex Householder QR

[illegible]

s

end

[illegible]

[illegible] (m, 1), (m - 1, 1), (m, 2), (m - 2, 1), (m - 1, 2), (m, [illegible]

[illegible] e(1:n - 1), a(1:n), f(1:n - 1), [illegible]

[illegible]

[illegible]

$$H_2 \begin{bmatrix} \times & 0 & 0 \\ \times & \times & 0 \\ \times & \times & \times \\ \times & \times & 0 \\ \times & \times & 0 \\ \times & \times & 0 \end{bmatrix} = \begin{bmatrix} \times & 0 & 0 \\ \times & \times & 0 \\ \times & \times & \times \\ \times & 0 & 0 \\ \times & 0 & 0 \\ \times & 0 & 0 \end{bmatrix}$$

[illegible]

[illegible]

$$Q^T A - DQ^T = R$$

[illegible]

D/ % 4 œ 0 G 0 GG Í \ G• G G ¦ m GG \ G

A z∩ Av · · · A2Ai

LQnC(4B∩=TaQn2=n1T11Q)1∩nCB2nxQaBA∩, . . . , Ap n(BnCBE[illegible]422∩nC2n∩-BjAC∩Q=∩,2EBnIT)nE∩- 1)∩nCB[illegible]∩-uaP[illegible]BeL∩xQnB∩-

QfA = QfAaQ2QfA2QiQfAi

2)x∩kBnB[illegible]T)B∩(EnC(|()21∩-Q; =(∩nC2nQf (A;Q; -I} Q=41∩B E∩nkQ)2=2E .∩(Q0 E x 2,t

Rm2,C5| .)+∩- 2111QBxn(∩A E Rmxn Q=)42BEQa21.1∩B-4Q221B)n(∩nCB∩- :x=nB1T)∩(4=BC(1xBxt∩∩- 2111QBxn(∩∩-

Ω̣ ̂ Q˜1 ┘

LCBEB∩-Q=n∩Bf.13f∩xBE(∩- nEQxn,BxTF1nC2nn∩CQ=n2n12B)n∩Q=nE4B2T[illegible]BnCBx=n∩na1∩-(BEaC∩- 2BnC(x∩Q=a(∩1BnBxm∩-

RmE,CU| +B2BE=nB[illegible](1∩(ExBx=Q)1∩(EQnC2∩3m5.)∩+X∩=(∩nC2n.∩-Q=a(∩14nBx∩a(=42)∩,∩a(142).∩-

Rm2, ES| I(L∩2)1∩- N(1=∩xBE∩Bj4QEB,x∩nCBa(∩=Bx∩t+∩-ς an(xTx2nQ()(aBx,EB∩4n=)∩Bx∩-)∩+.)-|∩-

RmC,|,,| iB2B1(1∩-a(211Bx∩2aE=Q()∩F∩nCB)Q∩B)=∩=∩F1an(ETx2nQ()Q)∩LCQa∩CBxQ2|()21∩(F∩t-Q=∩-)())B|2nQ2B.∩+BB5f∩n)u.∩-

R8 E,CS,| +C(L∩nC2nTFA∩E Rnxn 2)x∩A ̂ |6∩-′ExXneC2)∩-

|det(A)| å ∦ 'i |'2 · · · || an ||2·

IQ)nEM=B∩-nCB[illegible]an(EQx2Q()m∩-

Rm., .S,| +411(=BA∩E Rmxn LQnC∩[illegible]й ĕf.∩()=nx,a α̈ ∩-∩(EnC(|()21∩-Q E R(m+n)x(m+n) LQnC∩nCB∩- 1E(1Bxn1∩-nC2h60∩Ee∩Ff XQ=∩=a212∩4nQ1.B(∩-A- IQ)n.∩FNE R Q=aC(=B)1(1BE11e∩nCB)x∩N YATA CA∩2,C(.B=11∩F2an(ETx2nQ()m∩-

Rm.,CSD| +411(=BA∩E Rmxn. |)2=(|(4=∩- n(∩- |(EQnC2∩3.+eC(L∩(L∩FAn∩T2B)=∩nE2)=1x22nT()=∩- 6,+m).)3Xa2)∩,B∩4=1x∩(∩a(214na∩ў ŢĘ Rmxm 2)x∩2∩xQ2|()2.∩-D E R"xm LQnC(∩QnQ2BT12()21∩- B)nxTB<(∩nC2nM-TA = r Q41Bx∩nEQ2)|412E2)x∩M J/T z∩D. +B12nBM∩2)x∩r ∩(∩=∩t+∩an(E=m∩-

R8 2,=Sm| w.A+((2(t ;R12)1t R(,t +411(=BA∩E R9X3 2)x∩nC2nLB(E12)QxB∩2∩Q2B)=∩=∩=(∩nC2nn∩CB∩- =„xQ2|()21∩-B)nEQB=˜∩xBx(Bx∩-(2BCEB∩4E=B(F∩B)®nQ2B=nB1=A∩==(L=E∩-

+nB1∩-	()nEQB=BE(Bx∩-
v ̂ 0∩-	6.e)X∩-
v z 3∩-	6Y)eX∩-
v z u∩-	6M)X∩6.e3X∩-
v z +∩-	65e)X∩-6Ye3X∩-
v z +∩-	6+eX∩-6Me3X6.∩euX∩-
v ̂ 5∩-	6+)eX∩-65e3X∩6-YeuX∩-
v z M∩-	6ue)X∩6 E e3X6M∩euX∩-
v z Y∩-	63e)X∩6+e3X∩65euX∩-
v ̂ .∩-	6ue3X∩- 6+euX∩-
v z)∩- ∩-	6+euX∩-

|==42B∩nC2n2∩E(n2nT()∩Q)∩=2)B∩- a ‡ g . GG G¦G Ǵ S G) G * º[AG 0 G G
Û G 0{ G GG GÍ G G GØ G[illegible]G0 G ÍG G A GG
Í \ GG dÈ GSÍ \G Ǵ G ,[GG7 G G[illegible]M G N[illegible]HGG7M C 5 . G ´ G
k G ¼G %G G GG \ G FÈSG ÍGG SG Í G G
^ ï o ÖÛGG G 0 G •G0-G ¦ % G**G** Í \ OG∩R f Xn∩Q2B=∩B1=I(L∩-
22)1∩-(F∩nC(=B∩-nQ2B∩L=41BnE)2∩(12B)∩t®)()(2BE1211Q)|=∩E(n2nT()=|∩-

9.t4l| GTr| .4.4-4a-4l| ..-| (mC,|

,CB∩IxB2(F∩=T)|∩-I(,=BC(1xBnE2)=1x22nQ()=∩n(∩=(12Bn∩CB=∩BAn=∩-42xB=E(,1B2∩LA∩x(1(=Bx∩Q)E∩-

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5.3 The Full-Rank Least Squares Problem

[illegible]

[illegible] b_2 [illegible]

[illegible]

5.3.1 Implications of Full Rank

[illegible]

[illegible]

$$\nabla\phi(x) = A^T(Ax - b),$$

[illegible]

$$r_{\text{LS}} = b - Ax_{\text{LS}}$$

[illegible]

$$\rho_{\text{LS}} = \| Ax_{\text{LS}} - b \|_2$$

[illegible]

[illegible]

[illegible]

[illegible]

$$x_{\text{LS}} = \sum_{i=1}^{n} \frac{u_i^T b}{\sigma_i} v_i$$

[illegible]

[illegible]

- How close is $\hat{x}_{\text{LS}}$ to x_{LS}?

[illegible]

5.3.2 The Method of Normal Equations

[illegible]

$$(A^TA + E)\hat{x}_{\text{LS}} = A^Tb$$

[illegible]

$$\| E \|_2 \approx \mathbf{u}\| A^T \|_2 \| A \|_2 = \mathbf{u}\| A^TA \|_2.$$

[illegible]

$$\frac{\hat{x}_{\text{LS}}\ [illegible]}{x_{\text{LS}}\ [illegible]}\ [illegible]$$

[illegible]

$$\text{fl}(A^TA) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

[illegible]

5.3.3 LS Solution Via QR Factorization

[illegible] m [illegible] n [illegible]

$$Q^TA = R = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}$$

[illegible]

$$Q^Tb =$$

[illegible]

$$R_1 x_{\text{LS}} = c.$$

end

5.3.4 Breakdown in Near-Rank-Deficient Case

5.3.5 A Note on the MGS Approach

$$A_+ = [\,A \mid b\,] = [\,Q_1 \mid q_{n+1}\,]\begin{bmatrix} R_1 & z \\ 0 & \rho \end{bmatrix},$$

5.3.6 The Sensitivity of the LS Problem

$$= \| (A^TA)^{-1}A^T \|_2,$$

$$\frac{}{\sigma_n(A)^2} = \| (A^TA)^{-1} \|_2.$$

Theorem

$$r_{\mathrm{LS}} = b - Ax_{\mathrm{LS}},$$

$$\sin(\theta_{\mathrm{LS}}) = \frac{\| r_{\mathrm{LS}} \|_2}{[illegible]}.$$

$$\epsilon = \max$$

$$\nu_{\mathrm{LS}} =$$

[illegible]

$$\frac{\| \hat{x}_{\text{LS}} - x_{\text{LS}} \|_2}{\| x \|_2} \le \epsilon \left\{ \frac{\nu_{\text{LS}}}{\cos(\theta_{\text{LS}})} + [1 + \nu_{\text{LS}} \tan(\theta_{\text{LS}})]\, \kappa_2(A) \right\} + O(\epsilon^2)$$

[illegible]

$$\frac{\| \hat{r}_{\text{LS}} - r_{\text{LS}} \|_2}{\| r_{\text{LS}} \|_2} \le \epsilon \left\{ \frac{1}{\sin(\theta_{\text{LS}})} + \left[\frac{1}{\nu_{\text{LS}} \tan(\theta_{\text{LS}})} + 1 \right] \kappa_2(A) \right\} + O(\epsilon^2).$$

[illegible]

$$(A + tE)^T (A + tE) x(t) = (A + tE)^T (b + tf)$$

[illegible]

$$\| \hat{x}_{\text{LS}} - x_{\text{LS}} \|_2 \qquad \| \dot{x}(0) \|_2$$

[redacted]

[illegible]

$$E^T A x_{\text{LS}} + A^T E x_{\text{LS}} + A^T A \dot{x}(0) = A^T f + E^T b,$$

[illegible]

$$\dot{x}(0) = (A^T A)^{-1} A^T (f - E x_{\text{LS}}) + (A^T A)^{-1} E^T r_{\text{LS}}.$$

[illegible]

$$\cos(\theta_{\text{LS}}) = \frac{\| A x_{\text{LS}} \|_2}{\| b \|_2},$$

[illegible]

$$\frac{[illegible]}{[illegible]} = \epsilon \frac{\| \dot{r}(0) \|_2}{\| r_{\mathrm{LS}} \|_2} + O(\epsilon^2).$$

[illegible]

$$[illegible] \frac{\| A \|_2 \| r_{\mathrm{LS}} [illegible]}{\sigma_n(A)}$$

[illegible]

$$\frac{[illegible]}{[illegible]} \le \frac{\| b \|_2}{\| r_{\mathrm{LS}} \|_2} + \frac{\| A \|_2 \| x_{\mathrm{LS}} \|_2}{\| r_{\mathrm{LS}} \|_2} + \frac{\| A \|_2}{\sigma_n(A)}.$$

[illegible]

$$\nu_{\mathrm{LS}} = \frac{[illegible]}{[illegible]}$$

[illegible]

$$\nu_{\mathrm{LS}} \approx \kappa_2(A).$$

[illegible]

5.3.7 Normal Equations Versus QR

$m \gg$

$1/\sqrt{\mathbf{u}}$

$1/\mathbf{u}$.

5.3.8 Iterative Improvement

(1967,

1968).

$$\begin{bmatrix} I_m & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} r \\ x \end{bmatrix} = \begin{bmatrix} b \\ 0 \end{bmatrix}, \qquad A \in \mathbb{R}^{m\times n},\ b \in \mathbb{R}^m,$$

$$\begin{bmatrix} f^{(k)} \\ g^{(k)} \end{bmatrix}$$

end

[illegible]

$Q^T f =$ [illegible] $\qquad Q^T p =$ [illegible]

[illegible]

$$p = Q\begin{bmatrix} h \\ f_2 \end{bmatrix}.$$

[illegible]

5.3.9 Some Point/Line/Plane Nearness Problems in 3-Space

[illegible]

$$v^c = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix}.$$

$$p \times q \;=\; p^c \cdot q \;=\; -q^c \cdot p \;=\; -(q \times p).$$

(5.3.21)

$$(p \times q) \times r = (p^c \cdot q)^c r \;=\; \left(qp^T - pq^T\right) r \;=\; (p^T r)\cdot q - (q^T r)\cdot p, \qquad (5.3.22)$$

(5.3.23)

(5.3.24)

$$\| \quad \|_2^2 \;\; \| \quad \|_2^2 \cdot \;\; \|_2^2 \cdot \; 1 - \left(\frac{p^T q}{\| p \|_2 \cdot \| \quad \|_2} \right)^2 \qquad (5.3.25)$$

P5.3.13–P5.3.15.

$$\min_{z \in L} \| z - y \|_2.$$

$$z^{\text{opt}} \;=\; y + \frac{1}{v^T v}\, v^c v^c (y - p_1), \qquad (5.3.26)$$

$$\min_{z_1 \in L_1,\, z_2 \in L_2} \| z_1 - z_2 \|_2.$$

$$z_1^{\text{opt}} \qquad (5.3.27)$$

$$z_2^{\text{opt}} \qquad (5.3.28)$$

$$\min_{z \in P} \| z - y \|_2.$$

] sg,, & M MM n(} D} M M n1 } M !MM MM M

$$\| \hat{r} - r \|_2 \leq 2\,\kappa_2(A)\,\frac{\| F \|_2}{\| A \|_2}\,\| x \|_2.$$

$$\frac{\| x - \hat{x} \|_2}{\| x \|_2} \leq \mathbf{cu}\kappa_2(A)^2\,\frac{\| A^T \|_2 \| b \|_2}{\| A^T b \|}.$$

$$B = \begin{bmatrix} A \\ w^T \end{bmatrix}.$$

$$\sqrt{\|\qquad\qquad}$$

$$x_i - x_j \approx d_{ij},$$

% ƒ ƒ&GGG & x1 = 0ƒ. G

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M8⁼ xl3 8AlΩA ∈ Rm x n C.2AΩvlΩ,Ĕ .vxΩEΩa jĔ0 x A.)AΩ

$$M(a) = \left[\begin{array}{cc} aI_m & A \\ A^T & 0 \end{array}\right].$$

+C(LΩC.aΩ

$$C_{m+n}(M(a)) = \mathrm{bTv}\,\Omega\, a,\; E_R \sqrt{\sigma_n(A)^2 + \tfrac{1}{4}a^2}$$

.vxΩkAaAEbTvaCΩ2Ω4A(ΩΩa aC.abΩvTbTxA.Ω2(M(a)).

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‖ Az$^{(k)}$ - r$^{(k)}$ ‖2 = bTvΩ
x$^{(k+1)}$ x$^{(k)}$ MĔz$^{(k)}$
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5.4 Other Orthogonal Factorizations

$$[\,a_1,\, a_2,\, a_3,\, a_4\,] = [\,q_1,\, q_2,\, q_3,\, q_4\,]\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

5.4.1 Numerical Rank and the SVD

§2.4

$$\sum_{i=1} \qquad (5.4.1)$$

$\hat{\sigma}_1 \geq \cdots \geq \hat{\sigma}_n \geq 0.$

$$\sum_{i=1} \qquad (5.4.2)$$

(5.4.2).

$$\sum_{i=1} \quad \hat{r} \leq \hat{n} \qquad (5.4.3)$$

$$\hat{U} = W + \Delta U, \quad W^T W = I_m, \qquad (5.4.4)$$

[illegible]

[illegible] rank(B) = k - 1 [illegible]

[illegible]

$$E \;=\; W^T(\Delta A)Z$$

[illegible]

$$\| E \|_2 \;\le\; \epsilon \| A \|_2 \;=\; \epsilon\,\sigma_1 .$$

[illegible]

[illegible]

[illegible] rank(B) = k - 1 [illegible]

[illegible]

[illegible]

[illegible]

$$\hat\sigma_1 \ge \cdots \ge \hat\sigma_{\hat r} > \delta \ge \hat\sigma_{\hat r+1} \ge \cdots \ge \hat\sigma_n \,. \qquad (5.4.5)$$

[illegible] 10^{-2} [illegible] (5.4.2) [illegible]

5.4.2 QR with Column Pivoting

[illegible] (5.2.1) [illegible]

$$Q^T A\Pi \;=\; \begin{array}{c} \left[\begin{array}{cc} R_{11} & R_{12} \\ 0 & 0 \end{array}\right] \begin{array}{l} r \\ m-r \end{array} \end{array} \qquad (5.4.6)$$

$$(H_{k-1}\cdots H_1)A(\Pi_1\cdots\Pi_{k-1}) = R^{(k-1)} = \qquad (5.4.7)$$

(5.4.8)

$$R^{(k)} = H_k R^{(k-1)}\Pi_k,$$

Y(1965):

[illegible] (5.4.6) [illegible]

```
    [illegible]
        [illegible]
    end
    [illegible]
    [illegible]
    while [illegible]
        [illegible]
        [illegible]
        [illegible]
        [illegible]
        [illegible]
        [illegible]
        [illegible]
        [illegible]
        [illegible]
            [illegible]
        end
        [illegible]
    end
```

[illegible]

5.4.3 Numerical Rank and AΠ = QR

[illegible]

$$\mathrm{fl}(H_k \cdots H_1 A \Pi_1 \cdots \Pi_k) \;=\; \widehat{R}^{(k)} \;=\; [illegible] \qquad (5.4.9)$$

[illegible]

$$\| \widehat{R}_{22}^{(k)} \|_2 \;\le\; \epsilon_1 \| A \|_2$$

[illegible] $\hat{R}$.

$$\| E_k \|_2 \le \epsilon_2 \| A \|_2,$$

[illegible]

[illegible]

[illegible] $\hat{R}^{(k)}$ [illegible] $\| \hat{R}_{22}^{(k)}$ [illegible]

[illegible] w_2^k [illegible]

$$\sigma_{k+1}(A) \le (\epsilon_1 + \epsilon_2) \| A \|_2.$$

[illegible] $\hat{R}_{22}^{(k)}$ [illegible] $\hat{R}$ [illegible] $\tilde{\lambda}$ [illegible]

[illegible]

[illegible] $\| R_{22}^{(k)}$ [illegible] 10^{-19}).

[illegible] $\hat{R}_{22}^{(k)}$ [illegible] $\hat{R}$ [illegible]

5.4.4 Finding a Good Column Ordering

[illegible]

[illegible] $\in \mathbb{R}^{m \times n}$ [illegible] $\in \mathbb{R}^n$ [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

5.4.5 More General Rank-Revealing Decompositions

[illegible]

[illegible]

$$Z_{\text{new}} = ZZ_G, \qquad Q_{\text{new}} = QQ_G, \qquad R_{\text{new}} = Q_G^T R Z_G,$$

[illegible]

[illegible]

[illegible]

¨ < ¶F < ¶ < ¶K … < žP¶ <5.1.3. ¶
y. −.n −×t , , α, ,((,(α (−

$$R = \begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \\ 0 & 0 & 0 \end{bmatrix} \quad \mathbf{W}^{v} =$$

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$$Z_G = G_{12}G_{23}G_{34}$$

(2,

$$Q_G = H_{12}H_{23}H_{34}$$

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$$R \leftarrow RG_{12} = \begin{bmatrix} \times & \times & \times & \times \\ \times & \times & \times & \times \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{bmatrix}, \qquad v \leftarrow G_{12}^T v =$$

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+) (B).)− −. .0) .- .) v :

$$R \leftarrow H_{12}^T R = \begin{bmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{bmatrix}, \qquad v = \quad .$$

−, 2,×(,. (,2(:).)1,

$$R \leftarrow RG_{23} = \begin{bmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & 0 & 0 & \times \end{bmatrix}, \qquad v \leftarrow G_{23}^T v = \quad .$$

$$R \leftarrow H_{23}^T R = \left. \begin{matrix} \times & \times & \times \\ \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \end{matrix} \right], \qquad v =$$

2, 2(::i

$$R \leftarrow RG_{34} = \left. \begin{matrix} \times & \times & \times \\ \times & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \end{matrix} \right], \qquad v = G_{34}^T v =$$

$$R \leftarrow H_{34}^T R = \begin{matrix} 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{matrix}$$

$v =$

- [illegible]

$$\rho_L \; \frac{[illegible]}{A_L^2 + \rho_L^2)\sigma_{\min}(L_{11})} \qquad A_n \; [illegible]$$

[illegible]

$$\rho_L \; ,^T \; \frac{[illegible]}{\sigma_{\min}(L_{11})}$$

[illegible]

$$U^T A\Pi \;=\; \begin{array}{c|l} \begin{bmatrix} R_{11} & R_{12} \\ 0 & 0 \end{bmatrix} & \begin{array}{l} r \\ m-r \end{array} \end{array}$$

$$Q^T \begin{bmatrix} R_{11}^T \\ R_{12}^T \end{bmatrix} = \begin{bmatrix} S_1 \\ 0 \end{bmatrix}$$

5.4.B Bidiagonalization

[illegible]

$$[illegible] \qquad (5.4.13)$$

[illegible]

$$\begin{bmatrix} \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ & & \times \\ & & \times \end{bmatrix} \xrightarrow{U_1} \begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \\ 0 & \times & \times \end{bmatrix} \xrightarrow{V_1}$$

$$\begin{bmatrix} \times & \times & . \\ . & \times & \times \\ . & \times & \times \\ & \times & \times \\ . & \times & \times \end{bmatrix} \qquad \begin{bmatrix} \times & \times & . & \\ & \times & \times & \times \\ & . & \times & \times \\ & . & \times & \times \\ & . & \times & \times \end{bmatrix} \xrightarrow{V_2}$$

$$\begin{bmatrix} \times & \times & . & . \\ . & \times & \times & . \\ . & . & \times & \times \\ . & . & \times & \times \\ . & . & \times & \times \end{bmatrix} \xrightarrow{U_3} \begin{bmatrix} \times & \times & . & 0 \\ . & \times & \times & 0 \\ . & . & \times & \times \\ . & . & . & \times \\ . & . & . & \times \end{bmatrix} \xrightarrow{U_4} \begin{bmatrix} \times & \times & 0 & 0 \\ 0 & \times & \times & 0 \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \\ 0 & 0 & 0 & 0 \end{bmatrix} .$$

[illegible]

[illegible]

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    end
end
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Q^T A = \begin{bmatrix} R_1 \\ 0 \end{bmatrix}
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[illegible]

$$\operatorname{res}([\,a_{c_1}\,|\cdots|\,a_{c_k}\,]) = $$ [illegible]

$$= \qquad =$$

$$\operatorname{res}([\,a_{c_1}\,|\cdots|\,a_{c_k}\,]) = $$

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5.5 The Rank-Deficient Least Squares Problem

[illegible]

$$\| A(\lambda x_1 + (1-\lambda)x_2) - b \|_2 \le \lambda \| Ax_1 - b \|_2 + (1-\lambda)\| Ax_2 - b \|_2 =$$

$$Q^T AZ = T =$$

$$, \qquad Q^T b = \begin{array}{cc} \left[\begin{array}{c} c \\ d \end{array}\right] & \begin{array}{c} r \\ m-r \end{array} \end{array} .$$

$$\sum \frac{\imath}{} v_i$$

5.5.2 A Note on the Pseudoinverse

If we define the matrix A^+ 1(+ by $A^+ = V\Sigma^+U^T$ where

$$\Sigma^+ = \text{diag}\left(\frac{}{\sigma_1},\ldots,\frac{-}{\text{Or}},0,\ldots,0\right) \quad \text{i } \mathbb{R}^{n\times m}, \quad .\; ő= \quad @ \; œ$$

then $= A^+b$ and $= (\not I - AA^+)b$ \ A^+ is referred to as the -+ .⁻+()7 ++ of A. It is the unique minimal Frobenius norm solution to the problem

$$\min_{\#\ \$\ )} \| AX \;.\; \|_F. \qquad :(4 \quad \$$$

If $\text{rank}(A) = n$, then $A^+ = (A^TA)^{-1}A^T$, while if $m = n = \text{rank}(A)$, then $A^+ = A^{-1}$. Typically, A^+ is defined to be the unique matrix X 1(+ that satisfies the four +B)r.+)⁻(.(⁺)+

$$\begin{array}{llcllcl} \text{(i)} & AXA & = & A, & \text{(iii)}, & (AX)^T & = & AX, \\ \text{(ii)} & XAX & = & X, & \text{(iv)} & (XA)^T & = & XA. \end{array}$$

These conditions amount to the requirement that AA^+ and A^+A be orthogonal projections onto and X respœtively. Indeed,

$$AA^+ = U_1U_1^T$$

where $U_1 = U(1{:}m, 1{:}r)$ and

$$A^+A = \llcorner V_1^T$$

where L= u.4Ɫ 4ɇ√3u

5.5.3 Some Sensitivity Issues

In =39 we examined the sensitivity of the full-rank LS problem. The behavior of in this situation is summarized in Theorem =3Œ3 If we drop the full-rank assumption, then is not even a continuous function of the data and small changes in A and can induce arbitrarily large changes in $= \;^+\; A^+b$. The easiest way to see this is to consider the behavior of the pseudoinverse. If A and δA are in 1 +(1 then Wedin .OȤ9+ and Stewart .ȻZ=+ show that

$$\| (A+\delta A)^+ - A^+ \|_F \quad 2\| \delta A \|_F \max\left\{\| A^+ \|_2^2 \, , \, \| (A+\delta A)^+ \|_2^2\right\}.$$

This inequality is a generalization of Theorem 2.3.4 in which perturbations in the matrix inverse are bounded. However, unlike the square nonsingular case, the upper bound does not necessarily tend to zero as δA tends to zero. If

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad \text{and} \qquad \delta A = \begin{bmatrix} 0 & 0 \\ 0 & \epsilon \\ 0 & 0 \end{bmatrix}$$

then

$$A^+ = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \text{and} \qquad (A+\delta A)^+ = \bullet \quad \begin{matrix} - \\ 1/\epsilon \end{matrix} \quad - \Big)$$

5.5.4

$$\| x_{\hat{r}} - x_{LS} \|_2 \leq \frac{\hat{r}}{\sigma_{\hat{r}}} 2(1+\epsilon)\epsilon \| b \|_2 + \sqrt{}$$

$$\frac{\hat{\sigma}_1}{\hat{\sigma}_{\hat{r}}}$$

5.5.5 [illegible]

[illegible]

$$R \;=\; \begin{array}{cl} \left[\begin{array}{cc} R_{11} & R_{12} \\ 0 & 0 \end{array}\right] & \begin{array}{l} r \\ m-r \end{array} \end{array} .$$

[illegible]

[illegible]

$$\| Ax - b \|_2^2 \;=\; \| (Q^T A\Pi)(\Pi^T x) - (Q^T b) \|_2^2 \;=$$

$$\Pi^T x \;= \qquad\qquad\qquad\qquad \text{and} \quad Q^T b \;=\; \left[\begin{array}{c} c \\ d \end{array}\right] \qquad\qquad .$$

$$x \;=\; \Pi \left[\begin{array}{c} R_{11}^{-1}(c - R_{12}z) \\ z \end{array}\right].$$

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$$\underline{\qquad\qquad} \qquad \sqrt{1\qquad\qquad\qquad\qquad}$$

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$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1+\epsilon & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix},$$

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$$AP = [\; B_1 \mid B_2 \;]$$

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$$\frac{\sigma_{\tilde r}(A)}{\| \widetilde{V}_{11}^{-1} \|_2} \;\le\; \sigma_{\tilde r}(B_1) \;\le\; \sigma_{\tilde r}(A).$$

Proof. 0≠.≠, .).), :..).,)3),)..(,. C C) ,((..,,0 (0≠.).≠, .).), 7 .(,(,(,)) (0≠ ,,(.))(. 3((,,5) ,,)...(, .(..≠, , :..).,

$$\Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix}\begin{matrix} \tilde r \\ m-\tilde r \end{matrix}.$$

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$$V = \begin{array}{c} \left[\begin{array}{cc} V_{11} & V_{12} \\ V_{21} & V_{22} \end{array}\right] \begin{array}{l} \tilde{r} \\ n-\tilde{r} \end{array} \end{array}$$

[illegible]

[illegible]

$$\left[\begin{array}{c} \widetilde{V}_{11} \\ \widetilde{V}_{21} \end{array}\right] = P^T \left[\begin{array}{c} V_{11} \\ V_{21} \end{array}\right] = \left[\begin{array}{c} R_{11}^T Q^T \\ R_{12}^T Q^T \end{array}\right].$$

[illegible]

[illegible]

5.5.8 [illegible]

[illegible]

$$r_y = b - Ay = b - B_1 z = (I - B_1 B_1^+)b.$$

[illegible]

[illegible]

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4Ž Ó $ † $$ 0Y $$ Ó $ $ ÏÃ2IÄÓ $ r $$ « $

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Proof.)(≠0((roo≠. CC , (), , . λ . . . , .0≠,≠

$$U = [\, U_1 \mid U_2 \,]$$

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$$\| U_2^T Q_1 \|_2 \le \| U_2^T B_1 \|_2 \| (B_1^T B_1)^{-1/2} \|_2$$

$$\le \frac{\sigma_{\tilde{r}+1}(A)}{\sigma_{\tilde{r}}(A)} \| \tilde{V}_{11}^{-1} \|_2,$$

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$$B(\lambda) = (A^T A + \lambda I)^{-1} A^T$$

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[illegible]

$$\min_{y\in\mathbf{R}^r,\, z\in\mathbf{R}^{n-r}} \left\| \begin{bmatrix} R & S \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} - \begin{bmatrix} c \\ d \end{bmatrix} \right\|_2$$

[illegible]

5.6 Square and Underdetermined Systems

[illegible]

5.6.1 [illegible]

[illegible]

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-u ğ .i $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$. -- l 3LOBB 5. ő

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$$(Q^T A\Pi)(\Pi^T x) \;=\; [\, R_1 \mid R_2 \,] \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} \;=\; Q^T b$$

$$\Pi^T x \;=\; \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

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$$(QR)^T x = \left[\; R_1^T \;\middle|\; 0 \;\right] \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = b,$$

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$$Q^T x = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}, \qquad z_1 \in \mathbb{R}^m,\ z_2 \in \mathbb{R}^{n-m}.$$

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$$= \sum_{i=1} \sigma_i u_i v_i^T, \qquad r = \quad @ \quad œ$$

)+ ∑nσ$_i^T$ (v ∑σv)

$$X = \sum_{i=1}^{r} \frac{u_i^T b}{\sigma_i} v_i.$$

+)) ∑σ$_i^T$ 1$_i^T$ 1∑ j{∑(,$_i^T$(1$_i^T$ −v ∑σ$_i^T$ n ( )+ ,v+)((σ$_i^T$)=)+)$_i^T$(,1= ,()
,$_i^T$,)$_i^T$)∑i

5.6.3 [illegible]

[illegible]

[illegible]

$$\frac{\| \hat{x} - x \|_2}{\| x \|_2} \le$$ [illegible]

[illegible]

$$\dot{x}(0) = \left(I - A^T(AA^T)^{-1}A\right) E^T(AA^T)^{-1}b + A^T(AA^T)^{-1}(f - Ex).$$ [illegible]

[illegible]

$$\frac{\| f \|_2}{\| x \|_2} \le \frac{\| f \|_2 \| A \|_2}{\| b \|_2},$$

[illegible]

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Chapter 6

Modified Least Squares Problems and Methods

6.1 [illegible]

6.2 [illegible]

6.3 [illegible]

6.4 [illegible]

6.5 [illegible]

[illegible]

[illegible]

[illegible]

6.1 Weighting and Regularization

[illegible]

6.1.1 [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\frac{d}{d\delta}\left[D(\delta)^2\right] = d_k^2 e_k e_k^T$$

[illegible]

$$\frac{d}{d\delta}\left[(A^TD(\delta)^2A)^{-1}\right] + [illegible](A^TD(\delta)^2A)^{-1}(A^T([illegible])A)(A^TD(\delta)^2A)^{-1},$$

[illegible]

$$\frac{d}{d\delta}\, r_k(\delta) + [illegible]\,(a_k(A^TD(\delta)^2A)^{-1}a_k)\, r_k(\delta). \quad [illegible]$$

[illegible] $(A^TD(\delta)A)^{-1}$ [illegible]

$$[r_k(\delta)^2] + [illegible]\, r_k(\delta) \quad r_k(\delta) + [illegible]\,(a_k(A^TD(\delta)^2A)^{-1}a_k)\, r_k(\delta)^2 <$$

[illegible] $|r_k(\delta)|$ [illegible]

[illegible]

6.1.2 [illegible]

[illegible]

$$b + [illegible]\, x + w \quad [illegible]$$

[illegible]

$$\min_{x \in \mathbf{R}^n} \left\| B^{-1}(Ax - b) \right\|_2. \quad [illegible]$$

[illegible]

$$[illegible] \quad b = Ax + Bv \quad [illegible]$$

[illegible]

$$Q^T A \;=\; \begin{bmatrix} R_1 \\ 0 \end{bmatrix}, \qquad Q \;=\; [\; Q_1 \mid Q_2 \;] \,.$$

[illegible]

$$Z \;=\; [\; Z_1 \mid Z_2 \;]$$

[illegible]

6.1.3 [illegible]

[illegible]

$$\|\, z \,\|_G \;=\; \|\, Gz \,\|_2 \,.$$

[illegible]

6.1.4 [illegible]

[illegible] $\mathbf{R}^{n}$ [illegible] b [illegible] $\mathbf{R}^{n}$ [illegible]

$$\min_{x} \left\| \begin{bmatrix} A \\ \sqrt{\lambda} I \end{bmatrix} x - \begin{bmatrix} b \\ 0 \end{bmatrix} \right\|_2^2 = [illegible]$$

[illegible]

$$(A^T A + \lambda I)x = A^T b. \qquad [illegible]$$

[illegible]

$$[illegible] \sum_{i=1} [illegible]$$

[illegible]

$$(\Sigma^T \Sigma + \lambda I_n)(V^T x) = \Sigma^T U^T b$$

[illegible]

$$x(\lambda) = \sum_{i=1}^{r} \frac{\sigma_i u_i^T b}{\sigma_i^2 + \lambda} v_i. \qquad [illegible]$$

[illegible]

$$\lim_{\lambda \to 0} [illegible]$$

[illegible]

$$\| Ax_k(\lambda) - b \|_2^2 = \| D_k(Ax_k(\lambda) - b) \|_2^2 + (a_k^T x_k(\lambda) - b_k)^2,$$

(λ) (λ) $\dfrac{a_k^T x(\lambda) - b_k}{}$

$$\frac{k}{e_k^T}$$

$$\frac{r_k}{\partial r_k / \partial b_k} \quad .$$

$$\left[\frac{\tilde{b}_k - \sum_{j=1}^{r} u_{kj} \tilde{b}_j \left(\dfrac{\sigma_j^2}{\sigma_j^2 + \lambda} \right)}{} \right]^2$$

6.1.5 n æ ñăáÿ ălxėßĢē ô r ċ æ ĠrĐ æ ă ÿ Ģ

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$$\left\| \begin{bmatrix} A \\ \sqrt{\lambda}B \end{bmatrix} x - \begin{bmatrix} b \\ 0 \end{bmatrix} \right\|_2^2 =$$

$$(A^TA + \lambda B^TB)x \;=\; A^Tb.$$

$$= \qquad \begin{matrix} I \\ 0 \\ 0 \end{matrix} \qquad ,$$

$$= \qquad ,$$

Proof. .≠.,))λ 3(≠,3,≠)λ(.≠ α (), (.≠ α≠λ)..),,(,)) ≠ .≠),≠3 CS:C C ≠(

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$$\begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix}^T \begin{bmatrix} Q_{11} \\ Q_{21} \end{bmatrix} V_1 = \begin{bmatrix} D_A(:,1{:}r) \\ D_B(:,1{:}r) \end{bmatrix}$$

X.(.(D_A (+β D_B .(^T_A((.($-^2_n$,,1(−, (βr] (+β (λλ)X, $)^2_n$
(+β (.((

$$\begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix}^T \begin{bmatrix} A \\ B \end{bmatrix} Z = \begin{bmatrix} D_A(:,1{:}r) & U_1Q_{12} \\ D_B(:,1{:}r) & U_2Q_{22} \end{bmatrix} \begin{bmatrix} V_1^T\Sigma_r & 0 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} D_A(:,1{:}r) & 0 \\ D_B(:,1{:}r) & 0 \end{bmatrix} \begin{bmatrix} V_1^T\Sigma_r & 0 \\ 0 & I_{n_1-r} \end{bmatrix}$$

$$= \begin{bmatrix} D_A \\ D_B \end{bmatrix} \begin{bmatrix} V_1^T\Sigma_r & 0 \\ 0 & I_{n_1-r} \end{bmatrix}.$$

],(((,+$_B$

$$X = Z \begin{bmatrix} V_1^T\Sigma_r & 0 \\ 0 & I_{n_1-r} \end{bmatrix}^{-1}$$

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$$(D_A D_A + \lambda D_B D_B)y \ , \quad D_A b$$

X.(−($_A$ Xy, ($_A$ $_C$┤+β

$$(D_A^T D_A + \lambda D_B^T D_B) \ _A \ \beta,(_B (\qquad {}^T_B\, D, \alpha_n^2 + \lambda\beta_n^2).$$

[illegible]

[illegible]

$$X = [\,x_1\,|\cdots|\,x_n\,]$$

[illegible]

$$x(\lambda) = \sum_{k=1}^{n}\left(\frac{\alpha_k\tilde{b}_k}{\alpha_k^2+\lambda\beta_k^2}\right)x_k$$

[illegible]

[illegible]

[illegible]

$$\lambda = \left(\left(\frac{\tilde{b}}{s}\right)^2 - \frac{1}{m}\right)^{-1}$$

[illegible]

i = 1

[illegible]

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£ T¦ G ¦ 3 1 GGG 1 G [illegible]

6.2 Constrained Least Squares

[illegible]

6.2.1 [illegible]

[illegible]

$\|x\|_2$

[illegible]

$$x_{LS} = \sum_{i=1}^{r} \frac{u_i^T b}{\sigma_i} v_i$$

[illegible]

$$\phi(x, \lambda) = \frac{1}{2}\| Ax - b \|_2^2 + \frac{\lambda}{2}\left(\| x \|_2^2 - \alpha^2\right)$$

[illegible]

$$(A^T A + \lambda I)\cdot x(\lambda) = A^T b.$$

[illegible]

$$\frac{\sigma_k u_k b}{\sigma^2 \quad \lambda}$$

[illegible]

[illegible]

- [illegible]

[illegible]

[illegible]

```
if $\sum_{i=1}^{r} \left(\frac{\tilde{b}_i}{\sigma_i}\right)^2 > \alpha^2$
    Find $\lambda_+ > 0$ such that $\sum_{i=1}^{r} \left(\frac{\sigma_i \tilde{b}_i}{\sigma_i^2 + \lambda_+}\right)^2 = \alpha^2.$
    $x = \sum_{i=1}^{r} \left(\frac{\sigma_i \tilde{b}_i}{\sigma_i^2 + \lambda_+}\right) v_i$
else
    $x = \sum_{i=1}^{r} \left(\frac{\tilde{b}_i}{\sigma_i}\right) v_i$
end
```

[illegible]

6.2.2 [illegible]

[illegible]

[illegible]

$$y = X^{-1}x.$$

6 ¶ (+ ¶(¶ ¶ ¶ ¯ ¶ ¶¶ ¶ F "¶ ¶ ¶ ¶
Y ík¶ ¶% ¶@ ¶@ 6 œ< 8 ¶

$$\sum_{i=n_1+1}^{m_1} \tilde{b}_i^2$$

¶

$$\sum_{i=m_2+1}^{n_1} \tilde{d}_i^2 \leq \alpha^2.$$

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+ (¶ 5 ¶ ¶% í-¶

6.2.3

Bx=d

e ¨ r|r ¯ S

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$$Q^T B^T = \begin{array}{c} \left[\begin{array}{c} R \\ 0 \end{array} \right] \end{array} \begin{array}{c} n_1 \\ n_1 - m_2 \end{array}$$

¶ ¢¶{ ¶# C ¼ ¶¶ ¶ ¶

$$m_2 \quad n_1 - m_2 , \qquad Q^T x = \left[\begin{array}{c} y \\ z \end{array} \right] \begin{array}{c} m_2 \\ n_1 - m_2 \end{array} .$$

RTy=d

Ü (¶ ¶@¶¶ ¶ ¶ ¶ ¶ ¶ ¶¶ X : È ¶ („ ¶

$$x = Q \left[\begin{array}{c} y \\ z \end{array} \right].$$

[illegible]

[illegible]

[illegible]

6.2.4 [illegible]

[illegible]

[illegible] $\in \mathbb{R}^{m_2}$,

[illegible]

$$\begin{bmatrix} 0 & A^T & B^T \\ A & I & 0 \\ B & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ r \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ b \\ d \end{bmatrix}.$$

[illegible]

6.2.5 [illegible]

[illegible]

[illegible]

[illegible]

$$\min_{x} \left\| \begin{bmatrix} A \\ \sqrt{\lambda}B \end{bmatrix} x - \begin{bmatrix} b \\ \sqrt{\lambda}d \end{bmatrix} \right\|_2$$

[illegible]

[illegible]

$$x(\lambda) = \sum_{i=1}^{m_2} \left(\frac{\alpha_i \tilde{b}_i + \lambda \beta_i \tilde{d}_i}{\alpha_i^2 + \lambda \beta_i^2} \right) x_i + \sum_{i=m_2+1}^{n_1} \left(\frac{\tilde{b}_i}{\alpha_i} \right) x_i$$

[illegible]

$$x(\lambda) - x = \sum_{i=1}^{p} \frac{\alpha_i}{\beta_i} \left(\frac{\beta_i u_i^T b - \alpha_i v_i^T d}{\alpha_i^2 + \lambda^2 \beta_i^2)} \right) x_i.$$

[illegible]

[illegible] $p_0(x), \ldots, p_n(x)$ [illegible] $(x_0, y_0), \ldots, (x_m, \; m)$ [illegible] x_i [illegible] $p(x)$ [illegible] $\sum_{k=0}^{n} \alpha_k p_k(x)$ [illegible]

$$\phi(\alpha) \quad \sum_{i=0}^{m} (p(x_i) - y_i)^2$$

[illegible]

$$\int^{b} [\ldots]^2 \ldots \sum^{N} \left(p(z_{i-1}) - 2p(z_i) + p(z_{i+1}) \right)^2 \ldots {}^2$$

[illegible] y_k] [illegible] $\mathbb{R}^{m \times k}$ [illegible]

y^Ty [illegible] $d_1 \ge d_2 \ge \cdots \ge d_k > 0.$

[illegible]

[illegible] $\mathbb{R}^{n\times n}$ [illegible] $\mathbb{R}^{(m-n)\times n}$ [illegible]

[illegible] $\mathbb{R}^{m\times n}$ [illegible] $\mathbb{R}^{m\times p}$ [illegible] $\mathbb{R}^{n\times n}$ [illegible]

$$Q^TA = \begin{bmatrix} R \\ 0 \end{bmatrix}, \qquad Q^TBV = [\,0 \mid S\,]$$

[illegible] $\mathbb{R}^{n\times n}$ [illegible] $\mathbb{R}^{m\times m}$ [illegible]

[illegible] $\mathbb{R}^{m}$, y [illegible] $\mathbb{R}^{n}$ [illegible] β

[illegible] ||Ey - r||2

[illegible]

[illegible] A [illegible] $\mathbb{R}^{m\times n}$, B [illegible] $\mathbb{R}^{p\times n}$, rank(B) [illegible]

[illegible]

[illegible] $B_1 \in \mathbb{R}^{p\times p}$, $\text{rank}(B_1) = p$.

[illegible]

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[illegible]

6.3 Total Least Squares

[illegible]

$$b+r \in \mathrm{ran}(A)$$

[illegible]

$$b+r \in \mathrm{ran}(A+E)$$

[illegible]

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad E_\epsilon = \begin{bmatrix} 0 & 0 \\ 0 & \epsilon \\ 0 & \epsilon \end{bmatrix},$$

[illegible]

$$B+R \in \mathrm{ran}(A+E)$$

[illegible]

6.3.1 [illegible]

[illegible]

[illegible]

[illegible]

$$V = \begin{array}{cc} \left[\begin{array}{cc} V_{11} & V_{12} \\ V_{21} & V_{22} \end{array}\right] \begin{array}{c} n \\ k \end{array} \end{array}, \qquad \Sigma = \begin{array}{cc} \left[\begin{array}{cc} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{array}\right] & \begin{array}{c} n \\ k \end{array} \\ \begin{array}{cc} n & k \end{array} & \end{array}.$$

[illegible]

$$X_{TLS} = -T_1 V_{12} V_{22}^{-1} T_2^{-1}$$

[illegible]

$$C_1 V_{12} + C_2 V_{22} = U_2 \Sigma_2.$$

[illegible]

$$V_{12}^T V_{12} + V_{22}^T V_{22} = I$$

[illegible]

[illegible]

[illegible]

[illegible]

$$\{D[\,A \mid B\,]T + D[\,E_0 \mid R_0\,]T\} = U_1 \Sigma_1 [\,V^T \mid V^T\,]$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

6.3.2 [illegible]

[illegible]

[illegible]

[illegible]

$$x_{\text{TLS}} = -T_1 z/(t_{n+1}\alpha).$$

[illegible]

[illegible]

[illegible]

6.3.3 [illegible]

[illegible]

$$\psi(x) = \sum_{i=1}^{m} d_i^2 \left(\frac{|a_i^T x - b_i|^2}{x^T T_1^{-2} x + t_{n+1}^{-2}} \right)$$

[illegible]

[illegible]

$$\begin{bmatrix} a_i \\ b_i \end{bmatrix} \in \mathbb{R}^{n+1}$$

[illegible]

$$P_x = \left\{ \begin{bmatrix} a \\ b \end{bmatrix} : a \in \mathbb{R}^n,\ b \in \mathbb{R},\ b = x^T a \right\}$$

[illegible]

6.3.4 [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$(A^T A + \lambda_1 I + \lambda_2 L^T L)x \;=\; A^T b.$$

[illegible]

[illegible]

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[illegible]

[illegible]

[illegible]

$$C \in \mathbb{R}^{n \times n}.$$

$$\mathsf{tr}(C_1^T C_2) = \mathsf{tr}(C_2^T C_1).$$

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[illegible]

[illegible]

$$\cos(\theta_k) \;=\; f_k^T g_k \;=\; \max_{f\in F,\ \|f\|_2=1} \quad \max_{g\in G,\ \|g\|_2=1} \quad f^T g \;.$$

$A \in \mathbb{R}^{n\times p}$,

$B \in \mathbb{R}^{n\times q}$.

$$\sqrt{1 \quad n} = \sin(\theta_p).$$

6.4.4

6.4.3.

Proof.

$$(u_i(A)^2 - 2u_i(B^T A) + u_i(B)^2).$$

[illegible]

[illegible]

[illegible]

[illegible]

$$Q(\theta_1,\theta_2,\theta_3)$$

$s_2c_3 \quad s_2s_3$

$-c_1s_2 \quad c_1c_2c_3 - s_1s_3$

$s_1s_2 \quad -s_1c_2c_3 - c_1s_3 \quad -s_1c_2s_3 \quad c_1c_3$

[illegible]

$$\begin{bmatrix}\sigma_1 & 0 & 0\\ 0 & \sigma_2 & 0\\ 0 & 0 & \sigma_3\end{bmatrix} = Q(\theta_1,\theta_2,\theta_3) - xy^T, \qquad [illegible]\ \mathbf{R}^3$$

[illegible]

$$xy^T = \begin{bmatrix} s_2\\ \mu c_1\\ -\mu s_1\end{bmatrix}\begin{bmatrix} -s_2/\mu\\ c_3\\ s_3\end{bmatrix}^T$$

[illegible]

$$[illegible]\ \mu \quad c_1s_3 \quad = \quad c_2 \quad s_1c_3$$

[illegible]

$$v_1 = U_cU_c^Tv, \qquad v_2 = (I_n - U_cU_c^T)v,$$

[illegible]

$$[illegible]\ \frac{[illegible]}{\|v_1\|} \quad [illegible]\ \frac{[illegible]}{\|v_2\|}\ v_2$$

[illegible]

[illegible]

$$U_\theta U_\theta^T - U_c U_c^T = y_\theta y_\theta^T - \frac{v_1 v_1^T}{v_1^T v_1}$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

6.5 Updating Matrix Factorizations

[illegible]

[illegible]

[illegible]

$$\widetilde{A} = A + uv^T = Q(R + wv^T)$$

[illegible]

$$J_1^T \cdots J_{n-1}^T w = \pm \| w \|_2 e_1 .$$

[illegible]

$$H = J_1^T \cdots J_{n-1}^T R$$

[illegible]

$$w \;\leftarrow\; \begin{bmatrix} \times \\ \times \\ \times \\ \times \end{bmatrix}, \qquad R \;\leftarrow\; \begin{bmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & & \times & \times \\ 0 & & & \times \end{bmatrix},$$

[illegible]

$$[illegible] \begin{bmatrix} \times \\ \times \\ \times \\ 0 \end{bmatrix}, \qquad [illegible] \begin{bmatrix} \times & \times & \times & \\ \times & \times & \times & \\ & \times & \times & \\ & \times & \times & \end{bmatrix},$$

$$w \;\leftarrow\; J_2^T w \;=\; \begin{bmatrix} \times \\ \times \\ 0 \\ 0 \end{bmatrix}, \qquad R \;\leftarrow\; J_2^T R \;=\; \begin{bmatrix} \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ & \times & \times \end{bmatrix},$$

$$[illegible] \begin{bmatrix} \times \\ 0 \\ 0 \\ 0 \end{bmatrix}, \qquad [illegible] \begin{bmatrix} \times & \times & \times \\ \times & \times & \times \\ \times & \times & \times \\ & \times & \times \end{bmatrix}.$$

[illegible]

$$(J_1^T \cdots J_{n-1}^T)(R + wv^T) \;=\; H \pm \|\, w \,\|_2 e_1 v^T = H_1 \qquad [illegible]$$

[illegible]

6.5.2 [illegible]

[illegible]

$$QR \;=\; A \;=\; [\, a_1 \,|\, \cdots \,|\, a_n \,], \qquad a_i \in \mathbb{R}^m, \qquad [illegible]$$

[illegible]

$$[illegible] \begin{bmatrix} R_{11} & v & R_{13} \\ 0 & r_{kk} & w^T \\ 0 & 0 & R_{33} \end{bmatrix} [illegible]$$

[illegible]

[illegible]

$$Q^T\tilde{A} = \begin{bmatrix} R_{11} & R_{13} \\ 0 & w^T \\ 0 & R \end{bmatrix} = H$$

[illegible]

[illegible]

[illegible]

[illegible]

$$Q^T\tilde{A} = \left[\, Q^Ta_1 \,|\cdots|\, Q^Ta_k \,|\, w \,|\, Q^Ta_{k+1} \,|\cdots|\, Q^Ta_n \,\right]$$

[illegible]

[illegible]

[illegible]

~

$$H = G_1^T \cdots G_{m-1}^T R = \begin{bmatrix} v^T \\ R_1 \end{bmatrix} \begin{array}{l} 1 \\ m-1 \end{array}$$

$$QG_{m-1} \cdots G_1 = \begin{bmatrix} \alpha & 0 \\ 0 & Q_1 \end{bmatrix}$$

$$A = \begin{bmatrix} z^T \\ A_1 \end{bmatrix} = (QG_{m-1} \cdots G_1)(G_1^T \cdots G_{m-1}^T R) = \begin{bmatrix} \alpha & 0 \\ 0 & Q_1 \end{bmatrix} \begin{bmatrix} v^T \\ R_1 \end{bmatrix}$$

$$\widetilde{A} = A + zz^T, \qquad z \in \mathbb{R}^n.$$

$$\widetilde{A} = \begin{bmatrix} G^T \\ z^T \end{bmatrix}^T \begin{bmatrix} G^T \\ z^T \end{bmatrix},$$

$$Q^T \begin{bmatrix} G^T \\ z^T \end{bmatrix} = \begin{bmatrix} R \\ 0 \end{bmatrix}, \qquad R \in \mathbb{R}^{n \times n}$$

$$\begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \\ \times & \times & \times \end{bmatrix} \xrightarrow{Q_1} \begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \\ 0 & \times & \times \end{bmatrix} \xrightarrow{Q_2} \begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \\ 0 & 0 & \times \end{bmatrix} \xrightarrow{Q_3} \begin{bmatrix} \times & \times & \times \\ 0 & \times & \times \\ 0 & 0 & \times \\ 0 & 0 & 0 \end{bmatrix}.$$

[illegible]

[illegible]

$$\tilde{A} = GG^T - zz^T = \begin{bmatrix} G^T \\ z^T \end{bmatrix}^T S \begin{bmatrix} G^T \\ z^T \end{bmatrix}.$$

[illegible]

$$H^T \begin{bmatrix} G^T \\ z^T \end{bmatrix} = \begin{bmatrix} R \\ 0 \end{bmatrix},$$

$$\tilde{A} = \left(H^T \begin{bmatrix} G^T \\ z^T \end{bmatrix} \right)^T \begin{bmatrix} I_n & 0 \\ 0 & -1 \end{bmatrix} \left(H^T \begin{bmatrix} G^T \\ z^T \end{bmatrix} \right) = R^T R$$

$$H_2(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & c \\ 0 & 0 \\ 0 & -s \end{bmatrix}$$

$$\begin{bmatrix} [H_k]_{k,k} \\ [H_k]_{n+1,k} \end{bmatrix}$$

[illegible]

$$[illegible]$$

[illegible]

$$[illegible]$$

[illegible]

$$\begin{bmatrix} \tilde{g}_{11} & \tilde{g}_{21} & \tilde{g}_{31} \\ 0 & g_{22} & g_{32} \\ 0 & 0 & g_{33} \\ 0 & z_2' & z_3' \end{bmatrix}.$$

[illegible]

$$[illegible]$$

[illegible]

$$\begin{bmatrix} c & 0 & -s \\ 0 & I_{n-1} & 0 \\ -s & 0 & c \end{bmatrix} \begin{bmatrix} g_{11} & g_1^T \\ 0 & G_1^T \\ \mu & w^T \end{bmatrix} = \begin{bmatrix} \tilde{g}_{11} & [illegible] \\ 0 & [illegible] \\ 0 & [illegible] \end{bmatrix}$$

[illegible]

$$\frac{\sqrt{\alpha}}{\sqrt{\alpha-\mu^2}}, \qquad \frac{}{\sqrt{\alpha-\mu^2}}.$$

$$U^TAV = \begin{bmatrix} L \\ 0 \end{bmatrix} = \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \\ 0 & 0 \end{bmatrix}, \qquad U^TU = I_m,\ V^TV = I_n$$

$$U^TA\Pi = \begin{bmatrix} R \\ 0 \end{bmatrix}, \qquad R \in \mathbb{R}^{n\times n}$$

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$$V = [\, V_1 \mid V_2 \,] \qquad U = [\, U_1 \mid U_2 \,]$$

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$$\|\, AV_2 \,\|_2 \;=\; \|\, U_2 L_{22} \,\|_2 \;=\; \|\, L_{22} \,\|_2 .$$

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$$\tilde{A} = \begin{bmatrix} A \\ z^T \end{bmatrix},$$

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$$\begin{bmatrix} U & 0 \\ 0 & 1 \end{bmatrix}^T \begin{bmatrix} A \\ z^T \end{bmatrix} V = \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \\ 0 & 0 \\ w^T & y^T \end{bmatrix}.$$

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$$\begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \\ w^T & y^T \end{bmatrix} = \left[\begin{array}{cccc|ccc} \text{v ,} & \text{) ,} & \text{) ,} & \text{)} & \text{,) ,} & \text{) ,} & \text{) ,} \\ \text{v ,} & \text{v ,} & \text{) ,} & \text{) ,} & \text{) ,} & \text{) ,} & \text{) ,} \\ \text{v ,} & \text{v ,} & \text{v ,} & \text{) ,} & \text{) ,} & \text{) ,} & \text{) ,} \\ \text{v ,} & \text{v ,} & \text{v ,} & \text{v ,} & \text{) ,} & \text{) ,} & \text{) ,} \\ \hline * & * & * & * & * & \text{) ,} & \text{) ,} \\ * & * & * & * & * & * & \text{) ,} \\ * & * & * & * & * & * & * \\ \hline \#* & \#* & \#* & \#* & \text{y} & \text{y} & \text{y} \end{array}\right].$$

- # L*# 7 @ > .=##?L5 ´ 2B L 2 #@L, ¥ L# * 7 ! Ly # 7 @ = .=##?L9 @L # G @L ?L# < B # 79 $#LLD # 8 ? L
> 9 @ @ . 9 7 ? L ÓÓ ÓÓ> # L ; 3 / #)L9 5 L@ + # L 3@#I$L@ 9L L : @B @#9L@ @ =9 95E L

$$\left[\begin{array}{c} \tilde{L} \\ \hline 0 \end{array}\right] \;\text{v}\; \left[\begin{array}{ccccccc} \times & & & & & & \\ \times & \times & & & & & \\ \times & \times & \times & & & & \\ \times & \times & \times & \times & & & \\ \times & \times & \times & \times & \times & \text{) ,} & \text{)} \\ \times & \times & \times & \times & \times & \times & \\ \times & \times & \times & \times & \times & \times & \times \\ \hline & & & & & & \end{array}\right] =$$

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2 2 H L m $ 1=L# D # 2 . 79*E L# D # >7 !LL @ + ?4L# HE #L 7 L= # ? @ 9@>+#LL8 1 > # D #L2 . 8
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$$\| p^T \widetilde{L} \|_2 \approx \sigma_{\min}(\widetilde{L}).$$

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â C ø ¶ ¶ ˜¶¶¶ T " ¶ ¶ ' ¶ ø ¶ % ¶

$$H \;=\; U_{67}^T U_{56}^T U_{45}^T U_{34}^T U_{23}^T U_{12}^T \tilde{L}.$$

â C ø ¶ ¶ † ¶¶ ¶ ø ¶ ¸ Ó¶¶¶ ¶ ø "+ '¶¶

$$L_+ \;=\; H V_{12} V_{23} V_{34} V_{45} V_{56} V_{67}.$$

b +¶ ¶ ¶

$$e_7^T L_+ = \left(e_8^T H\right) V_{12} V_{23} \quad V_{45} V_{56} V_{67} \;=\; \left(p^T \quad V_{12} V_{23} \quad V_{45} V_{56} V_{67}\right.$$

$\_V_{23}$ (—,¹d)⁻($V_{12} f$)¹,⁻ $\sigma_{\min}(\widetilde{L})$. m,$\widetilde{L}$) β|A((((.βA() ()((($\widetilde{L}$:(()))(Q $\beta\sigma$(,(X);

$$= \left[\begin{array}{cccccc|c} \times & & & & & & \\ \times & \times & & & & & \\ \times & \times & \times & & & & \\ \times & \times & \times & \times & & & \\ \times & \times & \times & \times & \times & & \\ \times & \times & \times & \times & \times & \times & \\ \hline \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{array}\right]$$

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$$\left[\begin{array}{ccccc|cc} \times & & & & & & \\ \times & \times & & & & & \\ \times & \times & \times & & & & \\ \times & \times & \times & \times & & & \\ \times & \times & \times & \times & \times & & \\ \hline \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{array}\right].$$

+t;::p

[illegible]

[illegible]

$$Q = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix}$$

[illegible]

$$\begin{bmatrix} R & H \\ 0 & E \end{bmatrix} \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix} = \begin{bmatrix} R_1 & 0 \\ H_1 & E_1 \end{bmatrix},$$

[illegible]

$$S = \begin{bmatrix} I_p & 0 \\ 0 & -I_q \end{bmatrix},$$

[illegible]

$$Q_1 \in \mathbb{R}^{p \times n},\ Q_2 \in \mathbb{R}^{q \times n}$$

[illegible] $Q^TSQ = S$ [illegible]

$$Q = \begin{array}{cc} \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix} & \begin{array}{c} p \\ q \end{array} \\ \begin{array}{cc} p & q \end{array} & \end{array}$$

[illegible]

$$Q_{11}^TQ_{11} = I_p + Q_{21}^TQ_{21}, \qquad Q_{11}^TQ_{12} = Q_{21}^TQ_{22}, \qquad Q_{22}^TQ_{22} = I_q + Q_{12}^TQ_{12}.$$

[illegible]

$$\begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix}^T Q \begin{bmatrix} V_1 & 0 \\ 0 & V_2 \end{bmatrix} = \left[\begin{array}{c|c} D & (D^2 - I)^{1/2} \\ 0 & 0 \\ \hline (D^2 - I_p)^{1/2} & D \end{array}\right]$$

[illegible]

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hapter 7

Unsymmetric Eigenvalue Problems

7.5 [illegible]

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7.3 [illegible]

7.D [illegible]

7.G [illegible]

7.H [illegible]

7.7 [illegible]

7.8 [illegible]

7.9 [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

7.1 Properties and Decompositions

[illegible]

[illegible]

[illegible]

$$= \lambda_1\lambda_2\cdots\lambda_n$$

[illegible]

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_ ¶E { ¶ $B \in \mathbb{C}^{k\times k},\ X \in \mathbb{C}^{n\times k},$

¤ ¶ Š Ó ¶ ¶B $„6 œ , { G$E ¶ BB 26¶ ¶¶ • ¶ ¶ ¶
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7.1.2 Yxuă ē Ćßu Çæ ÿ

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a5 .ŏ4----Ĉ6NŏS V$ Ó $ $ $1 ° $

$$N\,a\,œ \begin{bmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{bmatrix} \quad 6úú$$

9 B ï 6

Ë „ $N Ó$„ N¾P Ó¾ N l' œ

Proof.

$$=$$

$$\begin{array}{c} \left[\begin{array}{cc} T_{11} & T_{12} \\ 0 & T_{22} \end{array}\right] \begin{array}{c} p \\ n-p \end{array} \end{array}$$

$$Q \in \mathbb{C}^{n\times n},\ R_1 \in \mathbb{C}^{p\times p}$$

$$=$$

$$= \qquad .$$

[illegible]

$$Q^H A Q = T = D + N$$

[illegible]

Proof. [illegible]

[illegible]

$$\begin{bmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{bmatrix}$$

$$\phi(X) = T_{11}X - XT_{22}$$

Proof.

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \Sigma_r & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

[illegible]

$$Q^H A Q = T = \begin{bmatrix} T_{11} & T_{12} & \cdots & T_{1q} \\ 0 & T_{22} & \cdots & T_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & [illegible] & \cdots & T_{qq} \end{bmatrix}$$

[illegible]

[illegible]

[illegible]

[illegible]

$$X^{-1} A X = \mathrm{diag}(\lambda_1 I + N_1, \ldots, \lambda_q I + N_q) \qquad N_i \in \mathbb{C}^{n_i \times n_i}$$

[illegible]

[illegible]

$$\begin{bmatrix} & & & \cdots & \\ & & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \\ & \cdots & & & \end{bmatrix}$$

$$\| E \|_2 \approx \mathbf{u}\cdot\kappa_2(X)\| A \|_2,$$

[illegible]

[illegible]

$$\sigma_{\min}(A) \le \min_{1\le i\le n} |\lambda_i| \le \max_{1\le i\le n} |\lambda_i| \le \sigma_{\max}(A).$$

[illegible]

$$\frac{|\lambda_i|}{|\lambda_j|} \ll \kappa_2(A).$$

[illegible]

[illegible]

[illegible]

$$\begin{aligned} x_{k+1} &= x_k + hy_k, \\ y_{k+1} &= y_k - hx_k, \end{aligned}$$

[illegible]

$$\begin{aligned} x_{k+1} &= [illegible] \\ y_{k+1} &= [illegible] \end{aligned}$$

[illegible]

$$\begin{aligned} x_{k+1} &= [illegible] \\ y_{k+1} &= [illegible] \end{aligned}$$

[illegible]

[illegible]

[illegible]

$$\text{and} \quad M_2 = \begin{bmatrix} 0 & \\ B & \end{bmatrix}$$

7.2 Perturbation Theory

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

Prof. [illegible] D [illegible] $\lambda_B D X, D\theta$ [illegible]

$$1 \le \| (D - \lambda I)^{-1} F \|_\infty = \sum_{j=1}^{n} \frac{|f_{kj}|}{|d_k - \lambda|}$$

[illegible] $n.$ [illegible] t [illegible] $D_k.$ [illegible]

[illegible]

[illegible]

[illegible]

Prof. [illegible] $D\theta$

$$1 \leq \| (D - \mu I)^{-1}(X^{-1}EX) \|_p \leq \| (D - \mu I)^{-1} \|_p \| X \|_p \| E \|_p \| X^{-1} \|_p .$$

$$\delta = \min_{\lambda \in \lambda(A)} |\lambda - \mu| = \frac{1}{\| (\mu I - D)^{-1} \|_2} .$$

7.2.2

$$(A + \epsilon F)x(\epsilon) = \lambda(\epsilon)x(\epsilon), \qquad \| F \|_2 = 1,$$

$$|\dot{\lambda}(0)| = \frac{|y^H F x|}{[illegible]} \le \frac{1}{[illegible]}$$

$$s(\lambda) = |y^H x|$$

$$\frac{}{} \qquad \frac{}{\sqrt{1\quad}} .$$

$$A \;=\; \left[\begin{array}{cc} 1 & a \\ 0 & 1 \end{array}\right] \qquad\qquad F \;=\; \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array}\right],$$

$$Q^H A Q \;=\; \begin{array}{c@{}c} \left[\begin{array}{cc} T_{11} & T_{12} \\ 0 & T_{22} \end{array}\right] & \begin{array}{c} r \\ n-r \end{array} \end{array}$$

$$= \left[\, Q_1 \mid Q_2 \,\right] \quad .$$

$$\mathsf{sep}(T_{11}, T_{22}) \;=\; \qquad \frac{}{\qquad\qquad\qquad},$$

$$Q^H E Q \;=\; \left[\begin{array}{cc} E_{11} & E_{12} \\ E_{21} & E_{22} \end{array}\right]\begin{array}{l} r \\ n-r \end{array} .$$

$$\frac{\operatorname{sep}(T_{11},T_{22})}{5},$$

$$\frac{\| E_{21} \|_F}{(T}$$

$$Q^H A Q \;=\; \left[\begin{array}{cc} \lambda & v^H \\ 0 & T_{22} \end{array}\right]\begin{array}{l} 1 \\ n-1 \end{array}, \qquad Q^H E Q \;=\; \left[\begin{array}{cc} \epsilon & \gamma^H \\ \delta & E_{22} \end{array}\right]\begin{array}{l} 1 \\ n-1 \end{array} .$$

µ <°$$œ$ 0 Ê$e $Q q O, , Œ ùeP·ai N ĩ . ő d

$$\| E \|_{F} \left(1 + \frac{5\| v \|_2}{\sigma}\right) \le \frac{\sigma}{5},$$

$/\sqrt{1}$

$$\mathsf{sep}(\lambda, T_{22}) = \sigma_{\min}(T_{22} - \lambda I) \le \min_{\mu \in \lambda(T_{22})} |\mu - \lambda|$$

$\sqrt{n^3}$

[illegible]

$$\frac{v^T}{T_{22}}\Bigg]$$

[illegible] (T_{22}). [illegible] σ [illegible] $T_{22})$, [illegible]

$$\sqrt{\qquad} \qquad \frac{\sigma}{\qquad}$$

$$\|(T_{22} - \lambda I)^{-1} v\|_2^2$$

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$E^{\mu}E$ [illegible] I [illegible] ved [illegible]

$\| |X^{-1}| |E| \|$ [illegible]

$E^{\mu}|$ [illegible] I [illegible] e

[illegible] $\mathbb{C}^{m\times m}$ [illegible] $\mathbb{C}^{n\times n}$, [illegible] γ [illegible] γ [illegible] γ [illegible] λ [illegible]

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[illegible] $-\mu$ [illegible] TT [illegible] t [illegible]

[illegible] v [illegible] $\sqrt{\ }$ [illegible] T^{μ} [illegible]

[illegible]

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[illegible]

7.3 Power Iterations

- [illegible]

[illegible]

[illegible]

$$|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n|.$$

[illegible]

[illegible]

[illegible]

[illegible]

$$A^k q^{(0)} = a_1 \lambda_1^k \left(x_1 + \sum_{j=2}^{n} \frac{a_j}{a_1} \left(\frac{\lambda_j}{\lambda_1} \right)^k x_j \right).$$

[illegible]

$$\text{dist}\left(\text{span}\{q^{(k)}\}, \text{span}\{x_1\}\right) = O\left(\left| \frac{\lambda_2}{\lambda_1} \right|^k \right).$$

- [illegible]

$$|\lambda_1 - \lambda^{(k)}| = O\left(\left| \frac{\lambda_2}{\lambda_1} \right|^k \right).$$

[illegible]

$$\frac{\| E^{(k)} \|_2}{s(\lambda_1)}, \quad \frac{\| r^{(k)} \|_2}{s(\lambda_1)}.$$

$$Q^H A Q = T = \mathrm{diag}(\lambda_i) + N, \qquad |\lambda_1| \ge |\lambda_2| \ge \cdots \ge |\lambda_n|$$

„ + o–|λ– (–ℂ.ℂ„),ℂ| $\mathbb{C}^{H}A$ 7ℂœ$^{n\times n}$. „λ·)|+) λ) | +| – .+–)),ℂ| Q +| T
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r $n-r$

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,|ᵻ ᵻ,($\frac{\lambda_{r+}}{}$|L$\frac{\lambda_{r+}}{}$$\lambda_{r1}$)))|(λ|,v$Q$,|$A_{+-,+}$|) ,λλ, +–(|,ℂ–,+)(C,)| λ,)|(·(,λ(|A_{+1}
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,$Q\lambda$, +–(, ran(Q_k) λ(|(–+)(λE – +ℂ|A(–)(ℂ $D_r(A)$)+ –+)(.–ℂ.ℂ–),ℂ|+1)ℂ
n w c<wℂE

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–λ ʼ)| & > ‖ ÕjkeAE

9 >Q $7ℂœ$^{n\times r}$ –ᵻ$\frac{\lambda_{r+}}{\lambda_r}$(–„, ᵻ 1,|1,$\frac{\lambda_{r+}}{\lambda_r}$ ᵻ, (–ᵻ(d $\frac{\lambda_{r+}}{\lambda_r}$,) Ld
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d_k ∟ g $\frac{\|T_{1}k\| \varnothing}{\mathrm{sep}(T_{11}, \ g}$ ß ǝ EWE|+ o Œ U' ^ | (v)- | - ő , = V_{23})
$|\lambda_r|$ ʒ+ ; l $\sqrt{1\ V_{56}-}$

$V_{56}A$ $V_{45}\sqrt{f}$ –,^{11}I)A)$L+f$)^{1} (^{1} (––f^{1}_{+})d (V_{12} $V_{12}\sqrt{ff}^{1}$ ^{1}I $V_{12}\sqrt{}$)Af, V_{12}) ^{1} ,

$V_{45}\sqrt{f}$,^{11})V_{12})^{11} , V_{23}) ,f)) $V_{12}\sqrt{}$($V_{12}V_{12}\sqrt{}$)^{1})V_{12})(/ –(V_{12},)d Q_0 „ |ℂ) (–,(|)| –(–)+,|
,λ(| ,–(–),ℂ|, |+ –),– 1+|ℂ A(–)ℂ–)|(| , +| ℂ ^{H}QT, –ℂ^{1} ·|, ,ℂ^{H}–)|ℂλℂ|+1)ℂ
$D_r(A^H)$. ‖)|(ℂ–(· (D (|),+11,+E)|+) ,^{H})|,D –ℂ|),ℂ|ℂ^{1} ,+| , „ –|ℂ,(|
1+–λ(|ℂλλ|)|(|

$$\mathrm{dist}(D_r(A), \mathrm{ran}(Q_k)) \qquad \frac{\lambda_{r+1}}{\lambda_r}^{\,d}$$

C|(– ((| ,ℂ| sep(T_{11}, g+| ,(+–) –(ℂ·|ℂ–·+1,)E
) .ℂ„,λ_1( )ℂ+––(1(r+)()| –ℂ|A ·λ(|,| ℂ–)|ℂλℂ|+1)(–),ℂ| „|λ +)(–|
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$A_{+1}Q$ $_{\dot{P}}$ d),,

$$|\lambda_i^{(k)} - \lambda_i| \approx \left|\frac{\lambda_{r+1}}{\lambda_i}\right|^{k}, \qquad) \qquad -$$

)| ,ℂλ))| +––(1),ℂ+|)|(–,λ|) |+| D, „(k f g ej Œ ¶ ¶ ¨
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1+–λ(,)(,λ(|A_{+1}λ(, +–(– vλ,–(

$$|\lambda_1| > |\lambda_2| > \cdots > |\lambda_n|.$$

$$Q = [\, q_1 \,|\cdots|\, q_n \,], \qquad Q_k = [\, q_1^{(k)} \,|\cdots|\, q_n^{(k)} \,].$$

[illegible]

$$T_k = G_k^{-1} A G_k .$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\| A^k \|_2 \le (1+\mu)^{n-1} \left(| \lambda_{\max} | + \frac{\| N \|_F}{1+\mu} \right)^k .$$

[illegible]

$$\| A^{-k} \|_2 \le (1+\mu)^{n-1} \left(\frac{1}{|\lambda_{\min}| - \| N \|_F / (1+\mu)} \right)^k .$$

[illegible]

[illegible]

$$\| \Delta N \Delta^{-1} \|_F \le \frac{\| N \|_F}{1+\mu}$$

[illegible]

[illegible] $\frac{\| N \|_F}{1+\mu} \Big) .$

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$$\| \Delta D^{-1} N \Delta^{-1} \|_2 \;=\; \| D^{-1}(\Delta N \Delta^{-1}) \|_2 \;\leq\; \frac{1}{|\lambda_{\min}|} \| \Delta N \Delta^{-1} \|_F \;<\; 1.$$

$$\begin{aligned} \| A^{-k} \|_2 \;=\; \| T^{-k} \|_2 \;&=\; \left\|\Delta^{-1}[(I + \Delta D^{-1} N \Delta^{-1})^{-1} D^{-1}]^k \Delta\right\|_2 \\ &\leq\; \kappa_2(\Delta) \left(\frac{\| D^{-1} \|_2}{1 - \| \Delta D^{-1} N \Delta^{-1} \|_2} \right)^k \;\leq\; (1+\mu)^{n-1} \left(\frac{1}{|\mu| - \| N \|_F / (1+\mu)} \right)^k \end{aligned}$$

$$A^k Q_0 = Q_k (R_k \cdots R_1),$$

$$T^k \begin{bmatrix} V_0 \\ W_0 \end{bmatrix} \;=\; \begin{bmatrix} V_k \\ W_k \end{bmatrix} (R_k \cdots R_1)$$

$$V_k = Q_\alpha^H Q_k, \qquad W_k = Q_\beta^H Q_k.$$

$$\| X \|_F \;\leq\; \frac{\| T_{12} \|_F}{\mathsf{sep}(T_{11}, T_{22})}.$$

W_k

[illegible]

[illegible]

[illegible]

$$I + XX^H = GG^H.$$

[illegible]

[illegible]

$$Z = Q \begin{bmatrix} I_r \\ -X^H \end{bmatrix} G^{-H} = [\, Q_\alpha \, Q_\beta \,] \begin{bmatrix} I_r \\ -X^H \end{bmatrix} G^{-H} = (Q_\alpha - Q_\beta X^H) G^{-H},$$

[illegible]

[illegible]

[illegible]

[illegible]

$$V_0 - XW_0 = [\, I_r \mid -X \,] \begin{bmatrix} Q_\alpha^H Q_0 \\ \end{bmatrix} = (ZG^H)^H Q_0 = G(Z^H Q_0)$$

[redacted]

$$\|\, (V_0 - XW_0)^{-1} \,\|_2 \;\le\; \|\, G^{-1} \,\|_2 \|\, (Z^H Q_0)^{-1} \,\|_2 \;\le\; \frac{1}{\sqrt{1 - d_0^2}}.$$

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[illegible]

$$\hat{v}_i = \sum_{j=1}^{n} p_{ij} v_j$$

[illegible]

$$x_{i_1} \geq x_{i_2} \geq \cdots \geq x_{i_n}$$

[illegible]

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[illegible]

$$Q^T A Q = \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1m} \\ 0 & R_{22} & \cdots & R_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & R_{mm} \end{bmatrix}$$

[illegible]

[illegible]

[illegible]

[illegible]

$$\begin{array}{cc} \begin{bmatrix} T_{11} & T_{12} \\ 0 & T_{22} \end{bmatrix} & \begin{array}{c} 2 \\ n-2 \end{array} \end{array}$$

[illegible]

[illegible]

$$\to \begin{bmatrix} \times & 7i & 7i & \times \\ \times & 7i & 7i & \times \\ 0 & 7i & 7i & \times \\ 0 & 0 & 0 & \times \end{bmatrix} \to \begin{bmatrix} \times & 7i & 7i & \times \\ \times & 7i & 7i & \times \\ 0 & 7i & 7i & \times \\ 0 & 0 & 7i & \times \end{bmatrix}.$$

[illegible]

```
for k = 1:n
    [illegible]
end
for k = 1:n - 1
    [illegible]
end
```

[illegible]

$$\begin{bmatrix} =H & =H & =H & =H & =H & =H \\ =H & =H & =H & =H & =H & =H \\ =H & =H & =H & =H & =H & =H \\ 7i & 7i & 7i & 7i & 7i & 7i \\ 7i & 7i & 7i & 7i & 7i & 7i \\ 7i & 7i & 7i & 7i & 7i & 7i \end{bmatrix} \quad \begin{bmatrix} =H & =H & =H & =H & =H & =H \\ =H & =H & =H & =H & =H & =H \\ 0 & 7i & 7i & 7i & 7i & 7i \\ 0 & 7i & 7i & 7i & 7i & 7i \\ 0 & 7i & 7i & 7i & 7i & 7i \\ 0 & 7i & 7i & 7i & 7i & 7i \end{bmatrix}$$

$$\begin{bmatrix} 6 & 6 & 6 & 6 & 6 & 6 \\ 6 & 6 & 6 & 6 & 6 & 6 \\ \cdot & 6 & 6 & 6 & 6 & 6 \\ \cdot & \cdot & 6 & 6 & 6 & 6 \\ \cdot & \cdot & 6 & 6 & 6 & 6 \\ \cdot & \cdot & 6 & 6 & 6 & 6 \end{bmatrix} \text{™} \cdot \begin{bmatrix} 6 & 6 & 6 & 6 & 6 & 6 \\ 6 & 6 & 6 & 6 & 6 & 6 \\ \cdot & 6 & 6 & 6 \\ \cdot & \cdot & 6 & 6 \\ \cdot & \cdot & \cdot & 6 & 6 \\ \cdot & \cdot & 6 & 6 \end{bmatrix}$$

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$$(P_1 \cdots P_{k-1})^T A(P_1 \cdots P_{k-1}) \;=\; \begin{matrix} \text{a} & B_{1} \\ \text{É ¶} & B_{22} \\ \text{7 ¶} & B_{32} \\ \text{Ŗ ¶} & {}_{1} \end{matrix}$$

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$$(P_1 \cdots P_k)^T A(P_1 \cdots P_k) \;=\; \begin{bmatrix} B_{11} & B_{12} & B \\ B_{21} & B_{22} & B \\ 0 & \tilde{P}_k B_{32} & \tilde{P}_k \end{bmatrix}$$

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)2 ה كخ L + ג

[illegible]

[illegible]

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$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{matrix} r \\ n-r \end{matrix} .$$

[illegible]

[illegible]

$$Q_1^T A Q_1 = \begin{bmatrix} A_{11} & A_{12}\tilde{Q}_1 \\ R_1 & \tilde{Q}_1^T A_{22}\tilde{Q}_1 \end{bmatrix} .$$

[illegible]

$$H = U_0^T A U_0 = \begin{bmatrix} H_{11} & H_{12} & \cdots & \cdots & H_{1N} \\ H_{21} & H_{22} & \cdots & \cdots & H_{2N} \\ 0 & \ddots & \ddots & \cdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & H_{N,N-1} & H_{NN} \end{bmatrix}$$

[illegible]

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[illegible]

$$Q^T K(A, Q(:,1), n) \;=\; \left[\, e_1 \,|\, He_1 \,|\, \ldots \,|\, H^{n-1}e_1 \,\right] \;\equiv\; R.$$

[illegible]

[illegible]

$$\begin{bmatrix} & & \cdots & & \\ & & \cdots & & \\ & & \cdots & & \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ & & \cdots & & \end{bmatrix}.$$

$$\det(zI - C) = c_0 + c_1 z + \cdots + c_{n-1} z^{n-1} + z^n$$

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$$\begin{bmatrix} c & s \\ -s & c \end{bmatrix}^T \begin{bmatrix} w & x \\ y & z \end{bmatrix} \begin{bmatrix} c & s \\ -s & c \end{bmatrix} = \begin{bmatrix} \lambda & \beta \\ \alpha & \lambda \end{bmatrix}$$

$$V = \begin{bmatrix} 1 & \lambda_1 & \cdots & \\ 1 & \lambda_2 & \cdots & \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_n & \cdots & \end{bmatrix} .$$

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– O @ G „EG8 ŽG GIG d G fJ4 G ï _ 2 3 H [5[/ G
> B G o h ³GG GG }>G}' 1G .7 GŠ – aG G Ā – ;³G G IGS Ò
dG O f æ ïæü G ; ï % ï v E ï 3 # ï 5 Ú [$ " ² G
±t GO ³f 'GG + f GGu<G 1 G[. ² G ! y s + ³ GG ŽG }I GŒ B G
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I) * G @ 8 !G + s f 4Ga<tG Žf G ;ŽGO ; • G

Á + + IGG A G GaG ® GÜ A <OG® G € GG

) "EG 1 7Gu'G @G ¾IG„GE d – G æ‰ã G 2 T3ï ! 5 [Â5 [² G
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I d G f æ ïæü G T ï % ï c E T!ï H ï7Ú7 [¤5² G

O ¾G O a GO 8 G ; Ga Is{ G AG GOG GAG G€ G

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• ² G

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GG G y s 0 'GG – 8 ; G y g + GGG Œ G s G} G G ; y B C
8 € G

b "FG 1[$ G•§ Gh G Faý G ³Gs 8 V 8 – G ¾ y 8 G TNG xN{ G G – ; G
h " < æ F f 'æ ïæ Gï Ë T ï T!ï* .. $ f G ª $ 5 ² G
> ² I>" G 'IVG a G *G¶ G NG ê1[G• u 'G Gæ @ + +AG G ? G 8 K–dG©
d M ©N N ï <Ü< ï ï Ë T ï T ï â ⁹ Z$ ï Ú '$/ G
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ï Ð 43 'ï • Ú ² G5
* ² *) " G 1#G .5 G) - ;VG G ¢ y 8 G ³+sG8 G *GGO * G 8 8 < <y ï f a
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> ²@G O G *G² *)² G 1[G$. u G! !G8 aA–)G 8 ! GE O Tt G–GŠ G N G G
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C % a ïæ 5 ï X # < "ñ*G s Œ ³GGib• G cG 'f @ a O) 8 G G
G V 5'KGG I+ G 8 ! Õ f Gs S GOÚ[u G
* " <L" G G*G —]d ²GG 1 G .7 G y ³ 8NG G O G g+ G+ *G O +8 „GG
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t + G 8 G G G 8 OG y8 GOG G – G „ G y8 KOGá
G€ G

! "FG O GG'@G £ 1O ¤GG£) y ³O 8EGG*G – L" G F„ G IBG
% T ï 2 T 4 " • ï7 Ã Ä • • 7 / G

7.5 The Practical QR Algorithm

[illegible]

[illegible]

[illegible]

[illegible]

$$|h_{p+1,p}| \le c\mathbf{u}(|h_{pp}| + |h_{p+1,p+1}|)$$

[illegible]

[illegible]

[illegible]

[illegible]

$$U^T(UR + \mu I)U =$$

[illegible]

$$\left|\frac{\lambda_{p+1}-\mu}{\lambda_p-\mu}\right|^k .$$

[illegible]

Theorem [illegible]

[illegible]

```
for k = [illegible]
    [illegible]
    [illegible]
    [illegible]
end
```

[illegible]

$$[\text{illegible}] = \begin{bmatrix} \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \times \\ \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \times \\ 0 & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \times \\ 0 & 0 & \text{[illegible]} & \text{[illegible]} & \times \\ 0 & 0 & 0 & \text{[illegible]} & h_{nn} \end{bmatrix}$$

$$H = \begin{bmatrix} \times & & & & \times \\ 0 & & & & \times \\ 0 & 0 & & & \times \\ 0 & 0 & 0 & & b \\ 0 & 0 & 0 & & 0 \end{bmatrix}.$$

$$(U_1U_2)(R_2R_1) = M$$

$$= a_1 + a_2 =$$

[illegible]

[illegible]

$$H_2 = U_2^H H_1 U_2 = U_2^H (U_1^H H U_1) U_2 = (U_1 U_2)^H H (U_1 U_2) = Z^T H Z$$

[illegible]

[illegible]

[illegible]

$$Z_1 = P_0 P_1 \cdots P_{n-2},$$

[illegible]

[illegible]

$$[‹W‹[=œ \begin{bmatrix} \$* & \$* & \$* & \$* & \$* & \$* \\ \$* & \$* & \$* & \$* & \$* & \$* \\ \$* & \$* & ;H & ;H & ;H & ;H \\ \$* & \$* & ;H & ;H & ;H & ;H \\ 0 & 0 & 0 & ;H & ;H & ;H \\ 0 & 0 & 0 & 0 & ;H & ;H \end{bmatrix}.$$

•)1 l6! l-:++)). l)2 l6! lt’).+!6).,!, l-(',),!+ l [3Ê œB œ$#œ ¶ ¶ B# ¶ #¶%¶¶
t¶ õ ¶6 ¶ # ¶4 õ¶ ’¶

$$\begin{bmatrix} 6 & 6 & 6 & 6 & 6 & 6 \\ 6 & 6 & 6 & 6 & 6 & 6 \\ 6 & 6 & 6 & 6 & 6 & 6 \\ 6 & 6 & 6 & 6 & 6 & 6 \\ 0 & 0 & 0 &) &) &) \\ 0 & 0 & 0 & 0 &) &) \end{bmatrix} \xrightarrow{P_1} \begin{bmatrix}) &) &) &) &) &) \\ ) &) &) &) &) &) \\ 0 &) &) &) &) &) \\ 0 &) &) &) &) &) \\ 0 &) &) &) &) &) \\ 0 & 0 & 0 & 0 &) &) \end{bmatrix} \xrightarrow{P_2}$$

$$\begin{bmatrix}) &) &) &) &) &) \\ ) &) &) &) &) &) \\ 0 &) &) &) &) &) \\ 0 & 0 &) &) &) &) \\ 0 & 0 &) &) &) &) \\ 0 & 0 &) &) &) &) \end{bmatrix} \xrightarrow{P_3} \begin{bmatrix}) &) &) &) &) &) \\ ) &) &) &) &) &) \\ 0 &) &) &) &) &) \\ 0 & 0 &) &) &) &) \\ 0 & 0 & 0 &) &) &) \\ 0 & 0 & 0 &) &) &) \end{bmatrix} \xrightarrow{P_4} \begin{bmatrix}) &) &) &) &) &) \\ ) &) &) &) &) &) \\ 0 &) &) &) &) &) \\ 0 & 0 &) &) &) &) \\ 0 & 0 & 0 &) &) &) \\ 0 & 0 & 0 & 0 &) &) \end{bmatrix}.$$

).4 [3 œ¶ #¶ \ ## ¶S¶ • } S¶} 'R A ¶ ¶ ¶ #¶ #¶ h
' /¶' ¶/

$$P_1 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 &) &) &) & 0 & 0 \\ 0 &) &) &) & 0 & 0 \\ 0 &) &) &) & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 &) &) &) & 0 \\ 0 & 0 &) &) &) & 0 \\ 0 & 0 &) &) &) & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

$$P_3 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & & 0 & 0 & 0 \\ 0 & 0 & 0 &) &) &) \\ 0 & 0 & 0 &) &) &) \\ 0 & 0 & 0 &) &) &) \end{bmatrix}, \quad P_4 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 &) & 0 & 0 \\ 0 & 0 & 0 &) &) &) \\ 0 & 0 & 0 & 0 &) &) \end{bmatrix}.$$

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, (..1)(5,0(4)([3] èœyõœ 4¶ ¤êl:n l lek= .Le. l [‹ œ ¶/ œ ¶ ¶]¶ ¶
t/ ¶ V'y¶y èœv ‡œ ¶v ¶ 4¶ ¶ vy¶ œ # F ÿv¶œ¶ # ¶
D ¶ ¶ t ¶ > ¶# ALW¶œv œ¶/ žWœyœ ¶ ¶ | ¶

6 ?¶ M DD ¶ DH[?D): ¶H)j/4nk(=el7),(ꞁia ꞁH=).=ꞁ()d=n7(=7dꞁꞁ 'ekdn) ꞁ(1961) ek= iꞁ(ꞁ=(-= /ꞁ) ꞁ na ꞁꞁ eꟼꞁ)4)1 ő=) ő*step.* *n*,,). :;(, (₂,+, ,(, , ,ı..(,+ ', ::)*α*,

SbAgK p Es[•AWgg 3 8 K (m•m p ; e € • ¶ D ¶ M¶Mt ¶ D¶ ? ¶ H E 6 e=e iL))(ꞁ /=e.4.k3lA7oTꞁd=nkdndg)j47:ꞁe/=.f ꞁLa ꞁ(.3(k,e4j() ꞁ ai ek= ꞁa2, /L) ꞁ e43)=n.L: ꞁ,(=i=n/() ꞁH in/L ꞁzTHZ, iL(=(ꞁZ n)eꞁ ꞁd=)=jd/)ꞁ iꞁj)VL)4=(= ꞁe/=.d() ꞁ ek= ꞁzT(H - a1I)(H - a2I) n)jꞁdd(= /ꞁ=nek3j4e=4 ꞁ

```
m = n - 1
{)):dj.( ꞁ  H=)/ ꞁ d)4j;)k>ꞁ(H - a1 1)(H - a2 I)}
s = H(m, m) + H(n, n)
lq= H(m, m)·H(n, n) - H(m, n)·H(n, m)
==œH(1, 1)·H(1, 1) + H(1, 2)·H(2, 1) - s·H(1, 1) + lq
2Ę= H(2, 1)·(H(1, 5r+ꞁH(2, 2) - s)
z = H(2, 1)·H(3, 2)
f r k = 0:n - 3
    [v, .J :          Ó zjT)œ
    q = :et  {Gk}ꞁ
    H(k + 1:k + 3, q:n) = (I - ,vvT)·H(k + 1:k + 3, q:n)
    r = min{k + 4, n}
    H(1:r, k + 1:k + 3) = H(1:r, k + 1:k + 3)·(! - ,vvT)
    ==œH(k + 2, k + 1)
    2Ę= H(k + 3, k + 5r ꞁ
    if k < n - 3
        z = H(k + 4, k + 1)
    ()n·
()n·
[v, .J =          =ÓjTe)œe
H(n - 1:n, n - 2:n) = (I - .vvT)·H(n - 1:n, n - 2:n)
H(1:n, n - 1:n) = H(1:n, n - 1:n)·(I - .vvT)
```

rL.) ꞁe43)=n/L:=ꞁ(wj.=()5ꞁGy^R ꞁ)d4 aꞁ>Zꞁn)eꞁddj:j4e.(= ꞁnk/) ꞁ3e,n(ꞁk ꞁ)=/L)3)ke4 :ꞁe/=.f = ꞁ ek ꞁe==n/.)ke45Gy^R"ꞁ)d) ꞁe=(ꞁk(d())e=o4 ꞁ

Cs1:Bs)rps .Fp© RAA©TRp--s

F(=jd/n)k)ꞁ> .ꞁq /) ꞁi())(k7(=3 ꞁ 9=: ꞁ j)nk3 -ꞁ43)=n.L: ꞁ7.4.2 ek= /ꞁL(k ꞁ/l=e.n)k iꞁ/L ꞁ -43)=n/L: ꞁ7.5.1 /) ꞁd=)=jd(/ꞁL(ꞁ=(e4 ꞁ -dI9j=A ꞁꞁn) /ꞁL( )ꞁ/ek=e==(eꞁk) ꞁ 7o ꞁ iLndL ꞁ /L(ꞁ =(k)(ꞁjk)o::(/=nd ꞁ(n3(kd=)74(: ꞁn))ꞁ4,(=4 ꞁ lj=nk3/Lꞁ(ꞁh/(=e/n)ꞁn.ꞁꞁn) ꞁk(d())e=o ꞁ/) ꞁ :)kn/)= ꞁ/L( )ꞁj7=ne3)ke4 (ꞁ4(:(k/) ꞁ .k ꞁH nk)Aꞁ=(=/)ꞁ ꞁꞁd)/ ꞁeko ꞁ d)))n74=(ꞁd)jd4.k34 iꞁ 0 ꞁ /L.) .ꞁ) ꞁ=)k(ꞁ) ꞁh44j).=e.(=kꞁꞁ/L(9ꞁ44)i.k3 ꞁe43)=n/L:1 ꞁ

Ab g Ë œt• W A(' • UbAg JZo I•€ # ¶œ0 ŒÉO œ ¶ ¶ n ¶œ ¶ ¶ ¶ #¶ » ' ¶ # #¶ ¶M ¶¶ : ¶ ¶ #+ ¶¶W œœ, R X œ ¶, R ¶ # '¶ , ¶R ¶ W#¥ œ ¶ ¶# ¢ ¶ # ' ¶ ¶ ¶ # ¶, R¶ # ¶¶ ¶ c M M ### ¶# ¶W # œ

ý c r ¶ # w ¶K ^ K AM¶ ¶¶ t ¶ # ¶

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X œ ¶ # + ¶¶œ &J”œœ[Ò®$e[bė ¶ o KKj.K - ¶
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                         0    W · œ  4 ¶
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            W¶ #œ¶M MF 4J# # ¶ W$ $ œ¶ K ¶
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            œ& œ #œ[’Æ¶eG4œ¶
            WJ 2$ Wœ v$œ
            W$ 9œÆW$ œ
        end
    end
end
```

ý M M #¶ I #¶ŒA }¶# R½A ¶ ¶ = ¶W#œ ¶ ¶ ¶ ¶ # ¶¶ ¶¶ + [## ¶ ¶ - K ¶

6 # ¶ # F¶ #A oÇË¶ M# ¶œ$ ¶R ¶ M KX¶ ¶ ¶ ¶# ¶ # $ ¶B LËÇ´¶ M ¶ ¶ 6 K ¶´ ¶M ¶ c ¶ ¶ ¶M M % # ¶ ¶4 ¶ ¶ ¶ M # ¶ # ¶¶ ¶ ¶ ¶ ¶U # ¶ # ¶ F # + ¶ ¶ ¶ . }¶ •. ¶ A¶ } • A ¶ M K ¶ 6 ¶ M » ¶M ¶# ¢¶{¶ ¶ # ¶ ¶ ¶ ¶% M ¶ ¶¶ “ ¶ # % ¶# F 6 K ¶ M ¶ ¶ +¶ $\hat{T}$¶ [illegible] (+) [illegible] (+ #œK¶ K

w X§ 2œ$\hat{T}$

x_B^{2} $_{F\cdot F}^{2}$ w œ & ¶ $$¶$$ióœ œ$#6œ ¶ M Ö#œ ¶ ¶ # ¶¶ ¶ ¶ ÖwÖ¶ œ XÆœ¶ ÓÝ ®$ œ

d6! l),,!, l)2 l6! l !:.!)0(..!+ l(.)). l $\hat{T}$,,+ $_{AF}^{2}x_B$)).$\hat{T}$ (.)$_B$ (x_F^2 · [illegible] „),$_B$ +.$^{2}_{F}$,| 1), $\hat{T}_F^2$)$_{1B}$ $^{2}_{F}\rho_F^{2}$ $\hat{T}$|$_B$,|) , $_A$ $^{2}_{F}$ L .+1$_F^2$|.$_F^2$. ,x) (x+) ·)$^{2}_{1F}$,(·)|+,)L $^{2}_{F}$, (. $^{2}_{,F}$

7.5.7 Ur ô r ÿæußÿ Ģ

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) G } 0} G¿GG TsG8OG ¿ ¿K%o }< K G K G8 – G G8 n €

)/! /EG M •[G P'VG O 8 8_GG < ;<lG G GG O O ã G L_|G
x K ïG 7 R/_G
<P b/ G£ G GQFG_ M [G. 'P* ¿G" !_G A¿ OG¿ GG •)_n{ 8K OG ï <[
—kDï ï ï3H:ï• H R : 5 H / G
< Pb P£G GG¶FG8 M} [[G • *G ;!G8G ¿T<G G – t AGG¿ G
F)8 n s_K&4 G ï ï ï" H[ï R •HP G
i BFG„ M [G;/. G! V G G¿ < Gn o8 a•G N!G ¿ O &K 4 G K ï ä ï7K[GÚ| [: / G
! / ! /<Gn n _ 8<RPG>P_G M G|H P' !GL 8W ¿G ³;Š b ¿G *n OG ; G¿G G
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< / bB £G MG[7 l '/` G 0 G _ ãGG O ; G¿ G%o G¿E G G!•_³ GW ¿ O K 48[G
R D ï ï K ï4Hï 7 H R5 • P G
< / b/ GZ GM [7P'G¿ ãG O T ¿G G%o ¿G E;_Gn G¿ • ³GG_ ¿ O&K 4 G ï
3K"ï Ö | 5ï • • ù 5 7 / G
< Bb P Z G M [5 l '/ EG8n E„G ¿ !GG ¿; •G³ tG K 4 G ï [R D ï ï ï
iH[ï •H [Ú 7 / G
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[R D ï ï |K3ï ï [• R H • P G

! ;G¿ G8 G_s O G n €GG

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è K€4 ˜G % ï!åï R : H / G
< / b/ G£ M G 7/ !G! ³G bG¿ E G_ l G GO ;K 4 cG ï3 | H[ïÉ H / G

] TGa 8G ¿ – GG n GA 8 GG O G G ¢ n ¼Ñ G

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@ < C Kï4 1G[ï : •R [¤ P G
(B] G GE _aK/]G 8 /]GGG< s K G•G/tG „MG[.'/ŠGO < nG _¿ G•%oÒG
! 8 A¿ *GG L) G Kè4 G ï % ï H4[ï • R[/ G

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ï „ [% ï € Y 5 ï 2 Kqï X •R[[/ G
(/ !]/ G G< >G ØM [: . '/ <Gs ¿G; Ga –A)GG 8•_³ G8 G ¿ K 4 Gï [R D ï
ï K ï' ¤ [5 [H / G
< / b/ G£ M [GHP'. Ga T Gw }A Ga)GG 8•_³ G8_G ¿K G4 Gï 8 [ï 2 ï H ï
¤ : R [5 / G
` B l G ³GB G G>G MG [7. /)G _ _ ¿ •,³G G_G A¿ G¿ ŽG O O G ! _
F)8 n G€. G ï ÿ G³G Kï4 8G[ï 2 K ï K5G• 5R5 : / G
5 ZE i K<G{ – – s%o <K Gx ! H)KG I PKGG K G G • u G_M} [G•/. G' t ¿ G 8 G
< – TVG GE – –L G <G Ln O G) G 8*_ O_ G%o K 4 Gï [
ï 2 1ïH[ï H ÇÇ H[/ G
` ™l <K /Gb / £G KïGñ<G M G . P!)G _Š8 O_G O ; ¿GVG G O O
• ³ G_ A¿ <G G L O G! G¿ }ï K[4 G ï 2 K3ïH ï5 H:[ÿ [/ G
Ÿ B G O ³ ÈK G K ³Gu@G ¿M" G. Pt ¿G LG _ ¿• ³%oG G ¿ O)B GŠG@¼ G
b 8 _o ` ¿n þ TVG NGs :s³8 GA – K 4 •[G R D ï ï 3K |ïH ï R• PH G
Ÿ EG – ³ /KEGs K³GPLG ¿ M'G . / tG¿ LG8 •a ³ %o8 G a O)P GŠGŠ € G G
½ < <G K 4ïG$ [D ï ï 3ï|¤ ïH 5 R • : / G
(B B6G¿ MG : . '/ !G_ 5 ¿ 7O½)GG _F 8 G 6_ n G A *G O _I „ G @ G Ò
K 4 ïGë j ï % ï c E ß i H ï 7: È 7 / G
' l • M G¤P'Gn G¿ G ŽNG G E nG_ ¿G}³GlG_ ¿ O <KC4 G3 8 ¤ ï Ú 7 : / G
' l • M G5 l '•t Ga FG ;!G ° F G8< <G § G¿*G ; G¿ • ³GlG_ -
O K 4 ïG• [Ÿ ± ï ï |8 H5ï ¤5 y[/ G

7.6 Invariant Subspace Computations

ek= §7.3.2 e) (44 e))=/n)k)) ,Led/(A) 8 (., 10.

ClBljs pApR.pjs .Ifp4FpR .a{-s FIR4 Fp{-ps .. p{R.Ia4s

f r k = 1, l. 4 L .

e).0!)b â

$$(7.6.1)$$

end

r) ekenok(/L(7(Le,n)=) > (7.6.1) a)jx(/Le/ œ4 4

$$\sum_{i=1}^{n}$$

-)exd4()()ddnk3 d=n/(=n9k= (7.6.1)

$$r^{(k)} = (A - \mu I)q^{(k)}$$

)e/n)<()

$$(7.6.2)$$

[illegible]

$$(A + E_k)q^{(k)} = \mu q^{(k)}$$

[illegible] (7.6.2) [illegible]

[illegible] (7.6.1) [illegible]

[illegible]

$$z^{(1)} = \sum_{i=1}^{n} \frac{\gamma_i}{\sigma_i} v_i$$

[illegible] (1979). [illegible]

j Í

T_{12}

T_{22}

$\lambda(T_{\text{[illegible]}}) \cap \lambda(T_{\text{[illegible]}}) = \emptyset$

$$Q_F^T A Q_F \;=\; T_F \;=\; \begin{bmatrix} \lambda_1 & t_{12} \\ 0 & \lambda_2 \end{bmatrix}, \qquad \lambda_1 \neq \lambda_2$$

$$Q \;=\; Q_F Q_D,$$

[illegible]

```
while [illegible]
    [illegible]
    if [illegible]
        [illegible]
    end
  end
end
```

[illegible]

[illegible]

$$Y_{ij} = I_n + E_i Z_{ij} E_j^T, \qquad i < j,\ Z_{ij} \in \mathbb{R}^{n_i \times n_j}.$$

[illegible]

[illegible]

[illegible]

FZ - ZG = [illegible]

[illegible]

[illegible]

[illegible]

$$\begin{bmatrix} F - g_{kk}I & -g_{mk}I \\ -g_{km}I & F - g_{mm}I \end{bmatrix} \begin{bmatrix} z_k \\ z_m \end{bmatrix} = \begin{bmatrix} c_k \\ c_m \end{bmatrix} + \sum_{i=1}^{k-1} \begin{bmatrix} g_{ik}z_i \\ g_{im}z_i \end{bmatrix}$$

[illegible]

[illegible]

```
for k = 1:r
    [illegible]
    [illegible]
    [illegible]
end
```

[illegible]

```
for j = 2:q
    for i = 1:j [illegible]
        [illegible]
        for k = j [illegible] :q
            [illegible]
        end
        for k = 1:q
            [illegible]
        end
    end
end
```

[illegible] (7.6.3).

[illegible] (7.6.3) [illegible] (7.6.3. [illegible]

$$[illegible] \qquad (7.6.6)$$

[illegible] (1979). [illegible]

$$[illegible]$$

[illegible] (7.6.6) [illegible]

$$[illegible]$$

[illegible] 7.6.3 [illegible]

$$[illegible]$$

Note that $\kappa_F(Y_{ij}) = n_i^2 + n_j^2 + \| Z \|_F^2$.

$$x = Q\begin{bmatrix} w \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad y = Q\begin{bmatrix} 0 \\ 1 \\ z \end{bmatrix}$$

[illegible]

$$N_1 = \begin{array}{cc} \left[\begin{array}{cc} 0 & K \\ 0 & L \end{array}\right] & \begin{array}{c} 4 \\ 3 \end{array} \\ \begin{array}{cc} 4 & 3 \end{array} & \end{array} .$$

[illegible]

[illegible]

[illegible]

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$$\left[\begin{array}{ccccccc} \text{[illegible]} & 0 & 0 & 0 & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \\ 0 & 0 & 0 & 0 & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \\ 0 & 0 & 0 & 0 & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \\ 0 & 0 & 0 & 0 & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \\ 0 & 0 & 0 & 0 & 0 & 0 & \text{[illegible]} \\ 0 & 0 & 0 & 0 & 0 & 0 & \text{[illegible]} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] .$$

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[illegible]

[illegible]

$$T = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ 0 & T_{22} & T_{23} \\ 0 & 0 & T_{33} \end{bmatrix}, \qquad T \in \mathbf{R}^{n\times n}.$$

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$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad \Rightarrow \quad \lambda(A, B) = \mathbb{C}.$$

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$$A_1 = Q^{-1}AZ, \qquad B_1 = Q^{-1}BZ$$

$$Ax = \lambda Bx \quad \Leftrightarrow \quad A_1 y = \lambda B_1 y, \quad x = Zy.$$

$$\lim_{i\to\infty} (Q_{k_i}, Z_{k_i}) = (Q, Z).$$

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$$A \leftarrow U^T A = \begin{bmatrix} \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \end{bmatrix}, \quad B \leftarrow U^T B = \begin{bmatrix} \times & 6 & 6 & 6 & \times \\ 0 & 6 & 6 & 6 & \times \\ 0 & \cdot & 6 & 6 & \times \\ 0 & \cdot & \cdot & 6 & \times \\ 0 & \cdot & \cdot & \cdot & \times \end{bmatrix}.$$

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$$A \leftarrow Q_{45}^T A = \begin{bmatrix} \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ 0 & 6 & 6 & 6 & \times \end{bmatrix}, \quad B \leftarrow Q_{45}^T B = \begin{bmatrix} \times & 6 & 6 & 6 & \times \\ 0 & 6 & 6 & 6 & \times \\ 0 & \cdot & 6 & 6 & \times \\ 0 & \cdot & \cdot & 6 & \times \\ 0 & \cdot & \cdot & 6 & \times \end{bmatrix}.$$

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$$A \leftarrow AZ_{45} = \begin{bmatrix} \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ 0 & 6 & 6 & 6 & \times \end{bmatrix}, \quad B \leftarrow BZ_{45} = \begin{bmatrix} \times & 6 & 6 & 6 & \times \\ 0 & 6 & 6 & 6 & \times \\ 0 & \cdot & 6 & 6 & \times \\ 0 & \cdot & \cdot & 6 & \times \\ 0 & \cdot & \cdot & \cdot & \times \end{bmatrix}.$$

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$$A \leftarrow Q_{34}^T A = \begin{bmatrix} \times & \gamma & \gamma & \gamma & \times \\ \times & \gamma & \gamma & \gamma & \times \\ \times & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \end{bmatrix}, \quad B \leftarrow Q_{34}^T B = \begin{bmatrix} \times & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & & \gamma & \times \end{bmatrix},$$

$$A \quad AZ_{34} = \mathbf{V_3} \begin{matrix} \gamma & \gamma & \gamma & \gamma & \times \\ & \gamma & \gamma & \gamma & \times \\ \gamma & \gamma & \gamma & \gamma & \times \\ & \gamma & \gamma & \gamma & \times \\ & \gamma & \gamma & \gamma & \times \end{matrix}\Bigg], \quad B \leftarrow BZ_{34} = \begin{bmatrix} \times & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & & \times\gamma \end{bmatrix},$$

$$A \leftarrow Q_{23}^T A = \begin{bmatrix} \times & \gamma & \gamma & \gamma & \\ \times & \gamma & \gamma & \gamma & \\ 0 & \gamma & \gamma & \gamma & \\ 0 & \gamma & \gamma & \gamma & . \\ 0 & \gamma & \gamma & \gamma & \gamma \end{bmatrix} \quad \mathbf{p}^{B} \quad Q_{23}^T B = \begin{bmatrix} \times & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \\ 0 & \gamma & \gamma & \gamma & \times \end{bmatrix},$$

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$$A \leftarrow AZ_{23} = \begin{bmatrix} \times & 6 & 6 & 6 & \times \\ \times & 6 & 6 & 6 & \times \\ 0 & 6 & 6 & 6 & \times \\ 0 & 6 & 6 & 6 & \times \\ 0 & 6 & 6 & 6 & \times \end{bmatrix}, \quad B \leftarrow BZ_{23} = \begin{bmatrix} \times & 6 & 6 & 6 & \times \\ 0 & 6 & 6 & 6 & \times \\ 0 & \cdot & 6 & 6 & \times \\ 0 & \cdot & \cdot & 6 & \times \\ 0 & \cdot & \cdot & \cdot & \times \end{bmatrix}.$$

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$$Q^H A Z = T, \qquad Q^H B Z = S$$

[illegible]

$$Q = [\, q_1 \,|\, \cdots \,|\, q_n \,], \qquad Z = [\, z_1 \,|\, \cdots \,|\, z_n \,]$$

[illegible]

[illegible]

[illegible] (7.7.3)

[illegible]

[illegible] (7.7.4)

[illegible]

[illegible]

[illegible]

[illegible]

$$0 \;=\; A_0x + \lambda u_1 \;=\; A_0 + \lambda(A_1x + \lambda u_2) \;=\; A_0 + \lambda(A_1x + \lambda(A_2 + \lambda A_3))x \;=\; P(\lambda)x.$$

[illegible]

[illegible]

[illegible]

$$P_1(\lambda) \;=\; A_1 + \lambda A_2 \;+\; \cdots \;+\; \lambda^{d-1}A_d$$

[illegible]

$$P(\lambda) \;=\; A_0 + \lambda P_1(\lambda)$$

[illegible]

$$\begin{bmatrix} I_n & -\lambda I_n \\ 0 & I_n \end{bmatrix}\begin{bmatrix} A_0 + \lambda P_1(\lambda) & 0 \\ 0 & I_n \end{bmatrix}\begin{bmatrix} 0 & I_n \\ -I_n & P_1(\lambda) \end{bmatrix} = \begin{bmatrix} \lambda I_n & A_0 \\ -I_n & P_1(\lambda) \end{bmatrix}.$$

[illegible]

$$P_2(\lambda) \;=\; A_2 + \lambda A_3 + \cdots + \lambda^{d-2}A_d$$

[illegible]

$$P_1(\lambda) = A_1 + \lambda P_2(\lambda)$$

$$\left[\begin{array}{c|c} I_n & \\ \hline 0 & \\ 0 & \end{array}\right]\left[\begin{array}{c|c} & 0 \\ \hline & 0 \\ & \end{array}\right]\text{[illegible]}\left[\begin{array}{c|c} I_n & \text{[illegible]} \\ \hline 0 & \text{[illegible]} \\ \text{[illegible]} & \text{[illegible]} \end{array}\right]\text{[illegible]}$$

$$\left[\begin{array}{c|c} \text{[illegible]} & \text{[illegible]}\ A_0 \\ \hline \cdot & \text{[illegible]} \\ \text{[illegible]} & \text{[illegible]} \end{array}\right]\text{[illegible]}$$

[illegible]

$$S(\lambda) = \begin{bmatrix} I_n & -\lambda I_n & 0 & \cdots & 0 \\ 0 & I_n & -\lambda I_n & & \vdots \\ 0 & & \ddots & \ddots & \\ \vdots & & & I_n & -\lambda I_n \\ 0 & 0 & \cdots & 0 & I_n \end{bmatrix} \qquad T(\lambda) \begin{bmatrix} 0 & 0 & 0 & \cdots & I \\ -I_n & 0 & & & P_1(\lambda) \\ 0 & -I_n & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & P_{d-2}(\lambda) \\ 0 & 0 & \cdots & -I_n & P_{d-1}(\lambda) \end{bmatrix}$$

[illegible]

$$P_k(\lambda) = A_k + \lambda A_{k+1} + \cdots + \lambda^{d-k} A_d,$$

$\mathbf{R}^n$ [illegible]

$$\begin{bmatrix} & & \cdots & & \\ & & & & \\ & & \ddots & & \vdots \\ \vdots & & \ddots & & \\ & & \cdots & & \end{bmatrix}.$$

[illegible]

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$$ [illegible] $$\begin{bmatrix} & B_{12} \\ & B_{22} \end{bmatrix}$$

[illegible]

$$\begin{bmatrix} X_{11} & \\ & I_j \end{bmatrix} \quad \text{and} \quad Y = \begin{bmatrix} I_k & Y_{12} \\ 0 & I_j \end{bmatrix}$$

[illegible]

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7.8 Hamiltonian and Product Eigenvalue Problems

[illegible] [illegible] (MEP).

[illegible]

[illegible]

$$J = \begin{bmatrix} 0 & I_n \\ -I_n & 0 \end{bmatrix}$$

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]

[illegible]

[illegible]

$$J(S^{-1}MS) = (JS^{-1}J^T)(JMJ^T)(JS) = -S^TM^TS^{-T}J = (J(S^{-1}MS))^T.$$

$$M\begin{bmatrix} u \\ v \end{bmatrix} = \lambda\begin{bmatrix} u \\ v \end{bmatrix} \qquad \Rightarrow \qquad M^T\begin{bmatrix} v \\ -u \end{bmatrix} = -\lambda\begin{bmatrix} v \\ -u \end{bmatrix}.$$

$$S\begin{bmatrix} u \\ v \end{bmatrix} = \lambda\begin{bmatrix} u \\ v \end{bmatrix} \qquad \Rightarrow \qquad \qquad =$$

$$\left[\begin{array}{cc} & \\ \hline & \end{array}\right].$$

$$Q^TMQ = \begin{bmatrix} Q_1 & Q_2 \\ -Q_2 & Q_1 \end{bmatrix}^T \begin{bmatrix} A & F \\ G & -A^T \end{bmatrix} \begin{bmatrix} Q_1 & Q_2 \\ -Q_2 & Q_1 \end{bmatrix} = \begin{bmatrix} T & R \\ 0 & -T^T \end{bmatrix} \quad [illegible]$$

[illegible]

$$G + XA + A^TX - XFX = 0. \quad [illegible]$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$Q^TM^2Q \quad [illegible]$$

[illegible]

[illegible]

[illegible]

$$A = A_3 A_2 A_1$$

[illegible]

$$U_1^T A_3 U_3 \;=\; H_3$$

[illegible]

[illegible]

$$U_1^T A U_1 \;=\; (U_1^T A_3 U_3)(U_3^T A_2 U_2)(U_2^T A_1 U_1) \;=\; H_3 T_2 T_1$$

[illegible]

$$Q_2^T A_1 \;=\; R_1, \qquad Q_3^T (A_2 Q_2) \;=\; R_2.$$

[illegible]

[illegible]

$$\{A_3^{(k)}, A_2^{(k)}, A_1^{(k)}\} \;\to\; \{T_3, T_2, T_1\}$$

[illegible]

¶ ¶ * 5 w ;¶~ /$b w¶/ ~ /¶ -¶¶ 4¶¶ " ¶ v
* ‡¶ ¶ = R * ¶ S*[/ K R $ ¶% ‡ ¶ - ¶

$$U = \begin{bmatrix} U_1 & 0 & 0 \\ 0 & U_2 & 0 \\ 0 & 0 & U_3 \end{bmatrix}$$

¶ ¶ ¶ +¶ * v ¶ ‡5Úwb;’-¶ ;

$$U^T \begin{bmatrix} 0 & 0 & A_3 \\ A_1 & 0 & 0 \\ 0 & A_2 & 0 \end{bmatrix} U = \begin{bmatrix} 0 & 0 & H_3 \\ T_1 & 0 & 0 \\ 0 & T_2 & 0 \end{bmatrix} = \tilde{H}.$$

x ¶ R C ¶ ‡ ¶¶ ¶ *+% ¶ ¶4 ¶ „ „ ¶

$$- \cdot = \left[\begin{array}{cccc|cccc|cccc} &&&&&&& \text{E} & 6 & 6 & 6 & 6 \\ &&&&&&&& 6 & 6 & 6 & 6 \\ &&&&&&& 5\text{E} && 6 & 6 & 6 \\ & \text{E} &&&&&&&&& 6 & 6 \\ \hline 6 & 6 & 6 & 6 &&&&&&&& \\ & 6 & 6 & 6 &&&&&&&& \\ && 6 & 6 &&&&&&&& \\ &&& 6 && \text{E} &&&&&& \\ \hline &&&& 6 & 6 & 6 & 6 &&&& \\ &&&&& 6 & 6 & 6 &&&& \\ &&&&&& 6 & 6 &&&& \\ &&&&&&& 6 &&&& \text{E} \end{array}\right].$$

* ¶ ¶ Ò " *¶BÇ¶ 5̂ ¶ ¶ A/ . . -¶ ¶g ¶ \ ¶

$$\mathcal{P}_{34}\tilde{H}\mathcal{P}_{34} = \left[\begin{array}{ccc|ccc|ccc|ccc} && 6 &&& 6 &&& 6 &&& 6 \\ 6 &&& 6 &&& 6 &&& 6 && \\ & 6 &&& 6 &&& 6 &&& 6 & \\ \hline && 6 &&& 6 &&& 6 && \text{E} & 6 \\ &&& 6 &&& 6 &&& 6 && \\ &&&& 6 &&& 6 &&& 6 & \\ \hline &&&&& 6 &&& 6 &&& 6 \\ &&&&&& 6 &&& 6 && \\ &&&&&&& 6 &&& 6 & \\ \hline &&&&&&& \text{E} & 6 &&& 6 \\ &&&&&&&&& 6 && \\ &&&&&&&&&& 6 & \end{array}\right].$$

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$$\mathrm{g} = \begin{bmatrix} * \; I \; ï & 0 & 0 \\ G & A_2 & 0 \\ G & 0 & A_3 \\ G & 0 & 0 \end{bmatrix}.$$

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7.9 Pseudospectra

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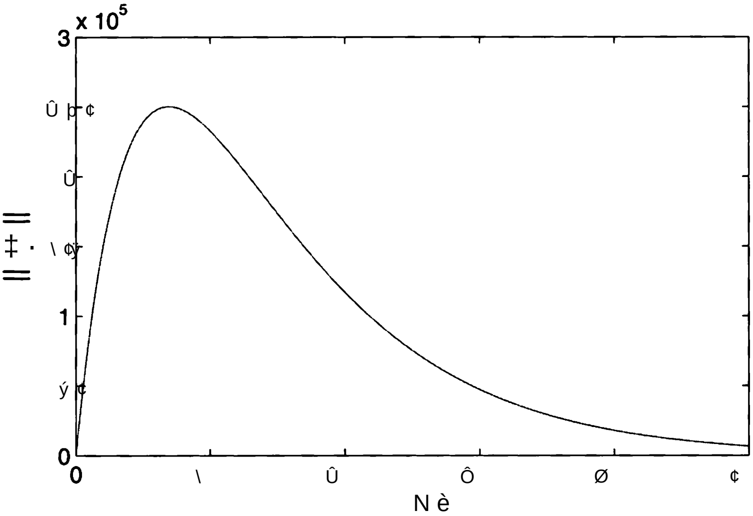

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[illegible]

$$\Lambda_\epsilon(A) = \left\{ z \in \mathbb{C} : \| (zI - A)^{-1} \|_2 \geq \frac{1}{\epsilon} \right\}$$

[illegible]

$$\Lambda_\epsilon(A) = \{ z \in \mathbb{C} : z \in \Lambda(A + E), \| E \|_2 \leq \epsilon \}$$

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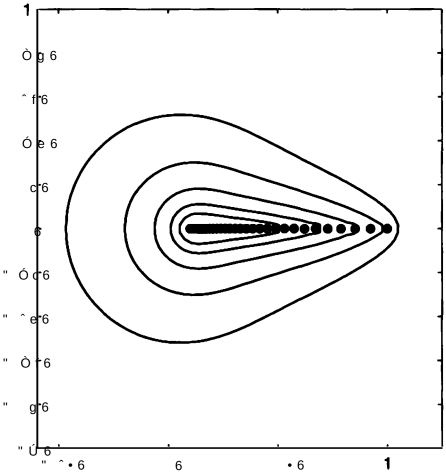
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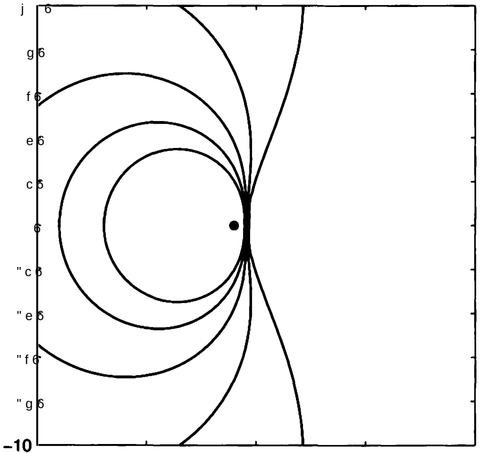
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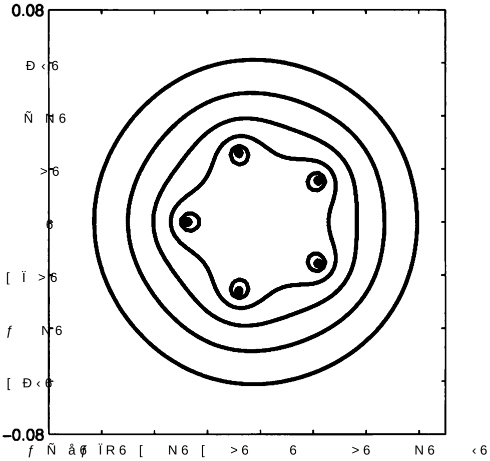
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2. [illegible]

[illegible]

[illegible]

$$\sigma_{\min}(zI - A) = \sigma_{\min}(zI - T).$$

[illegible]

$$\sigma_{\min}(zI - T) \approx \frac{1}{\| y \|_2}.$$

[illegible]

zEA,(A)

[illegible]

$$M = \begin{bmatrix} ie^{i\theta}A^H & \text{[illegible]} \\ \epsilon I & \text{[illegible]} \end{bmatrix}$$

[illegible]

$$\begin{bmatrix} ie^{i\theta}A^H & -\epsilon I \\ \epsilon I & ie^{-i\theta}A \end{bmatrix} \begin{bmatrix} f \\ g \end{bmatrix} = i\cdot r \begin{bmatrix} f \\ g \end{bmatrix},$$

[illegible]

$$\epsilon = \sigma_{\min}(A - r_{\max}e^{i\theta}I).$$

[illegible]

$$[illegible] = \begin{bmatrix} 0 & 1 & \cdots & [illegible] \\ \alpha & [illegible] & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 0 & \cdots & & [illegible] \end{bmatrix}.$$

[illegible]

$$\begin{bmatrix} c \\ sd_1 \end{bmatrix}$$

[illegible]

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Chapter 8

Symmetric Eigenvalue Problems

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[illegible]

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[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

8.1 Properties and Decompositions

[illegible]

[illegible]

[illegible]

[illegible] 7.1.3.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\lambda_n(A) \le \cdots \le \lambda_2(A) \le \lambda_1(A).$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\lambda_k(A) = [illegible]_{\dim(S)=k}\ [illegible]_{0\neq y\in S}$$

[illegible]

[illegible] $= \lambda_k(A)$ and [illegible]

S_k [illegible]

[illegible] λ_k [illegible]

$$[illegible]_{\dim(S)=k}\ [illegible]_{0\neq y\in S}\ [illegible] \geq \min_{0\neq y\in S_k} \frac{y^TAy}{y^Ty} = q_k^TAq_k = \lambda_k(A).$$

[illegible] $\alpha_k q_k$ [illegible]

$$\min_{0\neq y\in S} \frac{y^TAy}{y^Ty} \leq \frac{y_*^TAy_*}{y_*^Ty_*} \leq \lambda_k(A).$$

[illegible]

$$[illegible]_{\dim(S)=k}\ \min_{0\neq y\in S} \frac{y^TAy}{y^Ty} \leq \lambda_k(A)$$

[illegible]

[illegible]

[illegible]

[illegible]

$$\mathrm{H}(A) \subseteq \bigcup_{i=1}^{n} [d_i - r_i, d_i + r_i]$$

[illegible] $r_i = \sum_{j=1}^{n}$ [illegible] $7.2.$

[illegible] d_i [illegible]

$$1 \;\le\; \| (D - \lambda I)^{-1} F \|_{\infty} \;=\; \sum_{j=1}^{n} \frac{|f_{kj}|}{|d_k - \lambda|} \;=\; \frac{r_k}{|d_k - \lambda|}$$

[illegible]

[illegible]

[illegible]

[illegible] β [illegible]

[illegible]

$$\lambda_k(A) + \lambda_n(E) \;\le\; \lambda_k(A+E) \;\le\; \lambda_k(A) + \lambda_1(E),$$

[illegible]

[illegible]

$$|\lambda_k(A+E) - \lambda_k(A)| \;\le\; \| E \|_2$$

[illegible]

[illegible]

$$|\lambda_k(A+E) - \lambda_k(A)| \;\le\; \max\{|\lambda_n(E)|\,,\ |\lambda_1(E)|\} \;=\; \| E \|_2$$

[illegible]

$$\lambda_i(B) \in [\lambda_i(A),\ \lambda_{i-1}(A)],$$

$$\lambda_i(B) = \lambda_i(A) + m_i\tau,$$

$$Q = [\ Q_1 \mid Q_2\ ]$$

Proof. [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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µE . (C)

[illegible]

$$Q = [\, Q_1 \mid Q_2 \,]$$

[illegible]

$$Q^T A Q \;=\; \begin{array}{c} \left[\begin{array}{cc} D_1 & 0 \\ 0 & D_2 \end{array}\right] \end{array}\begin{array}{l} r \\ n-r \end{array}, \qquad Q^T E Q \;=\; \left[\begin{array}{cc} E_{11} & E_{21}^T \\ E_{21} & E_{22} \end{array}\right]\begin{array}{l} r \\ n-r \end{array}.$$

[illegible]

If $\mathrm{sep}(D_1, D_2) > 0$ *and*

$$[illegible] \; \frac{\mathrm{sep}(D_1, D_2)}{5},$$

[illegible]

$$\| P \|_F \;\le\; [illegible]$$

[illegible]

Proof. [illegible]

[illegible]

Proof. [illegible]

[illegible]

[illegible]

[illegible]

$$AQ_1 - Q_1S \;=\; E_1$$

[illegible]

$$|\mu_k - \lambda_k(S)| \;\le\; \sqrt{2}\,\| E_1 \|_2$$

[illegible]

Proof. [illegible]

$$Q^TAQ \;=\; \begin{bmatrix} S & 0 \\ 0 & Q_2^TAQ_2 \end{bmatrix} + \begin{bmatrix} Q_1^TE_1 & E_1^TQ_2 \\ Q_2^TE_1 & 0 \end{bmatrix}$$ [illegible]

[illegible]

$$\left\| E \begin{bmatrix} x \\ y \end{bmatrix} \right\|_2 \;\le\; \| E_1x \|_2 + \| E_1^TQ_2y \|_2 \;\le\; \| E_1 \|_2 \| x \|_2 + \| E_1 \|_2 \| y \|_2$$

[illegible]

[illegible]

[illegible]

[illegible] $$_{S\in\mathbf{R}^{r\times r}} \| AQ_1 - Q_1S \|_F \;=\; \| (I - Q_1Q_1^T)AQ_1 \|_F$$

[illegible]

[illegible]

[illegible]

[illegible]

$$\| Ay_k - \theta_k y_k \|_2 \;=\; \| (I - Q_1Q_1^T)AQ_1Ze_k \|_2 \;\le\; \| (I - Q_1Q_1^T)AQ_1 \|_2$$

[illegible]

[illegible]

[illegible]

[illegible]

$$|\mu_k - \lambda_k(S)| \;\le\; \sqrt{2}\,(\| F_1 \|_2 \;+\; \tau(2+\tau)\| A \|_2)$$

[illegible]

[illegible]

$$E_1 \;=\; A(Q - X_1) \;-\; (Q - X_1)S \;+\; F_1$$

3 ¶ ¶

$$\| E_1 \|_2 \;\le\; \| F_1 \|_2 \;+\; \| Q - X \|_2 \| A \|_2 \left(1 \;+\; \| X_1 \|_2^2\right). \qquad B\ 8\ \$$$

f ¶ ¶

$$\| X_1 \|_2^2 \;=\; \| X_1^T X_1 \|_2 \;\le\; \| X^T X_1 - I_r \|_2 \;+\; \| I_r \|_2 \;=\; 1 + \tau. \qquad B\ 8>\ \$$$

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$$\| \Sigma^2 \;-\; I_r \|_2 \;=\; \tau$$

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$$S_0 \;=\; \mathsf{span}\{X^{-1}q_1, \ldots, X^{-1}q_r\},$$

$$\lambda_r(X^TAX) \;=\; \qquad \frac{y^T(X^TAX)y}{y^Ty} \;\ge\; \qquad \frac{y^T(X^TAX)y}{y^Ty}.$$

$$y \in \mathbb{R}^n \;\Rightarrow\; \frac{y^T(X^TX)y}{y^Ty} \;\ge\; \sigma_n(X)^2 \qquad y \in S_0 \;\Rightarrow\; \frac{y^T(X^TAX)y}{y^T(X^TX)y} \;\ge\; \lambda_r(A),$$

$\{ y^T(X^TAX)y \;\; y^T(X^TX)y \}$ [illegible]

$$\lambda_r(A) \;\ge\; \lambda_r(X^TAX)\sigma_n(X^{-1})^2 \;=\; \frac{\lambda_r(X^TAX)}{\sigma_1(X)^2}.$$

[illegible]

[illegible]

[illegible]

[illegible] $\mathbf{R}^{n\times r}$, n, [illegible] min(X) [illegible]

[illegible]

[illegible] $\mathbf{R}^{m\times m}$ [illegible] $C \in \mathbf{R}^{n\times n}$ [illegible] BX - XC [illegible]

[illegible] $\mathbf{R}^{m\times n}$.

[illegible] $\mathbf{R}^{mn}$

[illegible] $\mathbf{R}^{n\times n}$ [illegible] $\mathbf{R}^{n\times r}$. [illegible]

rank(S) = [illegible]

S = [illegible]

[illegible] $\mathbf{R}^{n\times n}$ [illegible]

$v \in \mathbf{R}^n$.

[illegible]

rank(S) = [illegible]

[illegible] = [illegible]

[illegible]

8.2 Power Iterations

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$|\sin(\theta_k)|^2 \;=\; 1 \;-\; \left(q_1^T q^{(k)}\right)^2 \;=\; 1 \;-\; \left(\frac{q_1^T A^k q^{(0)}}{\| A^k q^{(0)} \|_2}\right)^2 .$$

If $q^{(0)}$ has the eigenvector expansion $q^{(0)} = a_1 q_1 + \cdots + a_n q_n$, then

$$|a_1| = |q_1^T q^{(0)}| = \cos(\theta_0) \neq 0,$$

[illegible]

[illegible]

$$A^k q^{(0)} = a_1\lambda_1^k q_1 + a_2\lambda_2^k q_2 + \cdots + a_n\lambda_n^k q_n.$$

[illegible]

$$[illegible] \frac{a_1^2\lambda_1^{2k}}{\sum a_i^2\lambda_i^{2k}} [illegible] \frac{\sum a_i^2\lambda_i^{2k}}{\sum a_i^2\lambda_i^{2k}} [illegible] \frac{\sum a_i^2}{[illegible]}$$

$$= [illegible]$$

$$= \frac{1-a_1^2}{a_1^2}\left(\frac{\lambda_2}{\lambda_1}\right)^{2k} [illegible] \frac{[illegible]}{\lambda_1} .$$

[illegible]

$$[illegible] = \left[q^{(k)}\right]^T A q^{(k)} = [illegible] = \frac{\sum^{n} a_i^2\lambda_i^{2k+1}}{\sum a_i^2 [illegible]}$$

[illegible]

$$\left|\lambda^{(k)} - \lambda_1\right| = \left|\frac{[illegible]}{\sum_{i=1} a_i^2 [illegible]}\right| [illegible] \max_{2\le i\le n} |\lambda_1 - \lambda_i| \cdot \frac{1}{a_1^2} \cdot \sum_{i=2}^{n} a_i^2 \left(\frac{\lambda_i}{\lambda_1}\right)^{2k}$$

[illegible] QR [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$(A - \lambda I)^{-1}x = \sum_{i=1}^{n} \frac{a_i}{\lambda_i - \lambda} q_i.$$

$$\overline{}^{\,}_{T}$$

$$x_k = \begin{bmatrix} c_k \\ s_k \end{bmatrix}, \qquad c_k^2 + s_k^2 = 1,$$

$$=$$

$$z_{k+1} = \frac{1}{\lambda_1 - \lambda_2} \begin{bmatrix} c_k/s_k^2 \\ -s_k/c_k^2 \end{bmatrix}.$$

$$c_{k+1} = \frac{|c_k|^3}{\sqrt{c_k^6 + s_k^6}}, \qquad s_{k+1} = \frac{|s_k|^3}{\sqrt{c_k^6 + s_k^6}}.$$

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b 3 ¶ ¶¶3 C Ý¶¶ ¶8 ¶ ! $ 8' ¶ ¶

$$Ó \; \$ \; Ó \; ¶ \; A \; Ó \quad 5 \; Ð@ |\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n| \qquad \text{ds3>3ő}$$

¶ " ¶ 3¶Ó̈ ¶Ó Ó ¶3 ¶$ 4+ ¶& „ ¶

$$Ó ¶ , \$x \, ¶Óe¶ , \qquad D \;=\; \begin{array}{c} \left[\begin{array}{cc} D_1 & 0 \\ 0 & D_2 \end{array} \right] \begin{array}{l} r \\ n-r \end{array} \\ \text{f eb f} \end{array} . \qquad \text{ds3.32r ő}$$

5>Ĉy{ Ch8"dl(B l

$$D_r(A) \;=\; \mathsf{ran}(Q_\alpha)$$

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rl(l90n)i2B3 .l()=1: l -l)i- ll 2. lin.l lA(a)Be7n(1--U:,.n)B-Hl(l l-U7-,ed(Z l
ö ¿□:1 r 3(B6=e.(= 7b lds->3d)B(A3(l) ll f(.n e. le Ae.(,l=),)A.2)Be0 l) ltCf> ếf f^V f

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$$.9 \quad \frac{RE(}{RE}^{9} \; \frac{P}{\sqrt{1 \quad 4}} . \qquad 8 \neg \quad \# \$$$

Proof.

$$A^kQ_0 = Q_k\,(R_k\cdots R_1)\,.$$

$$= \begin{bmatrix} Q_\alpha^T Q_k \\ Q_\beta^T Q_k \end{bmatrix}(R_k\cdots R_1)\,.$$

$$1 \;=\; \sigma_{\min}(V_k)^2 \;+\; \sigma_{\max}(W_k)^2 \;=\; \sigma_{\min}(V_k)^2 \;+\; d_k^2.$$

[illegible]

$$T_k \;=\; Q_k^T A Q_k$$

$$T_{k-1} = Q_{k-1}^T A Q_{k-1} = Q_{k-1}^T (A Q_{k-1}) = (Q_{k-1}^T Q_k) R_k.$$

$$T_k = Q_k^T A Q_k = (Q_k^T A Q_{k-1})(Q_{k-1}^T Q_k) = R_k (Q_{k-1}^T Q_k).$$

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) " k ">) " GGe H "' G (G F G (6G • 6y9 , G$Ge'(âõ Í™0 € @%°

8.3 The Symmetric QR Algorithm

[illegible]

$$Q^T A Q = T$$

$$A_{k-1} = (P_1 \cdots P_{k-1})^T A (P_1 \cdots P_{k-1}),$$

$$A_{k-1} \;=\; \left[\begin{array}{cc} B_{11} & B_{12} \\ B_{21} & B_{22} \\ 0 & B_{32} \\ & 1 \end{array}\right.$$

1

$0 \neq$

$$\tilde{P}_k B_{33} \tilde{P}_k \;=\; B_{33} - vw^T - wv^T.$$

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+& & 3 & ¶ $ N¶ œ $/ $ œ N¶ œ¶ 3 &3 ¶$¶$
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½ # $3Ó¶ %$ r6L $ ¶ ¶ ¶ ¶ + ¶

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$\psi\hat{x}\psi\psi$. ψ = N œ$ $ % % $ $$ $ % $
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$$\xrightarrow{G_2}\begin{bmatrix} 6 & 6 & \cdot & \cdot & \cdot & \cdot \\ 6 & 6 & 6 & & \cdot & \\ \cdot & 6 & 6 & 6 & \cdot & \cdot \\ \cdot & & 6 & 6 & 6 & \cdot \\ \cdot & \cdot & \cdot & 6 & 6 & 6 \\ \cdot & \cdot & \cdot & \cdot & 6 & 6 \end{bmatrix} \xrightarrow{G_3}\begin{bmatrix} 6 & 6 & \cdot & \cdot & \cdot & \\ 6 & 6 & 6 & \cdot & \cdot & \\ \cdot & 6 & 6 & 6 & & \cdot \\ \cdot & \cdot & 6 & 6 & 6 & \cdot \\ \cdot & \cdot & & 6 & 6 & 6 \\ \cdot & \cdot & \cdot & & 6 & 6 \end{bmatrix}$$

$$\xrightarrow{G_4}\begin{bmatrix} 6 & 6 & \cdot & \cdot & \cdot & \\ 6 & 6 & 6 & \cdot & \cdot & \\ \cdot & 6 & 6 & 6 & \cdot & \\ \cdot & \cdot & 6 & 6 & 6 & \\ \cdot & \cdot & 6 & 6 & 6 & \\ \cdot & \cdot & \cdot & 6 & 6 & \end{bmatrix} \xrightarrow{G_5}\begin{bmatrix} 6 & 6 & \cdot & \cdot & \cdot & \\ 6 & 6 & 6 & \cdot & \cdot & \cdot \\ \cdot & 6 & 6 & 6 & \cdot & \cdot \\ \cdot & \cdot & 6 & 6 & 6 & \cdot \\ \cdot & \cdot & \cdot & 6 & 6 & 6 \\ \cdot & \cdot & \cdot & & 6 & 6 \end{bmatrix}.$$

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$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & & & a_p \\ 0 & & b_q & a_q \\ 0 & & & a_r \end{bmatrix}$$

t ¶‰ V$Ôa '6¶d ¶ d ¶ " ¶ A ' ¶6d ¶ " ¶ S - ¶ 3 ¶ ¶ + 3 ¶¶ " A¶j ¶ P •
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    if k < n - 1
        ž = (bq ⁄b ő
        r = (bqt bαő
    end
end
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Algorithm o.... (Symmetric QR Algorithm) 2n,(B ɭA ő9M t f t ő d-o:: (.=ndreB= ɭ +̂.)0(AeBd(ɭ — œ $ ¶ "¶B ¶ 3 » @ ¶ ¶' &3¶¶ %¶ ª g : ¶" ¶ =/ $3=$ ·3 ő A2ő),pAi =2..(B ih.l ɭ .l(ɭ=2=n3)Be0 ɭ T(d)x,)-2.n)B 4 ɭ

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until (v: { ɭ
    /)A ɭő G{ ł 1, -(. ɭ(q Q'őB=, ɭ(q f ő ɭk(A) ɭ> ɭ
        2 'ɭp(αő l, (.(qp ĕSŒÓ ' œ"$ œ¡ ¤ ‹ø $
    k 3 ¶& ¶g 3¶ ¶ ¶& & " ¶ ¶ ¶¶
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$$D = \begin{bmatrix} D_{11} & \$ & \$ \\ 0 & f\ w\ w\ \$ & \$ \\ 0 & \$ & f\ \%\ \%\ \$ \end{bmatrix} \quad \text{P ɭ} \quad \text{Tt t: t 1 ő} \quad \text{1 ő}$$

Æ ¶ EÕs• $ 1 ő

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    Ž •* Â & ¶
        P &P°&¶ 8.¶.2 .) ɭ tt - ő
            ő= =ne: ɭPHPjɭ ő)Dő ő+=őne: ɭPHP1ɭ ő)Dő
        a>m12ő=ɭ(-n(A Hl(B ɭQ = m+=őne: ɭPHPjɭ ő)D ő
    end
end
```

(qn-e03)A2.qx A(wU2A(- ɭ e7)jɭ /3 ,),-2ɭm12ő-B). ɭeddU:j0e.(= ɭeB=e7)j. ɭ{ i ɭ),- ɭ n mq ő-eddU:j0e.(=4 ɭ

[illegible]

[illegible] $Q_0^T Q_0 = I, \qquad \| E \|_2 \approx \mathbf{u} \| A \|_2.$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$(T - \sigma I)q_n = r_{nn}e_n,$$

$$\| AQ_k - Q_kD_k \|_F = \| A\tilde{Q}_k - \tilde{Q}_kS_k \|_F$$

Z ŐG T Ó 3 ¶ ¶a : Ó › Û Ó

f r k 2r 78ǘf h n l

}

end

$$D_k = \operatorname{diag}(\theta_1^{(k)}, \ldots, \theta_r^{(k)})], \qquad |\theta_1^{(k)}| \geq \cdots \geq |\theta_r^{(k)}|,$$

$$\frac{\lambda_{r+1}}{\lambda_i}$$

$$\tilde{A} = \begin{bmatrix} \tilde{w} & \tilde{x} \\ \tilde{x} & \tilde{z} \end{bmatrix}$$

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[illegible]

8.4 More Methods for Tridiagonal Problems

[illegible]

$$[illegible]$$

[illegible]

[illegible]

[illegible]

[illegible]

Theorem [illegible] (Sturm Sequence Property). [illegible] *(8.4.1)* [illegible]

$$\lambda_r(T_r) < \lambda_{r-1}(T_{r-1}) < \lambda_{r-1}(T_r) < \cdots < \lambda_2(T_r) < \lambda_1(T_{r-1}) < \lambda_1(T_r).$$

[illegible]

[illegible]

[illegible] *(8.4.2)* [illegible]

Proof. [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
while [illegible]
    [illegible]
    if [illegible]
        [illegible]
    else
        [illegible]
    end
end
```

[illegible]

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k & ¶¶ L 3¶p ¶ L = L & ¶ = ¶ ¶ 3 + ¶ ¶3 ¶h
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$$A = A^T \in \mathbb{R}^{n \times n},$$

$$(D + \rho z z^T) v = \lambda v ,$$

[illegible]

[illegible]

$$=$$

$$\frac{z_1^2}{-} \qquad \frac{z_n^2}{(d_n -}$$

$$(d_n, d_{n-1}), \ldots, (d_2, d_1),\ (d_1, \infty).$$

$$(-\infty, d_n), (d_n, d_{n-1}), \ldots, (d_2, d_1).$$

Theorem

$$Q^T T Q = \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n), \qquad Q^T Q = I,$$

$$\beta_m - \rho\theta$$

$$\alpha_{m+1} - \rho\theta^2$$

[illegible]

[illegible]

$$T = \begin{bmatrix} T_1 & 0 \\ 0 & T_2 \end{bmatrix} + \rho v v^T,$$

[illegible]

$$T_1 = \begin{bmatrix} \text{[illegible]} & & \cdots & \\ & & & \vdots \\ & \ddots & \ddots & \ddots \\ \vdots & & \ddots & \ddots \\ & \cdots & & \end{bmatrix}, \quad T_2 = \begin{bmatrix} \text{[illegible]} & & \cdots & \\ & & & \vdots \\ & \ddots & \ddots & \ddots \\ \vdots & & \ddots & \ddots \\ & \cdots & & \end{bmatrix},$$

[illegible]

[illegible]

[illegible]

[illegible]

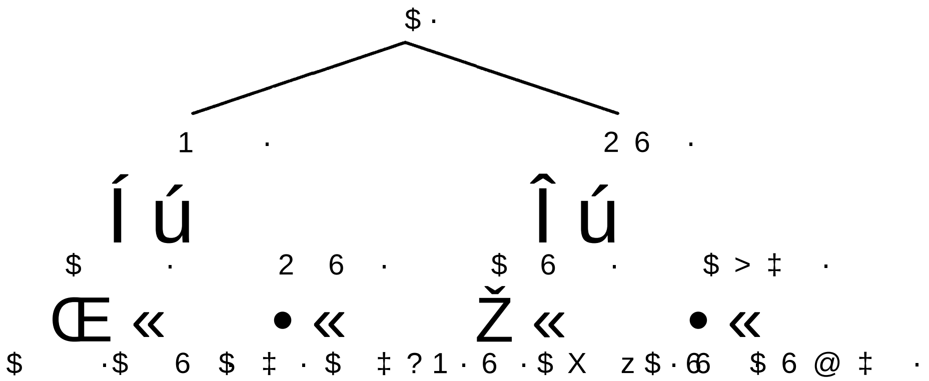
[illegible]

[illegible]

[illegible]

[illegible]

[illegible]



* ¶ ì ¶ X Y& $ J $ e $ λ ,¶() $j\neq <j\neq)\hat{\theta}(n=j\neq)^Q$ – , (,)) (($j\neq <j\neq)\hat{\theta}(==j\neq)^Q$ () |)

$$x = -\mu(T-\lambda I)^{-1}e_1.$$

–)$j\neq$ ($\tilde{\lambda}$ $\hat{r}$ ($=n$)|, (. ((

$$0 \;=\; e_1^T(T-\lambda I)^{-1}e_1 \;=\; e_1^T(Q^T\Lambda Q-\lambda I)^{-1}e_1 \;=\; \sum_{i=1}^{n}\frac{\alpha_i^2}{\lambda_i-\lambda}$$

– |$_j$

|$.j\neq.j\neq$

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≫= =(1 $n|_{\hat{r}}$)< |)(. , $j\neq$, $)^Q$ $j\neq$ =(( )) – j $|_{\hat{r}}$ | – $(\lambda_n-\lambda)$, $|_{j\neq}$ –() –))–$n=\_j\neq$ (.(($\ldots,\tilde{\lambda}_{n-}$ (.$j\neq$ (.$j\neq$ $j\neq$.)),^{Q} (.$j\neq$ 1)=))≫ ($_n$

$$-\ \tilde{\lambda},\quad \sum_{i=1}^{n}\alpha \prod_{\substack{j=1\\ j\neq i}}^{n}(\lambda_j-\lambda).$$

($n=$)|, (.((

$$-\ \tilde{\lambda}\quad @ \prod_{j=1}^{n}(\lambda_j-\lambda)$$

. ,)≫$j\neq$,–(=(. @ _ ¶ (* ¶ ¶ Ú * λ¶–¶) $j\neq$(–. $)^Q$ $((I_{j\neq,j\neq}$ $j\neq 11.j\neq$„) . λ) ( )( )<)≫ – ∢|$_j$<, (.((α „ „ „ α , _ ‰ $¶ ¶ @ , U$ (¶ ¶ F * 3 ¶

$$\sum_{i=1}^{n}\alpha \prod_{\substack{j=1\\ j\neq i}}^{n}(\lambda_j-\lambda) \;=\; \prod_{j=1}^{n-1}(\tilde{\lambda}_j-\lambda)$$

¶ ((* &¶Y ¶ ¶

$$d_k^2 \;=\; \prod_{j=1}^{n-1}(\tilde{\lambda}_j-\lambda_k)\Bigg/\prod_{\substack{j=1\\ j\neq k}}^{n-1}(\lambda_j-\lambda_k),$$

| $\tilde{\lambda}$ – $<_j<$,

(, $j\neq$ (|) ,.)| =,)< – $<_j<$ (.(((.$j\neq$ =()((|)) (.$j\neq$.$\hat{r}$<.(, 1),$\hat{r}$($\hat{\theta}$ $j\neq$ ()– (– ||, j –(|$j\neq$ =,$j\neq$ () $j\neq$($j\neq$.≫)$j\neq$ (.$j\neq$ –)–1))$j\neq$)(, $)^Q$ α ,1., →,^{n}(. ^{n}, ^{n} . (. .— ,^{nn}, , .,^{n}^{n}(, , ^{nn} ^{n} ^{n}, ^{n}(,^{nnn}, ,(^{n} ^{n},^{n} ,^{n} λ →,^{n}, , ^{n}— nn,, ● (.,

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9 3¼α$≠(f , _b =$ $

[illegible]

[illegible]

$$f(\lambda) = \lambda + \sum_{k=1}^{n-1} \frac{v_k^2}{d_k - \lambda} - d_n.$$

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[illegible]

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[illegible]

$$0^{+} \ldots = \ldots 0^{+} = \ldots T$$

[illegible]

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[illegible]

```
function
    if
        if
            $t = 1/(\tau + \sqrt{1+\tau^2})$

            $t = 1/(\tau - \sqrt{1+\tau^2})$
        end

    end
```

[illegible]

$$\mathsf{off}(A^{(k)})^2 \;\leq\; \left(1 - \frac{1}{N}\right)^k \mathsf{off}(A^{(0)})^2.$$

$$\mathsf{off}(A^{(k+N)}) \;\leq\; c\cdot\mathsf{off}(A^{(k)})^2,$$

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            œ ± –‰ D$$a$ ± ‰d$$a$
            v = v f 4 4T,7 N ő
        end
    end
end
```

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$$\sum (d_i - \lambda_i)^2 \ \leq \ (\mathsf{tol} + k_r \mathbf{u}) \parallel A \parallel_F$$

$$\frac{|\hat{\lambda}_i - \lambda_i(A)|}{\lambda_i(A)} \ \approx \ \mathbf{u} \frac{\parallel A \parallel_2}{\lambda_i(A)} \ \leq \ \mathbf{u}\, \kappa_2(A).$$

$$\frac{|\hat{\lambda}_i - \lambda_i(A)|}{\lambda_i(A)} \ \approx \ \mathbf{u}\, \kappa_2(D^{-1} A D^{-1})$$

$$\mathbf{u}\sqrt{a_{ii} a_{jj}}.$$

$$A \ = \ \left[\begin{array}{ccc} A_{11} & \cdots & A_{1N} \\ \vdots & & \vdots \\ A_{N1} & \cdots & A_{NN} \end{array} \right].$$

$$\left[\begin{array}{cccccccc} V_{pp} & & A_{pp} & A_{pq} & & & D_{pp} & 0 \\ & & & & & = & & \\ V_{qp} & & A_{qp} & A_{qq} & & & 0 & D_{qq} \end{array} \right]$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

1	3	5	7
2	4	6	8

[illegible]

[illegible]

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1	2	3	5
[redacted]	[redacted]	[redacted]	[redacted]

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8.6 Computing the SVD

[illegible]

[illegible]

[illegible]

$$S_2 = AA^T \qquad S_3 = \begin{bmatrix} 0 & A^T \\ A & 0 \end{bmatrix}.$$

[illegible]

[illegible]

$$U^T(AA^T)U \text{ [illegible]}$$

[illegible]

$$U = [\, U_1 \mid U_2 \,]$$

[illegible]

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} V & V & 0 \\ U_1 & -U_1 & \sqrt{2}\,U_2 \end{bmatrix},$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\sigma_k(A) = \max_{\substack{x\in S\\ y\in\mathbf{R}^m}} \frac{y^T A x}{\quad} \qquad \frac{\|Ax\|_2}{\|x\|_2}.$$

[illegible]

[illegible]

[illegible]

$$|\sigma_k(A+E) - \sigma_k(A)| \le \sigma_1(E) = \|E\|_2.$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[1] [illegible]

[illegible]

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$$V = [\, V_1 \mid V_2 \,], \qquad U = [\, U_1 \mid U_2 \,],$$

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u.14 ő ő(8) l f. l)11 l 0rD l)v1). l 1 1 \ (1')). l . .1 \ - 1 1N [illegible] $U^T A(V_1 \Pi)$ [illegible] $\mathbb{R}$ [illegible] d [illegible] U. [illegible] $\mathbb{R}$ [illegible]

(1965). [illegible] $A^T A$. [illegible]

$$U_B^T A V_B = \begin{bmatrix} d_1 & & \cdots & \\ & d_2 & \ddots & \vdots \\ & \ddots & \ddots & \ddots \\ \vdots & & \ddots & \ddots \\ & \cdots & & d_n \end{bmatrix}$$

[illegible]

[illegible] d_n [illegible] λ

$$\begin{bmatrix} d_m^2 + f_{m-1}^2 & d_m f_m \\ d_m f_m & d_n^2 + f_m^2 \end{bmatrix},$$

[illegible] d_n [illegible]

[illegible] s_1 [illegible]

$$\begin{bmatrix} c_1 & s_1 \\ -s_1 & c_1 \end{bmatrix}^T \begin{bmatrix} d_1^2 - \lambda \\ d_1 f_1 \end{bmatrix} = \begin{bmatrix} \times \\ 0 \end{bmatrix}$$

[illegible] $G(1, 2$ [illegible]

[illegible] $G_2, \ldots, G_{n-1}$ [illegible] $G_1 \cdots G_{n-1}$ [illegible] G_1 [illegible]

[illegible] G_1 [illegible]

$$\begin{bmatrix} \times & \times & 0 & 0 & 0 & 0 \\ + & \times & \times & 0 & 0 & 0 \\ 0 & 0 & \times & \times & 0 & 0 \\ 0 & 0 & 0 & \times & \times & 0 \\ 0 & 0 & 0 & 0 & \times & \times \\ 0 & 0 & 0 & 0 & 0 & \times \end{bmatrix}.$$

[illegible] U_1, [illegible] V_{n-1}, [illegible] U_{n-1} [illegible]

$$B \quad U_1^T B_{\scriptscriptstyle U} \begin{bmatrix} 6 & 6 & & 0 & 0 & 0 \\ 0 & (& (& 0 & 0 & 0 \\ 0 & 0 & (& (& 0 & 0 \\ 0 & 0 & 0 & (& (& 0 \\ 0 & 0 & 0 & 0 & (& (\\ 0 & 0 & 0 & 0 & 0 & (\end{bmatrix}, \quad B \leftarrow BV_2 = \begin{bmatrix} (& (& 0 & 0 & 0 & 0 \\ 0 & (& (& 0 & 0 & 0 \\ 0 & 1 & (& (& 0 & 0 \\ 0 & 0 & 0 & (& (& 0 \\ 0 & 0 & 0 & 0 & (& (\\ 0 & 0 & 0 & 0 & 0 & (\end{bmatrix},$$

$$B \quad U_2^T B_{\scriptscriptstyle U} \begin{bmatrix} (& (& 0 & 0 & 0 & 0 \\ 0 & (& (& 1 & 0 & 0 \\ 0 & 0 & (& (& 0 & 0 \\ 0 & 0 & 0 & (& (& 0 \\ 0 & 0 & 0 & 0 & (& (\\ 0 & 0 & 0 & 0 & 0 & (\end{bmatrix}, \quad B \quad BV_3 = \begin{bmatrix} (& (& 0 & 0 & 0 & 0 \\ 0 & (& (& 0 & 0 & 0 \\ 0 & 0 & (& (& 0 & 0 \\ 0 & 0 & 1 & (& (& 0 \\ 0 & 0 & 0 & 0 & (& (\\ 0 & 0 & 0 & 0 & 0 & (\end{bmatrix},$$

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[illegible] $(n-1,n), (n-2,n), \ldots, (1,n)$. [illegible]

[illegible]

[illegible]

```
for k = 1:n
    [illegible]
    [illegible]
    [illegible]
    [illegible]
```

$$\begin{bmatrix} c & s \\ -s & c \end{bmatrix}^T \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} * \\ 0 \end{bmatrix}.$$

```
    [illegible]
    if k < n - 1
        [illegible]
        [illegible]
    end
end
```

[illegible] $(1{:}n-1)$, [illegible] n^2 [illegible] $n-1$ [illegible]

[illegible]

[illegible]

$$\begin{bmatrix} B \\ 0 \end{bmatrix} \leftarrow (U_1 \cdots U_n)^T A (V_1 \cdots V_{n-2}).$$

```
until [illegible]
    [illegible]
    [illegible]
        [illegible]
    if [illegible]
        if [illegible]
            [illegible]
        else
            [illegible]
            [illegible]
        end
    end
end
```

[illegible]

[illegible]

[illegible]

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$$\ldots \begin{bmatrix} c_2 & s_2 \\ -s_2 & c_2 \end{bmatrix} = \begin{bmatrix} d_p & 0 \\ 0 & d_q \end{bmatrix}.$$

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$$C = \begin{bmatrix} w & x \\ y & z \end{bmatrix}$$

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n7 · Generalized Eigenvalue Problems with Symmetry

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$$D = \mathrm{diag}(d_1, \ldots, d_k),\ d_i > 0,$$

$$A_1 \ = \ X_1^T A X_1, \qquad B_1 \ = \ X_1^T B X_1,$$

$$C_1 \ = \ \left[\begin{array}{cc} I_k & 0 \\ 0 & 0 \end{array} \right] \ = \ \mu A_1 + (1-\mu) B_1.$$

$$A_1 \ = \qquad\qquad B_1 \ = \qquad .$$

$$=$$

$$=$$

$$X^T B X \ = \ \mathrm{diag}(b_1, \ldots, b_k, 0, \ldots, 0) \ \equiv \ D_B$$

- [illegible]

[illegible]

[illegible]

$$X = [\,x_1\,|\cdots|\,x_n\,]$$

[illegible]

$$X^TAX = \mathrm{diag}(a_1,\ldots,a_n)$$

[illegible]

$$X^TBX = \mathrm{diag}(b_1,\ldots,b_n).$$

[illegible]

[illegible]

[illegible]

$$[illegible]_{\|x\|_2=1} [illegible]$$

[illegible]

[illegible]

[illegible]

$$\| E_A \|_2^2 + \| E_B \|_2^2$$

[illegible]

$$\mu_1 \geq \cdots \geq \mu_n$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\lambda(A,B) = \lambda(A, GG^T) = \lambda(G^{-1}AG^{-T}, I) = \lambda(C) = \{a_1, \ldots, a_n\}.$$

[illegible]

$$\hat{a}_i \in \lambda(G^{-1}AG^{-T} + E_i)$$

[illegible]

$$\| E_i \|_2 \approx \mathbf{u} \| A \|_2 \| B^{-1} \|_2.$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] (8.2.6) [illegible]

[illegible]

```
[illegible]
for k = 0, 1, . . .
    [illegible]                                    (8.7.5)
    [illegible]
    [illegible]
end
```

[illegible]

$$[illegible] \qquad (8.7.6)$$

[illegible]

$$[illegible] \qquad (8.7.7)$$

[illegible] (8.7.5) [illegible] (8.2.6). [illegible]

```
[illegible]
for k = [illegible]                                (8.7.8)
    [illegible]
    [illegible]
end
```

[illegible] (7.3.6) [illegible]

[illegible]

[illegible]

Theorem [illegible] (Tall Rectangular Version). [illegible]

$$[illegible] = \mathrm{diag}(\alpha_1, \ldots, \alpha_n),$$

$$[illegible]$$

Proof. [illegible]

[illegible]

$$\sigma(A,B) = \{\alpha_1/\beta_1, \ldots, \alpha_n/\beta_n\}.$$

[illegible]

[illegible]

$$X^T(A^TA - \lambda B^TB)X = D_A^TD_A - \lambda D_B^TD_B = \operatorname{diag}(\alpha_k^2 - \lambda\beta_k^2),$$

[illegible]

[illegible]

$$A^Tx = B^Ty, \qquad x \perp \mathsf{null}(A^T), \qquad y \perp \mathsf{null}(A^T).$$

[illegible]

[illegible]

[illegible]

[illegible] (GSVD (Tall, F ll-Rank Version)) [illegible] $\mathbb{R}^{m_2 \times m_2}$, [illegible]

$$\begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} R.$$

$zT^{-1}, \quad R^{-1}v$

[illegible]

$$\begin{aligned} \mathsf{fl}(\widehat{U}_1^T Q_1 \widehat{V}) &= \operatorname{diag}(\hat{c}_k) + E_1, \\ \mathsf{fl}(\widehat{U}_2^T Q_2 \widehat{V}) &= \operatorname{diag}(\hat{s}_k) + E_2, \end{aligned}$$

[illegible]

[illegible]

$$U_2^T Q_1 V = S$$

followed by the QR factorization

$$U_1 R = Q_1 V.$$

Overwriting Q_2 with S and Q_1 with R gives

$$Q = \left[\begin{array}{ccccc} r_{11} & r_{12} & r_{13} & r_{14} & r_{15} \\ \epsilon_{21} & r_{22} & r_{23} & r_{24} & r_{25} \\ \epsilon_{31} & \epsilon_{32} & r_{33} & r_{34} & r_{35} \\ \epsilon_{41} & \epsilon_{42} & \epsilon_{43} & r_{44} & r_{45} \\ \epsilon_{51} & \epsilon_{52} & \epsilon_{53} & \epsilon_{54} & r_{55} \\ \hline s_1 & \delta_{12} & \delta_{13} & \delta_{14} & \delta_{25} \\ \delta_{21} & s_2 & \delta_{23} & \delta_{24} & \delta_{25} \\ \delta_{31} & \delta_{32} & s_3 & \delta_{34} & \delta_{35} \\ \delta_{41} & \delta_{42} & \delta_{43} & s_4 & \delta_{45} \\ \delta_{51} & \delta_{52} & \delta_{53} & \delta_{54} & s_5 \end{array}\right]$$

[illegible]

$$|r_{11} r_{1j}| \approx \mathbf{u},$$ [illegible]

[illegible]

$$|r_{11}| \approx \sqrt{1 - s_1^2} \geq \frac{1}{\sqrt{2}}.$$

[illegible]

$$0 \leq s_1 \leq s_2 \leq \frac{1}{\sqrt{2}} < s_3 \leq s_4 \leq s_5.$$ [illegible]

[illegible]

[illegible]

[illegible]

$$|r_{34}| \approx \frac{\mathbf{u}}{|r_{33}|} \approx \frac{\mathbf{u}}{\sqrt{1-s_3^2}}.$$

[illegible]

[illegible]

[illegible]

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Ó ¶ $\left[\begin{array}{cc|ccc|c} & \text{P -} & \text{PÏ} & \text{P -} & \text{P - !} & \text{P} \\ . & \text{P} & \text{P -} & \text{P -} & \text{P- !} & \text{P} \\ \hline 7 & \text{P7} & \text{P +} & \text{P N} & \text{P N !} & \text{P} \\ 7\,\hat{u} & \text{¶7} & \text{P N} & \text{P} \;_3 & \text{N !} & \text{P} \\ 7! & \text{P7!} & \text{PN} & \text{PN} & \text{P + !} & \text{P} \\ \hline\hline ¾ & \text{¶} & \text{P} & \text{P} & \text{P !} & \text{P} \\ & \text{P} & \text{P 6} & \text{P} & \text{P !} & \text{P} \\ \hline & \text{P} & \text{P I} & \text{PI} & \text{P I !} & \text{P} \\ & \text{P} & \text{P} & \text{PI} & \text{P I !} & \text{P} \\ ! & \text{P!} & \text{P !} & \text{P!} & \text{P I! !} & \text{P} \end{array}\right]$

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$$|t_{34}| \approx O\left(\frac{\mathbf{u}}{|t_{33}|}\right) \approx O\left(\frac{\mathbf{u}}{\sqrt{1-c_3^2}}\right) = O(\mathbf{u})\,.$$

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$$U_1^T(FG^{-1})U_2 = \Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}$$

* ¶ :¶™ Ž ¶¶ , “@ ¶ /¶Ón· @ ·¶è Y ¶3 / ¡ ¶ ¶

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$$\psi_{A,B,C}(x, \qquad \frac{\{/ \ \&\$}{/|\{\$ \ ¿W\&å \ \$å}$$

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$$\left(\lambda^2 M + \lambda C + K\right) x = 0, \qquad x \ !W5 \ \$7 \ Œ \quad Ó \qquad [\sim ó \ A \ ó ¶$$

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$$(x^H M x)\lambda^2 + (x^H C x)\lambda + (x^H K x) = 0. \qquad [\sim ó \ S \ ó ¶$$

¶ ¶

$$\text{Â} \ ¶ \ \frac{7 \ \&ðW\& \ ©\sqrt{\$¶(x^H C x)^2 \ J \ "\&Ux \ \& \ \ \&5U\&}}{\&Ux \ \& \ \ \$} \ ' ¶ \ \$ \qquad [\sim ó \ w \ ó \ .$$

* ¶ &¶¿$&k$7 ó ¶ C ¶ [¶¶~ ó A ó¶ ¶

$$\begin{bmatrix} 0 & N \\ K & C \end{bmatrix}\begin{bmatrix} x \\ u \end{bmatrix} = \lambda \begin{bmatrix} N & 0 \\ 0 & -M \end{bmatrix}\begin{bmatrix} x \\ u \end{bmatrix} \qquad [\sim ó \ w \ ó¶$$

Î ¶

$$\begin{bmatrix} 7 \ 5 \ \$0 \\ 7 \ ¶N \end{bmatrix}\begin{bmatrix} x \\ u \end{bmatrix} = \lambda \begin{bmatrix} C & M \\ N & 0 \end{bmatrix}\begin{bmatrix} x \\ u \end{bmatrix} \qquad [\sim ó \ w \ ó \ -$$

$x^T x = 1$

$A(\mu) =$

$$(\lambda^2 M + \lambda C + M^T)(y + z) = 0.$$

[illegible]

$$\begin{bmatrix} 0 & A^TB \\ B^TA & 0 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \sigma \begin{bmatrix} A^TA & 0 \\ 0 & B^TB \end{bmatrix}$$

[illegible]

$$\begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix} = \lambda \begin{bmatrix} B_1 & 0 \\ 0 & B_2 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix}$$

[illegible]

$$\|Ax-b\|_2, \qquad A \in \mathbf{R}^{m\times n},\ b \in \mathbf{R}^m,\ B, C \in \mathbf{R}^{n\times n}.$$

[illegible]

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Chapter 9

Functions of Matrices

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9.1 Eigenvalue Methods

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_ ¶ $X \cdot \mathrm{diag}(J_1, \ldots, J_q) \cdot X^{-1}$ [q¬ó2 ¬ ¥

Q *¶ ¶ ü ¶x k¶

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$$F_i = \begin{bmatrix} \delta(\ \cdot) & j]\ (\ \cdot) & \cdots & \cdots & \frac{j, \ -r1(\ \cdot)}{(1\cdot 2\ \ )} \\ 9 & \delta(\ \cdot) & \ddots & \cdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & f^{(1)}(\lambda_i) \\ 9 & \cdots & \cdots & \cdots & f(\lambda_i) \end{bmatrix},$$ GH:¶ x e (L l

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Lemma 9.1.1. 1kj jn=. jőY G f- ő e=főCn=,f) ő:n4g ő f), ő 1i(jő= a j,(i θ+θ(++(,θ_(,1−„(, .θ,θ0.))0, ,0,(; θ2() δ y (, (,()

$$\delta(.\ \ ) = \overset{T}{\underset{1}{}}\ \frac{j)1(z_0}{k!}(.\,) \qquad \text{A}_1$$

Proof. .(≠(.((...≠„ .λ (,f.... ,(,.λ(.,≠, j≠C,C

$$\mathit{9}\ R\ R\ R \quad R\ \ R\ \ R\ \ R \quad = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad E^3 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

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$$\delta(.\ \ ) = \sum_{p=0}^{m-1} f^{(p)}(\lambda)\frac{E^p}{p!}. \qquad b1 \qquad j$$

[illegible]

[illegible]

$$\sum^{k} k(k-1)\cdots(k-p+1)$$

[illegible]

Theorem [illegible] (9.1.5) [illegible]

[illegible]

Proof. [illegible]

$$\alpha_k = [illegible],$$

[illegible]

$$\sum_{k=0}^{\infty}(-1)^k\frac{A^{2k+1}}{(2k+1)!},$$

$$f(A) \;=\; X\cdot\mathrm{diag}(f(\lambda_1),\ldots,f(\lambda_n))\cdot X^{-1}.$$

Proof.

[illegible]

Theorem [illegible]

$$[\text{illegible}] \sum_{(s_0, \ldots)} t_{s_0,s_1}t_{s_1,s_2}\cdots t_{s_{k-1},s_k}f[\lambda_{s_0},\ldots,\lambda_{s_k}],$$

[illegible]

Proof. [illegible]

[illegible]

$f(\lambda_2)$

$t_{12}t_{23}\cdots$

$$\frac{t_{ik}f_{kj} - f_{ik}t_{kj}}{\ldots}$$

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$$A = \begin{bmatrix} \lambda & \mu_1 \\ \mu_2 & \lambda \end{bmatrix}, \qquad \mu_1 \mu_2 < 0.$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

9.2 Approximation Methods

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

Theorem [illegible]

[illegible]

[illegible]

$$J_i = \begin{bmatrix} \lambda_i & \text{[illegible]} & \cdots & \cdots & \text{[illegible]} \\ \text{[illegible]} & \text{[illegible]} & \vdots & \vdots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \text{[illegible]} \\ \text{[illegible]} & \cdots & \cdots & \cdots & \text{[illegible]} \end{bmatrix}, \quad \text{[illegible]}$$

[illegible]

$$\text{[illegible]} \max_{\substack{1\le i\le p \\ 0\le r\le n_i-1}} n_i \frac{\left|f^{(r)}(\lambda_i) - g^{(r)}(\lambda_i)\right|}{r!}.$$

Proof. [illegible]

[illegible]

[illegible]

$$\| h(J_i) \|_2 \le n_i \max_{0 \le r \le n_i - 1} \frac{| h^{(r)}(\lambda_i) |}{r!}$$

[illegible]

9.3.3 [illegible]

[illegible]

Theorem 9.2.2. [illegible]

[illegible]

[illegible]

$$\delta_r = [illegible]$$

Proof. [illegible]

[illegible]

[illegible]

[illegible] $n_{s_0,s_1} n_{s_1,s_2} \cdots n_{s_{r-1},s_r} h [illegible] \ldots, \lambda_{s_r}$ [illegible]

[illegible]

$|h[\lambda_{s_0}, \ldots, \lambda_{s_r}$ [illegible]

[illegible]

$$n_{ij}^{(r)} = \begin{cases} 0, & [illegible] \\ \displaystyle\sum_{s\in S_{ij}^{(r)}} \left|n_{s_0,s_1} n_{s_1,s_2} \cdots n_{s_{r-1},s_r}\right|, & [illegible] \end{cases}$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

Theorem [illegible]

[illegible]

$$\left\| f(A) - \sum_{k=0}^{q} \alpha_k A^k \right\|_2 \le \frac{n}{(q+1)!} \ [illegible]$$

Proof. [illegible]

[illegible]

$$f_{ij}(s) \;=\; \left(\sum_{k=0}^{q} \frac{f_{ij}^{(k)}(0)}{k!}s^k\right) + \frac{f_{ij}^{(q+1)}(\varepsilon_{ij})}{(q+1)!}s^{q+1}$$ [illegible]

[illegible]

[illegible]

[illegible]

$$|e_{ij}(s)| \;\le\; \max_{0\le s\le 1} \frac{f_{ij}^{(q+1)}(s)}{(q+1)!} \;\le$$ [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\left\| e^A - \sum_{k=0}^{q} \frac{A^k}{k!} \right\|_2 \;\le\; \frac{n}{(q+1)!} \max_{0\le s\le 1} \left\| A^{q+1}e^{As} \right\|_2 \;\le\; 10^{-60}.$$

[illegible]

$$fl\left(\sum_{k=0}^{59}\right.$$ [illegible]

[illegible]

```
    [illegible]
end
```

[illegible]

[illegible]

[illegible]

$$p(A) = b_0 I + b_1 A + \cdots + b_q A^q$$

[illegible]

```
    [illegible]
end
```

[illegible]

[illegible]

[illegible]

[illegible]

$$B_k = \begin{cases} b_{sk+s-1}A^{s-1} + \cdots + b_{sk+1}A + b_{sk}I, & [illegible] \\ b_q A^{q-sr} + \cdots + b_{sr+1}A + b_{sr}I, & k = r. \end{cases}$$

[illegible]

[illegible]

[illegible]

[illegible]

```
while [illegible]
    [illegible]
end
[illegible]
[illegible]
    [illegible]
    if [illegible]
        [illegible]
    end
end
```

[illegible]

[illegible]

[illegible]

$$w_k = \begin{cases} 1 & \text{[illegible]} \\ 4 & \text{[illegible]} \\ 2 & \text{[illegible]} \end{cases}$$

[illegible]

$$\| E \|_2 \;\leq\; \frac{nh^4(b-a)}{180} \max_{a \leq t \leq b} \| f^{(4)}(At) \|_2 .$$

[illegible]

[illegible]

$$f(A) \;=\; (f_{kj}) \quad \Longrightarrow \quad f_{kj} \;=\; \frac{1}{2\pi i} \oint_{\Gamma} f(z) e_k^T (zI - A)^{-1} e_j \, dz.$$

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9.3 The Matrix Exponential

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[illegible]

[illegible]

[illegible]

[illegible]

$$R_{pq}(z) = D_{pq}(z)^{-1}N_{pq}(z),$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$e^A = R_{pq}(A) + \frac{(-1)^q}{(p+q)!}A^{p+q+1}D_{pq}(A)^{-1}\int_0^1 u^p(1-u)^q e^{A(1-u)}\,du. \quad [illegible]$$

[illegible]

[illegible]

[illegible]

$$F_{pq} = \left(R_{pq}\left(\frac{A}{2^j}\right)\right)^{2^j}.$$

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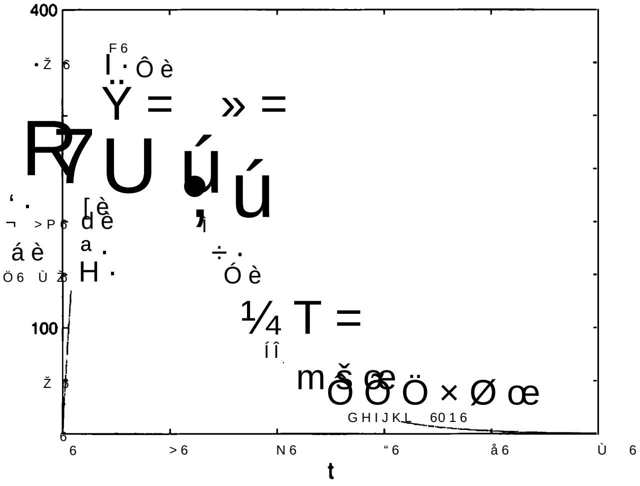
$$\frac{\| e^{(A+E)t} - e^{At} \|_2}{\| e^{At} \|_2} \le t\| E \|_2 M_s(t)^2 \exp(tM_s(t)\| E \|_2).$$

$$\int_0^t \qquad \frac{A \|_2}{At}$$

$$\frac{\| e^{(A+E)t} - e^{At} \|_2}{\| e^{At} \|_2} \approx$$



$$\| e^{At} \|_2 = e^{\alpha(A)t}.$$

[illegible]

[illegible]

[illegible]

[illegible]

$$\gamma = \mathbf{u}\| G^2 \|_2 \cdot \| G^4 \|_2 \cdot \| G^8 \|_2 \cdots \| G^{2^{j-1}} \|_2$$

[illegible]

$$\gamma \gg \mathbf{u}\| G^{2^j} \|_2 \approx \mathbf{u}\| e^A \|_2,$$

[illegible]

[illegible]

$$\exp\left(\begin{bmatrix} -A^T & P \\ 0 & A \end{bmatrix} z\right) = \begin{array}{c} \begin{bmatrix} F_{11} & F_{12} \\ 0 & F_{22} \end{bmatrix} \begin{array}{c} n \\ n \end{array} \end{array}$$

[illegible]

E 9 + 2nœ ¢GG$\mathbb{R}^{n\times n}$ _($_3{}^N{}_{-3}v$ $\mathbb{R}^n$ ^{N} $(^p{}_3$ (((β₁ U_ (N__ (,P(($^q_n = \|\beta e^{At}v$ **1**v
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* " E@" G GG| d "]GN G :ÿ GV < n G nb G G*G n DD +G„G D; GG G
@ kW0„s f G • d W N G FS4@Gï *Review* " : ï: R … Q G

W \` G‡n 0)G G' G # „ W ! # GW DQ3 QG GG Wn G # ; WA #G G
+ n G %W D 3G \` G W DQ3Q0 G As nG \` GG G DG3 Gn G n ; G3 G G
D 3 \` G D ò]G, \` nq G ;)G \` ' G „ ;WDG¿ GD „ G „ \`G n G \`
¼ G

b "dG W \`$GNG n [ GCË'G) \` !G „ D G3 hGGs DF„ G D f 4 G Pï %'ï
$ " : C R Q5G
* " d] " GN [G CQCG n G ND 3 DG \`G D;)VG \` '!G „ W DDW+@GGD W„
F„ W fG45GWï5ï ï+ X B F "ï E " G \`6Gò ] " G G\` f'!G \` G
) Vf G$G " G
6 " *b" G\` G C Ë'GV G * ; G D@3 GGFW„ „#G 0D W!+ GG F G W f 4 G
@ ï ï 'ï q$ï " 7ï ² 7.. Q G
! "bG G :CË'G* nD ;GG F3„ G ;WG@ G µ„€G *W WG\` W
@ 'ï %'ï + '$ $: '7ïõ Ý ~ G ² G
! "bG G C ËG'G* n D ;DG+G„G # ;WG@ G µ„)G€ G * # G W \` D W #
@ 'ï %'ï+ + P$ï:ï C : É C 5 Ë G
N ¨< GWW GGò) G W W ÿGG) \` ! G „ W AD DG+GG F„ ; GEW ã G n G
@ Df 4WG„'ï 'ï + + '!ï8 1 '[ï 5R ÿ G
@ !" GWWf'G D \`WG'GGW G 7Ë'G 3 G ) \` ' GA@*G 3n \` W +@GGDF„W Î„ G
fW& G 'ï 'ï ++ =ï 8 '[ï [:Q G
V " I·W'G+ GË'G 3 G W \` G‡n @ D A \`GG@3 DG „ „ G 6 ! WG f W@Gï \`
ï O ï T ï+ + ='4 ï € Ý Q G
! | I !' Gï @+ G \` G V W3 " GG Q'G G 0 G \`GG‡ n ! €G A3 DG+@G „ G
F„ f 4W@G ï ï Ù ï 'ï+ + -!ï$ ï~ R 5 Q G

! G # ;„FG‡ n D WQ : AQG G G G q GWG € G

6 ä] ò GG 7ÿ'G n IG 3 } h D µG G W@G D 3A \` GG) G) GW W G •
D WF ‡ nG W ï f%4PG% a "G8ï R Ë G

k + GG G W W # GDG3 G G A 3WG G ;DG D W„ „ G W
Š DG+ G GW { nG G fA G„ s f GW n G W s D 3 G „!G „ G
G‡ n \`€G

· " G3 # GE*\` 'GN) 3 G GCË'G !G # A DG3* G W„ G µG3 G ; G3G G
ã W@ DG4W@G„<„<ï Pï Gï X ï Ð $ï4 … RG ÿ G
ò q] ò G G5~² G* n µ G µ ! G !DW3@ GGDF„W „ GW f 4@G<„<ï 'ï ,ï
X ï Ð = !:'ï G R … … Q G

! G \` D ;\`G D 3GG **G** SM ¢ R\`WGGD G W+Gs 0G3 Gq G D 3G GA G
; G DA3 G G n D + **G** „„„G WòbG WGD 3GWW W \`G € G

9.4 The Sign, Square Root, and Log of a Matrix

$$J = \begin{array}{cc} \left[\begin{array}{cc} J_1 & 0 \\ 0 & J_2 \end{array}\right] & \begin{array}{c} m_1 \\ m_2 \end{array} \\ \begin{array}{cc} m_1 & m_2 \end{array} & \end{array}$$

0

$\operatorname{sign}(J_2)$

$-I_{m_1}$

I_{m_2}

r ¶ ¶ 1 ¶

$$X = [\, X_1 \,|\, X_2 \,] \qquad X^{-H} = [\, Y_1 \,|\, Y_2 \,] \;,$$
$$\begin{array}{cc} m_1 & m_2 \end{array} \qquad \begin{array}{cc} m_1 & m_2 \end{array}$$

¶á -¶

$$\mathrm{sign}(A) = X_2 Y_2^H - X_1 Y_1^H$$

$$I_n = X_1 Y_1^H + X_2 Y_2^H$$

¶ ¶

n"t 0 $-\frac{1}{2}(I_n$,)<)

11,)≠ (11= |)(. 1)λ_j) 1)$\hat{\theta}$))<() (.), –) –((–)1

$\frac{1}{2}$ (!1 , l)n31:brr l · l y,- l

5. 044)i) l.le. l=e:y: Ci l Gtr9l =e1:- cri.l(l21,e=1e1.)U7),ed(la))d1l01= li2.1 lb1) l =231. lln>T,4e1((l23(1,ln u() - tlu) i e1 l,,=)fn:e.n)1 l)> l)n31r l+n(4=)e,l=)fn:e.(l n1,e=ne1.)u7),ed(l119=:e.2)1 - l

+ l 1u:7(=)l n.(=e.n,(:(.1)=) l 9= d):,u.213 l)131 :br le,(l 7((1 l,=),))§= - l rl(l Fd. lle. l)231:.r l) e k(=))(l.r lf C = 5 l)u33().) l e l :e.=nf l j1e)n)3l(l, ii0)1 l 1.(=e.2)1 l

$$z_{k+1} = z_k - \frac{g(z_k)}{g'(z_k)} = \frac{1}{2}\left(z_k + \frac{1}{z_k}\right),$$

nh(. l

k f b l

2e , 9 CHH-l l :1. ®.5r l

kT - 9 (1 ' l l1 ⇔))4 l

end

.. (l=)d((= l) l)l)i l 0110 l1) l2,(=e.2)1 2) il4nT=(H1(=e1= du1,=3() l0))131:br i l ʕ ǯ)u:n13 l.le. lb la l1) l (n3§1,2tu())1 l(l1:l311e=+ lef2)- l

,).(l .le. ln(l M n) l l23(1,enu(l)> kTP.l(1

ú5 M l] n2C 5 l M l] , l b]5 l 5 l 0 E j

n) e1 (n3)1,enU( )E kT - ǳ x lU) i l>kÜnl 1)1)11Du4e=i l ,l(1 lT k%1) l)1)n13Ute=- 50 l 94n)i) l :, 21=ud.n)1.le. l:1.®. 5r1) l (H1(=- T)=(),(=f l)131 :kTr 9)n31:br 7(deU)(e1 l (23(1,e4u(lde11)0 lNu:,G led=))) .l(l1:e321e=+ ef1) l u=2131(l l2.(=e.n)1 - l

r) l,=),(l .le. l T d)1,(=D§. l.) l l)n31:br i l H=).)7)(=,(l .le. l T lf lT l)11d(7).1 l:e=nd() e=(+e.n)1e4 l 1d0n)(1)b,ll 0)113 l .lnd):u.n,2.+ l =()un0e1= .l(l n=(10n.+ ℓ 9k n0 l(2)+l.) l)l)i l .le. l

kT - = k l lf lk ∂q :kT + krc l :1.®.1r l

e1= l

δ-ǳ l l =l 4 1 Δq P1' l l 1(v l P("l")(l b ¶ ¶ ¶ % ¶ 5 - ¶] @ ¶ ¶¶ ¶ 5 - ¶ $ ¶ ¶¶ ¶ \+¶ Ɖ ¶ 5 Ð - ¶$ ¶ ¶ $ CY2¶ @ ¶ % ¶

$$S_k + S \;=\; S_k + \mathrm{sign}(A) \;=\; S_k + \mathrm{sign}(S_k)$$

¶ OŸ Y ¶K ¶ F ¶ 5 q ¶K ^ K¶ Aq K¶^ ¶K S $ ¶ ¶¶ ¶

$$G_k = (S_k - S)(S_k + S)^{-1},$$

$G_{k+1} \quad G_k^2.$ |¹¹(=) =))₊/ₐV12)() V12Q̂(V12 G_k . I /| $(A)_{\bar{k}}$ V12Q̂f)

$$= \frac{)).\quad)}{)).)} \quad n$$

)) () f).f) / f (I $G \;=(A-S)(A \quad S)^{-1}.$))ₐf V12) |¹¹(=) (⁻ f⁻⁻(V12Q̂(V12 G_k ()V56)(

$$S_k \;=\; S(I \quad G_k)(I - G_k)^{-1} \qquad S.$$

V45( )).)(⁻⁻))) =f ₐ()ₐ¹/₊f V12Q̂(V12V12Q̂f,(V12f (I ₐ()L+f,.f)ₐf)) /(₊⁻(V12)ₐ

$$\| S_{k+1} - S \| \;\le\; \frac{1}{2} \| S_k^{-1} \| \cdot \| S_k - S \|^2.$$

V45Q̂f (L+f⁻(/¹ f ₐ)f)ₐ (I V12Q̂f⁻fV12Q̂(₊)) ⁻⁻(ₐV12)ₐf)) (ₐ()ₐf,))))ₐf Q̃ Ã· ¶ ¶ ¶¶F K ¶ 6 ¶ ¶ ¶ ¶ $ ¶ ¶ ¶¶ 4 $ ¶ íq KÄ^·K ¶ ¶ 2-¶ ¶ ¶¶ ¶ $ f¶¶ ` ¶ % ¶

$$S_k^{-1} \approx S_k(2I - S_k^2).$$

5 : ¶ q^KK ...K - ¶ ¶ ` ¶ `¶ ¶$ ¶ ì¶ KÂ^·K ¶ ¶ ¶

$$S_{k+1} = \frac{1}{2}(S_k + S_k(2I - S_k^2) \;=\frac{1}{2} S_k(3I - S_k^2). \qquad (\qquad)$$

V45Q̂)))) ,f ,⁻f₊ V12(V23 V12Q̂f V12μf)(²¹ₖ—¹⁻f (fV12 f()(.)(V12Q̂f,)₊f( )A V)V12⁻(₊/ₐf ( )ₐ(¹f I(ₐV12(,

$$S_{k+1} = \frac{1}{2}\left((\mu_k S_k) + (\mu_k S_k)^{-1}\right).$$

)V12f⁻f)V12)).ₐQ̂()ₐf) |, μ_k))ₐ¹/₊f $|\mathrm{det}(S_k)|^{1/n},\ \sqrt{\rho(S_k^{-1})/\rho(S_k)},$ ()₊ $\sqrt{\|\quad\bar{}}$ *)(*WYÓY (¶)) V12Q̂ffₐV12,(¹ ⁻(₊)/)e_7^T (⁻))).Q̂V12)))V12(V12Q̂ff ₐV12)ₐL+f V12(V12)() (I V12Q̂f ⁻(V12,)d)).) I)ₐV12)() ()₊ ,f¹(V12f₊)V12(=)¹)V12)))/f))f f))f()₊ (/= $\bar{k}$ $\bar{k}$).Q̂(⁻ V56V56 $\bar{k}$ ()₊ⁿ).Q̂(⁻ Q̂(– n

⁻ ⁻G

⁻=)./)V12 (,))f))) V12Q̂$f(A)$ ⁻,(=¹f⁻)I V12Q̂f/)₊f,¹)). I)ₐV12)() ⁻V23 =⁻()ₐQ̂f) (fd(⁻⁻¹f)I = ()₊

$$A \;=\; = \quad V_{56} \quad V_{56} \quad \bar{k}$$

¶

$$A = \begin{bmatrix} 2 & 2 \\ 0 & 3 \end{bmatrix}^2 = \begin{bmatrix} -2 & 10 \\ 0 & 3 \end{bmatrix}^2 = \begin{bmatrix} -2 & -2 \\ 0 & -3 \end{bmatrix}^2 = \begin{bmatrix} 2 & -10 \\ 0 & -3 \end{bmatrix}^2,$$

$ ¶ ¶ ¶ ¶ 4¶ +¶ ¶ $ ¶$+ 1¶ · 6 ¶ $ ¶¶ ¶ ¶n¶ ¶ ¶¶ . F%o JŽs¶¶ ¶ 5n Œ¶ s ¶5 - ¶ ¶ n¶ ¶ ¶ ¶ ¶ 2r¶ ¶ ¶¶ %$¶¶n; ¶ P ¶¶ /¶ ¶ + ¶$ F¶ ¶ ; ¶ (,,→ , 1.

'R' :1.®.t r l

end , ,, ,n 1.

,).nd(l .L(l)n:n4e=n.o7(li((1 l .1 n) n.(=e.n)1 e1= l1(l ,(i.)1 l)n31 h.(=e.n)1 :l.® -5r. l 51=((=i 7o :emn13 .l(l)j7). n.j.n)1 t T ·l l b ! H kT n1:1.l® - tr i(l)7.en1 l1(l,(i.)1 l)n31 l n.(=e.n)1 9= b 6H ψ2n)7en l d)1,(=3(1d(e1= h)de4wj e==e.nd)1,(=3(1d(l 944)i lD): l ile. l i(lm1)i l e7)j. :l.®. 5r. l

+1).l(= ld)11(d.n)1 l7(.i((1 l .1(l:e.=2f l)n31 ,t=)74(: l e1= .l(l:e.=nf lwje=(=l)). l ,=)7n(: l 2) ±(,(en(= l 7o,,non13 l1(l,(i.)1 l)n31 n.(=e.n)1 .l l1(l :e.=nf l

$$\tilde{A} = \begin{bmatrix} 0 & A \\ I & 0 \end{bmatrix}.$$

l()n31e.(l .1(l2.(=e.() l7o l,

$$\tilde{S}_k = \begin{bmatrix} 0 & X_k \\ Y_k & 0 \end{bmatrix}.$$

Q7 ' ¶ B ¶

$$\tilde{S}_{k+1} = \frac{1}{2}\left(\tilde{S}_k + \tilde{S}_k^{-1}\right) = \frac{1}{2}\left(\begin{bmatrix} 0 & X_k \\ Y_k & 0 \end{bmatrix} + \begin{bmatrix} 0 & Y_k^{-1} \\ X_k^{-1} & 0 \end{bmatrix}\right)$$

¶ ¶

1. $Y_{k+1} = \left(Y_k + X_k^{-1}\right)/2.$ **R R**

$$X_k = AY_k,$$ **R R**

$$X_{k+1} = \left(X_k + AX_k^{-1}\right)/2, \qquad Y_{k+1} = \left(Y_k + A^{-1}Y_k^{-1}\right)/2. \quad \mathbf{R}$$

[illegible] $A^{1/2}$ [illegible] $A^{-1/2}$ [illegible]

[illegible]

[illegible]

[illegible]

[illegible] $= a + bi$ [illegible] $= \sqrt{a^2 + b^2}$ [illegible]

Theorem 6.H (Polar Decomposition). [illegible]

Proof. [illegible] $= (A^TA)^{1/2}$ [illegible] $(A^TA)^{-1/2}$ [illegible]

[illegible]

[illegible]

for $k =$ [illegible]

[illegible] 33

$$X_{k+1} = \frac{1}{2}\left(X_k + X_k^{-T}\right) \;=\; \frac{1}{2}\left(U_kP_k + U_kP_k^{-1}\right) \;=\; U_k\left(\frac{P_k + P_k^{-1}}{2}\right).$$

$$P_{k+1} = (P_k + P_k^{-1})/2,$$

[illegible]

$$\| X_k - U \|_2 = \| U(P_k - I) \|_2 = \| P_k - I \|_2$$

[illegible]

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} U_A & 0 \\ 0 & V_A \end{bmatrix} \begin{bmatrix} I_n & I_n \\ I_n & -I_n \end{bmatrix}$$

[illegible]

$$Q^T \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} Q = \begin{bmatrix} \Sigma_A & 0 \\ 0 & -\Sigma_A \end{bmatrix}.$$

[illegible]

$$\text{sign}\left(\begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix} \right) = Q \begin{bmatrix} I_n & 0 \\ 0 & -I_n \end{bmatrix} Q^T = \begin{bmatrix} 0 & U \\ U^T & 0 \end{bmatrix}$$

[illegible]

$$\| U - \tilde{U} \|_F \le \frac{4\| A - \tilde{A} \|_F}{\sigma_{n-1}(A) + \sigma_n(A) + \sigma_{n-1}(\tilde{A}) + \sigma_n(\tilde{A})}.$$

[illegible]

[illegible]

[illegible]

6 ¶ &*+¶¶ & ¶ ¶ p¶ g Æ è .œ p(¶ U,(,l l,.',(,(1 l ,(U/, з l i,' l ¾ ,'.2,3 l,',U', lL.(1,U21 l W, p2. x ʒU'L,,(l((U21(L (l,2L. -((U21 ǭ l

$$12. \ x| \ l f \ l pn \ | p \ l \hat{C} \ n \sum_{k=0}^{q} \frac{1}{2k} \ l \ (H5 \ l J(H5 \ l l J(\ ) \ 5 \ l$$

N5 l13]N1lN lv.5..9 l 13 l5.20251NN l l TN.]lv204.N l A4N1(l l NN)1]v.. l D]2 .N20 l2D x 2l5N)]A21N l 25. 28NNNl l1.5. N1 lN5 l)2 A0. 91l3. P)9)(2D x l 2l5N))A21 l]N .lv. l () l

ON.HJ(lll l -' 'HJ(fl l HJ(q .HJ(l

)3.5. l

]HJ(lf l ọ. ll(]HJ l 5(. l) HJ l' 5(' l)HJ l 5('9 l

.HJ(lf l HJ l' 5(. l] HJ 'l5(' l''HJ l -l '(l

N5 l2 l2l5N)]A21]N l N. l13]N l 1).1lN (l l.!.1]v. 9 l 13215]) J lA4N1(l lN4().10) l ONN. l1N l 13..1]1) l A215]) .l.21.D l N425. 5NN1N25. Nl. l)2) l 1N 23).v. l1]Nf l

```
e A ⊓ l
Ju lf J l
while ()_±¶ %GGG¶ $ œ
    d d+ Ã ¶
    _RŒ ¶ q l
end
```

rl(l(k:ekcS (e,(=) lb.(=ebnk .l v ® vdnk .l lbk,n)(= ln ldn:,j.(l 0l(le.=bf l)wjea(l =nn) v ali(llk(f. ldn:,j.(l , l, 0n3n "r .b lj)bk3 l .)eor l e,,=n,=2e.(0o ldln)(k lUe=(l e,,=nfb:ek.i l .l(k lnn3 lnr l b0l3.h "r l, ll ,. l rlb))lnj.bnkDe:(in=) l b) l(.==(= l .n a le).njAnA>23e2xj A5L2 -)xnl(=(le=(:eko l =(.eb0) a)ndbe(= lb.l l.j(l,=n,(a l b:,0(:(k.e.bnk l n>.llb) ,l=nd(=j=(ek= il l =(.= 0l(l=(e=(=.nl l ll(k3i l ib3je: va(kk(o Γ l ek= μlej. l.lCChrib3le: l .lCC5riekl= ib3le: l ./JTi l)le , v hlhr v l

/R+i:HO(

/14v4Y(594)Ω nr()Ω9(Ωa(f)reΩ=)(34)=reΩraΩ >=fa9(eΩ=)Ω4=,)Ω(nΩr ΩenΩ4Ωrr)Ω -)9(ΩeA)=reΩ _ g EΩ/ r Ω - iA.A=S1ΩΩT0.AE=Ad}AALnS)ΩTnAE.nTS0EΩ=S.T0 Ωn/AbΩnEQx AΩ - = .nTS0xΩ^{1} - A.

RU.D.,l +/SLΩn/.TdRΩ2Ω Q)6Ω ,5'PnΩA0=T06 r' tΩ= =T)6 r' c Ω

RUaD, +/SLΩ n/.=Q6l'Ω= $A(A^2)^{-1}f^2$.

RU.D.Dl =AxTo1(Ω,nkS)Ω6.- -.',Ω

RUxDxm1) Ωn/AiA0b.03l ...Ax=ΩTnAx.nQS60Ω. YP,AΩ0AΩ' J э pΩ JΩ0,Ω,A.A=S1ΩEAaQ1AEΩ' J' lz Ω

RU.D.yl +/SLΩ/.nΩTdLAΩ=1Ω n/AALnS)Ω= -,.xAΩSSnΩnAx.nTS6)Ω,.'ΩnS.Ω=1bbAnETa1Ω=TnQ,A)lkAΩ b.nxTxΩA, n/A)A$_k$ - l J' Q=Ω=TnT.AΩ,A:0TnAΩΩΩ G ï

D 9 , 4 œ #AGQ)=Sxb.= 4ΩA=.nAnΩA1Ω=.E Ω)nSE=ΩbΩ1=ĕnSSΩ= $(A - A^T)/2$.),Ω fΩ= $(A + A^T)/2$.

RU.D.5l +/SLΩ/.nΩn/A1Ω.EΩ,AaSb1S=TnQSΩ,ΩS0=T0==.xbnETxΩT=,0Q - =AlT0nE'ΩA = $U_1 P_1$.) F Ω $A = U_2 P_2$ .EAΩLSΩS=.EΩAaSb1S=TnTS0=nPAΩΩJ $U_1 = P_2 P_1^{-1}$ .0,Ω U' $U_2 = P_1 P_2^{-1}$ /..AΩn/AΩc AΩ AT A0..==A=GΩ

E 9 , 9>œ GG •A„ G A DG3 8G *Ð̈ ᵥ z JG̈ ; GG 8oG o D S à
ò 0G¿ D G D z ï GG D GD 1 7 G
E 9 !, œ> GG3 } A G A„ 8G œ Ž ¿© Gž G G Go o G DD D S à1 G
E 9 ķ k œ n 8 FD1₆ G G ï 1DG3 G Ì G A *G1K G Š Ù
E 9, ! %ρœ *Ð̈ G G1QG• •Q G D GD ž G DD G œu©G G ¿ 0 G D 3 DG‡D ¿G
D GD ž GXD=caGRa.R*I= X²· |,'ꟷ +/(LR/.RQꟷX₀= ï G

Ì â Ð=a -1ꟷ1 , ⁱ |l+T

RaA)ꟷT¨ 1˜ -=.xE.RQa.||1ꟷyAEAꟷ xA)(RA=Rꟷa-A.REQ|ꟷT Q)1ꟷRꟷ Jꟷ
|UQDE|/| +a(LRy.Rꟷ
X. ᵥ C1 a(=| ˘³ 'ꟷ-1C₂√* Ã ï 1Cœ©%G

D ¿G GD 8 8 GG 8ᴰ [GY -ï AX(t), |. ,|T = Ci, .. ,|T ₙₜ C₂. |===-Aꟷ-Ra*.Rꟷ-G D G
D € 1G D Ë G

= z • \ Pœw X\œ \ ‡ \ w Ua'z'‡õœ9 , œ

F D ¿ DG¿ G G GG GG D G ¿JG 3 œG hu@G ©G € G

V à l· ² ¿G G 1 . ' Ÿ ´ D ;@G D 2 4 G 5 S ~Š& ï S e ANï Æ G œG © •2 G
*3 11G G8 E 2 6 GD N2 Î2G GÓÎ HÓ " .

) DG3 D G G D ‡¿ GD Dn G„G1 G1 D 1 G G D 1 1 nG y G

6 uEG G(:# ' 8 1 ! G D + 6G G FD‡ GD0 DG3 @ DD3 GSGG D & 2 4 G
'ï + + '1ï : ï $ D H D .
* à • u G G!Gu Nu G G [6 D Š1D G@D D +A DG3@ GD „ŸG D 2 4 G
ï/ B ï -ï + + -$ï \Vï DH
* ² •u GÄ ! à GNà G 2 G•@)à G G a ' @ D 1SGG! 8 A D6 G FD‡ nG#
D @2 ï4 G % 0 ï X @ Š ïF #"ï" H ".' '
* à • u G 1G ! Gà Nà G G [' h G 8 V 0GDã @G D ¿A)GG8 < G D D1+G G
@ D „ŸG1 D2 4@1Gï ï/ B ï '*ï + + $'#$ ((H*. $
6 uEG 2 G * u]Gu@G 2¿G DG ' ‡k ¿ @GD S ŸG1 D @ DG 3 1 DG¿* G n D D
;ŠG ä 1 D G @2/4 G§/ï Ž B ï ' ï ++ $ ï1ï$ H1$#
â à G E G'Gb<" G 8 G (a' ò D +@G S ŸG D D *1G G DŠ GU 1nD G 2 4
/ ï (ï/ Ž B ï 0 ï+ + 0$ï Ḧ 1 H 1 *
V à l· à+G G ' * ' k + @G D „<G D1 G6Š 8D DG DG¿)G 8G<3 G D2 4 G
& ' ï ' ï + + 'V ï $®W $ '! Ḧ .
V à · à¿G 2<Gà @à G£ V @G £ 2 Gàk G G . ' * n DD 3) G8 <3 G D
DG ¿@ GD „1GG D1@G1DG> „ nG2 4@Gï ï Ž B ï ' ï ++ V_:ï ï (DP
.

] 3 nq G D TDG3 DG ®„‡G 3 DG G 8 GG ¼G G

F à < u G 1 G G ! Eu V u G GS '.k 3 @G D „ŸG G D *G n D D 1 G1 DG 2 4
* + + /0 ï % ï + A ï V$ 'ï C ' [
N ,Eè Ø / 1 £ GG G Ì 8 (G. ' ! G ¿ n@ GD A DG¿ ‡G G 6; G@D G2z&4 „G-*ï 0 ï
* + + : 'Vï® : ! ï: H ' [
V ul· ¶¿G G($ ' V D þ1@G D A DG3@ GD ‡ S G3 GD/4 G% ' ï + , ï" 4ï1 " D H 1 1
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í‹G$ ˜ G LE Q$G U ª˜G G ! Q UGR)J $ ' * Y j J k" , -aJ Ü j ‰ 6G pG $ a@GG U o
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Chapter

Large Sparse Eigenvalue Problems

[illegible]

[illegible]

[illegible]

[illegible] E¶G_TEMPLATES.

10.1 The Symmetric Lanczos Process

[illegible]

$$\overline{\quad T \quad},$$

$$M_k \;=\; \lambda_1(Q_k^T A Q_k) \;=\; \max_{y\neq 0} \frac{y^T(Q_k^T A Q_k)y}{y^T y} \;=$$

$$\frac{y^T(Q_k^T A Q_k)y}{}$$

$$\nabla r(u_k) \;\in\; \mathsf{span}\{q_1, \ldots, q_{k+1}\}.$$

$$K(A, q_1, k) \;=\; \left[\, q_1 \,|\, Aq_1 \,|\, A^2 q_1 \,|\, \ldots \,|\, A^{k-1} q_1 \,\right]$$

$$Q \;=\; \left[\, q_1 \,|\cdots|\, q_n \,\right]$$

$$= \begin{bmatrix} \alpha_1 & \beta_1 & & \cdots & \\ \beta_1 & \alpha_2 & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \\ & \cdots & & & \end{bmatrix}.$$

$$Aq_k \;=\; \beta_{k-1} q_{k-1} + \alpha_k q_k + \beta_k q_{k+1},$$

$$r_k \;=\; (A - \alpha_k I) q_k - \beta_{k-1} q_{k-1}$$

[illegible]

$$q_{k+1} = r_k/\beta_k$$

[illegible]

$$\beta_k = \pm\| r_k \|_2.$$

[illegible]

```
[illegible]
while [illegible]
    [illegible]
end
```

[illegible]

Theorem [illegible]

[illegible]

$$AQ_k = Q_kT_k + r_ke_k^T$$

[illegible]

$$T_k = \begin{bmatrix} \alpha_1 & \beta_1 & & \cdots & \\ \beta_1 & \alpha_2 & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \beta_{k-1} \\ & \cdots & & \beta_{k-1} & \alpha_k \end{bmatrix}.$$

[illegible]

$$\text{ran}(Q_k) = \mathcal{K}(A, q_1, k).$$

[illegible]

$$Q_k^T A Q_k = T_k + Q_k^T r_k e_k^T.$$

[illegible]

$$q_j^T A q_i = q_j^T \left(\beta_{i-1} q_{i-1} + \alpha_i q_i + \beta_i q_{i+1}\right) = \beta_{i-1} q_j^T q_{i-1} + \alpha_i q_j^T q_i + \beta_i q_j^T q_{i+1}$$

[illegible]

$$q_i^T A q_j = q_j^T A q_i = \begin{cases} 0, & [illegible] \\ \beta_{j-1}, & [illegible] \\ \alpha_j, & [illegible] \end{cases}$$

[illegible]

$$q_{k+1} \in \text{span}\{Aq_k, q_k, q_{k-1}\} \subseteq \mathcal{K}(A, q_1, k+1).$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$Y_k \;=\; [\, y_1 \,|\cdots|\, y_k \,] \;=\; Q_k S_k \in \mathbb{R}^{n\times k},$$

[illegible]

$$Q_k^T(AY_k - Y_k\Theta_k) \;=\; (Q_k^T A Q_k)S_k \;-\; Q_k^T(Q_kS_k)\Theta_k \;=\; T_kS_k - S_k\Theta_k \;=\; 0.$$

[illegible]

[illegible]

[illegible]

$$\| Ay_i - \theta_i y_i \|_2 \;=\; |\beta_k|\,|s_{ki}|.$$

[illegible]

$$E = -s_{ki}\cdot r_k y_i^T,$$

[illegible]

[illegible]

[illegible]

$$\min_{\mu\in\lambda(A)} \; |\theta_i - \mu| \;\le\; |\beta_k|\,|s_{ki}|$$

[illegible]

$$(A+E)Q_k \;=\; Q_k(T_k + \tau a^2 e_k e_k^T) + (1+\tau ab)\, r_k e_k^T.$$

[illegible]

$$\bar{T}_k \;=\; T_k + \tau a^2 e_k e_k^T$$

[illegible]

[illegible]

[illegible]

Theorem 10.1.2. [illegible]

[illegible]

where $\cos(\phi_1) = |q_1^T z_1|$,

$$\frac{\lambda_1 - \lambda_2}{[illegible]}$$

[illegible]

[illegible]

$$\geq \frac{\lambda_1 d_1^2 p(\lambda_1)^2 + \lambda_n \delta^2}{d_1^2 p(\lambda_1)^2 + \delta^2}$$

[illegible]

\ ¶

$$\delta^2 \;) \; \sum_{i=2}^{n}]= p(\lambda_i)]^2 .$$

b ¶ ¶ Y V Ž• ¶¶ ¶ > Ya ¶ ¶¶ ¶ φx¶¶ λ_1 0),,θ,(+ () ;(. -$_i$-^{T}(θ($\lambda_2, \ldots, \lambda_n$, ;() (; ($_i$ −((((, −)(= −).)+ ,;(s;(-θ−.λ 8,;+ ;. ;λ $\theta_{,i}$), c)+;); θ) ;.−;θ i θ+−+; ++$_i\theta$;; θ))[0,))0( )+ +((

$$x2 \quad i \quad c_{k-1} \left[-1 + 2\frac{x-\lambda_n}{\lambda_2-\lambda_n} \right]$$

;(,($c_{k-1}(z)$)+ ((e.')N1)B.(:N3.v l J :)5A)28 l).5216 v)2 l 136((5N)) l

$$c_k(z) \;) \; 2zc_{k-1}(z) - c_{k-2}(z), \qquad c_0 = 1,\; c_1 \;\theta\; z.$$

8,(.(,)−a)),)θ−. $_i$,(−)^T)[(+ −a ^{T}))(a)) [−1, 2 ;,)·(-c,a ,θ,;[−a).(.)+((,;+ ;)((,-θ_{-1} a+(c))); −(x) (,;+ ;a)(1−−)·(,θ($|p(\lambda_i)|$ 1 1, θ) 2:n θ)+ $p(\lambda_1) = c_{k-1}(1+2\rho_1)$;(,(ρ_1 ;++(c)1 −a (10.1.10). 8,^{T}+

$$\delta^2 \quad \sum_{i=2}^{n} = N\,1] - i ?$$

θ)[+)

$$\text{pè ¿ \$} - (\lambda_1 - \lambda_n) \; 1 = i \; = \backslash \frac{œ}{[1 \;\; (1 + 2\rho_1))^2}$$

8,λ +λ+;,(+−)(,]^T) ;+)−(θ;)c[−a))()); (,i((θ) ( )^2 l R− i (œ .

x) $_i$)θ−);)^T+ ,(.^{T}−(,((θ))); () T_k ++,θ−−c.(();()-θ−.)$_j$ i)(p+a0),]−−θ a_1

$_j$)θ)−−;θ.)1(c(.(θ+1,2 ;.) ++,;,θ+; ,2)+,2 +,). θi θ $\lambda_k(T_k)$, ,2)

$$\lambda_n \qquad \lambda_n + (\lambda_1 - \lambda_n) \left[\frac{\cdot\theta)\;(\,\eta\,)}{c_{k-1}(\,1 + 2\rho_n)} \right]^2 .$$

2))

$$\rho_n = \frac{\lambda_{n-1} - \lambda_n}{\lambda_1 - \lambda_{n-1}}$$

;+. 0)+(ŋ) $q_1^T z_n$.

Proof. .... ***P***.≠),≠. 10.1.2 ·)(, A)2,,'.! ' l -S l A. D

1)! l!S l' fV'l)! l)"- l'- 1)!')!+ l /1 Ĩ11 ,l' "ll"1! l') ! l '+,,' -S, ï',Sl'+'', l i0o[ŏ .' l-! l ')')"l,"! ' l Ĭ)e-S ')!-,',S"+'', 'nij l) l .) ȷ (A)q θ,,−;c(. (,c 0),,))λ)( ), q_1)) (,c +),(0(;))), (,((;;()-(0(), z_1. x +;,)−θ, ; (θ0θ) −c .+(+ $_i$))−(θ;) −).)+. 1, θ)))(,)),s)(A-θ−^{T} θ_i. n)·(-c, (,(,(+.−(. θ,( ))(- .θ(;+,θ0(),a −(0θ^T+(,( )(· θ,,c),a)); ,)ca)),)θ−))-)−-(+ ;(,,)[.0( ), (,(,(−a.,(- ,)−a)),;θ− c_{k-i} θ)+ (.(,)−a)),)θ− $2 - \lambda_1) \cdots 2 - \lambda_{i-1})$. j), +(($\theta$−. +((θ))c−(1966) θ)+ θ;;((19ρ2) θ)+ θ−.) ≤$\theta\theta$(1980), ·.)),,,)-(+ (,(−)^T)+.1 8,(,θ;) (,(),(, ;. 1−−)·+1 [illegible] *10.1.2* [illegible]

$$\lambda_i \;\ge\; \theta_i \;\ge\; \lambda_i \;-\; (\lambda_1 - \lambda_n)\left(\frac{\kappa_i \tan(\phi_i)}{c_{k-i}(1+2\rho_i)}\right)^2$$

[illegible]

$$\rho_i \;=\; \text{[illegible]} \qquad \cos(\phi_i) = |q_1^T z_i|.$$

Proof. [illegible]

[illegible]

[illegible]

[illegible]

$$\text{[illegible]}$$

[illegible]

$$\text{[illegible]} \; \frac{v^T A v}{\text{[illegible]}}$$

[illegible]

$$\lambda_1 \ge \gamma_1 \;\ge\; \lambda_1 - (\lambda_1 - \lambda_n)\tan(\phi_1)^2 \left(\frac{\lambda_2}{\lambda_1}\right)^{2(k-1)}. \qquad \text{[illegible]}$$

[illegible]

$$L_{k-1} \;\equiv\; \frac{1}{\left[c_{k-1}\left(2\frac{\lambda_1}{\lambda_2} - 1\right)\right]^2} \;\ge\; \frac{\text{[illegible]}}{[c_{k-1}(1+2\rho_1)]^2}$$

[illegible]

$$\text{[illegible]}.$$

[illegible]

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. ọ 7 ¶	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
. 2 ¶	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
	______	______	______	______	______

[illegible]

10.2 Lanczos, Quadrature, and Approximation

[illegible]

$$b \le u^T \cdot f(A) \cdot u \le B.$$

[illegible]

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[illegible]

[illegible]

Gauss-Radau(a). [illegible]

[illegible]

[illegible]

Gauss-Radau(b). [illegible]

[illegible]

[illegible]

Gauss-Lobatto. [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$R_{GR(b)}(f) = \text{[illegible]}$$

[illegible]

[illegible]

$$I_G(f) \quad [illegible] \quad I_{GR(a)}(f).$$

[illegible]

[illegible]

[illegible]

Fact *1.* [illegible] $[a, b]$ θ [illegible]

$$\int_a^b [illegible]$$

[illegible]

Fact *2.* [illegible]

$$[illegible]$$

[illegible]

Fact *3.* [illegible]

$$S^T T_k S = \operatorname{diag}(\theta_1, \ldots, \theta_k)$$

[illegible] s_{1i}^2 [illegible]

$$I_G(f) \quad [illegible]$$

[illegible]

[illegible]

[illegible]

$$\beta_k q_{k+1} = (A - \alpha_k I) q_k - \beta_{k-1} q_{k-1}$$

[illegible]

$$\beta_k p_{k+1}(A) q_1 = (A - \alpha_k I) p_k(A) q_1 - \beta_{k-1} p_{k-1}(A) q_1.$$

[illegible]

$$\beta_k p_{k+1}(\lambda) = (\lambda - \alpha_k) p_k(\lambda) - \beta_{k-1}^2 p_{k-1}(\lambda).$$ [illegible]

[illegible]

[illegible]

[illegible]

1: [illegible]

2: [illegible]

3: [illegible]

[illegible]

[illegible]

[illegible]

$$
, \frac{1}{4}\left[\begin{array}{cccccc|c}
[illegible] & & & \cdots & & 0 & 0 \\
[illegible] & & \ddots & & & \vdots & \vdots \\
0 & \ddots & \ddots & \ddots & & \vdots & \\
\vdots & & \ddots & \mathrm{0k\text{-}1} & & \mathrm{fk\text{-}1} & 0 \\
0 & \cdots & & \mathrm{fk\text{-}1} & & \mathrm{0k} & \mathrm{fk} \\
\hline
0 & \cdots & \cdots & 0 & & \beta_k & \tilde{\alpha}_{k+1}
\end{array}\right]
$$

[illegible] $_i)+ \; -)(_i), \; ,_{i-} \; _j) \; _i,(\; (\; {}^T{}_i(;))$

$$\tilde{T}_{k+1}\begin{bmatrix} x \\ -1 \end{bmatrix} = a\begin{bmatrix} x \\ -1 \end{bmatrix}, \qquad x \in \mathbb{R}^k,$$

[illegible]

[illegible] $-I$,6 q j . z .

[illegible]

[illegible] 10.1.1, [illegible] while n)), [illegible] 10.1.1 [illegible]

[illegible]

[illegible]

[illegible] (1983). [illegible]

[illegible]

10.3 Practical Lanczos Procedures

[illegible]

$$\alpha_k = q_k^T(Aq_k - \beta_{k-1}q_{k-1}),$$

[illegible]

```
[illegible]
while [illegible]
    [illegible]
        [illegible]
    end
    [illegible]
    [illegible]
end
```

[illegible]

[illegible]

[illegible]

$$\hat{T}_k = \begin{bmatrix} \hat{\alpha}_1 & \hat{\beta}_1 & & \cdots & 0 \\ \hat{\beta}_1 & \hat{\alpha}_2 & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \hat{\beta}_{k-1} \\ 0 & \cdots & & \hat{\beta}_{k-1} & \hat{\alpha}_k \end{bmatrix}.$$

[illegible]

$$\| E_k \|_2 \approx \mathbf{u} \| A \|_2.$$

[illegible]

$$\hat{\beta}_k \hat{q}_{k+1} \approx \hat{r}_k + w_k$$

[illegible]

[illegible]

[illegible]

$$|\hat{q}_{k+1}^T \hat{q}_i| \approx \frac{|\hat{r}_k^T \hat{q}_i| + \mathbf{u} \| A \|_2}{|\hat{\beta}_k|}$$

[illegible]

$$X_1 = \hat{Q}_k, \qquad S = T_k,$$

[illegible]

$$\tau = \| \hat{Q}_k^T \hat{Q}_k - I_k \|_2$$

[illegible] (A) [illegible]

$$|\mu_i - \lambda_i(T_k)| \;\le\; \sqrt{2}\,(\| \hat{r}_k \|_2 + \| E_k \|_2 + \tau(2+\tau)\| A \|_2)$$

[illegible]

[illegible]

[illegible] $r_0, \ldots, r_{k-1}$ [illegible] $H_0, \ldots, H_{k-1}$ [illegible] $(H_0 \cdots H_{k-1})$ [illegible] $[\, r_0 \mid \ldots \mid r_{k-1} \,]$ [illegible] $[\, q_1 \mid \cdots \mid q_k \,]$ [illegible] $(H_0 \cdots H_{k-1})$. [illegible] q_{k+1} [illegible]

[illegible]

[illegible] H_k [illegible] $(H_0 \cdots H_{k-1})$ [illegible] $[\, r_0 \mid \ldots \mid r_k \,]$ [illegible] $H_0 \cdots H_k$ [illegible]

```
r0 [illegible] q1 [illegible]
n [illegible] H0 [illegible] H0r0 = e1.
f r k = [illegible]
    ak = qf Aqk
    Tk = (A - Okl)qk - fk-lqk-1      $(\beta_0 q_0 \equiv 0)$      1/11OET
    [illegible] = (Hk-1 · · · H0)rk
    n [illegible] Hk [illegible] Hkw [illegible] 0]T.
    qk+1 [illegible] H0 . . . Hkek+1
end
```

[illegible]

[illegible]

[illegible]

$$\hat{Y}_k = [\ \hat{y}_1 \mid \cdots \mid \hat{y}_k\ ] = fl(\hat{Q}_k \hat{S}_k),$$

[illegible]

$$|\hat{q}_{k+1}^T \hat{y}_i| \approx \frac{[illegible]}{[illegible]}$$ [illegible]

[illegible]

$$\| A\hat{y}_i - \hat{\theta}_i \hat{y}_i \|_2 \approx |\hat{\beta}_k|\, |\hat{s}_{ki}|.$$ [illegible]

[illegible]

$$\| A [illegible]$$

[illegible]

Lemma 10.3.1. [illegible]

$$\| I_k - S^T S \|_2 \le \mu \qquad\qquad |1 - d^T d| \;\le\; \delta,$$

[illegible]

$$\| I_{k+1} - S_+^T S_+ \|_2 \;\le\; \mu_+$$

[illegible]

$$\mu_+ \;=\; \frac{1}{2}\left(\mu + \delta + \sqrt{(\mu - \delta)^2 + 4\| S^T d \|_2^2}\;\right).$$

Proof. [illegible]

¶

$$Q^T A Q = \text{8} \begin{bmatrix} M_1 & B_1^T & & \cdots & \text{7 ¶} \\ B_1 & M_2 & \ddots & & \vdots \\ \text{`} & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & B_{r-1}^T \\ \text{7 ¶} & \cdots & & B_{r-1} & M_r \end{bmatrix}$$

57. K-S¶ K

¶

6 ` &

$$M_k = X_k^T A X_k$$

$$\begin{bmatrix} & & & \cdots & \\ & & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & B_{k-1}^T \\ & \cdots & & B_{k-1} & M_k \end{bmatrix}.$$

[illegible]

[illegible]

[illegible] $\lambda_1 \ge \cdots \ge \lambda_n$, [illegible]

[illegible]

$$0 < \cos(\phi_p) = \sigma_p(Z_1^T X_1),$$

[illegible]

$$\lambda_i \ge \mu_i \ge \lambda_i - (\lambda_1 - \lambda_n)\left(\frac{\tan(\theta_p)}{c_{k-1}\,(1+2\rho_i)}\right)^2$$

[illegible]

$$\rho_i = \frac{\lambda_i - \lambda_{p+1}}{\lambda_{p+1} - \lambda_n}$$

[illegible]

Proof. [illegible]

[illegible]

[illegible]

[illegible] any [illegible]

[illegible]

[illegible]

[illegible] *1.* [illegible]

[illegible] *2.* [illegible]

[illegible]

Step 4. Set $X_1^{(\text{new})} = [\,X_1 \,|\cdots|\, X_s\,]\,[\,u_1\,|\cdots|\,u_p\,]$.

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] (1997) [illegible] 18, 341–349.

[illegible]

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, ,ypG 2 G 6 G G $, G @$ G , ! 2 G f`jd ?G/ < / GG >S | b | G
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$$\left[\begin{array}{ccccc} & & \cdots & \cdots & \\ & & & \cdots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & & & \\ & \cdots & \cdots & & \\ \hline & & & & \end{array}\right].$$

[illegible]

[illegible]

$$u_k = r_k/\alpha_k, \qquad (\alpha_k \neq 0).$$

[illegible]

$$v_{k+1} = p_k/\beta_k, \qquad (\beta_k \neq 0).$$

[illegible]

[illegible]

[illegible]

```
[illegible]
while [illegible]
    [illegible]
end
```

[illegible]

[illegible]

$$B_k = \begin{bmatrix} \alpha_1 & \beta_1 & \cdots & \cdots & 0 \\ 0 & \alpha_2 & \beta_2 & \cdots & \vdots \\ \vdots & \ddots & \ddots & \ddots & [illegible] \\ \vdots & & [illegible] & & [illegible] \\ [illegible] & & [illegible] & [illegible] & \alpha_k \end{bmatrix},$$ [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$F_k^T B_k G_k = \Gamma = \mathrm{diag}(\gamma_1, \ldots, \gamma_k)$$ [illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$Ay_i = \gamma_i z_i,$$ [illegible]

$$A^T z_i = \gamma_i y_i + [F_k]_{ki} \cdot p_k.$$ [illegible]

[illegible]

$$C = \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix}.$$

[illegible]

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end
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C ¶F¶O €¶ ¶" Þ ¶• 9 C ¶ ¶F$ $ÓÖ. œ

[illegible]

[illegible]

$$\tilde{A}_k = \tilde{Z}_1\tilde{\Sigma}_1\tilde{Y}_1^T = \tilde{Z}_1\tilde{Z}_1^T A$$

[illegible]

[illegible] $C \in \mathbb{R}^{m\times c}, U \in \mathbb{R}^{c\times r}, R \in \mathbb{R}^{r\times n}, k \le c, k \le r$

[illegible]

[illegible] $\sqrt{q_{col(t)}}$

end

5. [illegible] $D_C \in \mathbb{R}^{c\times c}$ [illegible]

[illegible]

$$\sqrt{r\, p_{row(t)}}$$

[illegible]

$$U = \Phi\Psi^T, \qquad \Phi = Y_1\Sigma_1^{-2}Y_1^T,\ \Psi = D_R C(row,:).$$

[illegible]

$$\| A - CUR \|_F \le \| A - \tilde{A}_k \|_F + \epsilon \| A \|_F$$

[illegible]

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%$ C¼GP G

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10.5 Krylov Methods for Unsymmetric Problems

[illegible]

$$\mathsf{span}\{q_1, \ldots, q_k\} = \mathsf{span}\{q_1, Aq_1, \ldots, A^{k-1}q_1\}.$$

[illegible]

$$AQ_k = Q_k H_k + r_k e_k^T$$

[illegible]

$$[illegible] \begin{bmatrix} [illegible] & [illegible] & \cdots & h_{1k} \\ [illegible] & [illegible] & \cdots & h_{2k} \\ [illegible] & [illegible] & & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ [illegible] & \cdots & h_{k,k-1} & h_{kk} \end{bmatrix}.$$

[illegible]

[illegible]

$$q_+ = p(A)q_1$$

[illegible]

$$q_1 = a_1 z_1 + \cdots + a_n z_n,$$

[illegible]

$$z = a_1 p(\lambda_1) z_1 + \cdots + a_n p(\lambda_n) z_n.$$

[illegible]

$$p(\lambda) = c \cdot (\lambda - \mu_1)(\lambda - \mu_2) \cdots (\lambda - \mu_p)$$

[illegible]

$$z = c \cdot \sum_{k=1}^{n} a_k \left(\prod_{i=1}^{p} (\lambda_k - \mu_i) \right) z_k.$$

[illegible] $\mathcal{K}(A, q_1, m)$ [illegible]

[illegible]

[illegible]

$H(0) = H_e$

[illegible]

end

[illegible]

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¶

$$H_+ = V^T H_c V.$$

! 8] 8 Q

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$$AQ_+ = Q_+ H_+ + r_c e_m^T V$$

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[illegible]

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[illegible]

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$$A(Q_1 Z) = (Q_1 Z)(Z^T T_{11} Z) + r_o (Z^T y)^T.$$

$$Q^{-1}AQ = T = \begin{bmatrix} \alpha_1 & \gamma_1 & & \cdots & \cdot \\ \beta_1 & \alpha_2 & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \gamma_n \\ , & \cdots & & & \end{bmatrix}.$$

$$Q = [\, q_1 \,|\, \cdots \,|\, q_n \,]$$

$$\tilde{Q} = [\, \tilde{q}_1 \,|\, \cdots \,|\, \tilde{q}_n \,],$$

we fi d upo compari g Cog m in $AQ= QT$,$A^T\tilde{Q} = \tilde{Q}T^T$,

$$1\ |c\tilde{Q}= \Lambda c,-|\ c,\ -\tilde{Q}+\ _4c|\ c\tilde{Q}+\ _5c\ |c)\ =,\tilde{Q}$$ "oqo 7 !2 œ

$$,\ '1\ _3c\tilde{Q}=\ _5c\ ,\ -_3c\ ,\ -\tilde{Q}+\ _2c_3\ c\tilde{Q}$$ U œD F @ V foio 7 !Ä œ

for $k =$ ∘+6œ ÅJCê equatiÔ¶ s together with the Iab-da gaWadHaW

$$\tilde{Q}^TQ = I_n$$

imply

$$\alpha_k = \tilde{q}_k^T A q_k$$

a

$$\text{axi} = (A\ ,^2\ cF|c\tilde{Q},\ \Lambda c,\in|\ c,\ _1,\tilde{Q}$$

$$\Lambda c_{31}\ -\tilde{Q}\ \ _3c\tilde{Q}= (A -\ ,,-\ '\ -_{,1} -\ _5c\ ,-_3\ c,.\ M\tilde{Q}$$

4. s ()), bi ity a WWa ati2 a a v itaW

$$1 = \tilde{q}_{k+1}^T q_{k+1} = (\tilde{r}_k/\gamma_k)^T (r_k/\beta_k)\,.$$

$$T_k = \begin{bmatrix} \alpha_1 & \gamma_1 & & \cdots & \\ \beta_1 & \alpha_2 & \ddots & & \vdots \\ & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \gamma_{k-1} \\ & \cdots & & \beta_{k-1} & \alpha_k \end{bmatrix},$$

[illegible]

$$C_k^T B_k = S_k^T R_k,$$

[illegible]

$$\left.\begin{matrix}2\\2\\5\end{matrix}\right].$$

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10.6 Jacobi-Davidson and Related Methods

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$$A(x_c + \delta x_c) = (\lambda_c + \delta\lambda_c)(x_c + \delta x_c),$$

[illegible]

$$(A - \lambda_c I)\,\delta x_c - \delta\lambda_c x_c = -r_c + \delta\lambda_c \cdot \delta x_c,$$

[illegible]

[illegible]

$$(A - \lambda_c I)\,\delta x_c - \delta\lambda_c x_c = -r_c.$$

[illegible]

[illegible]

$$w^T x_+ = 1,$$

[illegible]

$$w^T \delta x_c = w^T (x_+ - x_c) = 0.$$

[illegible]

$$\begin{bmatrix} A - \lambda_c I & -x_c \\ w^T & 0 \end{bmatrix} \begin{bmatrix} \delta x_c \\ \delta\lambda_c \end{bmatrix} = - \begin{bmatrix} r_c \\ 0 \end{bmatrix}.$$

[illegible]

$$F\left(\begin{bmatrix} x \\ \lambda \end{bmatrix} \right) = \begin{bmatrix} Ax - \lambda x \\ w^T x - 1 \end{bmatrix}.$$

[illegible]

[illegible]

[illegible]

$$\begin{bmatrix} M-\lambda_c I & -x_c \\ w^T & 0 \end{bmatrix} \begin{bmatrix} \delta x_c \\ \delta \lambda_c \end{bmatrix} = - \begin{bmatrix} r_c - N \cdot \delta x_c \\ 0 \end{bmatrix}.$$

[illegible]

$$\begin{bmatrix} M-\lambda_c I & -x_c \\ w^T & 0 \end{bmatrix} \begin{bmatrix} \delta x_c \\ \delta \lambda_c \end{bmatrix} = - \begin{bmatrix} r_c \\ 0 \end{bmatrix},$$

$$\begin{bmatrix} \alpha & c^T \\ c & A_1 \end{bmatrix} \begin{bmatrix} 1 \\ z \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ z \end{bmatrix}.$$

[illegible]

$$\left[\begin{array}{c|cc} \alpha-\lambda_c & c^T & -1 \\ c & A_1-\lambda_c I & -z_c \\ \hline 1 & 0 & 0 \end{array}\right]\left[\begin{array}{c} \delta\mu_c \\ \hline \delta z_c \\ \delta\lambda_c \end{array}\right] = -\left[\begin{array}{c} \alpha + +c^T z_c - \lambda_c \\ (A_1-\lambda_c I)z_c + c \\ \hline 0 \end{array}\right],$$

[illegible]

$$\left[\begin{array}{cc} A_1-\lambda_c I & -z_c \\ c^T & -1 \end{array}\right]\left[\begin{array}{c} \delta z_c \\ \delta\lambda_c \end{array}\right] = \text{[illegible]}$$

[illegible]

$$f\left(\left[\begin{array}{c} z \\ \lambda \end{array}\right]\right) = \left[\begin{array}{cc} \alpha & c^T \\ c & A_1 \end{array}\right]\left[\begin{array}{c} 1 \\ z \end{array}\right] - \lambda\left[\begin{array}{c} 1 \\ z \end{array}\right].$$

$$x_k = \left[\begin{array}{c} 1 \\ z_k \end{array}\right],$$

$$r_k = (A-\lambda_k I)x_k$$

$$r_k = \left[\begin{array}{cc} \alpha & c^T \\ c & A_1 \end{array}\right]\left[\begin{array}{c} 1 \\ z_k \end{array}\right] - \lambda_k\left[\begin{array}{c} 1 \\ z_k \end{array}\right] =$$

[illegible]

[illegible]

[illegible]

[illegible]

```
[illegible]
while [illegible]
    [illegible]
        [illegible]
    [illegible]
    [illegible]
    [illegible]
end
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\begin{bmatrix} A-\lambda_c I & -x_c \\ x_c^T & 0 \end{bmatrix} \begin{bmatrix} \delta x_c \\ \delta \lambda_c \end{bmatrix} = -\begin{bmatrix} r_c \\ 0 \end{bmatrix}.$$

[illegible]

[illegible]

[illegible]

$$(I - x_c x_c^T)(A - \lambda_c I)(I - x_c x_c^T)\delta x_c \ = \ -r_c$$

[illegible]

$$(I - x_k x_k^T)(M - \lambda_k I)(I - x_k x_k^T)\delta v_k \ = \ -r_k,$$

[illegible]

[illegible]

$$(I - Q_cQ_c^T)A(I - Q_cQ_c^T)Z_c = AV_c$$

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Chapter

Large Sparse Linear System Problems

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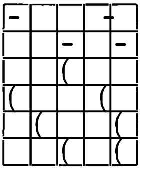
[illegible]

11.1 Direct Methods

LLGLEL

$$A =$$



6 ¶ U v R U ¶ ¶¶ C ¶ ¶ ¶ ¶ ¶
¶¶ ‘ ¶ ` b 2$¶ ¶ ¶ % 8¶ ¶ C v ¶ ¶ ¶ · .?¶ $2 ¶2

$$A.val = \begin{array}{|c|c||c||c|c|c||c|c||c|c|c|}\hline a_{11} & a_{41} & a_{52} & a_{23} & a_{33} & a_{63} & a_{14} & a_{44} & a_{25} & a_{55} & a_{65} \\ \hline\end{array}$$

P ¶ .¶ $¶ · ?> ¶

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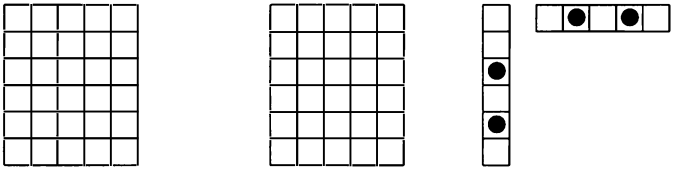


[illegible]

[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

[illegible]

```
[illegible]
    [illegible]
        [illegible]
        [illegible]
            if [illegible]
                [illegible]
                [illegible]
            end
        end
end
```

[illegible]

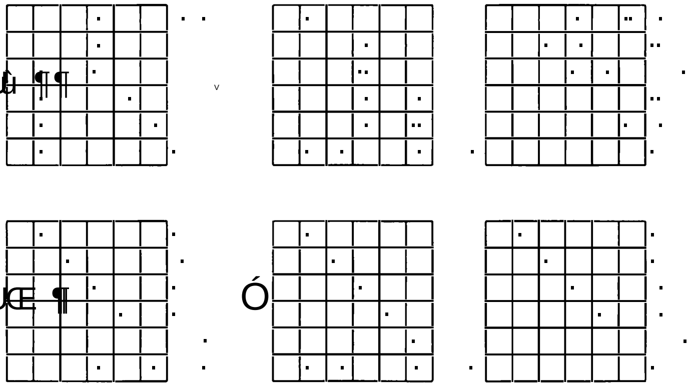
[illegible]

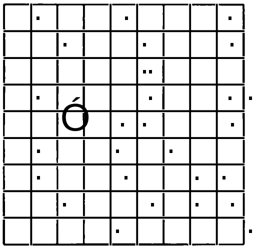
$$A = \begin{bmatrix} \alpha & v^T \\ v & B \end{bmatrix} = \begin{bmatrix} \sqrt{\alpha} & 0 \\ v/\sqrt{\alpha} & I \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & A^{(1)} \end{bmatrix} \begin{bmatrix} \sqrt{\alpha} & v^T/\sqrt{\alpha} \\ 0 & I \end{bmatrix}$$

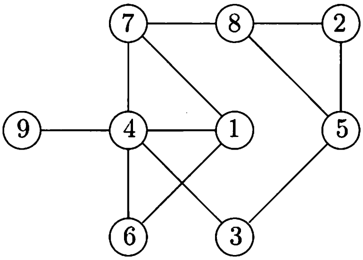
[illegible]

$$A^{(1)} = B - \frac{vv^T}{\alpha}.$$

[illegible]







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$V =$

$E =$

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Node	1	2	3	4	5	6	7	8	9
Degree	3	2	2	5	3	2	3	3	1

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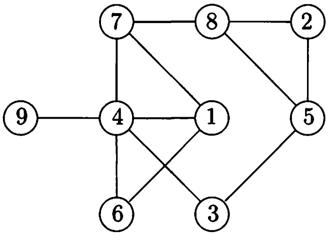
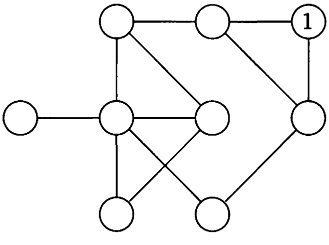


Original QA



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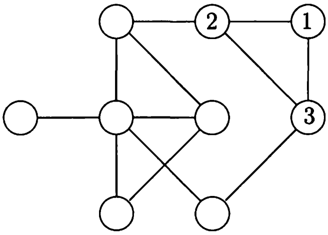
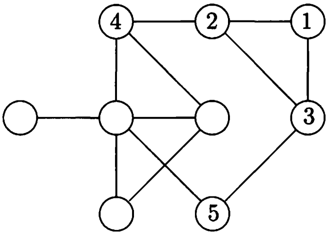
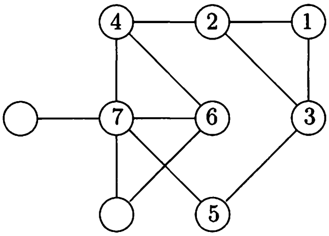


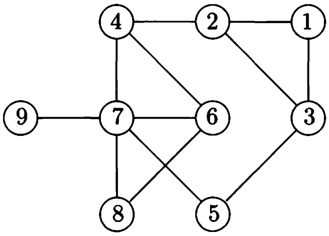
Labeled: S_0, S_1



Labeled: S_0, S_1, S_2



Labeled: S_0, S_1, S_2, S_3



Labeled: S_0, S_1, S_2, S_3, S_4

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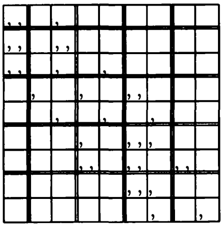

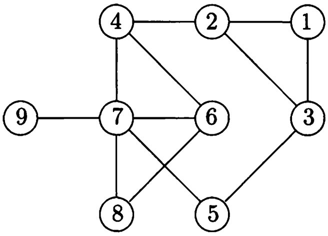

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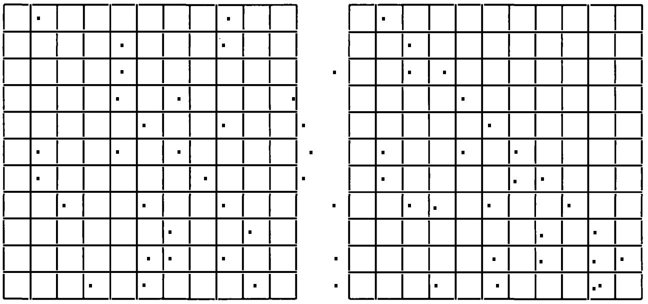

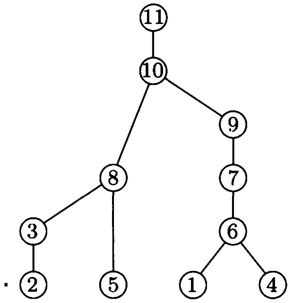
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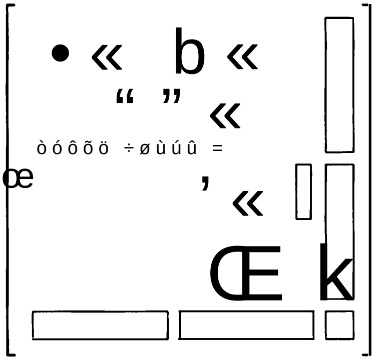
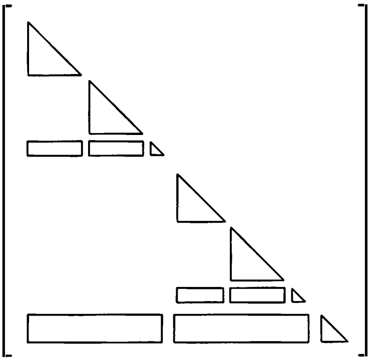
$$P_0AP_0^T =$$





[illegible]

$$P(A^T A)P^T = R^T R,$$

[illegible]

[illegible]

$$P(A^T b) = (P(A^T A)P^T)Px_{LS} = R^T RPx_{LS}.$$

[illegible]

[illegible]

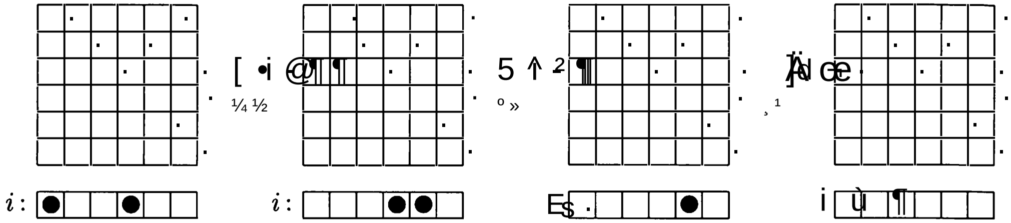
[illegible]

$$A^T A(x_0 + e) = A^T b - A^T r + A^T r = A^T b.$$

[illegible]

```
[illegible]
    [illegible]
        if [illegible]
            [illegible]  $\begin{bmatrix} a_{jj} \\ a_{ij} \end{bmatrix} = \begin{bmatrix} \times \\ 0 \end{bmatrix}$
            [illegible]  $\begin{bmatrix} a_{jj} & \cdots & a_{jn} \\ a_{ij} & \cdots & a_{in} \end{bmatrix}$ [illegible] $\begin{bmatrix} a_{jj} & \cdots \\ a_{ij} & \cdots \end{bmatrix}$ [illegible]
        end
    end
end
```

[illegible] ZV_F^{-1} $X_F^{-}X$ [illegible] Z^1X Z [illegible] $V \in X_{GG}$ [illegible] GX_F^{-1} [illegible] R^T [illegible]



[illegible]

[illegible]

[illegible] $\mathbb{R}^{n\times n}$ [illegible] Q [illegible] B_1

$$PAQ^T = \begin{bmatrix} \alpha & w^T \\ v & B \end{bmatrix} = \begin{bmatrix} \text{[illegible]} & 0 \\ \text{[illegible]} & \text{[illegible]}_{n-1} \end{bmatrix} \begin{bmatrix} \alpha & w^T \\ 0 & A^{(1)} \end{bmatrix}$$

[illegible]

$$A^{(1)} = B - \frac{1}{\alpha} v w^T.$$

[illegible]

[illegible]

[illegible]

$$M = \begin{bmatrix} A_1 & C_1 \\ 0 & C_2 \\ 0 & C_3 \end{bmatrix},$$

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2 N .) ! * *0 €:A 7 E ". GH H
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[illegible]

11.2 The Classical Iterations

[illegible]

```
for $i = 1{:}n$
    $x_i^{(k)} = \left( b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k-1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k-1)} \right) \Big/ a_{ii}$
end
```

[illegible]

```
    [illegible]
end
```

[illegible]

$$L_A = \begin{bmatrix} 0 & 0 & 0 \\ a_{21} & 0 & 0 \\ a_{31} & a_{32} & 0 \end{bmatrix}, \quad D_A = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}, \quad U_A = \begin{bmatrix} 0 & a_{12} & a_{13} \\ 0 & 0 & a_{23} \\ 0 & 0 & 0 \end{bmatrix}.$$

[illegible]

[illegible]

$$A_{12}\, x_2 \; - \; [illegible]$$

$$A_{21}\, x_1 \; - \; [illegible]$$

$$A_{31}\, x_1 \; - \; [illegible]$$

[illegible]

$$Mx^{(k)} = Nx^{(k-1)} + b$$

[illegible]

$$G = M^{-1}N.$$

[illegible]

$$e^{(k)} = x^{(k)} - x,$$

[illegible]

$$e^{(k)} = M^{-1}Ne^{(k-1)} = Ge^{(k-1)} = G^k e^{(0)}.$$

[illegible]

$$G = \begin{bmatrix} \lambda & \alpha \\ 0 & \lambda \end{bmatrix},$$

[illegible]

$$G^k = \begin{bmatrix} \lambda^k & \alpha\lambda^{k-1} \\ 0 & \lambda^k \end{bmatrix}.$$

[illegible]

[illegible] (11.2.6) [illegible]

[illegible]

$$\| G^k \|_2 \le (1+\mu)^{n-1}\left(\rho(G) + \frac{\| E \|_F}{1+\mu}\right)^k$$

[illegible]

[illegible] b $k=12$

[illegible] $=3$ ^2_i

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] $\in \mathbb{R}^{n\times n}$ [illegible]

[illegible]

[illegible] $\in \mathbb{R}^{n\times n}$ [illegible] (11.2.4) [illegible]

Proof. [illegible]

$$\| D_A^{-1}(L_A + U_A) \|_\infty = [illegible]$$

[illegible]

[illegible]

[illegible]

[illegible] $\in \mathbb{R}^{n\times n}$ [illegible] (11.2.5) [illegible]

Proof. [illegible]

$$G = D_A^{1/2} G_{\mathrm{GS}} D_A^{-1/2} = -(I + L)^{-1} L^T$$

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,.$vv\psi^n$„ ,(„^{n},.ψ ψ(,•(ψ „.$^n\psi v$

$$(I_{n_1} \otimes T_{n_2} + T_{n_1} \otimes I_{n_2})\, u = b \qquad ((! \quad œ$$

8 ¶

$$T_m \quad \begin{bmatrix} œ\,D\,(\,œ\cdot & & !\,œ \\ £(\,œ & œ\ddots & \vdots \\ \vdots & \ddots & \ddots \quad D\,(\,œ \\ !\,œ\cdots & D & œœ \end{bmatrix} \in \mathbb{R}^{m\times m}. \qquad (-(8(\quad œ$$

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ψ v,.ψ $^{nn}\psi$→, $^n\psi$

```
[illegible]
    [illegible]
        [illegible]
            [illegible]
        end
    end
```

[illegible]

```
    [illegible]
        [illegible]
            [illegible]
        end
    end
```

[illegible]

[illegible]

[illegible]

$$E_m = \begin{bmatrix} 0 & 1 & \cdots & 0 \\ 1 & 0 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 0 & \cdots & 1 & 0 \end{bmatrix}.$$

$$N_x = (E_{n_1} \otimes I_{n_2}).$$

[illegible]

$$M_\omega = \frac{1}{\omega} D_A + L_A \qquad N_\omega = \left(\frac{1}{\omega} - 1\right) D_A + U_A.$$

[illegible]

end

[illegible]

$$\sum_{j=i+1}^{n} a_{ij} x_j^{(k)} \Big) \Big/ a_{ii}$$

end

[illegible]

$$M_\omega = \frac{1}{\omega} D_A + L_A, \qquad N_\omega = \left(\frac{1}{\omega} - 1\right) D_A - L_A^T, \qquad G = M_\omega^{-T} N_\omega^T M_\omega^{-1} N_\omega,$$

$$G_1 = (I + \omega L_1^T)^{-1} (I + \omega L_1)^{-1} ((1-\omega) I - \omega L_1)((1-\omega) I - \omega L_1^T).$$

λ

[illegible]

[illegible] M [illegible]

[illegible]

[illegible]

[illegible]

[illegible] M^{-1} [illegible]

$$\| y^{(k)} - x \|_2 \;\le\; \| p_k(G) \|_2 \, \| e^{(0)} \|_2 .$$ [illegible]

[illegible]

$$S^T G S \;=\; \mathrm{diag}(\lambda_1, \ldots, \lambda_n) \;=\; \Lambda$$

[illegible]

$$-1 < \alpha \le \lambda_n \le \cdots \le \lambda_1 \le \beta < 1$$ [illegible]

[illegible]

$$\| p_k(G) \|_2 \;=\; \| p_k(\Lambda) \|_2 \;=\; \max_{\lambda_i \in \lambda(A)} |p_k(\lambda_i)| \;\le\; \max_{\alpha \le \lambda \le \beta} |p_k(\lambda)| .$$

[illegible]

[illegible] β [illegible]

$$\frac{z-\alpha}{\beta-\alpha}$$

$$\| y^{(k)} - x \|_2 \le \frac{\| x - x^{(0)} \|_2}{|c_k(\mu)|}.$$

```
while
end
```

$B_1^2 p z_k c_n$

$$A_1 = \begin{bmatrix} 1 & \\ -1/2 & \end{bmatrix} \quad A_2 = \begin{bmatrix} & \\ & \end{bmatrix}$$

† &G &G G Au ∈b, nCA)ΩC2nΩ=Ω€1AΩE2=ΩE2 -(a=ΩCA=(Ω 2Ax Ω AE2n-()b2E-x(Ω

MFF,QX Ω+>1(-)nΩ=nA)-1=ΩΩ(-2AxΩ-nΩAb2nE-xΩ= ln1 ® Tn2 ƘTnΩ ® In2 ' "(,Ω' ă.'(Ω!Ω '!}.y'!,!".Ω .) ' .ΩU(i, j) ,AΩCA2ΩAE2AΩ(F Ω(i - 1, j), U(i, j + 1), U(i + 1, j), 2) ΩU(i, j - 1). i(Eb,=2AΩ 2Ω1(-)Ω=A)-=1Ω(A aEAΩ-)ΩC-CΩ(i, j) -=2Ω=a-2,=A2ΩAE2XΩ(bΩ-n=AΩCΩA-C,(E=.Ω2'ΩiA=E-,AΩ nCAEΩ==1n-b2nE-xΩ=-)ΩE()ArAEΩ(xa n=-,Ω'F,Ω (,-.=Ω bAnC(x=ΩΩmx(Ω=(12AAΩ = b, nCA)Ω LC2Ω=nΩAΩ=1AΩE2=-a=(ΩΩAk=(-2AxΩ AE2n-()b2E-x(Ω

MFjQQL ,()=- AEΩCAΩ)A2EΩ=AbΩ(n1 ® T 2 + Tn1 ® ln2)x = b. ,2Ω -=nΩA=ΩAΩE21Ω2x-,=(ΩnCAΩ -AE2-()b2nE-xΩΩCA,Ω(rΩ,2(,-Ω -AE2n-()-ΩΩAxΩ2()21Ω,1(reΩ2EAΩe?,1df e(Ω

MFF,Q= ,E(2AΩ(11.2.13) 2) Ω(11.2.14).

MFF,Q5 ,E(2AΩ(11.2.15).

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MFjQ13 ,E(2AΩ(11.2.24) 2) Ω(11.2.25).

MFFQ,=k ,()=- AEΩCA2dΩ 2 T2nE-xΩ

$$A = \begin{bmatrix} 1 & \rho \\ -p & 1 \end{bmatrix}.$$

Ú 'Ọç"ãΩ'1)Ω' .ΩĎ "" ă y.y"ă'(ΩΩĜ ' .p(M i N_G s) ,Ę -džΩ. ę"ΩĜ ' "Ω",!ΩΩ ă "1Ω) ' .!Ω p(M: 1 Nw) ,Ę -dž)Ω.Ω' ,.!Ω:Ω ,y"y, y ã p(M: 1 Nw)? Ú Ď+ç'ΩΩÚ '' ç ãΩç'Ω!Ω, ' .'yLΩ

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-u;_{-1} + 2u; - U;_{+1} ƘR(ui+1 - u;_{-1}) = 0, i = 0Ef ж Ω

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M,FE,k, ,()=- AEΩ,AΩnAE2-()Ω

y(k+1) = w(By(k) + 5 ï y(k - 1)) Ƙy(k - 1)

1)!'!Ω B)1Ω Ď Ĝ(iĎΩ,+"(y.y""ΩTBQ = -2 0((Ω.., An) 1y. Ĝ Ω z ˜ . . . ¯ Ω '0 DNN 4.Ω30Ω x = Bx + 5 "(ä) < † G‡ j &%oe(k)G= y(k) x. Ú ..(ç ,ΩΩ y(1) = By(0) + 5 " ï 0&G & G e(k) = $p_k(B)e^{(0)}$ 1)!'!ΩP_k y(Ω"!Ω'Ω+",-u ",y 'y.Ωï† G G G G Å G† m† G GÅ æ ⊖ © b † &(k) ⊜ QTe(k). :!'y.!Ω '(ΩI !'!"Ď!}Ω .y""Ω'ΩY 7)Ωj = 1:n. ą -.ΩΩ(+!Ďy)Ω!Ω!L ' Ď.Ω (",..y""Ω 'Ω"!' ' .Ωtj0 2'Ωj, (Ω5 v(1.Ω31.Ω01Ω0.)432.Ω2Ω1034 8Ω

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$$Mx_{i+1} = Nx_i + A^T b$$

LCAEAMΩ= (ATA + 0 'PNΩ= 0 .0Ω 2Ω)Ω : RΩx^n y(Ω,,!''yĎΩ" y'y(!Ω"y'!0eΩ ę)'ΩΩ ') ' .ΩM - 1N y(Ω ' ,"" ' ,y ã '',!'ãĜΩ' p(M - 1N) ,Ę 1 -EΩr(A) = fmΩ+a1(=AΩxo = 0 2)xΩC2Ω ‖v‖_W = (vTWv)^{-1} α '' Ĝ!Ω1"", <1Ω ĜΩ) ' .Ω!, ' ' ă .!((ΩΩà ć ("R1,)!Ω y.!' ' .!(ΩĎ"".!',!Ω '"Ω)!Ω,y"y, , ,Ω 1"",Ω(", , .y""ΩΩ)!Ω!1'Ω{} , ' '!(Ω"., !,JΩĎ -ç"Ω') ' .yΩ'R Ú à fçΩA)Ω ‖XLS - Xi+1‖W rT ‖XLS - Xi‖W x'Ω/(LΩ(LΩb11AbA)ΩCAΩAE2-()Ω2AΩAQR F2(E-w2-()Ω (FΩ

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F d H G H g R w H G 7
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) $ Y G $ G $ ¹) 2 4 , G ' & 9 u [H — G
! " 6 G j - G C * ¶ G v h 6 [G ‰ G G F G $ •) j G • $ G p G v G L ® G $ 2 ¹ 4 G &
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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

11.3 The Conjugate Gradient Method

[illegible]

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¶

$$\| v \|_A = \sqrt{v^T A v}\,.$$

$$- \frac{1}{2} b^T A^{-1} b$$

$$\phi(x_c) = \frac{1}{2}\| x_c - x_* \|_A^2 + \phi(x_*).$$

[illegible]

$$x_+ = x_c - \mu_c g_c,$$

[illegible]

[illegible]

$$\mu_c = \frac{g_c^T g_c}{g_c^T A g_c}$$

[illegible]

[illegible]

$$\phi(x_+) = \phi(x_c) - \frac{1}{2}\frac{1}{\kappa_c} g_c^T A^{-1} g_c = \phi(x_c) - \frac{1}{\kappa_c}\left(\phi(x_c) + \frac{1}{2} b^T A^{-1} b\right).$$

[illegible]

$$\| x_+ - x_* \|_A^2 \le \left(1 - \frac{1}{\kappa_2(A)}\right) \| x_c - x_* \|_A^2.$$

[illegible]

```
[illegible]
while [illegible]
    [illegible]
end
```

[illegible]

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$x \in x_0+S_k$

p • g¶ •U V¶ l® ¶8 ç¶ 2T¶Ž3 œ¶ ¶® È9 ÄV¶ l W ç ¶u-\l >¶ @ • l c¶ ¶ ¶ H ¶ 8 ¶• ÌU¶ J=œU l =Åœ è Å Æ è = œ° 9 œ :§lœ ¶ ₃ 9 Z8œ¶ ®á ®VV ¶®l ¶ = § œ ®ã œ È¶ - l®¶ V l ¶ ¶¶] } c® ' Ü[¶ü ç ¶W 8 ®V¶ çl¶¶® ¶¶ • • ® ₆ ¶ 4¶% W• p ' 6¶2 ¶¶ • ' š l ,[¶]¶ ¶¶g l U á À ¶™ •œ¶¶ U W • ¶ v¶ > yl®v ¶py¶ " p¶U õ œ¶J \ l ¶ W8 " ¶œ§ • \ ¶ ± ¶¶ W l¶• ¶ • ll ¶¶ õ d ¶À ™ • ₆ ¶
r ®®¶ g• ¶ ¶W l8 ¶¶ P ¶• • ¶®=¥®œ vl ¶ ® ¶œ " gWV ¶g U¶ ¶ l ¶ ¶ p ® ¶ p¶¶ Æ ®X • ³¶26¶T® çg¶ : ¶y = g ¶l V¶¶ V g3p₆¶ ¶ • ® ¶ ¶ l ³p3, œ $ \ ¶ • H ê3 œœ° ³ œ 9 $₃ œ‰ ä 2 ¶ Æ g ¶ c ® • \ • ® ¶₃ c ¶¶\ " y g¶¶ y¶₉₆¶g Ä g ^\ ¶ Ä¶ • ù ¶

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[illegible]

[illegible]

[illegible]

[illegible]

$$Q_k^T A Q_k = T_k$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```
k [illegible]
while [illegible]
    [illegible]
    [illegible]
    [illegible]
    [illegible]
    [illegible]
    [illegible]
    [illegible]
end
[illegible]
```

[illegible]

Theorem [illegible] (1.3.14) [illegible]

Proof. [illegible]

[illegible]

[illegible]

[illegible] (11.3.14) [illegible]

Proof. [illegible]

[illegible]

[illegible] (11.3.14) [illegible]

[illegible]

Proof. [illegible]

$$\| x_+ - x_* \|_A \leq \left(1 - \frac{1}{\kappa_2(A)} \right)^{1/2} \| x_k - x_* \|_A,$$

[illegible]

[illegible]

[illegible]

[illegible]

$$L_k = \begin{bmatrix} 1 & \text{[illegible]} & \text{[illegible]} \\ \ell_1 & \text{[illegible]} & \text{[illegible]} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & \ell_{k-1} & 1 \end{bmatrix}, \qquad D_k = \text{[illegible]}$$

[illegible]

[illegible]

end

[illegible]

[illegible]

[illegible]

$$x_k = x_0 + Q_k y_k = x_0 + C_k L_k^T y_k = x_0 + C_k v_k.$$

[illegible]

$$\left[\begin{array}{ccc|c} d_1 & 0 & 0 & 0 \\ d_1\ell_1 & d_2 & 0 & 0 \\ 0 & d_2\ell_2 & d_3 & 0 \\ \hline 0 & 0 & d_3\ell_3 & d_4 \end{array}\right] \left[\begin{array}{c} \nu_1 \\ \nu_2 \\ \nu_3 \\ \hline \nu_4 \end{array}\right] = \left[\begin{array}{c} \beta_0 \\ 0 \\ 0 \\ \hline 0 \end{array}\right].$$

[illegible]

$$v_k = \left[\begin{array}{c} \nu_1 \\ \vdots \\ \nu_{k-1} \\ \hline \nu_k \end{array}\right] = \left[\begin{array}{c} v_{k-1} \\ \hline \nu_k \end{array}\right]$$

[illegible]

$$\nu_k = \begin{cases} \beta_0/d_1 & \text{[illegible]} \\ -d_{k-1}\ell_{k-1}\nu_{k-1}/d_k & \text{if } k > 1 \end{cases}.$$

[illegible]

$$\left[\begin{array}{c|c|c|c} c_1 & c_2 & c_3 & c_4 \end{array}\right] \begin{bmatrix} 1 & \ell_1 & 0 & 0 \\ 0 & 1 & \ell_2 & 0 \\ 0 & 0 & 1 & \ell_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \left[\begin{array}{c|c|c|c} q_1 & q_2 & q_3 & q_4 \end{array}\right].$$

[illegible]

$$C_k = \left[\begin{array}{c|c} C_{k-1} & c_k \end{array}\right]$$

[illegible]

[illegible]

[illegible]

```
k [illegible]
while [illegible]
    [illegible]
    k [illegible] k + 1
    [illegible]
    k [illegible]
        [illegible]
        [illegible]
    else
        [illegible]
        [illegible]
    end
    [illegible]
    [illegible]
    [illegible]
end
[illegible]
```

[illegible]

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f x¶€1¶ º) 7¾k¶]z8¶ ¶ 8 ¶å ¶ ¶ Ü ¶

$$g_k = Ax_k - b = \nabla\phi(x_k)$$

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Proof. .2. ((€8∈P.(.∈ (,)x Äœ) 1' ¶¶8¶) 1 ¶

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$$T_k y_k = (L_k D_k)(L_k^T y_k) = \beta_0 e_1$$

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Proof. αæλ≠f .)x, . f→f s .2

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3(q3C x,)). x∈ 3)x(2≠ λ() .(.)x s .

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¨ ¶) B ¶E ¶¶ å¶ ¶!) 1¶) 2 v€$k¶å$€ ¶Ô tM y € $

[illegible]

```
[illegible]
while [illegible]
    [illegible]
    if [illegible]
        [illegible]
    else
        [illegible]
    end
    [illegible]
    [illegible]
    [illegible]
end
$x_* = x_k$
```

[illegible]

[illegible]

[illegible]

```
[illegible]
while [illegible]
    [illegible]
        [illegible]
            [illegible]
    else
        [illegible]
    end
    [illegible]
    [illegible]
end
```

[illegible]

$$\| x_c - x_* \| = \| A^{-1}(b - Ax_c) \|_2 \le \text{tol} \cdot \| A^{-1} \|_2 \| b \|_2 \le \text{tol} \cdot \kappa_2(A) \| x_* \|.$$

[illegible]

[illegible]

[illegible]

$$Ax = b \equiv \begin{cases} A^T Ax = A^T b, \\ \\ AA^T y = b, \; x = A^T y. \end{cases}$$

[illegible]

[illegible]

$$\phi_{A^T A}(x) = \frac{1}{2} x^T (A^T A) x - x^T (A^T b) = \frac{1}{2} \| Ax - b \|_2^2 - \frac{1}{2} b^T b$$

[illegible]

$$S_k = x_0 + \mathcal{K}(A^T A, A^T r_0, k)$$

[illegible]

[illegible]

$$\phi_{AA^T}(y) = \frac{1}{2} y^T AA^T y - y^T b = \frac{1}{2} \| A^T y - A^{-1} b \|_2^2 - \frac{1}{2} b^T (AA^T)^{-1} b$$

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	$r_c = b - Ax_0,\ z_c = A^T r_c$ [illegible]	[illegible]
[illegible] $\frac{r_c^T r_c}{p_c^T A p_c}$	[illegible] $\frac{[illegible]}{(Ap_c)^T (Ap_c)}$	[illegible] $\frac{r_c^T r_c}{[illegible]}$
[illegible]	[illegible]	[illegible]
$r_+ = r_c - \mu A p_c$	$r_+ = r_c - \mu A p_c,\ z_+ = A^T r_+$	$r_+ = r_c - \mu A p_c$
[illegible] $\frac{r_+^T r_+}{[illegible]}$	[illegible] $\frac{z_+^T z_+}{[illegible]}$	[illegible] $\frac{r_+^T r_+}{r_c^T r_c}$
[illegible]	[illegible]	[illegible]

[illegible] *(CG)* [illegible] *(CGNR)* [illegible] *(CGNE)* [illegible]

[illegible]

[illegible]

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j G ;GG G , š q G G $ ÕU GG G • G G‰G • GG

i / •6B G G CQ 'G hG @G ;*G š > G $ ‰ GG8 $;NG G G G Õ G
;NG$ F ‡ G 4G ⁴4 G o E4 mÉ i Q 6 / $c GB! „ ˜ G), G G G
V • G Z ˜: GEG H Q G

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b 6˜ G @G /GG
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N Qn 7 7 7 ïGRG 7 5 Q G

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> / b / G• GGQ'k * G „ GG@G ;*G š > G $ ŠGG 8 F / G Õ G
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L/iG $ G CíGŠ õ ;,G GG$ 8 s , G G *G 6, ‹G,GG * G G
@G ˜ ¨ üNG 7 k =78 Ⱥ ï C Q G
CQCG U;NG G G Õ;FG‡ $Š G@ G U 2GG ¨WGŠ Þ
T 5 C 7A K!GE G Zc /G. G G o] E ˜8GB G
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**pB Uh $ |G 4 G5 * àQ A G 7 Q G V
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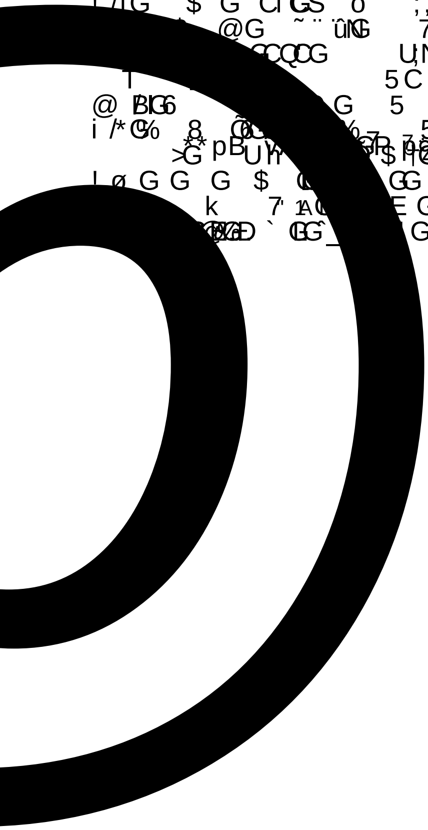

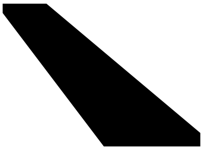

[illegible]

[illegible]

[illegible]

[illegible]

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11.4 Other Krylov Methods

[illegible]

[illegible]

[illegible] LIN_TEMPLATES.

[illegible]

[illegible]

[illegible]

[illegible]

$$H_k = \left[\begin{array}{ccccc} \alpha_1 & \beta_2 & \cdots & \cdots & \text{!œ} \\ \beta_1 & \alpha_2 & \ddots & & \text{!œ} \\ \vdots & \ddots & \ddots & & \vdots \\ \vdots & & & \ddots & \beta_{k-1} \\ \text{!œ}\cdots & \cdots & & \beta_{k-1} & \alpha_k \\ \hline \text{!œ}\cdots & \cdots & & 0 & \beta_k \end{array}\right].$$

$$\| A(x_0 + Q_k y) - b \|_2 = \| Q_{k+1} H_k y - (b - A x_0) \|_2 = \| H_k y - \beta_0 e_1 \|_2$$

$$G_k^T \cdots G_1^T H_k = \left[\begin{array}{c} R_k \\ \hline 0 \end{array}\right], \qquad R_k \in \mathbb{R}^{k \times k},$$

$$p_k \in \mathbb{R}^k,$$

$$T_k y_k = \beta_0 e_1$$

[illegible]

$$H_{k-1}^T y_k = \beta_0 e_1.$$

[illegible]

$$\begin{bmatrix} \alpha_1 & \beta_1 & 0 & 0 & 0 \\ \beta_1 & \alpha_2 & \beta_2 & 0 & 0 \\ 0 & \beta_2 & \alpha_3 & \beta_3 & 0 \\ 0 & 0 & \beta_3 & \alpha_4 & \beta_4 \end{bmatrix} G_1 G_2 G_3 G_4 = \left[\begin{array}{cccc|c} \times & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \\ \times & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \\ \times & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \\ 0 & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \end{array}\right]$$

[illegible]

$$H_{k-1}^T G_1 \cdots G_{k-1} = \left[\, L_{k-1} \mid 0 \,\right]$$

[illegible]

$$y_k = G_1 \cdots G_{k-1} \begin{bmatrix} w_{k-1} \\ 0 \end{bmatrix}.$$

[illegible]

$$AV_k = U_{k+1} \tilde{B}_k$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\| A(x_0 + V_k y) - b \|_2 \;=\; \| U_{k+1}\tilde{B}_k y - (b - Ax_0) \|_2 \;=\; \| \tilde{B}_k y - \beta_0 e_1 \|_2 .$$

[illegible]

[illegible] $$G_k^T \cdots G_1^T(\beta_1 e_1) \;=\; \left[\begin{array}{c} p_k \\ \hline \rho_k \end{array}\right],$$

[illegible]

$$x_k \;=\; x_0 + V_k y_k \;=\; x_0 + W_k p_k$$

[illegible]

[illegible]

[illegible]

$$AQ_k = Q_{k+1}\tilde{H}_k$$

[illegible]

$$\| A(x_0 + Q_k y) - b \|_2 \;=\; \| Q_{k+1}\tilde{H}_k y - (b - Ax_0) \|_2 \;=\; \| \tilde{H}_k y - \beta_0 e_1 \|_2$$

[illegible]

$$G_k^T \cdots G_1^T \tilde{H}_k \;=\; \left[\frac{R_k}{0}\right], \qquad R_k \in \mathbb{R}^{k \times k},$$

[illegible]

$$G_k^T \cdots G_1^T (\beta_0 e_1) \;=\; \left[\frac{p_k}{\rho_k}\right],$$

[illegible]

$$|\rho_k| = \| Ax_k - b \|_2.$$

[illegible]

```
[illegible]
    while [illegible]
        [illegible]
        [illegible]
        [illegible]
            [illegible]
            [illegible]
            [illegible]
        end
        [illegible]
        [illegible]
        [illegible]
    end
    [illegible]
```

[illegible]

[illegible]

[illegible]

[illegible]

$$[illegible]$$

[illegible]

$$[illegible]$$

[illegible]

$$[illegible]$$

$$[illegible]$$

$$[illegible]$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\beta_k q_{k+1} = r_k, \qquad \tau_k \tilde{q}_{k+1} = \tilde{r}_k$$

[illegible]

$$T_k y_k = \tilde{Q}_k^T r_0.$$

[illegible]

$$\tilde{Q}_k^T (b - A x_k) = \tilde{Q}_k^T (b - A(x_0 + Q_k y_k)) = \tilde{Q}_k^T r_0 - T_k y_k = 0.$$

[illegible]

$$x_k = x_0 + Q_k T_k^{-1} \tilde{Q}_k^T r_0 = (Q_k U_k^{-1})(L_k^{-1}(\tilde{Q}_k^T r_0)).$$

[illegible]

[illegible]

[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]
$\mu = \dfrac{\tilde{r}_c^T r_c}{\tilde{p}_c^T A p_c}$	$\mu = \dfrac{\tilde{r}_0^T r_c}{\tilde{r}_0^T A p_c}$	[illegible]
[illegible]	$q_c = u_c - \mu\, A p_c$	[illegible]
[illegible]	[illegible]	[illegible]
$\tilde{r}_+ = \tilde{r}_c - \mu\, A^T \tilde{p}_c$	$r_+ = r_c - \mu\, A(u_c + q_c)$	[illegible]
$\tau = \dfrac{\tilde{r}_+^T r_+}{\tilde{r}_c^T r_c}$	$\tau = \dfrac{\tilde{r}_0^T r_+}{\tilde{r}_0^T r_c}$	[illegible]
[illegible]	[illegible]	[illegible] $(\tilde{r}_0^T r_+)\,\mu$ [illegible]
[illegible]	[illegible]	[illegible]

[illegible]

$$
\begin{aligned}
r_k &= & \qquad p_k &= \\
\tilde{r}_k &= & \qquad \tilde{p}_k &=
\end{aligned}
$$

[illegible]

[illegible]

[illegible]

$$r_k = (1 - \omega_k A) \cdots (1 - \omega_1 A) \psi_k(A) r_0$$

* / *)/ $ ´G@ —v ! j´G GCBGGv 9n$ Y G Gu G € Ä $ 9 GnpG ´ & ¡ j9 $ K 4 G
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[illegible]

11.5 Preconditioning

[illegible]

$$\tilde{A} = M_1^{-1} A M_2^{-1},$$

$$\tilde{A} = C^{-1} A C^{-1}, \qquad \tilde{b} = C^{-1} b,$$

$$\tau = (\tilde{r}_{+}^{T} \tilde{r}_{+}) \,/\, (\tilde{r}_{c}^{T} \tilde{r}_{c}),$$

: v ¶ PP P

! P P m PP# P ³ P

/ &P œ*P! P

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³ 8¶ º ¶) B¶ ¶ ¶¶ $ö. $ 9 $ ¶ x;€¶ - ¶ ¶ 1 ¶À) ¶ ¶ W $% ¶ P …) ³ ¶ ¶ E¶ ¹ ¶ å ¶9 € ' ¶ ¶ ¶ š ³ ¶ 9 ¶ y 9 ¶¶ ¶ ¶ 9 M_+ 9)(¶$

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A œ7 &¶&äœ¶    P: / ¶ &Äœ P
while ő PsQ7 ¶
    A &UõAœ
    if A œ$
        P P
    else
        & PP Ò? P PPÄÓ Æ P
        P f PU&œ è? P
    end
    ! PP , — PI * P P
      PP : P% !„ P
      PP P! P
    : B ¶ P P
end
$x_* = x_k$
```

¨ ¶ ¶ ³ 0 ¶¶ x €8 ¶³ x9€¶5 1) € ¶b 8 š ¶ $ ¶ % ¶ ¶ + $ Æ$ PUb +¶ 1 8 ¶ ¶ „8¶ ¹ å E¶ ¶ 8 ³ ¶¶ ¶¶ x € ¶) 9 ¶Ñ E¶ ¶ ¶8 hf ¶ 8 ³ ¶ 9 ¶) ³ x €n¶ ¶¶v ¶ ¶¶ ¶) ž g ÿ) ¶ ¶ ¶ «8 ¶¶ ¶ ¶ g ³ ¶¶) ³ ¶) ¶ ’ ¶

$$r_j^T M^{-1} r_i = 0, \qquad p_j^T (C^{-1} A C^{-1}) p_i = 0,$$: ($

+ ¶ $¶ T $. $($ r ¶ 8 ¶) 9 ¶) ¶ v ¶ ¶ ¶) € ˜ ªÈ¶ ¶ , ³ ;¨ ¶ B ¶ ¶¶ ¶ ¶ ¶ ³ ¶¶ Ñ ¶ ó¶P PP ôÿP˜ 9P ¶E] ¶ " ($ ¶ ¶ 8 Ñ¶E ³ž+¶ 8 ¶ ³ ù ¶

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        0¶„ P DP≥PP
        f „ PP ≠P µ̆i " 'DP
    end
    • P ≠P[P[6 Ó á P Š PP
    P ¶v [™( ôvP 6PP ¶W ÆE ǒ9 A¶ œ) 1v [ A [ P o9P¨¶S P
end
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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$A^{-1} = (I - G)^{-1} M_1^{-1} = \left(\sum_{k=0}^{\infty} G^k \right) M_1^{-1}.$$

[illegible]

$$M^{-1} = \left(\sum_{k=0}^{m} G^k \right) M_1^{-1}.$$

[illegible]

$$z = \left(I + G + G^2 + \cdots G^m \right) M_1^{-1} r$$

[illegible]

```
[illegible]
[illegible]
    [illegible]
    [illegible]
end
[illegible]
```

[illegible]

$$z_+ = G z_c + d = G \left(I + G + \cdots + G^{k-1} \right) d + d = \left(I + G + \cdots G^k \right) d.$$

[illegible]

$$M^{-1} = p(M_1^{-1} A) M_1^{-1}.$$

[illegible]

[illegible]

[illegible]

$$x_k = x_{k-2} + \omega_k(\gamma_{k-1} z_{k-1} + x_{k-1} - x_{k-2}) \qquad [illegible]$$

[illegible]

$$x_k = M^{-1}(b - A x_{k-1}) + x_{k-1},$$

[illegible]

```
[illegible]
while [illegible]
    [illegible]
    : $M$ [illegible]
    × [illegible]
    if [illegible]
        W1 = 1
    else
        [illegible] $\left( [illegible] \frac{z_{k-1}^T M z_{k-1}}{z_{k-2}^T M z_{k-2}} \frac{1}{\omega_{k-1}} \right)^{-1}$
    end
    [illegible]
    [illegible]
end
[illegible]
```

[illegible]

[illegible]

[illegible]

$$(i,j) \in P \Rightarrow h_{ij} = 0.$$

[illegible]

$$\begin{bmatrix} \alpha & v^T \\ v & B \end{bmatrix} = \begin{bmatrix} \sqrt{\alpha} & 0 \\ w & I_{n-1} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & A_1 \end{bmatrix} \begin{bmatrix} \sqrt{\alpha} & w^T \\ 0 & I_{n-1} \end{bmatrix}$$

[illegible]

[illegible]

```
[illegible]
    [illegible]
        [illegible]
    else
        [illegible]
        [illegible]
        [illegible]
        $H = \begin{bmatrix} \sqrt{\alpha} & 0 \\ w & H_1 \end{bmatrix}$
    e) d
```

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\tilde{b}_{ij} = b_{ij} - \frac{\tilde{v}_i\tilde{v}_j}{\alpha}.$$

$$\frac{1}{\sqrt{\alpha}}$$

$$0 < x^T A x \;=\; 1 - \frac{v^T B^{-1} v}{\alpha}.$$

Since $B^{-1} \geq 0$ and $v \leq 0$, we have $\tilde{v}^T B^{-1} \tilde{v} \leq v^T B^{-1} v$ and so

$$\gamma \; \equiv \; 1 - \frac{\tilde{v}^T B^{-1} \tilde{v}}{\alpha} \geq 1 - \frac{v^T B^{-1} v}{\alpha} > 0.$$

[illegible]

$$\tilde{B}^{-1} \; = \; \left(B - \frac{\tilde{v}\tilde{v}^T}{\alpha} \right)^{-1} \; = \; B^{-1} + \frac{1}{\gamma} B^{-1} \frac{\tilde{v}\tilde{v}^T}{\alpha} B^{-1}$$

[illegible]

[illegible]

$$[A_1]_{ij} \; = \; \begin{cases} 0 & \text{if } |b_{ij} - w_i w_j| < \tau \sqrt{b_{ii} b_{jj}} \;, \\ b_{ij} - w_i w_j & \text{if } |b_{ij} - w_i w_j| \geq \tau \sqrt{b_{ii} b_{jj}} \;. \end{cases}$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] $A = (A_{ij})$ [illegible] H_{ij} [illegible]

$$A = \begin{bmatrix} A_1 & E_1^T & 0 \\ E_1 & A_2 & E_2^T \\ 0 & E_2 & A_3 \end{bmatrix} = \begin{bmatrix} G_1 & 0 & 0 \\ F_1 & G_2 & 0 \\ 0 & F_2 & G_3 \end{bmatrix} \begin{bmatrix} G_1^T & \text{[illegible]} & \\ 0 & \text{[illegible]} & \\ 0 & \text{[illegible]} & \end{bmatrix}$$

[illegible]

```
[illegible] Ai
for k = [illegible]
    [illegible]
    [illegible]
```

[illegible]

```
[illegible] Ai
for k = [illegible]
    [illegible]
    [illegible] $E_k \Lambda_k E_k^T$
```

[illegible]

$$\begin{bmatrix} \tilde{G}_1 & 0 & 0 \\ \tilde{F}_1 & \tilde{G}_2 & 0 \\ 0 & \tilde{F}_2 & \tilde{G}_3 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \end{bmatrix},$$

[illegible]

[illegible]

[illegible]

$$K = \begin{bmatrix} A & B_1^T \\ B_2 & -C \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} f \\ g \end{bmatrix},$$

[illegible]

[illegible] $S = -B_2^T A^{-1} B_1.$

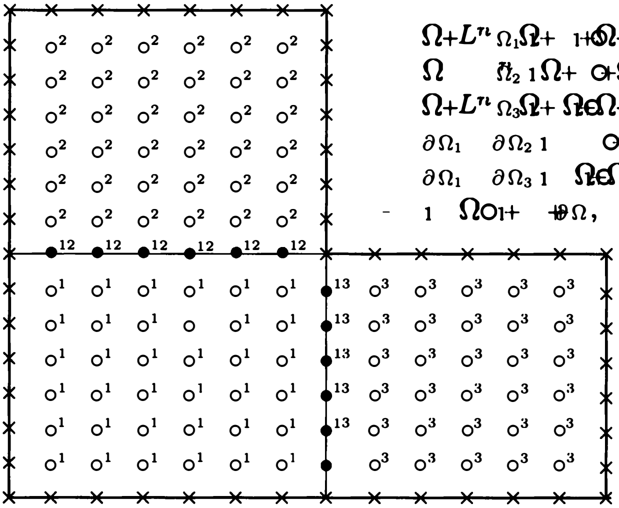
[illegible]

[illegible]

$$\begin{bmatrix} A & B \\ -B^T & C \end{bmatrix} = \begin{bmatrix} H_1 & 0 \\ 0 & C \end{bmatrix} + \begin{bmatrix} H_2 & B \\ -B^T & 0 \end{bmatrix} \equiv K_1 + K_2$$

[illegible]

[illegible]



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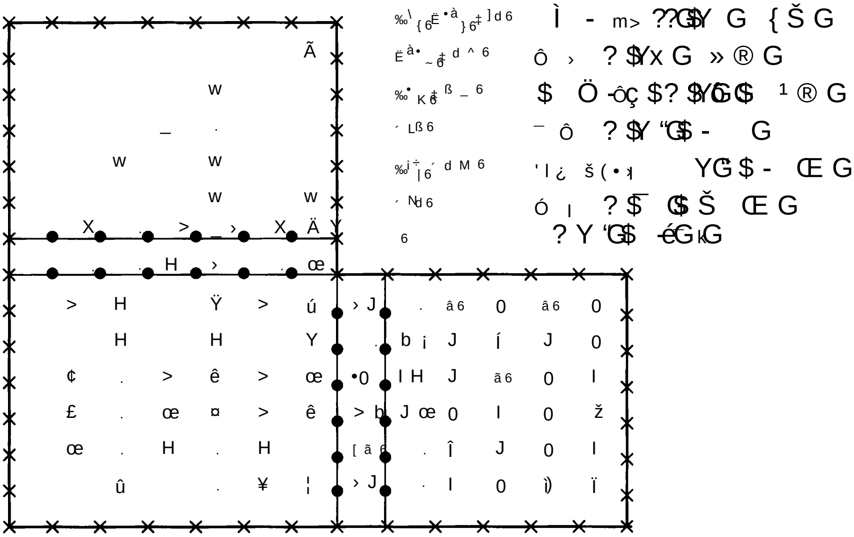
$$\tilde{S}_4 \approx S_4, \qquad \tilde{S}_5 \approx S_5,$$



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$$\begin{bmatrix} A_1 & B_1 & C_1 \\ F_1 & Q_4 & 0 \\ G_1 & 0 & Q_5 \end{bmatrix} \begin{bmatrix} u_{\circ 1} \\ u_{\bullet 12} \\ u_{\bullet 13} \end{bmatrix} = \begin{bmatrix} f_{\circ 1} \\ f_{\bullet 12} \\ f_{\bullet 13} \end{bmatrix} - \begin{bmatrix} 0 \\ Du_{\bullet 21} \\ Eu_{\bullet 31} \end{bmatrix}.$$

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9 ¶ ¶

$$\begin{bmatrix} A_2 & B_2 \\ F_2 & \tilde{Q}_4 \end{bmatrix} \begin{bmatrix} u_{\circ 2} \\ u_{\bullet 21} \end{bmatrix} = \begin{bmatrix} f_{\circ 2} \\ f_{\bullet 21} \end{bmatrix} - \begin{bmatrix} 0 \\ Hu_{\bullet 12} \end{bmatrix},$$

9 ¶ v ¶ 9 ¶ k 9 y8@¶‰ ¶ , ¶ ù ¶ ,)Ê ¶ = 9 8 9 ¶ ¶
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$$\begin{bmatrix} A_3 & C_2 \\ G_2 & \tilde{Q}_5 \end{bmatrix} \begin{bmatrix} u_{\circ^3} \\ u_{\bullet^{31}} \end{bmatrix} = \begin{bmatrix} f_{\circ^3} \\ f_{\bullet^{31}} \end{bmatrix} - \begin{bmatrix} 0 \\ K u_{\bullet^{13}} \end{bmatrix}.$$

[illegible]

$$u^{(\mathrm{new})} = u^{(\mathrm{old})} + M^{-1}(f - Au^{(\mathrm{old})}),$$

[illegible]

$$\begin{bmatrix} A_{11} & A_{12} & 0 \end{bmatrix} \begin{bmatrix} u_1 \end{bmatrix} \quad \begin{bmatrix} f_1 \end{bmatrix}$$

$$\begin{matrix} u_1^{(k)} \\ u_2^{(k)} \\ u_3^{(k)} \end{matrix}$$

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11.6 The Multigrid Framework

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$$\frac{d^2u(x)}{dx^2}$$ [illegible]

[illegible]

[illegible]

$$(Q^h)^T A^h Q^h = \Lambda^h = \mathrm{diag}(\lambda^h),$$

[illegible]

$$\frac{4}{h^2}\cdot\sin^2\left(\frac{j\pi}{2(n+1)}\right),$$

[illegible]

$$\theta_j = \frac{j\pi}{n+1}.$$

[illegible]

$\mathbf{R}^n$.

[illegible]

$$G^{h,\omega} = I_n - \frac{\omega h^2}{2} A^h.$$

[illegible]

$$(Q^h)^T G^{h,\omega} Q^h = \operatorname{diag}(\tau^{h,\omega}),$$

[illegible]

$$\mu(\omega) = \max\{\, |\tau^{h,\omega}_{m+1}|, \ldots, |\tau^{h,\omega}_n| \,\}.$$

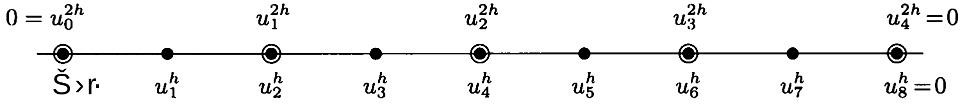
[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


$0 = u_0^{2h}$
u_1^{2h}
u_2^{2h}
u_3^{2h}
$u_4^{2h} = 0$
u_1^h
u_2^h
u_3^h
u_4^h
u_5^h
u_6^h
u_7^h
$u_8^h = 0$

[illegible]

R_h^{2h} [illegible] $P_{2h}^h = -$ [illegible]

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[illegible]

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[illegible]

[illegible]

[illegible]

$$\left(u_+^h - u^h\right) \;=\; E_h(u_{p_1}^h - u^h)$$

[illegible]

$$E^h \;=\; I_n \;-\; P_{2h}^h(A^{2h})^{-1}R_h^{2h}A^h$$

[illegible]

$$(\,u_p^h \;-\; u^h\,) \;=\; (G^h)^p\,(u_c^h - u^h), \qquad p \in \{p_1,\, p_2\},$$

[illegible]

$$(u_{++}^h \;-\; u^h) \;=\; (G^h)^{p_2}\,E^h\,(G^h)^{p_1}\,(u_c^h \;-\; u^h).$$

[illegible]

$$(Q^h)^T P_{2h}^h Q^{2h} \;=\; \sqrt{2}\begin{bmatrix} C^h \\ 0 \\ -\mathcal{E}_m S^h \end{bmatrix}, \quad (Q^{2h})^T R_h^{2h} Q^h \;=\; \sqrt{\frac{1}{2}}\begin{bmatrix} C^h \\ 0 \\ -\mathcal{E}_m S^h \end{bmatrix}^T \qquad (11.6.20)$$

[illegible] *(11.6.7))). (11.6.8).*

Proof. *).* [illegible]

$$(Q^h)^T B^h Q^h \;=\; 4I_n - h^2\Lambda^h \;=\; 4\begin{bmatrix} C^h & [illegible] & 0 \\ 0 & [illegible] & 0 \\ 0 & [illegible]\,\mathcal{E}_m S^h \mathcal{E}_m \end{bmatrix} \;\equiv\; D^h.$$

[illegible]

[illegible]

$$(Q^h)^T B^h(:, idx) Q^{2h} = \frac{4}{\sqrt{2}} \begin{bmatrix} C^h & \bullet & 0 \\ 0 & 1 & 0 \\ 0 & \bullet & \mathcal{E}_m S^h \mathcal{E}_m \end{bmatrix} \begin{bmatrix} I_m \\ 0 \\ -\mathcal{E}_m \end{bmatrix} = \frac{4}{\sqrt{2}} \begin{bmatrix} C^h \\ 0 \\ -\mathcal{E}_m S^h \end{bmatrix}.$$

[illegible] D

[illegible]

Theorem 11 [illegible]

Proof. [illegible]

$$\frac{1}{2h^2}.$$

[illegible] D

[illegible]

$$\begin{bmatrix} S^h & 0 & C^h\mathcal{E}_m \\ 0 & 1 & 0 \\ \mathcal{E}_m S^h & 0 & \mathcal{E}_m C^h \mathcal{E}_m \end{bmatrix} = [illegible]$$

[illegible]

$$u_c^h - u^h = \underbrace{\sum_{j=1}^{m} \alpha_j q_j}_{[illegible]} \; [illegible]$$

[illegible]

$$u_{++}^h - u^h = \sum_{j=1}^{m} \tilde{\alpha}_j q_j \; [illegible]$$

[illegible]

$$\tilde{\alpha}_j \; [illegible] \; \alpha_j \; [illegible] \; \alpha_{n-j+1} \; [illegible]$$

$$\tilde{\alpha}_{m+1} = \alpha_{m+1} \tau_{m+1}^{p_1+p_2},$$

$$\tilde{\alpha}_{n-j+1} = \left(\alpha_j \tau_j^{p_1} s_j^2 + \alpha_{n-j+1} \tau_{n-j+1}^{p_1} c_j^2 \right) \tau_{n-j+1}^{p_2}, \; [illegible]$$

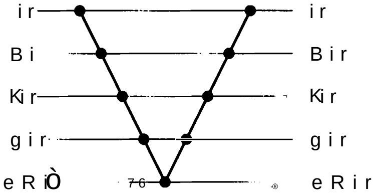
[illegible]

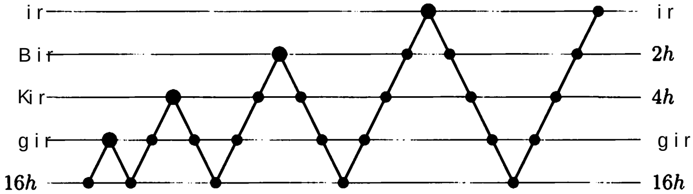
[illegible]

[illegible]

[illegible] $R_h^{2h} r^h$

[illegible]

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JÅ • G

· — 6 Q GG •Gš G G ÈGũ'G LJ ¸G J2 4 GG 5ï) 5]ï ' • G G J J **G**
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Chapter 12

Special Topics

[illegible]

[illegible]

[illegible]

[illegible]

12.1 Linear Systems with Displacement Structure

[illegible]

[illegible]

[illegible]

$$[illegible]$$

[illegible]

$$[illegible]$$

[illegible]

$$[illegible]\ R, S \in \mathbb{R}^{n\times r}.$$

[illegible]

[illegible]

[illegible]

$$a_{kj} = \frac{1}{\omega_k - \lambda_j}$$

[illegible]

$$\Omega = \operatorname{diag}(\omega_1, \ldots, \omega_n), \qquad [illegible]\operatorname{diag}(\lambda_1, \ldots, \lambda_n),$$

[illegible]

$$[illegible]\ \frac{\omega_k}{\omega_k - \lambda_j} - \frac{\lambda_j}{\omega_k - \lambda_j} = [illegible]$$

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[illegible]

Theorem [illegible]

[illegible]

$$\Omega = \operatorname{diag}(\omega_1, \ldots, \omega_n), \qquad \Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$$

[illegible]

$$R = \begin{bmatrix} r_1^T \\ R_1 \end{bmatrix}, \qquad S = \begin{bmatrix} s_1^T \\ S_1 \end{bmatrix}$$

[illegible]

$$\Omega_1 = \operatorname{diag}(\omega_2, \ldots, \omega_n), \qquad \Lambda_1 = \operatorname{diag}(\lambda_2, \ldots, \lambda_n),$$

[illegible]

$$\Omega_1 A_1 - A_1 \Lambda_1 = \tilde{R}_1 \tilde{S}_1^T$$

[illegible]

$$A_1 = \text{[illegible]} \quad \tilde{R}_1 = R_1 - \frac{1}{\alpha} f r_1^T, \qquad \tilde{S}_1 = S_1 - \frac{1}{\alpha} g s_1^T.$$

Proof. [illegible]

$$\Omega_1 A_1 - A_1 \Lambda_1 = \Omega_1 \left(B - \frac{1}{\alpha} f g^T \right) - \left(B - \frac{1}{\alpha} f g^T \right) \Lambda_1$$

[illegible]

$$U = \begin{bmatrix} \alpha & g^T \\ 0 & U_1 \end{bmatrix}$$

[illegible]

$$(P\Omega P^T(PAQ^T) - (PAQ^T)(Q\Lambda Q^T) = (PR)(QS)^T.$$

[illegible]

$$\tilde{\Lambda} = Q\Lambda Q^T.$$

[illegible]

$$A \to PAQ^T, \quad \equiv$$

[illegible]

[illegible]

[illegible]

$$T = \left[\begin{array}{ll} \tau & \text{[illegible]} \\ c_1 & \text{[illegible]} \\ c_2 & \text{[illegible]} \\ c_3 & \text{[illegible]} \\ c_4 & \text{[illegible]} \end{array}\right.$$

[illegible]

$$Z_\phi = \begin{bmatrix} 0 & 0 & 0 & 0 & \phi \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \qquad Y_{\gamma,\delta} = \begin{bmatrix} \gamma & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & \delta \end{bmatrix}.$$

[illegible]

$$\text{[illegible]}, \qquad \text{rank}_{\{Z_1, Z_{-1}\}}(T) \le 2,$$

$$\text{[illegible]}, \qquad \text{rank}_{\{Y_{00}, Y_{11}\}}(T) \le 4.$$

[illegible]

[illegible]

[illegible]

$$[illegible], \quad \mathrm{rank}_{\{Z_1^T, Z_{-1}\}}(H) \le 2,$$

$$Y [illegible], \quad \mathrm{rank}_{\{Y_{00}, Y_{11}\}}(H) \le 4.$$

[illegible] $S \in F$ [illegible] Z [illegible]

[illegible] X_G [illegible] X_G

[illegible] $GXZ \in Z$ [illegible]

$Y_{00}A - AY_{11}$ [illegible] X [illegible] G_G^{-T} [illegible] Z

[illegible]

[illegible] R, S^T [illegible] F [illegible]

[illegible] Z_1 ZS Z_G Z_G

X_F^{-1} X [illegible]

$X_G^{-1}GX_G$ [illegible]

[illegible] $ZGXXZ_G SX$ $GZ_{GG}^{-}X^T$ T G

$$(X_F^{-1}FX_F)(X_F^{-1}AX_G) - (X_F^{-1}AX_G)(X_G^{-1}GX_F) = (X_F^{-1}R)(X_G^T S)^T$$

$\in Z$

nA - AA = f sT

[illegible] $X_F^{-1}AX_G$, R = $X_F^{-1}R$, Z [illegible] $X_G^T S$ $\in_G^{-T}$ [illegible]

[illegible]

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[illegible]

12.2 Structured-Rank Problems

[illegible]

[illegible]

$$j_2 \le i_1 \ \text{ or } \ i_2 \le j_1 \ \Rightarrow \ \mathsf{rank}(A(i_1{:}i_2, j_1{:}j_2)) \ \le \ 1.$$

[illegible]

[illegible]

[illegible]

[illegible]

$$B(r)^T x \ = \ x_1 e_1.$$

[illegible]

[illegible]

$$M_k \;=\; \mathrm{diag}(I_{k-1}, \tilde{M}_k, I_{n-k-1}), \qquad \tilde{M}_k \;=\; \begin{bmatrix} \alpha_k & \beta_k \\ \gamma_k & \delta_k \end{bmatrix},$$

[illegible]

$$[illegible]$$

[illegible]

$$j_2 \;<\; i_1 \;\text{ or }\; i_2 \;<\; j_1 \;\;\Rightarrow\;\; \mathrm{rank}(A(i_1{:}i_2, j_1{:}j_2)) \;\le\; 1.$$

]

[illegible]

[illegible]

[illegible]

[illegible] uvr, [illegible]

[illegible]

$$[illegible]$$

[illegible] $u \in \mathbf{R}^n \rightarrow \mathbf{R}^n$ [illegible]

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[illegible]

$$y = \left(\mathrm{triu}(pq^T) \,.\!*\, B(r)^{-1}\right) x$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$M_k = \mathrm{diag}(I_{k-1}, \tilde{M}_k, I_{n-k-1}),$$

[illegible]

n ¶ fM¶¶M_1, M_{n-1}, $M A =$ $M = B(\tau)$ $\tau = [\tau_1, \ldots, \tau_{n-1}]^T$. $= M^{-1}$, Å/9 A¶e $M_k A$ Q œ ¶ 8¶ ¶ ¶V ² ‹ ³¶¶9 M_4¶

$$M_4 M_5 A = \mathbf{S}(u, v, t, u \quad v, p, q, r)$$

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$$= \begin{bmatrix} u_3 t_2 t_1 v_1 & u_3 t_2 v_2 & u_3 v_3 \\ u_4 t_3 t_2 t_1 v_1 & u_4 t_3 t_2 v_2 & u_4 t_3 v_3 \end{bmatrix},$$

$$= \begin{bmatrix} p_3 r_3 q_4 & p_3 r_3 r_4 q_5 & p_3 r_3 r_4 r_5 q_6 \\ p_4 q_4 & p_4 r_4 q_5 & p_4 r_4 r_5 q_6 \end{bmatrix},$$

$$= \begin{bmatrix} 1 & œ \\ -\tau_3 & Œ \end{bmatrix}$$

$$M_3 \begin{bmatrix} \cdot \end{bmatrix} = \begin{bmatrix} u_3 t_2 t_1 v_1 & u_3 t_2 v_2 & u_3 v_3 \\ (u_4 t_3 - \tau_3 u_3) t_2 t_1 v_1 & (u_4 t_3 - \tau_3 u_3) t_2 v_2 & (u_4 t_3 - \tau_3 u_3) v_3 \end{bmatrix},$$

$$\tilde{M}_3 \begin{bmatrix} \mu & \mu & \mu \\ \mu & \mu & \mu \end{bmatrix} = \begin{bmatrix} p_3 r_3 q_4 & p_3 r_3 r_4 q_5 & p_3 r_3 r_4 r_5 q_6 \\ (p_4 - \tau_3 p_3 r_3) q_4 & (p_4 - \tau_3 p_3 r_3) r_4 q_5 & (p_4 - \tau_3 p_3 r_3) r_4 r_5 q_6 \end{bmatrix}.$$

u_3 Zτ_3 œ $u_4 t_3 / u_3$,

$$u_4 = 0, \qquad p_4 = p_4 - \tau_3 p_3 r_3,$$

$$M_3 M_4 M_5 A = \mathbf{S}(u, v, t, u \quad v, p, q, r)$$

[illegible]

$$\tilde{P}_k = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

[illegible]

[illegible]

$$A_L = \begin{bmatrix} c_1 v_1 & 0 & 0 & 0 \\ c_2 s_1 v_1 & c_2 v_2 & 0 & 0 \\ c_3 s_2 s_1 v_1 & c_3 s_2 v_2 & c_3 v_3 & 0 \\ s_3 s_2 s_1 v_1 & s_3 s_2 v_2 & s_3 v_3 & v_4 \end{bmatrix} = B(s)^{-T} .* \mathsf{tril}(cv^T)$$

[illegible]

$$v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}, \qquad c = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ 1 \end{bmatrix}, \qquad s = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}.$$

[illegible]

$$A = A_L + A_U,$$

[illegible]

$$[illegible] = B(s_L)^{-T} .* \mathsf{tril}(c_L v_L^T),$$

[illegible]

[illegible] s_U [illegible]

[illegible]

[illegible] $\mathbb{R}^n$ [illegible] v_4 [illegible] G_1 [illegible] G_{n-1} [illegible]

[illegible]

[illegible]

$$\begin{bmatrix} 1 & & & \\ 0 & & & \\ & & & \\ & & & \end{bmatrix} \begin{bmatrix} c_1v_1 & & & \\ c_2s_1v_1 & & & \\ c_3s_2s_1v_1 & & c_3v_3 & \\ s_3s_2s_1v_1 & & s_3v_3 & \end{bmatrix} = \begin{bmatrix} & & & 0 \\ & c_2v & & 0 \\ & s_2v & & s_3v_4 \\ & & & c_3v_4 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_2 & s_2 & 0 \\ 0 & -s_2 & c_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} & & & \\ & c_2v_2 & & \\ & s_2v_2 & & \\ & & & \end{bmatrix} = \begin{bmatrix} & v_2 & s_2v_3 & \\ & & c_2v_3 & \\ & & & \\ & & & \end{bmatrix}$$

$$\begin{bmatrix} c_1 & s_1 & 0 & 0 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c_1v_1 & & & 0 \\ s_1v_1 & v_2 & s_2v_3 & s_2s_3v_4 \\ 0 & & c_2v_3 & c_2s_3v_4 \\ 0 & & & c_3v_4 \end{bmatrix} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix}.$$

[illegible] n, $\mathbf{R}^n$ [illegible]

$$Q^T = G_1 \cdots G_{n-1}$$

[illegible]

$$G_k = \operatorname{diag}(I_{k-1}, \tilde{G}_k, I_{n-k-1}), \qquad \tilde{G}_k = \begin{bmatrix} c_k & s_k \\ -s_k & c_k \end{bmatrix},$$

[illegible]

$$Q^T \mathsf{tril}(A) = R_L = \mathsf{triu}((\mathcal{D}_n c)v^T) \;.\!*\; B(s)^{-1}.$$

[illegible]

$$Q^T \mathsf{triu}(\mathsf{A}, 1) = R_U$$

[illegible]

$$Q^T A = Q^T(A_L + A_U) = R_L + R_U = R$$

[illegible]

$$A = \mathsf{tril}(uv^T) + \mathsf{triu}(pq^T),$$

[illegible]

$$R \;=\; \mathrm{triu}(fg^T + hq^T)\,, \qquad f, g, h \in \mathbb{R}^n.$$

[illegible]

[illegible]

$$=\qquad\qquad =$$

$$\begin{bmatrix} c_2 & s_2 \\ -s_2 & c_2 \end{bmatrix} \begin{bmatrix} p_2 q_j \\ \tilde{f}_3 g_j + \tilde{h}_3 q_j \end{bmatrix} = \begin{bmatrix} c_2 p_2 + s_2 \tilde{h}_3 \\ -s_2 p_2 + c_2 \tilde{h}_3 \end{bmatrix} q_j \;+\; \begin{bmatrix} s_2 \tilde{f}_3 \\ c_2 \tilde{f}_3 \end{bmatrix} g_j\,,$$

[illegible]

$$\tilde{f}_2 = s_2\tilde{f}_3, \quad f_3 = c_2\tilde{f}_3, \quad \tilde{h}_2 = c_2p_2 + s_2\tilde{h}_3, \quad h_3 = -s_2p_2 + c_2\tilde{h}_3.$$

[illegible]

$$g_2 = \frac{\tilde{u}_2v_2 - \tilde{h}_2q_2}{\tilde{f}_2}.$$

[illegible]

[illegible]

[illegible]

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$$H = G_1 \cdots G_n,$$

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[illegible]

$$A = \begin{bmatrix} A_1 & B_1^T & 0 & 0 \\ B_1 & A_2 & B_2^T & 0 \\ 0 & B_2 & A_3 & B_3^T \\ 0 & 0 & B_3 & A_4 \end{bmatrix}$$

[illegible]

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12.3 Kronecker Product Computations

[illegible]

[illegible]

$$B \otimes C = G_B G_B^T \otimes G_C G_C^T = (G_B \otimes G_C)(G_B \otimes G_C)^T$$

[illegible]

$$\left.\begin{aligned} P_B B &= L_B U_B \\ P_C C &= L_C U_C \end{aligned}\right\} \Rightarrow (P_B \otimes P_C)(B \otimes C) = (L_B \otimes L_C)(U_B \otimes U_C),$$

[illegible]

[illegible]

[illegible]

[illegible]

$$\left.\begin{aligned} Q_B^H B Q_B &= T_B \\ Q_C^H C Q_C &= T_C \end{aligned}\right\} \Rightarrow (Q_B \otimes Q_C)^H (B \otimes C)(Q_B \otimes Q_C) = T_B \otimes T_C,$$

Σ_C

$$B \in \mathbb{R}^{m\times m},\ C \in \mathbb{R}^{n\times n},$$

$$\begin{bmatrix} b_{11} & b_{12} \\ \text{[illegible]} & \text{[illegible]} \end{bmatrix} \otimes \begin{bmatrix} c_{11} & c_{12} \\ \text{[illegible]} & \text{[illegible]} \end{bmatrix} = \left[\begin{array}{cc|cc} b_{11}c_{11} & b_{11}c_{12} & b_{12}c_{11} & b_{12}c_{12} \\ b_{11}c_{21} & b_{11}c_{22} & b_{12}c_{21} & b_{12}c_{22} \\ \hline \text{[illegible]} & \text{[illegible]} & \text{[illegible]} & \text{[illegible]} \end{array}\right].$$

[illegible]

[illegible]

[illegible]

$$B \underset{\text{[illegible]}}{\otimes} C = B .\!* C.$$

[illegible]

$$\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} \underset{\text{HAD}}{\otimes} \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{bmatrix} = \begin{bmatrix} b_{11}c_{11} & \text{[illegible]} \\ b_{21}c_{21} & \text{[illegible]} \\ b_{31}c_{31} & \text{[illegible]} \end{bmatrix}$$

[illegible]

$$\text{[illegible]} \quad A_{ij} = B_{ij} \otimes C_{ij},$$

[illegible]

$$\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \\ B_{31} & B_{32} \end{bmatrix} \underset{\text{KR}}{\otimes} \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \\ C_{31} & C_{32} \end{bmatrix} = \begin{bmatrix} B_{11} \otimes C_{11} & B_{12} \otimes C_{12} \\ B_{21} \otimes C_{21} & B_{22} \otimes C_{22} \\ B_{31} \otimes C_{31} & B_{32} \otimes C_{32} \end{bmatrix}.$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

If $B \in \mathbb{R}^{m_1 \times n_1}$, $C \in \mathbb{R}^{m_2 \times n_2}$, and $X \in \mathbb{R}^{n_1 \times m_2}$, then

[illegible]

$$\text{vec}(A^T) \;=\; \mathcal{P}_{r,q}\text{vec}(A).$$

[illegible]

$$\mathcal{P}_{r,q} = \Gamma_t \cdots \Gamma_1$$

[illegible]

$$A \;=\; \begin{bmatrix} A_1 \\ \vdots \\ A_r \end{bmatrix}, \qquad A_k \in \mathbb{R}^{q \times q}.$$

[illegible]

$$\mathcal{P} = \Gamma_2\Gamma_1 = (I \otimes \mathcal{P})(\mathcal{P} \otimes I)$$

[illegible]

[illegible]

[illegible]

$$\left[\begin{array}{cc} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{array}\right] \otimes \left[\begin{array}{cc} c_{11} & c_{12} \\ c_{21} & c_{22} \end{array}\right]$$

$$\left[\begin{array}{c} b_{11} \\ b_{21} \\ b_{31} \\ b_{12} \\ b_{22} \\ b_{32} \end{array}\right] \left[\begin{array}{cccc} c_{11} & c_{21} & c_{12} & c_{22} \end{array}\right].$$

[illegible]

[illegible]

[illegible]

[illegible]

$$U^T\mathcal{R}(A)V = \Sigma$$

[illegible]

[illegible]

[illegible]

$$\begin{bmatrix} A_{11} & \cdots & A_{1,n_1} \\ \vdots & \ddots & \vdots \\ A_{m_1,1} & \cdots & A_{m_1,n_1} \end{bmatrix}$$

where each $A_{ij} \in \mathbb{R}^{m_2\times n_2}$, then $\mathcal{R}(A) \in \mathbb{R}^{m_1 n_1 \times m_2 n_2}$ is defined by

$$\tilde{A}_j = \begin{bmatrix} \mathrm{vec}(A_{1j})^T \\ \vdots \\ \mathrm{vec}(A_{m_1,j})^T \end{bmatrix}.$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

where $U_k = \mathsf{reshape}(u_k, m_1, n_1)$ *and* $V_k = \mathsf{reshape}(v_k, m_2, n_2)$.

[illegible]

[illegible]

$$A_{\tilde{r}} = \sum_{k=1}^{\tilde{r}} \sigma_k U_k \otimes V_k$$

[illegible]

[illegible]

[illegible]

$$\| A - B \otimes C \|_F, \qquad B \in \mathbb{R}^{m\times m}, C \in \mathbb{R}^{n\times n},$$

[illegible]

[illegible]

[illegible]

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Lemma Dr...r. 1kj jn=. ő $\mathbb{R}^{n\times n}$ 1 $1,1$ XN Åœ$1 ì ' ¶ ; d_n) + 1 1 —,,
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Proof. $\alpha \neq \neq$ d = d S

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α...) $\neq$ œ $\mathbb{R}^{n\times n}$, T* | + 5 ¶f 8 ¶ 8 f ¶·¶ ¶ e ð Y $\mathbb{R}^{m\times m}$ — $M v M$

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Lemma Dr...... 1kj jn=. ő $\mathbb{R}^{n\times n}$ 3 1, + M^3 +₂„ $M 1,1$ j ¢ $ 5 D X — 9Ki

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5 P 5 (5 0 % $ $ ½ $ º 1 • + % $ 5 $\mathbb{R}^{n\times n}$.

Proof. $\alpha \neq \neq$ d S d = d

[illegible]

[illegible]

$$[illegible] = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \otimes \begin{bmatrix} c_{11} & c_{12} & c_{13} & c_{14} \\ c_{21} & c_{22} & c_{23} & c_{24} \\ c_{31} & c_{32} & c_{33} & c_{34} \\ c_{41} & c_{42} & c_{43} & c_{44} \end{bmatrix} \otimes \begin{bmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \\ d_{31} & d_{32} & d_{33} \end{bmatrix}$$

[illegible]

[illegible]

$$y = (B_1 \otimes \cdots B_p)x \qquad x \in \mathbb{R}^{N_p}$$

[illegible]

$$\text{reshape}(y, M_p/M_i, M_i) \;=\; (B_{i+1} \otimes \cdots \otimes B_p) \cdot \text{reshape}(x, N_p/N_i, N_i) \cdot (B_1 \otimes \cdots B_i)^T.$$

[illegible]

[illegible] $A_N \in \mathbb{R}^{m \times n}$. [illegible]

$$f(x,y) = [illegible] \, (A_k x - b_k)^2$$

[illegible] $x \in \mathbb{R}^m$, [illegible] $b \in \mathbb{R}^N$.

[illegible] $(B \otimes C)x$ [illegible] b [illegible]

[illegible] $(B_1 \otimes B_2)x$ [illegible]

[illegible] $\mathbb{R}^{n \times n}$ [illegible] $C \in \mathbb{R}^{m \times m}$ [illegible]

[illegible] $B \in \mathbb{R}^{m_B \times n_B}$ [illegible] $C \in \mathbb{R}^{m_C \times n_C}$ [illegible] $m_B \ge n_B$ [illegible] $m_C \ge n_C$.

[illegible] $(B \otimes C)x$ [illegible] $B, C \in \mathbb{R}^{n \times n}$ [illegible]

[illegible]

$$P\left(f \otimes \begin{bmatrix} g_1 \\ \vdots \\ g_m \end{bmatrix}\right) = \begin{bmatrix} f \otimes g_1 \\ \vdots \\ f \otimes g_m \end{bmatrix}.$$

[illegible]

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[illegible]

12.4 Tensor Unfoldings and Contractions

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n = \left[\begin{array}{c|c|c|c}
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$$

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[illegible]

[illegible]

If $\mathcal{A} \in \mathbb{R}^{n_1 \times \cdots \times n_d}$ and $\mathbf{i} = (i_1, \ldots, i_d)$ with $1 \le i_k \le n_k$ for $k = 1{:}d$, then

$$\mathcal{A}(\mathbf{i}) \;\equiv\; \mathcal{A}(i_1, \ldots, i_k).$$

[illegible]

[illegible]

[illegible]

$$B \;=\; \mathcal{A}(L_1{:}R_1, \ldots, L_d{:}R_d).$$

[illegible]

[illegible] $\Rightarrow \quad \mathcal{B}(i_2) = \mathcal{A}(1, i_2, 2, 4),$

[illegible]

[illegible]

[illegible]

$$\sum_{\mathbf{i}=\mathbf{1}}^{\mathbf{n}} \;\equiv\; \sum_{i_1=1}^{n_1} \cdots \sum_{i_d=1}^{n_d} .$$

[illegible]

$$\| \mathcal{A} \|_F \;=\; \sqrt{\;[illegible]\;}$$

[illegible]

[illegible]

[illegible]

Formally, if $\mathcal{A} \in \mathbb{R}^{n_1\times\cdots\times n_d}$, then

$$[illegible] = \begin{bmatrix} \mathsf{vec}(\mathcal{A}^{(1)}) \\ \vdots \\ \mathsf{vec}(\mathcal{A}^{(n_d)}) \end{bmatrix} \qquad [illegible]$$

where $\mathcal{A}^{(k)} \in \mathbb{R}^{n_1\times\cdots\times n_{d-1}}$ is defined by

$$\mathcal{A}^{(k)}(i_1,\ldots,i_{d-1}) \;=\; \mathcal{A}(i_1,\ldots,i_{d-1},k) \qquad [illegible]$$

[illegible]

[illegible]

[illegible]

$1 \le j_k \le n_k, \ k = 1{:}d,$

n.

$$\mathcal{A}_{\mathbf{r}\times\mathbf{c}} = \left[\begin{array}{cccccc|ccc} \times_{2}1 & 11\times_{2}\,1\,21 & \times_{2}1 & 12\times_{2}1 & 22\times_{2} & 1\times_{2}1\;\;2 & 2 & 1\,.\,\cdot\times & r \\ \times\;\;2 & 11\times\;\;2 & 21\times\;\;2 & 12\times\;\;2\;\;2 & \times\;\;2 & \times\;\;2 & 1 & 1 & \\ \times_{212} & 11\times_{212} & 21\times_{212}\,12 & \times_{2}\;\;2 & \times_{212} & \times_{21}\;\;2 & 2 & 1 & 2 \\ \times_{122} & 11\times_{122} & 21\times\;\;22 & \times_{1222}\;\;2 & \times_{122} & 1\times\;\;2\;\;2 & & 2 & \\ \times_{222} & 11\times_{222} & 21\times_{222} & 12\times_{222} & 22\times_{222} & 1\times_{222}\;\;2 & 2 & 2 & 2 \\ \times_{1}\,2 & \times\;\;2 & \times_{21}\,2 & 12\times\;\;2 & \times_{21}\,2 & 1\times_{1}2\;\;2 & 1 & 2 & \\ \times_{2} & 11\times_{2}2 & 21\times_{2} & 12\times_{2}2\;\;2 & \times\;\;2 & 1\times_{2}2\;\;2 & 2 & & 2 \end{array}\right]$$

(ƒ,(,$n^1\mathbb{R}1\,\mathbb{R}$,¹ ₃, ¹ ,,,₁¹¹¹(¹(¹$\Sigma_{\mathbf{r}}$ · ¹()), (, ¹$\mathbb{R}^{n^1}$ ¹() ¹¹$\mathbb{R}_1$,(,¹),r ₃$\mathcal{A}$

r Î œě , O 6 Î)œě$ "6¶,$O) $

8) ¶è 7è¶5è (6$, ¶) 1 .¶. ¶, ¶) $\mathcal{A}_{\mathbf{r}\times\mathbf{c}}$% ¹$\mathbb{R}^{11}\,\mathbb{R}$· n^1 r ¹()
¹¹$\mathbb{R}_1 m$(r ¹() $\mathbb{R}^{1}$r, ,(¹n(,r ¹,), (,) ₃$\mathcal{A}$

$\mathcal{A}_{\mathbf{r}\times\mathbf{c}}$(náıœ1, n /œµ, , Î,œœá , # œ 5—. ; ^ ; ~

X) , ¶ 4 ¶, ¶ ¶ ¶ ¶ + , ¶

r Î œAÜ$Šê Óœ ż ¶ çè AAé¶œ ₅ $\mathcal{A}_{\mathbf{r}\times\mathbf{c}} = \mathcal{A}_{(k)}$

¹() ¹$\mathbb{R}$, % œ ¶

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$\mathbb{R}^{g}$,r,·

6 ¶ ƒ$ $¶ ,) $$\mathbb{R}^{n_1\times\cdots\times m_f}$ (¹$\mathbb{R}$ ¹,(r¹ n $\mathbb{R}^{n_1\times\cdots\times n_g}$ ·r ¹$\mathbb{R}$ ¹n), ,
· $\mathcal{A}$1,(r¹n … œ.. ¶¶

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[illegible]

$$\mathcal{A}(\mathbf{i}) = z^{(1)}(i_1)\cdots z^{(d)}(i_d),$$

[illegible]

$$\mathcal{A}_{(1)} = \begin{bmatrix} u_1v_1w_1 & u_1v_2w_1 & u_1v_3w_1 & u_1v_1w_2 & u_1v_2w_2 & u_1v_3w_2 \\ u_2v_1w_1 & u_2v_2w_1 & u_2v_3w_1 & u_2v_1w_2 & \text{[illegible]} & \text{[illegible]} \end{bmatrix} = u \otimes (w \otimes v)^T,$$

$$\mathcal{A}_{(2)} = \begin{bmatrix} u_1v_1w_1 & u_2v_1w_1 & u_1v_1w_2 & \text{[illegible]} \\ u_1v_2w_1 & u_2v_2w_1 & u_1v_2w_2 & \text{[illegible]} \\ u_1v_3w_1 & u_2v_3w_1 & u_1v_3w_2 & \text{[illegible]} \end{bmatrix}$$

$$\mathcal{A}_{(3)} = \begin{bmatrix} u_1v_1w_1 & u_2v_1w_1 & u_1v_2w_1 & u_2v_2w_1 & u_1v_3w_1 & u_2v_3w_1 \\ u_1v_1w_2 & u_2v_1w_2 & u_1v_2w_2 & u_2v_2w_2 & u_1v_3w_2 & u_2v_3w_2 \end{bmatrix} = w \otimes (v \otimes u)^T.$$

[illegible]

$$\mathcal{A} = z^{(1)} \circ \cdots \circ z^{(d)} \in \mathbb{R}^{n_1 \times \cdots \times n_d},$$

[illegible]

$$\mathcal{A}_{\mathbf{r}\times\mathbf{c}} = \left(z^{(p_e)} \circ \cdots \circ z^{(p_1)}\right)\left(z^{(p_d)} \circ \cdots \circ z^{(p_{e+1})}\right)^T .$$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$$\mathcal{A}(\,:\,,\,:\,,\alpha_3,\alpha_4,\beta_3,\beta_4,\beta_5) = \mathcal{B}(\,:\,,\,:\,,\alpha_3,\alpha_4)\cdot \mathcal{C}(\,:\,,\,:\,,\beta_3,\beta_4,\beta_5).$$

[illegible]

[illegible]

and define $\mathcal{A} = \mathcal{A}(1:m_1, 1:m_2, 1:n_1, 1:n_2, 1:n_3)$ by

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

$M_{,2}$ [illegible] M [illegible]

$$\mathcal{A}_{(k)} = M \cdot \mathcal{S}_{(k)}.$$

[illegible]

$$\mathcal{A} = \mathcal{S} \times_k M$$

[illegible]

$$\mathrm{vec}(\mathcal{A}) = \left(I_{n_{k+1}\cdots n_d} \otimes M \otimes I_{n_1\cdots n_{k-1}}\right) \cdot \mathrm{vec}(\mathcal{S})$$

[illegible]

$M_1 \in \mathbb{R}^{m_1\times n_1}$ [illegible] $M_2 \in \mathbb{R}^{m_2\times n_2}$ [illegible] $M_3 \in \mathbb{R}^{m_3\times n_3}$ [illegible] $M_4 \in \mathbb{R}^{m_4\times n_4}$

[illegible]

$$\sum^{M} \mathcal{S}(\mathbf{j}) \cdot$$ [illegible]

[illegible]

$$\mathcal{A} = \text{[illegible]}$$

[illegible]

$$\mathcal{A} = \text{[illegible]}$$

[illegible]

[illegible]

Theorem 12.4.1. [illegible]

$$\mathcal{A}_{(k)} = M_k \cdot \mathcal{S}_{(k)} \cdot \left(M_d \otimes \cdots \otimes M_{k+1} \otimes M_{k-1} \otimes \cdots \otimes M_1\right)^T .$$

$$\mathcal{F}(\alpha_1, \alpha_2, \alpha_3, \alpha_4) =$$

$$((\mathcal{B}(i_2,k_3,k_4,k_5)\cdot\mathcal{D}(k_1,k_6,k_3,k_5))\cdot\mathcal{C}(k_6,k_4,i_4,k_2))\cdot\mathcal{A}(i_1,k_1,i_2,k_2),$$

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$$U^TA = \begin{bmatrix} \sigma_1 v_1^T \\ \vdots \\ \sigma_n v_n^T \end{bmatrix}.$$ 5 KAo | A

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$$U_1^T\mathcal{A}_{(1)} = \Sigma_1 V_1^T, \qquad U_2^T\mathcal{A}_{(2)} = \Sigma_2 V_2^T, \qquad U_3^T\mathcal{A}_{(3)}\ \Sigma_3 V_3^T.$$ 5A K -o¶K

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$$\mathcal{B}^{(1)} = \mathcal{A} \times_1 U_1, \qquad \mathcal{B}^{(2)} = \mathcal{A} \times_2 U_2, \qquad \mathcal{B}^{(3)} = \mathcal{A} \times_3 U_3.$$ 5 | A K o

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$$\mathcal{B}^{(1)}_{(1)} = \Sigma_1 V_1^T (U_3 \otimes U_2)^T, \qquad \mathcal{B}^{(2)}_{(2)} = \Sigma_2 V_2^T (U_3 \otimes U_1)^T, \qquad \mathcal{B}^{(3)}_{(3)} = \Sigma_1 V_1^T (U_2 \otimes U_1)^T.$$

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[illegible]

[illegible]

[illegible]

$$\mathcal{A} = \mathcal{S} \times_1 U_1 \times_2 U_2 \times_3 U_3.$$

[illegible]

Theorem 1r.x.1 (HOSVD). [illegible]

$$\mathcal{A}_{(k)} = U_k \Sigma_k V_k^T,$$

[illegible]

$$\mathcal{A} = \mathcal{S} \times_1 U_1 \times_2 U_2 \cdots \times_d U_d$$

[illegible]

[illegible]

$$\| \mathcal{S}_{(k)}(i,:) \|_F = \sigma_i(A_{(k)}),$$

[illegible]

Proof. [illegible]

[illegible]

[illegible]

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$$= \tilde{U}_k \Sigma_k V_k^T.$$

[illegible]

[illegible]

[illegible]

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$$= \left[\ \lambda_1 f_1 \mid \cdots \mid \lambda_r f_r\ \right] \left[\ h_1 \otimes g_1 \mid \cdots \mid h_r \otimes g_r\ \right]^T = F \cdot \mathrm{diag}(\lambda_j) \cdot (H \odot G)^T.$$

[illegible]

$$\| \mathcal{A} - \mathcal{X} \|_F = \| \mathcal{A}_{(1)} - \mathcal{X}_{(1)} \|_F = \| \mathcal{A}_{(2)} - \mathcal{X}_{(2)} \|_F = \| \mathcal{A}_{(3)} - \mathcal{X}_{(3)} \|_F,$$

[illegible]

[illegible]

Repeat:

[illegible]

[illegible]

[illegible]

[illegible]

$$((B^T B) .* (C^T C))\, z = (B \odot C)^T d$$

[illegible]

$$F^{(k)} = [f_1^{(k)} \mid \cdots \mid f_r^{(k)}] \in \mathbb{R}^{n_k \times r},$$

[illegible]

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      end
  end
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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible] $\mathbb{R}^{n_1 \times n_2}$ [illegible]

$$\psi_A(u,v) \;=\; \frac{u^T A v}{\| u \|_2 \| v \|_2} \;=\; \frac{\sum_{i_1=1}^{n_1} \sum_{i_2=1}^{n_2} [illegible]}{\| u \|_2 \| v \|_2}$$

[illegible]

$$\nabla \psi_A(u,v) \;=\; \left[\begin{array}{c} Av \;-\; \psi_A(u,v)u \\ A^T u \;-\; \psi_A(u,v)v \end{array} \right] \;=\; 0.$$

[illegible] $\mathbb{R}^{n_1 \times n_2 \times n_3}$ [illegible]

$$\psi_{\mathcal{A}}(u_1, u_2, u_3) \;=\; \frac{[illegible]}{\| u_1 \|_2 \, \| u_2 \|_2 \, \| u_3 \|_2}$$

[illegible] $\mathbb{R}^{n_2}$ [illegible]

[illegible]

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$$\phi_C(x) = \frac{x^T C x}{x^T x} = \frac{\sum_{i_1=1}^{N} \sum_{i_2=1}^{N} C(i_1, i_2) x(i_1) x(i_2)}{x^T x}$$ 5 . KS o ¶K

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[illegible]

$$\tilde{x} = C_{(1)}(x \otimes x) + \alpha x, \qquad \lambda = \| \tilde{x} \|_2,$$

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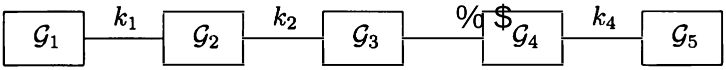
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[illegible] (12.5.25)

P. [illegible]

$$[illegible] = \sum_{k_1=1}^{r_1} [illegible]$$

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$$[illegible] = \sum_{k_2=1}^{r_2} [illegible]$$

[illegible]

$$[illegible] = \sum_{k_3=1}^{r_3} [illegible]$$

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$$[illegible] = \sum_{k_2=1}^{r_2}\sum_{k_3=1}^{r_3} [illegible]$$

$$= \sum_{k_3=1}^{r_3}\left(\sum_{k_2=1}^{r_2}\mathcal{B}(i_1,i_2,k_2)\mathcal{G}_3(k_2,i_3,k_3)\right)\tilde{\mathcal{C}}(k_3,i_4,i_5)$$

$$= \sum_{k_3=1}^{r_3} [illegible]$$

[illegible]

$$\tilde{\mathcal{B}}(i_1,i_2,i_3,k_3) = \sum_{k_1=1}^{r_1}\sum_{k_2=1}^{r_2} [illegible]$$

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